ThermalSense workbook: the register of every number and coordinate of the project

How to read it: yellow cells are inputs taken from the code or from the scene export; white cells are formulas.

Source: generated by tools/v10/build_cg_workbook.py from the scene export of the program and from its source files

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Primitives: vertex and triangle counts of the 16 mesh generators

Legend: yellow = input from the code; white = formula.

Source: src/primitives/.cpp (Library.h lists the scene's generator calls); exported counts and parameters: ts --export-docs 'primitives' (P01..P16).*

Primitive	Generator call	param a	param b	V formula	V	T formula	T	indices I = 3T	bytes indexed 24V+4I	bytes non-indexed 72T	V export	T export	dV	dT	instances (export)	scene triangles T export x inst
1 cube	createCube()	-	-	6 x 4	24	6 x 2	12	36	720	864	24	12	0	0	24	288
2 cuboid	createCube(), S non-uniform	-	-	6 x 4	24	6 x 2	12	36	720	864	24	12	0	0	2453	29436
3 sphere	createSphere(36, 18)	36	18	(k-1)s + 2	614	2s(k-1)	1224	3672	29424	88128	614	1224	0	0	810	991440
4 hemisphere	createHemisphere(36, 9)	36	9	s(k+1) + 2	362	2sk	648	1944	16464	46656	362	648	0	0	10	6480
5 cylinder	createCylinder(36)	36	-	4s + 2	146	4s	144	432	5232	10368	146	144	0	0	1359	195696
6 frustum	createCylinder(36, 0.03, 0.5, false)	36	-	2s	72	2s	72	216	2592	5184	72	72	0	0	3	216
7 cone	createCylinder(36, 0.5, 0, true)	36	-	3s + 1	109	2s	72	216	3480	5184	109	72	0	0	295	21240
8 torus	createTorus(0.35, 0.15, 36, 16)	36	16	uv	576	2uv	1152	3456	27648	82944	576	1152	0	0	97	111744
9 wedge	createWedge()	-	-	3 x 4 + 2 x 3	18	3 x 2 + 2	8	24	528	576	18	8	0	0	6	48
10 tapered box	createTaperedBox(0.8, 0.8)	-	-	6 x 4	24	6 x 2	12	36	720	864	24	12	0	0	11	132
11 pyramid	createPyramid()	-	-	4 + 4 x 3	16	2 + 4	6	18	456	432	16	6	0	0	2	12
12 capsule	createCapsule(16, 3, 0.25)	16	3	2cs + 2	98	4cs	192	576	4656	13824	98	192	0	0	349	67008
13 annulus	createAnnulus(0.43, 48)	48	-	4n	192	4n	192	576	6912	13824	192	192	0	0	2	384
14 helix	createHelix(0.475, 0.025, 5, 120, 6)	120	6	(n+1)m + 2(m+1)	740	2nm + 2m	1452	4356	35184	104544	740	1452	0	0	52	75504
15 grass blade	createGrassBlade()	-	3	2 x rows + 1	7	2(rows-1) + 1	5	15	228	360	7	5	0	0	0	0
16 terrain disc	createTerrainDisc(130, 384, 36)	384	130	1 + ks	49921	s(2k-1) - D	97440	292320	2367384	7015680	49921	97440	0	0	1	97440
Scene total (parts)															5474	1597068

dV = dT = 0: the closed-form counts match the exported meshes. Terrain: D = triangles dropped inside the entrance trench (export params.trench_dropped). Grass blade: 0 scene parts, 450000 GPU instances at EPIC.

Bytes: 24-byte vertex (position + normal, 6 floats), 4-byte indices; non-indexed = 3 vertices per triangle (72 bytes). Exported bytes_indexed / bytes_non_indexed agree.

Parametric equations (local space, unit size; export 'parametric' and 'axis')

Primitive	Point p(parameters)	Placement / axis
1 cube	corners: (0.5,-0.5,0.5) (0.5,-0.5,-0.5) (0.5,0.5,-0.5) (0.5,0.5,0.5) (-0.5,-0.5,-0.5) (-0.5,-0.5,0.5) (-0.5,0.5,0.5) (-0.5,0.5,-0.5)	centred, x, y, z in [-0.5, 0.5]
2 cuboid	corners: (0.5,-0.5,0.5) (0.5,-0.5,-0.5) (0.5,0.5,-0.5) (0.5,0.5,0.5) (-0.5,-0.5,-0.5) (-0.5,-0.5,0.5) (-0.5,0.5,0.5) (-0.5,0.5,-0.5)	the cube mesh, S per part
3 sphere	p = 0.5 (sin(theta) cos(phi), cos(theta), sin(theta) sin(phi)), theta = pi i / k, phi = 2 pi j / s; north half generated, south half by reflection y -> -y	poles on Y
4 hemisphere	p = (0.5 sin(theta) cos(phi), -0.5 + cos(theta), -0.5 sin(theta) sin(phi)), theta = (pi / 2) i / k, phi = 2 pi j / s; base disc y = -0.5	pole +Y, base disc at y = -0.5
5 cylinder	p = (r cos(phi), r sin(phi), +-0.5), phi = 2 pi j / s; caps: centre + own ring (normal +-Z)	axis Z
6 frustum	p = (r(z) cos(phi), r(z) sin(phi), z), r(-0.5) = r_bottom, r(+0.5) = r_top; side normal (cos, sin, r_bottom - r_top)	axis Z, open ends
7 cone	cylinder generator with r_top = 0: apex (0, 0, 0.5), base disc z = -0.5	axis Z, apex +Z
8 torus	p = ((R + r cos v) cos u, (R + r cos v) sin u, r sin v), n = (cos v cos u, cos v sin u, sin v)	axis Z, R + r = 0.5, z in [-r, r]
9 wedge	corners: (-0.5,-0.5,-0.5) (0.5,-0.5,-0.5) (0.5,-0.5,0.5) (-0.5,-0.5,0.5) (-0.5,0.5,-0.5) (0.5,0.5,-0.5)	triangle (y, z) = (-,-) (-,+) (+,-) extruded along X; slope faces +Y +Z
10 tapered box	corners: (-0.5,-0.5,-0.5) (0.5,-0.5,-0.5) (0.5,-0.5,0.5) (-0.5,-0.5,0.5) (-0.4,0.5,-0.4) (-0.4,0.5,0.4) (0.4,0.5,0.4) (0.4,0.5,-0.4)	bottom 1 x 1 at y = -0.5, top at y = +0.5
11 pyramid	corners: (-0.5,-0.5,-0.5) (0.5,-0.5,-0.5) (0.5,-0.5,0.5) (-0.5,-0.5,0.5) (0,0.5,0)	base 1 x 1 at y = -0.5, apex +Y
12 capsule	cylinder r = 0.5, |z| <= 0.5 - h, + half-ellipsoid caps p = (0.5 sin(theta) cos(phi), 0.5 sin(theta) sin(phi), +-(0.5 - h + h cos(theta))), theta = (pi / 2) i / c	axis Z
13 annulus	outer (0.5 cos(phi), 0.5 sin(phi), 0), inner (r_i cos(phi), r_i sin(phi), 0), phi = 2 pi j / n; front face +Z and back face -Z	XY plane (z = 0), normal +-Z
14 helix	c(t) = (R cos(2 pi k t), R sin(2 pi k t), L (t - 0.5)), L = 1 - 2 r; tube c(t) + r (cos(psi) N + sin(psi) B), N = (cos, sin, 0), B = T x N	axis Z
15 grass blade	strip x = +-0.625 at y = t - 0.5 (t = 0, 0.38, 0.72), tip (0, 0.5, 0); bent and placed per instance in the grass vertex shader	Y up, plane z = 0 (1.25 wide)
16 terrain disc	polar grid r = R i / k, phi = 2 pi j / s, y = terrainHeight(x, z), n = terrain normal	world units (radius R), Y up

Decomposition: objects as hierarchies of primitive parts

Legend: yellow = input from the code; white = formula.

Source: ts --export-docs 'components' (S, R_deg, T, parent, colour, specular, shininess) and 'objects'; objects/.cpp Builder. Rows below list every drawn part (leaf) of UAV-1 and of Soldier 1.*

1. Object-level primitive totals (whole scene, from the export)

Object	Cube	Wedge	Tapered box	Pyramid	Sphere	Hemisphere	Capsule	Cylinder	Cylinder (frustum)	Cone	Torus	Annulus	Helix	Terrain disc	parts (sum)	parts export	check	groups
Terrain														1	1	1	OK	1
Backdrop	4				24			9		86					123	123	OK	47
Environment	46				201		57	180		168					652	652	OK	89
Command Bunker	317	1			20	1	9	92							440	440	OK	18
Autonomous UAV	68				26			46	2	8	8				158	158	OK	34
Base	556			2	41		71	271					52		993	993	OK	33
Barracks hall	225				2			50							277	277	OK	5
UAV pads	4				8			4				2			18	18	OK	3
Target Rover	36				72			48							156	156	OK	12
Main Battle Tank	168	1	1		5			61			19				255	255	OK	29
Scout Helicopter	40				21			28	1	1					91	91	OK	19
AA Gatling Gun	19				4	1		54							78	78	OK	5
Air-defence radar	62				3			42		1					108	108	OK	4
SAM launcher	74				5			24		1	8				112	112	OK	9
MLRS rocket truck	45				9			52		6	24				136	136	OK	22
Main Battle Tank 2	168	1	1		5			61			19				255	255	OK	29
Main Battle Tank 3	168	1	1		5			61			19				255	255	OK	29
Ammunition bunker	17	2			10		8	6							43	43	OK	2
Camp camouflage	274				48		204	90		24					640	640	OK	21
Boom barrier	17				2			1							20	20	OK	2
Windsock	1							14							15	15	OK	7
Flag	40				2			5							47	47	OK	14
Soldier 1	16		1		37	1		20							75	75	OK	20
Soldier 2	16		1		37	1		20							75	75	OK	20
Soldier 3	16		1		37	1		20							75	75	OK	20
Soldier 4	16		1		37	1		20							75	75	OK	20
Soldier 5	16		1		37	1		20							75	75	OK	20
Soldier 6	16		1		37	1		20							75	75	OK	20
Soldier 7	16		1		37	1		20							75	75	OK	20
Soldier 8	16		1		37	1		20							75	75	OK	20
Sky Clock					1										1	1	OK	1
Scene total	2477	6	11	2	810	10	349	1359	3	295	97	2	52	1	5474	5474		595
Instances (export)	2477	6	11	2	810	10	349	1359	3	295	97	2	52	1	5474	5474		
check	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK		

3. UAV-1 (root UAV-00 'Autonomous UAV'): 79 drawn parts. M = T Ry Rx Rz S, world = parent_world M (the export's object 'Autonomous UAV' holds 2 UAVs).

ID	Part	Primitive	Parent (group)	Sx	Sy	Sz	Rx (deg)	Ry (deg)	Rz (deg)	Tx (m)	Ty (m)	Tz (m)	R	G	B	ks	shininess n
UAV-01	Fuselage	Cube	Autonomous UAV (UAV-00)	0.1600	0.0700	0.2600	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-02	Canopy	Sphere	Autonomous UAV (UAV-00)	0.1500	0.0800	0.2200	0.0000	0.0000	0.0000	0.0000	0.0400	0.0100	0.6200	0.6400	0.6600	0.8500	96.0000
UAV-03	Battery	Cube	Autonomous UAV (UAV-00)	0.1000	0.0400	0.1600	0.0000	0.0000	0.0000	0.0000	-0.0550	-0.0200	0.1400	0.1500	0.2000	0.4000	48.0000
UAV-04	GPS mast	Cylinder	Autonomous UAV (UAV-00)	0.0080	0.0080	0.0500	90.0000	0.0000	0.0000	0.0000	0.0700	-0.0700	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-05	GPS puck	Cylinder	Autonomous UAV (UAV-00)	0.0500	0.0500	0.0120	90.0000	0.0000	0.0000	0.0000	0.0980	-0.0700	0.6200	0.6400	0.6600	0.8500	96.0000
UAV-06	Arm Front-left	Cube	Autonomous UAV (UAV-00)	0.0250	0.0180	0.2200	0.0000	45.0000	0.0000	0.1061	0.0000	0.1061	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-08	Motor	Cylinder	Motor mount Front-left (UAV-07)	0.0450	0.0450	0.0350	90.0000	0.0000	0.0000	0.0000	0.0200	0.0000	0.4500	0.4600	0.5000	0.9000	128.0000
UAV-10	Prop hub	Sphere	Rotor (UAV-09)	0.0250	0.0250	0.0250	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.4500	0.4600	0.5000	0.9000	128.0000
UAV-11	Prop blade	Cube	Rotor (UAV-09)	0.2000	0.0040	0.0180	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.1000	0.1000	0.1100	0.3000	32.0000
UAV-12	Rotor blur disc	Cylinder	Motor mount Front-left (UAV-07)	0.2050	0.2050	0.0020	90.0000	0.0000	0.0000	0.0000	0.0450	0.0000	0.1000	0.1000	0.1100	0.2000	16.0000
UAV-13	Nav LED	Sphere	Motor mount Front-left (UAV-07)	0.0180	0.0180	0.0180	0.0000	0.0000	0.0000	0.0000	-0.0120	0.0000	1.0000	0.1000	0.0500	0.4000	64.0000
UAV-84	Motor cooling ring	Torus	Motor mount Front-left (UAV-07)	0.0500	0.0500	0.0333	90.0000	0.0000	0.0000	0.0000	0.0040	0.0000	0.1200	0.1200	0.1300	0.6000	64.0000
UAV-14	Arm Front-right	Cube	Autonomous UAV (UAV-00)	0.0250	0.0180	0.2200	0.0000	-45.0000	0.0000	-0.1061	0.0000	0.1061	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-16	Motor	Cylinder	Motor mount Front-right (UAV-15)	0.0450	0.0450	0.0350	90.0000	0.0000	0.0000	0.0000	0.0200	0.0000	0.4500	0.4600	0.5000	0.9000	128.0000
UAV-18	Prop hub	Sphere	Rotor (UAV-17)	0.0250	0.0250	0.0250	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.4500	0.4600	0.5000	0.9000	128.0000
UAV-19	Prop blade	Cube	Rotor (UAV-17)	0.2000	0.0040	0.0180	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.1000	0.1000	0.1100	0.3000	32.0000
UAV-20	Rotor blur disc	Cylinder	Motor mount Front-right (UAV-15)	0.2050	0.2050	0.0020	90.0000	0.0000	0.0000	0.0000	0.0450	0.0000	0.1000	0.1000	0.1100	0.2000	16.0000
UAV-21	Nav LED	Sphere	Motor mount Front-right (UAV-15)	0.0180	0.0180	0.0180	0.0000	0.0000	0.0000	0.0000	-0.0120	0.0000	0.1000	1.0000	0.2500	0.4000	64.0000
UAV-85	Motor cooling ring	Torus	Motor mount Front-right (UAV-15)	0.0500	0.0500	0.0333	90.0000	0.0000	0.0000	0.0000	0.0040	0.0000	0.1200	0.1200	0.1300	0.6000	64.0000
UAV-22	Arm Rear-left	Cube	Autonomous UAV (UAV-00)	0.0250	0.0180	0.2200	0.0000	135.0000	0.0000	0.1061	0.0000	-0.1061	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-24	Motor	Cylinder	Motor mount Rear-left (UAV-23)	0.0450	0.0450	0.0350	90.0000	0.0000	0.0000	0.0000	0.0200	0.0000	0.4500	0.4600	0.5000	0.9000	128.0000
UAV-26	Prop hub	Sphere	Rotor (UAV-25)	0.0250	0.0250	0.0250	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.4500	0.4600	0.5000	0.9000	128.0000
UAV-27	Prop blade	Cube	Rotor (UAV-25)	0.2000	0.0040	0.0180	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.1000	0.1000	0.1100	0.3000	32.0000
UAV-28	Rotor blur disc	Cylinder	Motor mount Rear-left (UAV-23)	0.2050	0.2050	0.0020	90.0000	0.0000	0.0000	0.0000	0.0450	0.0000	0.1000	0.1000	0.1100	0.2000	16.0000
UAV-86	Motor cooling ring	Torus	Motor mount Rear-left (UAV-23)	0.0500	0.0500	0.0333	90.0000	0.0000	0.0000	0.0000	0.0040	0.0000	0.1200	0.1200	0.1300	0.6000	64.0000
UAV-87	Status LED	Sphere	Motor mount Rear-left (UAV-23)	0.0160	0.0160	0.0160	0.0000	0.0000	0.0000	0.0000	-0.0120	0.0000	1.0000	0.5500	0.0800	0.2950	111.4300
UAV-29	Arm Rear-right	Cube	Autonomous UAV (UAV-00)	0.0250	0.0180	0.2200	0.0000	-135.0000	0.0000	-0.1061	0.0000	-0.1061	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-31	Motor	Cylinder	Motor mount Rear-right (UAV-30)	0.0450	0.0450	0.0350	90.0000	0.0000	0.0000	0.0000	0.0200	0.0000	0.4500	0.4600	0.5000	0.9000	128.0000
UAV-33	Prop hub	Sphere	Rotor (UAV-32)	0.0250	0.0250	0.0250	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.4500	0.4600	0.5000	0.9000	128.0000
UAV-34	Prop blade	Cube	Rotor (UAV-32)	0.2000	0.0040	0.0180	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.1000	0.1000	0.1100	0.3000	32.0000
UAV-35	Rotor blur disc	Cylinder	Motor mount Rear-right (UAV-30)	0.2050	0.2050	0.0020	90.0000	0.0000	0.0000	0.0000	0.0450	0.0000	0.1000	0.1000	0.1100	0.2000	16.0000
UAV-88	Motor cooling ring	Torus	Motor mount Rear-right (UAV-30)	0.0500	0.0500	0.0333	90.0000	0.0000	0.0000	0.0000	0.0040	0.0000	0.1200	0.1200	0.1300	0.6000	64.0000
UAV-89	Status LED	Sphere	Motor mount Rear-right (UAV-30)	0.0160	0.0160	0.0160	0.0000	0.0000	0.0000	0.0000	-0.0120	0.0000	1.0000	0.5500	0.0800	0.2950	111.4300
UAV-36	Strobe	Sphere	Autonomous UAV (UAV-00)	0.0220	0.0160	0.0220	0.0000	0.0000	0.0000	0.0000	0.0200	-0.1350	1.0000	1.0000	1.0000	0.5000	64.0000
UAV-37	Landing strut	Cylinder	Autonomous UAV (UAV-00)	0.0120	0.0120	0.0900	90.0000	0.0000	0.0000	-0.0600	-0.0800	-0.0700	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-38	Landing strut	Cylinder	Autonomous UAV (UAV-00)	0.0120	0.0120	0.0900	90.0000	0.0000	0.0000	-0.0600	-0.0800	0.0700	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-39	Landing skid	Cylinder	Autonomous UAV (UAV-00)	0.0140	0.0140	0.2400	0.0000	0.0000	0.0000	-0.0600	-0.1250	0.0000	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-40	Landing strut	Cylinder	Autonomous UAV (UAV-00)	0.0120	0.0120	0.0900	90.0000	0.0000	0.0000	0.0600	-0.0800	-0.0700	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-41	Landing strut	Cylinder	Autonomous UAV (UAV-00)	0.0120	0.0120	0.0900	90.0000	0.0000	0.0000	0.0600	-0.0800	0.0700	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-42	Landing skid	Cylinder	Autonomous UAV (UAV-00)	0.0140	0.0140	0.2400	0.0000	0.0000	0.0000	0.0600	-0.1250	0.0000	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-44	Gimbal yoke	Cube	Gimbal yaw (UAV-43)	0.0700	0.0300	0.0500	0.0000	0.0000	0.0000	0.0000	-0.0150	0.0000	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-46	Lamp housing	Cylinder	Gimbal pitch (UAV-45)	0.0600	0.0600	0.0800	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.3000	0.3100	0.3300	0.8000	96.0000
UAV-47	Lens	Sphere	Gimbal pitch (UAV-45)	0.0500	0.0500	0.0150	0.0000	0.0000	0.0000	0.0000	0.0000	0.0450	1.0000	0.9700	0.8500	1.0000	128.0000
UAV-48	Thermal camera	Cube	Gimbal pitch (UAV-45)	0.0300	0.0250	0.0350	0.0000	0.0000	0.0000	0.0000	0.0420	0.0000	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-83	Light cone	Cylinder (frustum)	Gimbal pitch (UAV-45)	4.8588	4.8588	9.7437	0.0000	0.0000	0.0000	0.0000	0.0000	4.9169	1.0000	0.9500	0.8000	0.0000	1.0000
UAV-94	Camera window	Sphere	Gimbal pitch (UAV-45)	0.0180	0.0180	0.0060	0.0000	0.0000	0.0000	0.0000	0.0420	0.0180	0.0500	0.0500	0.0600	0.9000	128.0000
UAV-50	Rail	Cube	Launcher rail L (UAV-49)	0.0120	0.0120	0.2200	0.0000	0.0000	0.0000	0.0000	0.0120	0.0000	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-51	Rail strut	Cube	Launcher rail L (UAV-49)	0.0080	0.0200	0.0080	0.0000	0.0000	0.0000	0.0000	0.0250	0.0500	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-53	Round body	Cylinder	Round (UAV-52)	0.0220	0.0220	0.1700	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.3200	0.3400	0.2800	0.5000	64.0000
UAV-54	Round nose	Cone	Round (UAV-52)	0.0220	0.0220	0.0400	0.0000	0.0000	0.0000	0.0000	0.0000	0.1050	0.0500	0.0500	0.0600	0.9000	128.0000
UAV-55	Round fin	Cube	Round (UAV-52)	0.0020	0.0300	0.0300	0.0000	0.0000	45.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-56	Round fin	Cube	Round (UAV-52)	0.0020	0.0300	0.0300	0.0000	0.0000	135.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-57	Round fin	Cube	Round (UAV-52)	0.0020	0.0300	0.0300	0.0000	0.0000	225.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000

UAV-58	Round fin	Cube	Round (UAV-52)	0.0020	0.0300	0.0300	0.0000	0.0000	315.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-60	Round body	Cylinder	Round (UAV-59)	0.0220	0.0220	0.1700	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.3200	0.3400	0.2800	0.5000	64.0000
UAV-61	Round nose	Cone	Round (UAV-59)	0.0220	0.0220	0.0400	0.0000	0.0000	0.0000	0.0000	0.0000	0.1050	0.0500	0.0500	0.0600	0.9000	128.0000
UAV-62	Round fin	Cube	Round (UAV-59)	0.0020	0.0300	0.0300	0.0000	0.0000	45.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-63	Round fin	Cube	Round (UAV-59)	0.0020	0.0300	0.0300	0.0000	0.0000	135.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-64	Round fin	Cube	Round (UAV-59)	0.0020	0.0300	0.0300	0.0000	0.0000	225.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-65	Round fin	Cube	Round (UAV-59)	0.0020	0.0300	0.0300	0.0000	0.0000	315.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-67	Rail	Cube	Launcher rail R (UAV-66)	0.0120	0.0120	0.2200	0.0000	0.0000	0.0000	0.0000	0.0120	0.0000	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-68	Rail strut	Cube	Launcher rail R (UAV-66)	0.0080	0.0200	0.0080	0.0000	0.0000	0.0000	0.0000	0.0250	0.0500	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-70	Round body	Cylinder	Round (UAV-69)	0.0220	0.0220	0.1700	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.3200	0.3400	0.2800	0.5000	64.0000
UAV-71	Round nose	Cone	Round (UAV-69)	0.0220	0.0220	0.0400	0.0000	0.0000	0.0000	0.0000	0.0000	0.1050	0.0500	0.0500	0.0600	0.9000	128.0000
UAV-72	Round fin	Cube	Round (UAV-69)	0.0020	0.0300	0.0300	0.0000	0.0000	45.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-73	Round fin	Cube	Round (UAV-69)	0.0020	0.0300	0.0300	0.0000	0.0000	135.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-74	Round fin	Cube	Round (UAV-69)	0.0020	0.0300	0.0300	0.0000	0.0000	225.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-75	Round fin	Cube	Round (UAV-69)	0.0020	0.0300	0.0300	0.0000	0.0000	315.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-77	Round body	Cylinder	Round (UAV-76)	0.0220	0.0220	0.1700	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.3200	0.3400	0.2800	0.5000	64.0000
UAV-78	Round nose	Cone	Round (UAV-76)	0.0220	0.0220	0.0400	0.0000	0.0000	0.0000	0.0000	0.0000	0.1050	0.0500	0.0500	0.0600	0.9000	128.0000
UAV-79	Round fin	Cube	Round (UAV-76)	0.0020	0.0300	0.0300	0.0000	0.0000	45.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-80	Round fin	Cube	Round (UAV-76)	0.0020	0.0300	0.0300	0.0000	0.0000	135.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-81	Round fin	Cube	Round (UAV-76)	0.0020	0.0300	0.0300	0.0000	0.0000	225.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-82	Round fin	Cube	Round (UAV-76)	0.0020	0.0300	0.0300	0.0000	0.0000	315.0000	0.0000	0.0000	-0.0700	0.2000	0.2100	0.2000	0.5000	64.0000
UAV-90	Telemetry antenna	Cylinder	Autonomous UAV (UAV-00)	0.0050	0.0050	0.0900	-35.0000	0.0000	0.0000	-0.0500	-0.0400	-0.1300	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-91	Telemetry antenna	Cylinder	Autonomous UAV (UAV-00)	0.0050	0.0050	0.0900	-35.0000	0.0000	0.0000	0.0500	-0.0400	-0.1300	0.0800	0.0800	0.0900	0.6000	64.0000
UAV-92	Battery strap	Cube	Autonomous UAV (UAV-00)	0.1060	0.0440	0.0120	0.0000	0.0000	0.0000	0.0000	-0.0550	-0.0700	0.5500	0.2200	0.0500	0.1000	3.0314
UAV-93	Battery strap	Cube	Autonomous UAV (UAV-00)	0.1060	0.0440	0.0120	0.0000	0.0000	0.0000	0.0000	-0.0550	0.0300	0.5500	0.2200	0.0500	0.1000	3.0314
UAV-95	Arming LED	Sphere	Autonomous UAV (UAV-00)	0.0120	0.0080	0.0120	0.0000	0.0000	0.0000	0.0400	0.0750	-0.0300	0.2000	0.6000	1.0000	0.2950	111.4300

4. Soldier 1 (root SLD1-00): 75 drawn parts (joint angles live in the parent groups).

ID	Part	Primitive	Parent (group)	Sx	Sy	Sz	Rx (deg)	Ry (deg)	Rz (deg)	Tx (m)	Ty (m)	Tz (m)	R	G	B	ks	shininess n
SLD1-02	Hips	Sphere	Pelvis (SLD1-01)	0.3778	0.2200	0.2400	0.0000	0.0000	0.0000	0.0000	0.0400	0.0000	0.2703	0.2726	0.1940	0.0520	2.0000
SLD1-03	Belt	Cylinder	Pelvis (SLD1-01)	0.3883	0.2500	0.0700	90.0000	0.0000	0.0000	0.0000	0.1200	0.0000	0.2700	0.2350	0.1550	0.1240	4.7240
SLD1-05	Hip joint	Sphere	Hip L (SLD1-04)	0.1500	0.1500	0.1500	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.2703	0.2726	0.1940	0.0520	2.0000
SLD1-06	Thigh	Cylinder	Hip L (SLD1-04)	0.1500	0.1500	0.4600	90.0000	0.0000	0.0000	0.0000	-0.2300	0.0000	0.2703	0.2726	0.1940	0.0520	2.0000
SLD1-08	Knee joint	Sphere	Knee L (SLD1-07)	0.1500	0.1500	0.1500	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.2703	0.2726	0.1940	0.0520	2.0000
SLD1-09	Shin	Cylinder	Knee L (SLD1-07)	0.1200	0.1200	0.4400	90.0000	0.0000	0.0000	0.0000	-0.2200	0.0000	0.2703	0.2726	0.1940	0.0520	2.0000
SLD1-11	Ankle joint	Sphere	Ankle L (SLD1-10)	0.1200	0.1200	0.1200	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0750	0.0550	0.0400	0.1750	12.1257
SLD1-12	Boot	Sphere	Ankle L (SLD1-10)	0.1300	0.1100	0.3000	0.0000	0.0000	0.0000	0.0000	-0.0250	0.0600	0.0750	0.0550	0.0400	0.1750	12.1257
SLD1-76	Calf	Sphere	Knee L (SLD1-07)	0.1300	0.2200	0.1400	0.0000	0.0000	0.0000	0.0000	-0.1300	-0.0150	0.2703	0.2726	0.1940	0.0520	2.0000
SLD1-77	Knee pad	Sphere	Knee L (SLD1-07)	0.1150	0.1300	0.0600	0.0000	0.0000	0.0000	0.0000	-0.0200	0.0600	0.1000	0.1000	0.0850	0.1000	3.0314
SLD1-75	Thigh muscle	Sphere	Hip L (SLD1-04)	0.1750	0.3100	0.1800	0.0000	0.0000	0.0000	0.0000	-0.1500	0.0100	0.2703	0.2726	0.1940	0.0520	2.0000
SLD1-14	Hip joint	Sphere	Hip R (SLD1-13)	0.1500	0.1500	0.1500	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.2703	0.2726	0.1940	0.0520	2.0000
SLD1-15	Thigh	Cylinder	Hip R (SLD1-13)	0.1500	0.1500	0.4600	90.0000	0.0000	0.0000	0.0000	-0.2300	0.0000	0.2703	0.2726	0.1940	0.0520	2.0000
SLD1-17	Knee joint	Sphere	Knee R (SLD1-16)	0.1500	0.1500	0.1500	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.2703	0.2726	0.1940	0.0520	2.0000
SLD1-18	Shin	Cylinder	Knee R (SLD1-16)	0.1200	0.1200	0.4400	90.0000	0.0000	0.0000	0.0000	-0.2200	0.0000	0.2703	0.2726	0.1940	0.0520	2.0000
SLD1-20	Ankle joint	Sphere	Ankle R (SLD1-19)	0.1200	0.1200	0.1200	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0750	0.0550	0.0400	0.1750	12.1257
SLD1-21	Boot	Sphere	Ankle R (SLD1-19)	0.1300	0.1100	0.3000	0.0000	0.0000	0.0000	0.0000	-0.0250	0.0600	0.0750	0.0550	0.0400	0.1750	12.1257
SLD1-81	Calf	Sphere	Knee R (SLD1-16)	0.1300	0.2200	0.1400	0.0000	0.0000	0.0000	0.0000	-0.1300	-0.0150	0.2703	0.2726	0.1940	0.0520	2.0000
SLD1-82	Knee pad	Sphere	Knee R (SLD1-16)	0.1150	0.1300	0.0600	0.0000	0.0000	0.0000	0.0000	-0.0200	0.0600	0.1000	0.1000	0.0850	0.1000	3.0314
SLD1-80	Thigh muscle	Sphere	Hip R (SLD1-13)	0.1750	0.3100	0.1800	0.0000	0.0000	0.0000	0.0000	-0.1500	0.0100	0.2703	0.2726	0.1940	0.0520	2.0000
SLD1-23	Abdomen	Sphere	Spine (SLD1-22)	0.3568	0.3000	0.2400	0.0000	0.0000	0.0000	0.0000	0.1200	0.0000	0.3003	0.3029	0.2156	0.0550	2.0000
SLD1-25	Torso	Tapered box	Chest (SLD1-24)	0.3988	0.3300	0.2300	0.0000	0.0000	180.0000	0.0000	0.1000	0.0000	0.3003	0.3029	0.2156	0.0550	2.0000
SLD1-26	Backpack	Cube	Chest (SLD1-24)	0.3200	0.3800	0.1600	0.0000	0.0000	0.0000	0.0000	0.0700	-0.2000	0.2902	0.2824	0.1825	0.0700	2.0000
SLD1-27	Bedroll	Cylinder	Chest (SLD1-24)	0.1100	0.1100	0.3400	0.0000	90.0000	0.0000	0.0000	0.3100	-0.2100	0.2902	0.2824	0.1825	0.0700	2.0000
SLD1-30	Rifle receiver	Cube	Rifle (held) (SLD1-28)	0.0600	0.1100	0.5200	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0700	0.0700	0.0750	0.4350	16.0000
SLD1-31	Rifle barrel	Cylinder	Rifle (held) (SLD1-28)	0.0350	0.0350	0.4000	0.0000	0.0000	0.0000	0.0000	0.0200	0.4500	0.0700	0.0700	0.0750	0.4350	16.0000
SLD1-32	Rifle magazine	Cube	Rifle (held) (SLD1-28)	0.0400	0.1600	0.0700	15.0000	0.0000	0.0000	0.0000	-0.1100	0.1000	0.0700	0.0700	0.0750	0.4350	16.0000
SLD1-89	Pistol grip	Cube	Rifle (held) (SLD1-28)	0.0350	0.1000	0.0450	-20.0000	0.0000	0.0000	0.0000	-0.0900	-0.1300	0.0700	0.0700	0.0750	0.4350	16.0000
SLD1-90	Handguard	Cube	Rifle (held) (SLD1-28)	0.0580	0.0680	0.2400	0.0000	0.0000	0.0000	0.0000	0.0000	0.3400	0.0700	0.0700	0.0750	0.4350	16.0000
SLD1-91	Optic	Cylinder	Rifle (held) (SLD1-28)	0.0420	0.0420	0.1300	0.0000	0.0000	0.0000	0.0000	0.0850	0.0200	0.0700	0.0700	0.0750	0.4350	16.0000
SLD1-33	Rifle receiver	Cube	Rifle (slung) (SLD1-29)	0.0600	0.1100	0.5200	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0700	0.0700	0.0750	0.4350	16.0000
SLD1-34	Rifle barrel	Cylinder	Rifle (slung) (SLD1-29)	0.0350	0.0350	0.4000	0.0000	0.0000	0.0000	0.0000	0.0200	0.4500	0.0700	0.0700	0.0750	0.4350	16.0000
SLD1-35	Rifle magazine	Cube	Rifle (slung) (SLD1-29)	0.0400	0.1600	0.0700	15.0000	0.0000	0.0000	0.0000	-0.1100	0.1000	0.0700	0.0700	0.0750	0.4350	16.0000
SLD1-92	Pistol grip	Cube	Rifle (slung) (SLD1-29)	0.0350	0.1000	0.0450	-20.0000	0.0000	0.0000	0.0000	-0.0900	-0.1300	0.0700	0.0700	0.0750	0.4350	16.0000
SLD1-93	Handguard	Cube	Rifle (slung) (SLD1-29)	0.0580	0.0680	0.2400	0.0000	0.0000	0.0000	0.0000	0.0000	0.3400	0.0700	0.0700	0.0750	0.4350	16.0000
SLD1-94	Optic	Cylinder	Rifle (slung) (SLD1-29)	0.0420	0.0420	0.1300	0.0000	0.0000	0.0000	0.0000	0.0850	0.0200	0.0700	0.0700	0.0750	0.4350	16.0000
SLD1-36	Carried crate	Cube	Chest (SLD1-24)	0.4600	0.3200	0.3600	0.0000	0.0000	0.0000	0.0000	-0.1000	0.4000	0.4200	0.3300	0.2000	0.1150	4.0000
SLD1-38	Neck	Cylinder	Neck (SLD1-37)	0.1250	0.1150	0.1100	90.0000	0.0000	0.0000	0.0000	0.0350	0.0000	0.3760	0.2629	0.1904	0.1750	12.1257
SLD1-40	Head	Sphere	Head (SLD1-39)	0.1800	0.2250	0.2050	0.0000	0.0000	0.0000	0.0000	0.0800	0.0000	0.3760	0.2629	0.1904	0.1750	12.1257
SLD1-41	Helmet	Hemisphere	Head (SLD1-39)	0.2450	0.1250	0.2650	-6.0000	0.0000	0.0000	0.0000	0.1822	-0.0165	0.2455	0.2644	0.1889	0.0550	2.0000
SLD1-42	Helmet brim	Cylinder	Head (SLD1-39)	0.2550	0.2800	0.0220	84.0000	0.0000	0.0000	0.0000	0.1380	-0.0050	0.1719	0.1851	0.1322	0.1980	9.1896
SLD1-61	Jaw	Sphere	Head (SLD1-39)	0.1300	0.0850	0.1150	0.0000	0.0000	0.0000	0.0000	0.0180	0.0370	0.3760	0.2629	0.1904	0.1750	12.1257
SLD1-62	Nose	Sphere	Head (SLD1-39)	0.0320	0.0500	0.0450	-15.0000	0.0000	0.0000	0.0000	0.0620	0.0980	0.3760	0.2629	0.1904	0.1750	12.1257
SLD1-63	Eye	Sphere	Head (SLD1-39)	0.0240	0.0160	0.0120	0.0000	0.0000	0.0000	0.0330	0.0930	0.0940	0.0350	0.0300	0.0250	0.2800	84.4485
SLD1-64	Ear	Sphere	Head (SLD1-39)	0.0220	0.0580	0.0420	0.0000	0.0000	0.0000	0.0890	0.0750	-0.0050	0.3760	0.2629	0.1904	0.1750	12.1257
SLD1-65	Chin strap	Cylinder	Head (SLD1-39)	0.0120	0.0120	0.1695	58.8041	-37.1309	0.0000	0.0615	0.0525	0.0250	0.0900	0.0850	0.0600	0.0850	2.2974
SLD1-66	Eye	Sphere	Head (SLD1-39)	0.0240	0.0160	0.0120	0.0000	0.0000	0.0000	-0.0330	0.0930	0.0940	0.0350	0.0300	0.0250	0.2800	84.4485
SLD1-67	Ear	Sphere	Head (SLD1-39)	0.0220	0.0580	0.0420	0.0000	0.0000	0.0000	-0.0890	0.0750	-0.0050	0.3760	0.2629	0.1904	0.1750	12.1257
SLD1-68	Chin strap	Cylinder	Head (SLD1-39)	0.0120	0.0120	0.1695	58.8041	37.1309	0.0000	-0.0615	0.0525	0.0250	0.0900	0.0850	0.0600	0.0850	2.2974
SLD1-69	Goggles	Cube	Head (SLD1-39)	0.1500	0.0320	0.0300	-35.0000	0.0000	0.0000	0.0000	0.2150	0.0920	0.0400	0.0500	0.0600	0.3465	194.0120
SLD1-70	NVG mount	Cube	Head (SLD1-39)	0.0400	0.0350	0.0250	-10.0000	0.0000	0.0000	0.0000	0.1630	0.1280	0.0700	0.0700	0.0750	0.4350	16.0000
SLD1-71	Trapezius	Sphere	Neck (SLD1-37)	0.2000	0.0850	0.1500	0.0000	0.0000	-20.0000	0.0850	-0.0200	-0.0200	0.3003	0.3029	0.2156	0.0550	2.0000
SLD1-73	Trapezius	Sphere	Neck (SLD1-37)	0.2000	0.0850	0.1500	0.0000	0.0000	20.0000	-0.0850	-0.0200	-0.0200	0.3003	0.3029	0.2156	0.0550	2.0000

SLD1-44	Shoulder joint	Sphere	Shoulder L (SLD1-43)	0.1300	0.1300	0.1300	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.3183	0.3211	0.2285	0.0550	2.0000
SLD1-45	Upper arm	Cylinder	Shoulder L (SLD1-43)	0.1300	0.1300	0.3000	90.0000	0.0000	0.0000	0.0000	-0.1500	0.0000	0.3183	0.3211	0.2285	0.0550	2.0000
SLD1-47	Elbow joint	Sphere	Elbow L (SLD1-46)	0.1300	0.1300	0.1300	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.3183	0.3211	0.2285	0.0550	2.0000
SLD1-48	Forearm	Cylinder	Elbow L (SLD1-46)	0.1100	0.1100	0.2600	90.0000	0.0000	0.0000	0.0000	-0.1300	0.0000	0.3183	0.3211	0.2285	0.0550	2.0000
SLD1-50	Hand	Sphere	Wrist L (SLD1-49)	0.0900	0.1200	0.0800	0.0000	0.0000	0.0000	0.0000	-0.0500	0.0000	0.1100	0.1000	0.0850	0.1000	3.0314
SLD1-78	Forearm muscle	Sphere	Elbow L (SLD1-46)	0.1200	0.1800	0.1200	0.0000	0.0000	0.0000	0.0000	-0.0700	0.0000	0.3183	0.3211	0.2285	0.0550	2.0000
SLD1-79	Elbow pad	Sphere	Elbow L (SLD1-46)	0.1000	0.1100	0.0550	0.0000	0.0000	0.0000	0.0000	-0.0100	-0.0550	0.1000	0.1000	0.0850	0.1000	3.0314
SLD1-72	Deltoid	Sphere	Shoulder L (SLD1-43)	0.1550	0.1700	0.1500	0.0000	0.0000	0.0000	0.0100	-0.0350	0.0000	0.3183	0.3211	0.2285	0.0550	2.0000
SLD1-52	Shoulder joint	Sphere	Shoulder R (SLD1-51)	0.1300	0.1300	0.1300	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.3183	0.3211	0.2285	0.0550	2.0000
SLD1-53	Upper arm	Cylinder	Shoulder R (SLD1-51)	0.1300	0.1300	0.3000	90.0000	0.0000	0.0000	0.0000	-0.1500	0.0000	0.3183	0.3211	0.2285	0.0550	2.0000
SLD1-55	Elbow joint	Sphere	Elbow R (SLD1-54)	0.1300	0.1300	0.1300	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.3183	0.3211	0.2285	0.0550	2.0000
SLD1-56	Forearm	Cylinder	Elbow R (SLD1-54)	0.1100	0.1100	0.2600	90.0000	0.0000	0.0000	0.0000	-0.1300	0.0000	0.3183	0.3211	0.2285	0.0550	2.0000
SLD1-58	Hand	Sphere	Wrist R (SLD1-57)	0.0900	0.1200	0.0800	0.0000	0.0000	0.0000	0.0000	-0.0500	0.0000	0.1100	0.1000	0.0850	0.1000	3.0314
SLD1-59	Wrench	Cylinder	Wrist R (SLD1-57)	0.0300	0.0300	0.2600	0.0000	0.0000	0.0000	0.0000	-0.0600	0.1000	0.6200	0.6300	0.6600	0.5110	48.5029
SLD1-83	Forearm muscle	Sphere	Elbow R (SLD1-54)	0.1200	0.1800	0.1200	0.0000	0.0000	0.0000	0.0000	-0.0700	0.0000	0.3183	0.3211	0.2285	0.0550	2.0000
SLD1-84	Elbow pad	Sphere	Elbow R (SLD1-54)	0.1000	0.1100	0.0550	0.0000	0.0000	0.0000	0.0000	-0.0100	-0.0550	0.1000	0.1000	0.0850	0.1000	3.0314
SLD1-74	Deltoid	Sphere	Shoulder R (SLD1-51)	0.1550	0.1700	0.1500	0.0000	0.0000	0.0000	-0.0100	-0.0350	0.0000	0.3183	0.3211	0.2285	0.0550	2.0000
SLD1-60	Plate carrier	Cube	Chest (SLD1-24)	0.3463	0.2700	0.2950	0.0000	0.0000	0.0000	0.0000	0.0700	0.0050	0.2700	0.2350	0.1550	0.1240	4.7240
SLD1-85	Magazine pouches	Cube	Chest (SLD1-24)	0.2500	0.1100	0.0600	0.0000	0.0000	0.0000	0.0000	0.0000	0.1750	0.2700	0.2350	0.1550	0.1240	4.7240
SLD1-86	Admin pouch	Cube	Chest (SLD1-24)	0.1300	0.0700	0.0350	0.0000	0.0000	0.0000	0.0000	0.1150	0.1660	0.2295	0.1998	0.1318	0.1240	4.7240
SLD1-87	Radio	Cube	Chest (SLD1-24)	0.0500	0.1500	0.0800	0.0000	0.0000	0.0000	0.1900	0.0400	-0.0500	0.1000	0.1100	0.0800	0.1710	5.2780
SLD1-88	Radio antenna	Cylinder	Chest (SLD1-24)	0.0080	0.0080	0.3693	-81.2925	153.4349	0.0000	0.2025	0.2975	-0.0950	0.0700	0.0700	0.0750	0.4350	16.0000

animated nodes
0
3
2
18
26
0
1
0
48
118
17
26
4
10
28
118
118
0
0
2
6
13
15
15
15
15
15
15
15
15
0
678

2. Parts per primitive (COUNTIFS over the part rows)

Primitive	UAV-1 COUNTIFS	UAV-1 export	check	Soldier 1 COUNTIFS	Soldier 1 export	check
Cube	34	34	OK	16	16	OK
Wedge	0	0	OK	0	0	OK
Tapered bo	0	0	OK	1	1	OK
Pyramid	0	0	OK	0	0	OK
Sphere	13	13	OK	37	37	OK
Hemispher	0	0	OK	1	1	OK
Capsule	0	0	OK	0	0	OK
Cylinder	23	23	OK	20	20	OK
Cylinder (f	1	1	OK	0	0	OK
Cone	4	4	OK	0	0	OK
Torus	4	4	OK	0	0	OK
Annulus	0	0	OK	0	0	OK
Helix	0	0	OK	0	0	OK
Total parts	79	79	OK	75	75	OK

UAV-1 export = controllable 'UAV-1 DRONE' (root UAV-00);
the object 'Autonomous UAV' below counts both UAVs.

Designs: every object design, its instances and its primitives

Legend: yellow = input from the code; white = formula.

Source: ts --export-docs 'designs' (src/report/Designs.cpp) and src/report/PrimitiveColors.h. Section 19 of the report draws one card per design.

1. Designs: parts by primitive over all instances (yellow: export), totals and checks (formulas)

Design	Family	x n	parts / inst.	parts total	n x parts	Cube	Cuboid	Wedge	Tapered box	Pyramid	Sphere	Hemisphere	Capsule	Cylinder	Cylinder (frustum)	Cone	Torus	Annulus	Helix	Terrain disc	sum	check
Soldier	people	8	75	600	600		128		8		296	8		160							600	OK
Main battle tank	vehicles and thei	3	255	765	765		504	3	3		15			183			57				765	OK
MLRS rocket truck	vehicles and thei	1	106	106	106	1	32				3			46			24				106	OK
Scout helicopter	vehicles and thei	1	91	91	91		40				21			28	1	1					91	OK
Autonomous UAV	vehicles and thei	2	55	110	110		36				26			38	2		8				110	OK
Cargo truck	vehicles and thei	2	31	62	62		30				4			28							62	OK
Target rover	vehicles and thei	6	26	156	156		36				72			48							156	OK
SAM launcher	vehicles and thei	1	107	107	107	1	71				4			23			8				107	OK
AA Gatling gun	vehicles and thei	1	58	58	58		19				4	1		34							58	OK
SAM missile	vehicles and thei	1	5	5	5		2				1			1		1					5	OK
MLRS rocket	vehicles and thei	6	5	30	30		12				6			6		6					30	OK
UAV round	vehicles and thei	8	6	48	48		32							8		8					48	OK
AA shell casing	vehicles and thei	20	1	20	20									20							20	OK
Command bunker	buildings	1	403	403	403		290	1			20	1	9	82							403	OK
Barracks hall	buildings	1	277	277	277		225				2			50							277	OK
Air-defence radar	buildings	1	108	108	108		62				3			42		1					108	OK
Motor pool	buildings	1	4	4	4		1				1			2							4	OK
Helipad	buildings	1	51	51	51		31				17			3							51	OK
Ammunition bunker	buildings	1	36	36	36		12	2			10		8	4							36	OK
Guard tower	buildings	4	37	148	148		44				8			96							148	OK
Watchtower	buildings	1	26	26	26		5				1			19		1					26	OK
Fuel depot	buildings	1	20	20	20		8				2			10							20	OK
Quonset hut	buildings	3	19	57	57		42				6			9							57	OK
Radio mast	buildings	1	20	20	20		5				2			13							20	OK
Camouflaged tent	buildings	1	14	14	14		3				4			7							14	OK
Container	buildings	2	11	22	22	12	6							4							22	OK
Guard booth	buildings	1	11	11	11		8							3							11	OK
UAV pad	buildings	2	9	18	18		4				8			4				2			18	OK
HESCO perimeter	fortifications	1	244	244	244		244														244	OK
Concertina wire	fortifications	1	102	102	102									50					52		102	OK
Perimeter fence	fortifications	1	86	86	86		1							85							86	OK
Outer gate	fortifications	1	50	50	50		7			2	3		34	4							50	OK
Camouflage net	fortifications	7	37..84	379	379		297							82							379	OK
Earth revetment	fortifications	5	42..56	246	246		42						204								246	OK
Radar surround	fortifications	1	51	51	51		2							49							51	OK
AA gun surround	fortifications	1	41	41	41		3						37	1							41	OK
Sandbag wall	fortifications	3	15..24	57	57								57								57	OK
Boom barrier	fortifications	1	20	20	20		17				2			1							20	OK
HESCO blocks	fortifications	2	2..6	8	8		8														8	OK
Tank trap	fortifications	2	3	6	6		6														6	OK
Jersey barrier	fortifications	6	2	12	12		12														12	OK
Broadleaf tree	vegetation	4	18	72	72						56			16							72	OK
Pine	vegetation	26	7..8	193	193									26		167					193	OK
Cut branches	vegetation	8	12	96	96						48			24		24					96	OK
Bush	vegetation	14	5..7	88	88						88										88	OK
Treeline pine	vegetation	43	2	86	86											86					86	OK
Flag	props	1	47	47	47		40				2			5							47	OK
Windsock	props	1	15	15	15		1							14							15	OK
Supply cache	props	7	4..7	41	41	10	13							18							41	OK

Boulder	props	13	3..4	46	46						46										46	OK
Lamp post	props	2	4	8	8		4				2			2							8	OK
Track sign	props	1	4	4	4		3							1							4	OK
Fire barrel	props	1	2	2	2						1			1							2	OK
Sky clock (sun)	props	1	1	1	1						1										1	OK
Moon	props	1	1	1	1						1										1	OK
Terrain disc	terrain and backd	1	1	1	1															1	1	OK
Far ground	terrain and backd	4	1	4	4		4														4	OK
Distant hills	terrain and backd	1	21	21	21						21										21	OK
Far radio tower	terrain and backd	1	12	12	12						3			9							12	OK
Road	terrain and backd	1	7	7	7		7														7	OK
Tyre tracks	terrain and backd	1	50	50	50		50														50	OK
Tank spot marking	terrain and backd	1	4	4	4		4														4	OK
Scene		247		5474	5474	24	2453	6	11	2	810	10	349	1359	3	295	97	2	52	1	5474	
Scene parts (export leaf_count)				5474	OK																	

2. The report's primitive colour key: sRGB inputs, greyscale lightness L* by formula

Primitive	Family	R	G	B	colour	Y (linear)	L*
Cube	box	0	70	144		0.0639	30.4
Cuboid	box	0	109	178		0.1415	44.4
Wedge	box	0	118	133		0.1465	45.2
Tapered box	box	80	101	147		0.1312	42.9
Pyramid	box	0	88	80		0.0756	33.0
Sphere	round	255	160	50		0.4663	74.0
Hemisphere	round	255	220	69		0.7288	88.4
Capsule	round	255	200	120		0.6392	83.9
Cylinder	cylinder	87	180	93		0.3546	66.1
Cylinder (frustum)	cylinder	148	180	70		0.3938	69.0
Cone	cylinder	51	174	141		0.3290	64.1
Torus	ring	210	71	144		0.2022	52.1
Annulus	ring	130	104	192		0.1845	50.0
Helix	ring	189	107	186		0.2488	57.0
Grass blade	special	149	147	142		0.2921	61.0
Terrain disc	special	175	137	103		0.2798	59.9

Each family has its own L band (box 30-45, ring 50-57, special 60-61, cylinder 64-69, round 74-88), so the families stay apart in greyscale.*

Transform_Trace: one real UAV vertex from local space to the window

Legend: yellow = input from the code; white = formula.

Source: ts --export-docs component UAV-14 'Arm Front-right' (S, R_deg, T, local_matrix, world_matrix, parent UAV-00 world_matrix); objects/Builder M = T Ry Rx Rz S; View_Projection sheet (V, P, viewport).

Part	UAV-14 Arm Front-right (Cube, createCube)

Parent	UAV-00 Autonomous UAV (root, R = (0, 180, 0), T = (0, 7.5, 12.5))

	x	y	z	w
S (scale)	0.0250	0.0180	0.2200	
R_deg (pitch x, yaw y, roll z)	0	-45	0	
T (m)	-0.1061	0	0.1061	
local vertex p (cube corner)	0.5000	0.5000	0.5000	1

createCube() vertex 3 = (0.5, 0.5, 0.5), +X face (mesh_samples in the export)

S = scale(Sx, Sy, Sz)

0.0250	0	0	0
0	0.0180	0	0
0	0	0.2200	0
0	0	0	1

Rx(pitch)

1	0	0	0
0	1.0000	0.0000	0
0	0.0000	1.0000	0
0	0	0	1

Ry(yaw)

0.7071	0	-0.7071	0
0	1	0	0
0.7071	0	0.7071	0
0	0	0	1

Rz(roll)

1.0000	0.0000	0	0
0.0000	1.0000	0	0
0	0	1	0
0	0	0	1

T = translate(Tx, Ty, Tz)

1	0	0	-0.1061
0	1	0	0.0000
0	0	1	0.1061
0	0	0	1

Ry Rx

0.7071	0.0000	-0.7071	0.0000
0.0000	1.0000	0.0000	0.0000
0.7071	0.0000	0.7071	0.0000
0.0000	0.0000	0.0000	1.0000

R = Ry Rx Rz

0.7071	0.0000	-0.7071	0.0000
0.0000	1.0000	0.0000	0.0000
0.7071	0.0000	0.7071	0.0000
0.0000	0.0000	0.0000	1.0000

R S

0.0177	0.0000	-0.1556	0.0000
0.0000	0.0180	0.0000	0.0000
0.0177	0.0000	0.1556	0.0000
0.0000	0.0000	0.0000	1.0000

M = T R S (local matrix)

0.0177	0.0000	-0.1556	-0.1061
0.0000	0.0180	0.0000	0.0000
0.0177	0.0000	0.1556	0.1061
0.0000	0.0000	0.0000	1.0000

M exported (local_matrix)

0.0177	0	-0.1556	-0.1061
0	0.0180	0	0
0.0177	0	0.1556	0.1061
0	0	0	1

\|M - M exported\|

0.0000	0.0000	0.0000	0.0000
0.0000	0.0000	0.0000	0.0000
0.0000	0.0000	0.0000	0.0000
0.0000	0.0000	0.0000	0.0000

Parent world (UAV-00, exported)

-1	0	0.0000	0
0	1	0	7.5000
0.0000	0	-1	12.5000
0	0	0	1

World = Parent M

-0.0177	0.0000	0.1556	0.1061
0.0000	0.0180	0.0000	7.5000
-0.0177	0.0000	-0.1556	12.3939
0.0000	0.0000	0.0000	1.0000

World exported (world_matrix)

-0.0177	0	0.1556	0.1061
0	0.0180	0	7.5000
-0.0177	0	-0.1556	12.3939
0	0	0	1

\|World - World exported\|

0.0000	0.0000	0.0000	0.0000
0.0000	0.0000	0.0000	0.0000
0.0000	0.0000	0.0000	0.0000
0.0000	0.0000	0.0000	0.0000

View V (View_Projection sheet)

0.8480	0.0000	-0.5299	0.0000
-0.2406	0.8910	-0.3850	-0.7128
0.4722	0.4540	0.7556	-34.3632
0.0000	0.0000	0.0000	1.0000

Projection P (View_Projection sheet)

1.2063	0.0000	0.0000	0.0000
0.0000	2.1445	0.0000	0.0000
0.0000	0.0000	-1.0002	-0.1000
0.0000	0.0000	-1.0000	0.0000

Vertex pipeline (column vectors)

stage	x	y	z	w
local p	0.5000	0.5000	0.5000	1.0000
world = World p	0.1750	7.5090	12.3073	1.0000
eye = V world	-6.3735	1.1973	-21.5719	1.0000
clip = P eye	-7.6882	2.5676	21.4755	21.5719
NDC = clip / w	-0.3564	0.1190	0.9955	1.0000
window (x px, y px top-left, depth, -)	617.8569	475.7275	0.9978	-

Checks

check	value
max |M - export|	1.80E-08
max |World - export|	2.09E-07
inside clip volume |x|,|y|,|z| <= w	yes
eye-space distance -z_eye (m)	21.5719

Exported matrices are float32: differences of ~1e-7 are rounding. Window: x = (ndc.x/2+0.5) W, y = (1-(ndc.y/2+0.5)) H, d = ndc.z/2+0.5 (glDepthRange 0..1)

Vertex_Groups: parts built from unit primitives by matrices (report Section 7.1)

Legend: yellow = input from the code; white = formula.

Source: ts --export-docs: BNK-11 'Retaining wall W' (S, R_deg, T, parent world), cube corners (primitives P01), v6.garrison.gait (thigh, shin, cycle), UAV gimbal parts; objects/Builder M = T Ry Rx Rz S; scene/Gait legFK.

A. Retaining wall W (BNK-11): unit cube -> world

	x	y	z
S	0.3500	3.8571	1.2450
R_deg	0	0	0
T	-1.6750	-1.8215	4.2625

M = T Ry Rx Rz S

0.3500	0.0000	0.0000	-1.6750
0.0000	3.8571	0.0000	-1.8215
0.0000	0.0000	1.2450	4.2625
0	0	0	1

W parent (export)

1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

W = W parent M

0.3500	0.0000	0.0000	-1.6750
0.0000	3.8571	0.0000	-1.8215
0.0000	0.0000	1.2450	4.2625
0.0000	0.0000	0.0000	1.0000

W exported

0.3500	0	0	-1.6750
0	3.8571	0	-1.8215
0	0	1.2450	4.2625
0	0	0	1
max |W - W exported|			0.0000

corner	x	y	z	w	world x	world y	world z
v0	0.5	-0.5	0.5	1.0	-1.5000	-3.7500	4.8850
v1	0.5	-0.5	-0.5	1.0	-1.5000	-3.7500	3.6400
v2	0.5	0.5	-0.5	1.0	-1.5000	0.1071	3.6400
v3	0.5	0.5	0.5	1.0	-1.5000	0.1071	4.8850
v4	-0.5	-0.5	-0.5	1.0	-1.8500	-3.7500	3.6400
v5	-0.5	-0.5	0.5	1.0	-1.8500	-3.7500	4.8850
v6	-0.5	0.5	0.5	1.0	-1.8500	0.1071	4.8850
v7	-0.5	0.5	-0.5	1.0	-1.8500	0.1071	3.6400

C. Leg FK: x = L1 sin h + L2 sin(h - k), y = -L1 cos h - L2 cos(h - k)

L1 thigh	0.4600	**L2 shin**	0.4400				
phase	**hip h**	**knee k**	**x FK**	**y FK**	**x export**	**y export**	**|diff|**
0.000	32.8	26.1	0.3000	-0.8238	0.3000	-0.8238	5.3E-07
0.125	30.1	42.2	0.1375	-0.8282	0.1375	-0.8282	3.3E-07
0.250	15.8	35.8	-0.0250	-0.8562	-0.0250	-0.8562	4.8E-07
0.375	1.6	28.6	-0.1875	-0.8518	-0.1875	-0.8518	4.9E-07
0.500	-6.5	36.1	-0.3500	-0.7810	-0.3500	-0.7810	3.8E-07
0.625	-21.1	24.2	-0.4788	-0.7384	-0.4788	-0.7384	1.0E-06
0.750	8.3	59.2	-0.2753	-0.7325	-0.2753	-0.7325	7.6E-07
0.875	41.9	59.9	0.1709	-0.7608	0.1709	-0.7608	2.9E-07
1.000	32.8	26.1	0.3000	-0.8238	0.3000	-0.8238	5.3E-07

D. UAV camera: view axis = R_UAV Ry(yaw) Rx(pitch) (0, 0, 1)

yaw (deg)	0	**pitch (deg)**	50

R_UAV (rotation of UAV-00 world, export)

-1	0	0.0000
0	1	0
0.0000	0	-1

gimbal axis
0.0000
-0.7660
0.6428

world view axis
0.0000
-0.7660
-0.6428

below horizon (deg)
50.0000

Change yaw / pitch (yellow): the axis turns; the pitch alone sets the angle below the horizon (the UAV flies level).

View_Projection: default orbit camera, lookAt basis, perspective and depth

Legend: yellow = input from the code; white = formula.

Source: scene/Camera.h (target, yaw 32, pitch 27, distance 34, fov 50), Camera.cpp compute() (orbit eye), Math3D.cpp lookAt / perspective, near 0.05 far 600, 24-bit depth (FrameBuffers.cpp).

Input / derived	x	y	z
target (m)	0.0000	0.8000	0.0000
up	0.0000	1.0000	0.0000
yaw (deg)	32		
pitch (deg)	27		
distance (m)	34		
fov y (deg)	50		
width (px)	1920		
height (px)	1080		
near (m)	0.0500		
far (m)	600		
near, comparison (m)	0.5000		
aspect = width / height	1.7778		
eye = target + d (cos p sin y, sin p, cos p	16.0535	16.2357	25.6910
eye - target	16.0535	15.4357	25.6910
n = normalize(eye - target)	0.4722	0.4540	0.7556
up x n	0.7556	0.0000	-0.4722
u = normalize(up x n)	0.8480	0.0000	-0.5299
v = n x u	-0.2406	0.8910	-0.3850
code: s = normalize(f x up), f = -n; |s - u|	0.0000		
u . v, u . n, v . n (orthogonal: 0)	0.0000	0.0000	0.0000

Depth mapping (24-bit fixed point: step = 1 / (2^24 - 1))

z_eye (m)	z_ndc (near 0.05)	depth d (near 0.05)	metres per depth step (near 0.05)	z_ndc (near 0.5)	depth d (near 0.5)	metres per depth step (near 0.5)
-0.0500	-1.0000	0.0000	2.98E-09	clipped	clipped	clipped
-0.1000	0.0001	0.5000	1.19E-08	clipped	clipped	clipped
-0.5000	0.8002	0.9001	2.98E-07	-1.0000	0.0000	2.98E-08
-1.0000	0.9002	0.9501	1.19E-06	0.0008	0.5004	1.19E-07
-2.0000	0.9502	0.9751	4.77E-06	0.5013	0.7506	4.76E-07
-5.0000	0.9802	0.9901	2.98E-05	0.8015	0.9008	2.98E-06
-10.0000	0.9902	0.9951	1.19E-04	0.9016	0.9508	1.19E-05
-34.0000	0.9972	0.9986	1.38E-03	0.9722	0.9861	1.38E-04
-50.0000	0.9982	0.9991	2.98E-03	0.9817	0.9908	2.98E-04
-100.0000	0.9992	0.9996	1.19E-02	0.9917	0.9958	1.19E-03
-200.0000	0.9997	0.9998	4.77E-02	0.9967	0.9983	4.76E-03
-400.0000	0.9999	1.0000	1.91E-01	0.9992	0.9996	1.91E-02
-600.0000	1.0000	1.0000	4.29E-01	1.0000	1.0000	4.29E-02

Summary

depth step at 100 m (m)	1.19E-02	1.19E-03
ratio near 0.05 / near 0.5 at 100 m	10.0075	
share of [0, 1] depth used by the first 1 m	0.9501	0.5004
share of depth used beyond 10 m	0.0049	0.0492

d = (F / (F - N)) (1 - N / \|z\|): precision falls with z^2.

Step size dz = step (F - N) z^2 / (F N): ten times the near plane = ten times finer steps.

Two surfaces closer than one step at 100 m share a depth value: z-fighting.

The code keeps near 0.05 m for the first-person views; 24-bit depth.

View V = lookAt(eye, target, up): rows u, v, n; last column

row 0 (u)	0.8480	0.0000	-0.5299	0.0000
row 1 (v)	-0.2406	0.8910	-0.3850	-0.7128
row 2 (n)	0.4722	0.4540	0.7556	-34.3632
row 3	0	0	0	1

t = 1 / tan(fov / 2)	2.1445

Perspective P (Math3D perspective, OpenGL clip z in [-w, w])

row 0	1.2063	0	0	0
row 1	0	2.1445	0	0
row 2	0	0	-1.0002	-0.1000
row 3	0	0	-1	0

Code (lookAt): f = normalize(target - eye), s = normalize(f x up), u' = s x f; rows s, u', -f. Since f = -n: s = u and u' = v, the same matrix.

Clipping: Cohen-Sutherland, Liang-Barsky, Sutherland-Hodgman (textbook worked example)

Legend: yellow = input from the code; white = formula.

Source: Worked textbook example (window and lines chosen for teaching). The code itself clips on the GPU (clip volume -w <= x, y, z <= w) and culls on the CPU with math/Frustum (6 planes, sphere test).

Window	xmin	0	xmax	10	ymin	0	ymax	8

Outcode bits T B R L = (y > ymax)(y < ymin)(x > xmax)(x < xmin); code = 8T + 4B + 2R + L

1. Cohen-Sutherland outcodes

line	x0	y0	x1	y1	T0	B0	R0	L0	code0	T1	B1	R1	L1	code1	bits 0	bits 1	AND of codes	decision
L1	2	2	8	6	0	0	0	0	0	0	0	0	0	0	0000	0000	0	accept (both inside)
L2	-2	-1	-1	9	0	1	0	1	5	1	0	0	1	9	0101	1001	1	reject (same side)
L3	-3	4	6	12	0	0	0	1	1	1	0	0	0	8	0001	1000	0	clip (subdivide)
L4	-2	3	12	5	0	0	0	1	1	0	0	1	0	2	0001	0010	0	clip (subdivide)

2. Liang-Barsky: x = x0 + t dx, y = y0 + t dy; p1 = -dx, q1 = x0 - xmin; p2 = dx, q2 = xmax - x0; p3 = -dy, q3 = y0 - ymin; p4 = dy, q4 = ymax - y0

line	dx	dy	p1	p2	p3	p4	q1	q2	q3	q4	t enter = max(0, q/p, p<0)	t exit = min(1, q/p, p>0)	parallel outside	visible	x0'	y0'	x1'	y1'
L1	6	4	-6	6	-4	4	2	8	2	6	0.0000	1.0000	no	yes	2.0000	2.0000	8.0000	6.0000
L2	1	10	-1	1	-10	10	-2	12	-1	9	2.0000	0.9000	no	no	-	-	-	-
L3	9	8	-9	9	-8	8	-3	13	4	4	0.3333	0.5000	no	yes	0.0000	6.6667	1.5000	8.0000
L4	14	2	-14	14	-2	2	-2	12	3	5	0.1429	0.8571	no	yes	0.0000	3.2857	10.0000	4.7143

3. Sutherland-Hodgman: triangle clipped against the 4 window edges in turn (output list -> next stage input)

Triangle	A	-2	3	B	7	-2	C	12	10

Stage 1: left x >= xmin

edge	Sx	Sy	Px	Py	S in	P in	Ix	Iy	output
1-2	-2.0000	3.0000	7.0000	-2.0000	0	1	0.0000	1.8889	I, P
2-3	7.0000	-2.0000	12.0000	10.0000	1	1	-	-	P
3-1	12.0000	10.0000	-2.0000	3.0000	1	0	0.0000	4.0000	I
out	(0, 1.89) (7, -2) (12, 10) (0, 4)								

4 vertices

Stage 2: right x <= xmax

edge	Sx	Sy	Px	Py	S in	P in	Ix	Iy	output
1-2	0.0000	1.8889	7.0000	-2.0000	1	1	-	-	P
2-3	7.0000	-2.0000	12.0000	10.0000	1	0	10.0000	5.2000	I
3-4	12.0000	10.0000	0.0000	4.0000	0	1	10.0000	9.0000	I, P
4-1	0.0000	4.0000	0.0000	1.8889	1	1	-	-	P
out	(7, -2) (10, 5.2) (10, 9) (0, 4) (0, 1.89)								

5 vertices

Stage 3: bottom y >= ymin

edge	Sx	Sy	Px	Py	S in	P in	Ix	Iy	output
1-2	7.0000	-2.0000	10.0000	5.2000	0	1	7.8333	0.0000	I, P
2-3	10.0000	5.2000	10.0000	9.0000	1	1	-	-	P
3-4	10.0000	9.0000	0.0000	4.0000	1	1	-	-	P
4-5	0.0000	4.0000	0.0000	1.8889	1	1	-	-	P
5-1	0.0000	1.8889	7.0000	-2.0000	1	0	3.4000	0.0000	I
out	(7.83, 0) (10, 5.2) (10, 9) (0, 4) (0, 1.89) (3.4, 0)								

6 vertices

Stage 4: top y <= ymax

edge	Sx	Sy	Px	Py	S in	P in	Ix	Iy	output
1-2	7.8333	0.0000	10.0000	5.2000	1	1	-	-	P
2-3	10.0000	5.2000	10.0000	9.0000	1	0	10.0000	8.0000	I
3-4	10.0000	9.0000	0.0000	4.0000	0	1	8.0000	8.0000	I, P
4-5	0.0000	4.0000	0.0000	1.8889	1	1	-	-	P
5-6	0.0000	1.8889	3.4000	0.0000	1	1	-	-	P
6-1	3.4000	0.0000	7.8333	0.0000	1	1	-	-	P
out	(10, 5.2) (10, 8) (8, 8) (0, 4) (0, 1.89) (3.4, 0) (7.83, 0)								

7 vertices

Each stage keeps the output-list rule: in-in -> P; in-out -> I; out-in -> I, P; out-out -> none. The final list is the clipped polygon (fan-triangulated afterwards).

Illumination: ambient + Lambert diffuse + Blinn-Phong specular at one roof point

Legend: yellow = input from the code; white = formula.

Source: scene/DayCycle (solar model, transmittance, 3.2 smoothstep(-1, 10, e), ambient mixLog); export 'lights'; ShaderSources.cpp brdf(); the command bunker's part 'Roof slab A'.

Quantity	Noon	Sunset	Night (moon)	UAV spot (beam axis)	UAV spot (roof point)
hour (clock)	12.00	17.50	23.00	23.00	23.00
H = 15 (hour - 12) (deg)	0.0000	82.5000	165.0000	165.0000	165.0000
sin e = sin phi sin d + cos phi cos d cos H	0.8878	0.0889	-0.9187	-0.9187	-0.9187
sun elevation e (deg)	62.6000	5.0981	-66.7312	-66.7312	-66.7312
daylight w = smoothstep(-8, 6, e)	1.0000	0.9881	0.0000	0.0000	0.0000
surface point P.x (m)	0.0000	0.0000	0.0000	0.0000	0.0000
surface point P.y (m)	-0.4500	-0.4500	-0.4500	-0.4500	-0.4500
surface point P.z (m)	-1.3000	-1.3000	-1.3000	5.8195	-1.3000
to light (unnormalised) x	0.0000	-0.9883	0.5598	0.0000	0.0000
to light (unnormalised) y	0.8878	0.0889	0.3868	7.8060	7.8060
to light (unnormalised) z	-0.4602	-0.1244	-0.7328	6.5505	13.6700
distance d = |to light| (m)	1.0000	1.0000	1.0000	10.1903	15.7417
L.x (unit)	0.0000	-0.9883	0.5598	0.0000	0.0000
L.y (unit)	0.8878	0.0889	0.3868	0.7660	0.4959
L.z (unit)	-0.4602	-0.1244	-0.7328	0.6428	0.8684
angle to spot axis (deg)	-	-	-	0.4138	20.2741
spot cone c^2, c = clamp((cos a - cos o)/(cos i - co	-	-	-	1.0000	0.0000
attenuation 1 / (1 + kl d + kq d^2)	-	-	-	0.2162	0.1084
air mass m (Kasten-Young)	1.1257	10.1423	-	-	-
light intensity (scalar)	3.2000	1.8600	0.9500	1.0811	0.0000
N . L (clamped at 0)	0.8878	0.0889	0.3868	0.7660	0.4959
L + V (H before normalising) x	0.4514	-0.5368	1.0112	0.5261	0.4514
L + V y	1.3570	0.5581	0.8560	1.3129	0.9651
L + V z	0.2988	0.6346	0.0262	1.2941	1.6274
N . H	0.9288	0.5574	0.6460	0.6848	0.4962
(N . H)^n (0 when N . L = 0)	0.8627	0.3107	0.4173	0.4690	0.2462

Per-channel terms (light colour = transmittance for the sun)

Term (channel)	Noon	Sunset	Night (moon)	UAV spot (beam axis)	UAV spot (roof point)
Il = light colour x intensity R	3.1760	1.0747	0.5225	1.0811	0.0000
Il = light colour x intensity G	3.1560	0.6804	0.5890	1.0271	0.0000
Il = light colour x intensity B	3.1166	0.2727	0.7600	0.8865	0.0000
ambient Ia R	0.3400	0.3330	0.0600	0.0600	0.0600
ambient Ia G	0.3600	0.3527	0.0650	0.0650	0.0650
ambient Ia B	0.4000	0.3927	0.0850	0.0850	0.0850
ambient kd Ia R	0.1020	0.0999	0.0180	0.0180	0.0180
ambient kd Ia G	0.1080	0.1058	0.0195	0.0195	0.0195
ambient kd Ia B	0.1120	0.1100	0.0238	0.0238	0.0238

Input (from the code)	x / R	y / G	z / B
latitude phi, declination delta (deg)	22.8000	-4.6000	
sun intensity, key above (deg)	3.2000	-1.0000	
optical depth tau (R, G, B)	0.0600	0.1100	0.2100
moon direction (travel)	-0.5598	-0.3868	0.7328
moon colour	0.5500	0.6200	0.8000
moon intensity	0.9500		
spot position (m)	0.0000	7.3560	12.3700
spot direction	0.0000	-0.7660	-0.6428
spot colour	1	0.9500	0.8200
spot intensity, inner, outer (deg)	5	9	14
spot attenuation linear, quadratic	0.0500	0.0300	
night ambient (LightRig)	0.0600	0.0650	0.0850
day ambient (kDayAmbient)	0.3400	0.3600	0.4000
roof albedo kd (Roof slab A colour)	0.3000	0.3000	0.2800
roof ks, shininess n	0.0760	2	
roof point P (top face, m)	0	-0.4500	-1.3000
normal N	0	1	0
viewer: orbit eye (View_Projection)	16.0535	16.2357	25.6910

Model (display-referred Blinn-Phong path), per channel:
I = kd Ia + kd Il (N.L) + ks Il (N.H)^n

Not modelled: shadows, AO, relief, HDR sky SH ambient, moon tint.

Spot cone (shader): c = clamp((cos a - cos outer) / (cos inner - cos outer)), weight c^2 (not smoothstep).

Roof point: 20 deg off the spot axis at t = 0, outside the 14 deg cone. Column E: the up-facing point on the beam axis at roof height.

Sun key direction is quantised to 1 deg in the code.

diffuse kd II (N.L) R	0.8459	0.0286	0.0606	0.2484	0.0000
diffuse kd II (N.L) G	0.8406	0.0181	0.0683	0.2360	0.0000
diffuse kd II (N.L) B	0.7748	0.0068	0.0823	0.1901	0.0000
specular ks II (N.H)^n R	0.2082	0.0254	0.0166	0.0385	0.0000
specular ks II (N.H)^n G	0.2069	0.0161	0.0187	0.0366	0.0000
specular ks II (N.H)^n B	0.2043	0.0064	0.0241	0.0316	0.0000
total I R	**1.1561**	**0.1539**	**0.0952**	**0.3050**	**0.0180**
total I G	**1.1555**	**0.1400**	**0.1065**	**0.2921**	**0.0195**
total I B	**1.0911**	**0.1232**	**0.1302**	**0.2455**	**0.0238**
luminance Y of total (Rec. 709)	**1.1510**	**0.1418**	**0.1058**	**0.2915**	**0.0195**

Ambient Ia = night^(1-w) day^w (DayCycle mixLog); spot columns: beams on at night only (sun below -2 deg). Linear HDR units (the sun's irradiance is 3.2).

Color_Interp: barycentric weights, Gouraud vs Phong, perspective-correct interpolation, sRGB blending, sky colour over the day

Legend: yellow = input from the code; white = formula.

Source: math/ExplorerMath edgeFunction / interpolateAffine / interpolatePerspective / srgbEncode (sections A-D: worked example values); section E: DayCycle solar model + export v6.day_cycle.grid (sky_zenith).

A. Screen triangle (worked example) with per-vertex data

vertex	x (px)	y (px)	clip w	tex u	N.x	N.y	N.z	I vertex
V0	100	100	1.0000	0.0000	-0.6012	-0.4008	0.6913	0.4148
V1	400	150	4.0000	1.0000	0.6007	-0.3004	0.7409	0.4445
V2	220	380	10.0000	0.5000	0.0000	0.6000	0.8000	0.4801
light	L = V = (0, 0, 1), H = (0, 0, 1)							
kd, ks, n	0.6000	0.6000	40					
2 x area (edge fn)	78000							

B. Barycentric weights at sample points: w0 = E12(p) / area, w1 = E20(p) / area, w2 = E01(p) / area

sample	px	py	w0	w1	w2	sum	inside
centroid	240	210	0.3333	0.3333	0.3333	1.0000	yes
near V1	360	160	0.0949	0.8410	0.0641	1.0000	yes
edge V0-V2 mid	160	240	0.5000	0.0000	0.5000	1.0000	yes
centre-left	200	180	0.5205	0.2359	0.2436	1.0000	yes
outside	420	330	-0.4744	0.7949	0.6795	1.0000	no

C. Gouraud (interpolate vertex I) vs Phong (interpolate N, normalise, then light)

sample	I Gouraud	N.x	N.y	N.z	|N|	I Phong	Phong - Gouraud
centroid	0.4465	-0.0002	-0.0337	0.7441	0.7448	1.1752	0.7288
near V1	0.4440	0.4482	-0.2522	0.7400	0.9011	0.4929	0.0489
edge V0-V2 mid	0.4474	-0.3006	0.0996	0.7457	0.8101	0.5740	0.1266
centre-left	0.4377	-0.1712	-0.1333	0.7295	0.7611	0.6851	0.2474

The highlight sits inside the triangle (interpolated N close to H): Phong finds it, Gouraud misses it.

D. Texture coordinate u: affine vs perspective-correct (clip w per vertex)

sample	u affine	u persp.	error
centroid	0.5000	0.2222	0.2778
near V1	0.8731	0.6852	0.1879
edge V0-V2 mid	0.2500	0.0455	0.2045
centre-left	0.3577	0.1178	0.2399

Perspective-correct: interpolate u / w and 1 / w linearly in screen space, then divide.

E. Blending two sRGB colours: naive (in sRGB) vs correct (in linear light)

colour	R (0-255)	G	B	R linear	G linear	B linear		
A	255	64	0	1.0000	0.0513	0.0000		
B	0	128	255	0.0000	0.2159	1.0000		
t	naive R	naive G	naive B	linear R	linear G	linear B	Y naive	Y linear
0.00	255.0	64.0	0.0	255.0	64.0	0.0	0.2493	0.2493
0.25	191.3	80.0	63.8	224.6	85.7	137.0	0.1721	0.2436
0.50	127.5	96.0	127.5	187.5	102.3	187.5	0.1446	0.2379
0.75	63.8	112.0	191.3	137.0	116.0	224.6	0.1644	0.2323
1.00	0.0	128.0	255.0	0.0	128.0	255.0	0.2266	0.2266

sRGB decode: c <= 0.04045 ? c/12.92 : ((c+0.055)/1.055)^2.4; encode = srgbEncode (ExplorerMath).

Y = relative luminance (Rec. 709). The naive mid-point is darker than the light-correct one.

F. Sky zenith colour over the day (solar model formulas; sky colour from the export) phi, delta: 22.8000 -4.6000

hour	e formula (deg)	e export (deg)	diff	daylight w	phase (no hysteresis)	zenith R	zenith G	zenith B	Y zenith	phase export
0.00	-71.8000	-71.8000	0.0000	0.0000	NIGHT	0.0300	0.0400	0.0750	0.0404	NIGHT
4.00	-29.3751	-29.3751	0.0000	0.0000	NIGHT	0.0300	0.0400	0.0750	0.0404	NIGHT
5.50	-8.6859	-8.6859	0.0000	0.0000	NIGHT	0.0300	0.0400	0.0750	0.0404	NIGHT
6.00	-1.7809	-1.7810	0.0000	0.4167	DAWN	0.0617	0.0753	0.1058	0.0746	DAWN
6.50	5.0981	5.0981	0.0000	0.9881	DAY	0.1885	0.2126	0.2329	0.2089	DAY
7.00	11.9319	11.9319	0.0000	1.0000	DAY	0.2206	0.2653	0.3436	0.2615	DAY
9.00	38.2196	38.2196	0.0000	1.0000	DAY	0.2852	0.4034	0.6962	0.3994	DAY
12.00	62.6000	62.6000	0.0000	1.0000	DAY	0.3089	0.4463	0.7643	0.4400	DAY
15.00	38.2196	38.2196	0.0000	1.0000	DAY	0.2852	0.4034	0.6962	0.3994	DAY
17.00	11.9319	11.9319	0.0000	1.0000	DAY	0.2206	0.2653	0.3436	0.2615	DAY
17.50	5.0981	5.0981	0.0000	0.9881	DAY	0.1885	0.2126	0.2329	0.2089	DAY
18.00	-1.7809	-1.7810	0.0000	0.4167	DUSK	0.0617	0.0753	0.1058	0.0746	DUSK
18.50	-8.6859	-8.6859	0.0000	0.0000	NIGHT	0.0300	0.0400	0.0750	0.0404	NIGHT
20.00	-29.3751	-29.3751	0.0000	0.0000	NIGHT	0.0300	0.0400	0.0750	0.0404	NIGHT
22.00	-55.7779	-55.7779	0.0000	0.0000	NIGHT	0.0300	0.0400	0.0750	0.0404	NIGHT

Day sky: Preetham analytic sky (math/SkyModel), blended with the night preset in log space (mixLog, weight w).

The export applies 0.25 deg hysteresis to the phase, so DAWN / DUSK can last a little longer.

Ray_Tracing: intersections, reflection, refraction, recursion (worked example)

Legend: yellow = input from the code; white = formula.

Source: Worked textbook example. The code: raytrace/Intersect (sphere / box / heightfield), Optics.h reflectDir, refractDir, fresnelDielectric; RayTracer.h whittedDepth = 5 (';' render modes).

1. Ray-sphere	x	y	z
ray origin o	0	2	-10
direction (unnormalised)	0.1000	-0.0500	1
d = normalize	0.0994	-0.0497	0.9938
sphere centre c	0	1	0
radius r	1.5000		
oc = o - c	0.0000	1.0000	-10.0000
a = d . d	1.0000		
b = 2 d . oc	-19.9755		
c' = oc . oc - r^2	98.7500		
discriminant b^2 - 4 a c'	4.0222		
t1 = (-b - sqrt(disc)) / 2a (near)	8.9850		
t2 = (-b + sqrt(disc)) / 2a (far)	10.9905		
hit p = o + t1 d	0.8929	1.5535	-1.0706
normal n = (p - c) / r	0.5953	0.3690	-0.7138
d . n	-0.6685		
reflection r = d - 2 (d . n) n	0.8953	0.4437	0.0395
check |r| (= 1)	1.0000		

2. Ray-plane (ground y = 0)	x	y	z
ray origin o	0	5	0
direction (unnormalised)	1	-1	0.5000
d = normalize	0.6667	-0.6667	0.3333
plane normal n	0	1	0
point on plane p0	0	0	0
n . d	-0.6667		
t = n . (p0 - o) / (n . d)	7.5000		
hit p = o + t d	5.0000	0.0000	2.5000
reflection r = d - 2 (d . n) n	0.6667	0.6667	0.3333

3. Snell refraction (air -> glass)	x	y	z
incident d (unnormalised)	0.5000	-1	0
d = normalize	0.4472	-0.8944	0.0000
normal n (facing against d)	0	1	0
n1, n2	1.0000	1.5000	
eta = n1 / n2	0.6667		
cos i = -d . n	0.8944		
k = 1 - eta^2 (1 - cos^2 i)	0.9111		
t = eta d + (eta cos i - sqrt k) n	0.2981	-0.9545	0.0000
incidence angle (deg)	26.5651		
refraction angle (deg)	17.3461		
Snell check n1 sin i - n2 sin t	0.0000		
Fresnel R (exact, fresnelDielectric)	0.0409		
Schlick R0 = ((n1 - n2) / (n1 + n2))^2	0.0400		

4. Total internal reflection (glass -> air, eta = 1.5)

incidence angle (deg)	k = 1 - eta^2 sin^2 i	TIR?	Fresnel R
critical angle asin(1/1.5)	41.8103		
20	0.7368	no	0.0417
30	0.4375	no	0.0552
41	0.0316	no	0.3798
42	-0.0074	yes	1.0000
60	-0.6875	yes	1.0000

5. Whitted recursion: weight of bounce k = kr^k

reflectance kr, depth limit (whittedDepth)	0.5000	5	
bounce k	kr^k	cumulative	share of the infinite sum
0	1.0000	1.0000	0.5000
1	0.5000	1.5000	0.7500
2	0.2500	1.7500	0.8750
3	0.1250	1.8750	0.9375
4	0.0625	1.9375	0.9688
5	0.0313	1.9688	0.9844
6	0.0000	1.9688	0.9844

Bounces past the limit (k > 5) contribute 0; the lost share of the infinite sum is kr^6.

Thermal_Kernels: Gaussian and Sobel kernels on a 5x5 thermal patch

Legend: yellow = input from the code; white = formula.

Source: math/ExplorerMath.cpp kernel(): Gaussian 5x5 = binomial [1 4 6 4 1] (x) [1 4 6 4 1] / 256 (sigma ~ 1), Sobel Gx / Gy, convolve() (correlation, kernel row 0 = row above, clamp-to-edge), applyGamma v^(1/g).

sigma	1.0000

size 5 (offsets -2..2)

threshold T on |G|	2.0000

Gaussian exp(-(x^2+y^2)/(2 s^2))

0.0183	0.0821	0.1353	0.0821	0.0183
0.0821	0.3679	0.6065	0.3679	0.0821
0.1353	0.6065	1.0000	0.6065	0.1353
0.0821	0.3679	0.6065	0.3679	0.0821
0.0183	0.0821	0.1353	0.0821	0.0183
sum	1.0000			

normalised (sum = 1)

0.0030	0.0133	0.0219	0.0133	0.0030
0.0133	0.0596	0.0983	0.0596	0.0133
0.0219	0.0983	0.1621	0.0983	0.0219
0.0133	0.0596	0.0983	0.0596	0.0133
0.0030	0.0133	0.0219	0.0133	0.0030
max |normalised - code|				0.0215

code: b_y b_x / 256

0.0039	0.0156	0.0234	0.0156	0.0039
0.0156	0.0625	0.0938	0.0625	0.0156
0.0234	0.0938	0.1406	0.0938	0.0234
0.0156	0.0625	0.0938	0.0625	0.0156
0.0039	0.0156	0.0234	0.0156	0.0039
sum	1.0000			

binomial
1
4
6
4
1

Gamma: out = v^(1/g)

v	g = 0.5	g = 1.0	g = 2.2
g:	0.5	1.0	2.2
0.0	0.0000	0.0000	0.0000
0.1	0.0100	0.1000	0.3511
0.2	0.0400	0.2000	0.4812
0.3	0.0900	0.3000	0.5785
0.4	0.1600	0.4000	0.6594
0.5	0.2500	0.5000	0.7297
0.6	0.3600	0.6000	0.7928
0.7	0.4900	0.7000	0.8503
0.8	0.6400	0.8000	0.9035
0.9	0.8100	0.9000	0.9532
1.0	1.0000	1.0000	1.0000

g > 1 lifts dark tones (applyGamma).

5x5 thermal patch (worked example, 0..1)

0.20	0.22	0.25	0.30	0.32
0.21	0.24	0.60	0.65	0.34
0.22	0.30	0.85	0.90	0.40
0.20	0.28	0.70	0.75	0.36
0.19	0.21	0.26	0.31	0.30

Sobel Gx

-1	0	1
-2	0	2
-1	0	1

Sobel Gy

-1	-2	-1
0	0	0
1	2	1

Gx step: patch x kernel at the centre

-0.2400	0.0000	0.6500
-0.6000	0.0000	1.8000
-0.2800	0.0000	0.7500
sum	2.0800	SUMPRODUCT:

2.0800

Gaussian 5x5 at the centre = SUMPRODUCT(patch, normalised kernel) 0.5545

code kernel: 0.5379

centre input: 0.8500

Gx (3x3 valid output)

1.4600	1.5000	-0.9000
2.1500	2.0800	-1.5000
1.7000	1.6400	-1.0900

Gy (3x3 valid output)

0.7800	1.8800	1.8800
0.1700	0.3400	0.3200
-0.8000	-1.8600	-1.8700

\|G\| = sqrt(Gx^2 + Gy^2)

1.6553	2.4051	2.0843
2.1567	2.1076	1.5338
1.8788	2.4798	2.1645

edge = \|G\| >= T

0	1	1
1	1	0
0	1	1

Valid region: output (i, j) uses patch rows i..i+2, cols j..j+2 (the code clamps at the border).

The Frame Explorer shows Gx, Gy with bias 0.5 (zero = mid-grey); Sobel \|G\| uses both kernels.

Tracking: Kalman predict / gate / update, two-view triangulation, time on target

Legend: yellow = input from the code; white = formula.

Source: vision/MultiTracker.cpp predict, mahalanobis2, kalmanUpdate, triangulate; ai/Squad.cpp roundFlightTime, planStrike. Worked example values in yellow; q, gate, flight model from --export-docs.

A. Predict: x' = F x, P' = F P F^T + Q (white-noise acceleration q)

dt (s)	0.10	q (m/s2)	1.50	gate	9.21	meas. sigma (m)	0.50

x = (x, z, vx, vz)
2.00
5.00
1.20
-0.40

F

1	0	0.1000	0
0	1	0	0.1000
0	0	1	0
0	0	0	1

P (before)

0.3000	0	0.1000	0
0	0.3000	0	0.1000
0.1000	0	0.5000	0
0	0.1000	0	0.5000

Q

0.0001	0	0.0011	0
0	0.0001	0	0.0011
0.0011	0	0.0225	0
0	0.0011	0	0.0225

F P

0.3100	0.0000	0.1500	0.0000
0.0000	0.3100	0.0000	0.1500
0.1000	0.0000	0.5000	0.0000
0.0000	0.1000	0.0000	0.5000

x' = F x

2.1200
4.9600
1.2000
-0.4000

P' = F P F^T + Q

0.3251	0.0000	0.1511	0.0000
0.0000	0.3251	0.0000	0.1511
0.1511	0.0000	0.5225	0.0000
0.0000	0.1511	0.0000	0.5225

B. Gate and update with a measurement m (ground point)

m = (x, z)	2.50	4.60
innovation nu	0.3800	-0.3600

S = H P' H^T + sigma^2 I

0.5751	0.0000
0.0000	0.5751

S^-1

1.7390	0.0000
0.0000	1.7390

K = P' H^T S^-1 (4x2)

0.5653	0.0000
0.0000	0.5653
0.2628	0.0000
0.0000	0.2628

x_new = x' + K nu

2.3348
4.7565
1.2999
-0.4946

d^2 = nu^T S^-1 nu	0.4765	inside gate

D. Time on target (roundFlightTime, planStrike)

v0	accel	ramp (s)	cruise	fuse (m)	
8.00	130.00	0.40	60.00	0.50	
UAV	**distance**	**d - fuse**	**ramp dist**	**T (s)**	**launch +s**
1	18.0	17.5000	13.6000	0.4650	0.0000
2	9.0	8.5000	13.6000	0.3053	0.1597

both rounds arrive together: launch delay = T_far - T_k

C. Two-view triangulation: minimise |o1 + s d1 - (o2 + t d2)|

	x	y	z
o1 (UAV 1)	-6.00	8.00	0.00
d1	0.55	-0.70	0.15
o2 (UAV 2)	7.00	9.00	5.00
d2	-0.50	-0.65	-0.32

a = d1.d1	0.8150	b = d1.d2	0.1320	c = d2.d2	0.7749	d = d1.w	-7.2000	e = d2.w	8.7500
s	10.9657	t	13.1597						

	x	y	z
p1 = o1 + s d1	0.0312	0.3240	1.6449
p2 = o2 + t d2	0.4201	0.4462	0.7889
midpoint	0.2256	0.3851	1.2169
residual |p1-p2|	0.9481		

Line_Follower: IR sensor array, PD steering, differential drive (one tick)

Legend: yellow = input from the code; white = formula.

Source: ai/HideController.cpp followLine (kSensorForward, kSensorSpacing, kLineHalfWidth, kKp, kKd, kMaxSpeed); objects/Rover.h kRoverTrackWidth. Line offset, previous error and dt ar

	spacing (m)	half width	err scale	Kp	Kd	v max	slow-down	omega max	track w	line offset	prev e	dt (s)
inputs	0.060	0.050	0.120	3.200	0.250	0.750	0.600	2.500	0.500	0.070	0.450	0.017

A. Sensors: lateral offset x_i = -2 spacing + i spacing (left > 0); on the line if |x_i - line| < half width

sensor i	x_i (m)	|x_i - line|	on line
0	-0.1200	0.1900	0
1	-0.0600	0.1300	0
2	0.0000	0.0700	0
3	0.0600	0.0100	1
4	0.1200	0.0500	0

B. PD steering and wheel speeds

active sensors	1.0000
error e	0.5000
de/dt	3.0000
omega (rad/s)	2.3500
speed v (m/s)	0.5250
left wheel	-0.0625
right wheel	1.1125
turn	left

Rate limits (kWheelAccel) and the line-lost search are not modelled here; see followLine.

Explosion: colour from temperature, particle motion, debris bounce

Legend: yellow = input from the code; white = formula.

Source: rendering/Effects.cpp bb(T), fire / spark colour; scene/Combat.cpp updateParticles (spark integrator); scene/Scene.cpp updateDebris (restitution). Yellow = constants copied from the code.

inputs

bb R exp	bb G exp	bb B gain	bb B exp	fire life exp	emit a	emit b	spark cool
0.45	1.90	0.85	4.50	0.60	0.50	5.00	0.90

A. Blackbody ramp bb(T) = (T^a, T^b, k T^c) and fire emission bb(T)(a' + b' T^2)

T	R	G	B	a'+b'T^2	R emit	G emit	B emit
0.0	0.0000	0.0000	0.0000	0.5000	0.0000	0.0000	0.0000
0.1	0.3548	0.0126	0.0000	0.5500	0.1951	0.0069	0.0000
0.2	0.4847	0.0470	0.0006	0.7000	0.3393	0.0329	0.0004
0.3	0.5817	0.1015	0.0038	0.9500	0.5526	0.0964	0.0036
0.4	0.6621	0.1754	0.0138	1.3000	0.8607	0.2280	0.0179
0.5	0.7320	0.2679	0.0376	1.7500	1.2811	0.4689	0.0657
0.6	0.7946	0.3789	0.0853	2.3000	1.8277	0.8714	0.1963
0.7	0.8517	0.5078	0.1707	2.9500	2.5126	1.4980	0.5037
0.8	0.9045	0.6544	0.3114	3.7000	3.3465	2.4214	1.1522
0.9	0.9537	0.8186	0.5291	4.5500	4.3393	3.7245	2.4073
1.0	1.0000	1.0000	0.8500	5.5000	5.5000	5.5000	4.6750

B. A flame core and a spark over their lives (t = age / life)

t	fire T	R	G	B	spark Tn	bb arg	R	G	B
0.0	1.0000	1.0000	1.0000	0.8500	1.0000	1.0000	1.0000	1.0000	0.8500
0.1	0.9387	0.9720	0.8868	0.6395	0.9100	0.9190	0.9627	0.8517	0.5812
0.2	0.8747	0.9415	0.7754	0.4653	0.8200	0.8380	0.9235	0.7148	0.3837
0.3	0.8073	0.9082	0.6659	0.3245	0.7300	0.7570	0.8823	0.5892	0.2429
0.4	0.7360	0.8712	0.5586	0.2140	0.6400	0.6760	0.8384	0.4752	0.1459
0.5	0.6598	0.8293	0.4538	0.1308	0.5500	0.5950	0.7916	0.3729	0.0822
0.6	0.5771	0.7808	0.3518	0.0716	0.4600	0.5140	0.7412	0.2824	0.0425
0.7	0.4856	0.7225	0.2535	0.0329	0.3700	0.4330	0.6862	0.2039	0.0197
0.8	0.3807	0.6476	0.1597	0.0110	0.2800	0.3520	0.6251	0.1375	0.0077
0.9	0.2512	0.5370	0.0724	0.0017	0.1900	0.2710	0.5557	0.0837	0.0024
1.0	0.0000	0.0000	0.0000	0.0000	0.1000	0.1900	0.4736	0.0426	0.0005

C. One spark, step by step: v_y -= g dt; v *= exp(-k dt); p += v dt; below ground: v_y = -e v_y, v_x *= f

	g	drag k	bounce e	keep f	dt (s)	v_x0	v_y0	y0
inputs	9.81	0.40	0.25	0.60	0.05	11.00	6.00	1.00

step	t (s)	x	y	v_x	v_y
0	0.0000	0.0000	1.0000	11.0000	6.0000
1	0.0500	0.5391	1.2700	10.7822	5.4004
2	0.1000	1.0675	1.5107	10.5687	4.8127
3	0.1500	1.5855	1.7225	10.3594	4.2366
4	0.2000	2.0932	1.9061	10.1543	3.6719
5	0.2500	2.5909	2.0620	9.9532	3.1184
6	0.3000	3.0787	2.1908	9.7561	2.5759
7	0.3500	3.5568	2.2930	9.5629	2.0441
8	0.4000	4.0255	2.3691	9.3736	1.5228
9	0.4500	4.4849	2.4197	9.1880	1.0119
10	0.5000	4.9352	2.4453	9.0060	0.5111
11	0.5500	5.3766	2.4463	8.8277	0.0202
12	0.6000	5.8093	2.4232	8.6529	-0.4610
13	0.6500	6.2333	2.3766	8.4816	-0.9327
14	0.7000	6.6490	2.3069	8.3136	-1.3950
15	0.7500	7.0565	2.2145	8.1490	-1.8482

D. Debris bounces (restitution e, no drag): apex = v^2 / 2g, flight = 2v / g

e	v0 (m/s)	rest below	g
0.30	5.00	0.25	9.81

bounce n	v_n (m/s)	apex (m)	flight (s)	moving?
0	5.0000	1.2742	1.0194	yes
1	1.5000	0.1147	0.3058	yes
2	0.4500	0.0103	0.0917	yes
3	0.1350	0.0009	0.0275	rest
4	0.0405	0.0001	0.0083	rest
5	0.0122	0.0000	0.0025	rest

total time		1.4169	

16	0.8000	7.4558	2.0998	7.9876	-2.2924
17	0.8500	7.8473	1.9634	7.8295	-2.7278
18	0.9000	8.2310	1.8057	7.6744	-3.1545
19	0.9500	8.6072	1.6271	7.5225	-3.5729
20	1.0000	8.9758	1.4279	7.3735	-3.9829
21	1.0500	9.3372	1.2087	7.2275	-4.3848
22	1.1000	9.6914	0.9698	7.0844	-4.7788
23	1.1500	10.0386	0.7115	6.9441	-5.1649
24	1.2000	10.3790	0.4343	6.8066	-5.5435
25	1.2500	10.7126	0.1386	6.6718	-5.9145
26	1.3000	11.0395	0.0200	3.9238	1.5695
27	1.3500	11.2319	0.0729	3.8461	1.0577
28	1.4000	11.4204	0.1007	3.7700	0.5559

Duty_Cycle: 24-hour roster of the 8 on-duty soldiers (2 shifts, not 6-hour shifts)

Legend: yellow = input from the code; white = formula.

Source: scene/Garrison.h kDayShiftStart 6.4, kDayShiftEnd 17.6, dayShift(); Garrison.cpp makePrograms() and shiftDelay = 1 + 2.5 rank s; rosters from ts --export-docs v6.garrison.soldiers.

day start (h)	6.40	day end (h)	17.60	day shift	06:24-17:36	hours	11.20	night hours	12.80	DayBreak, Dus	6.45	17.60

#	role	day roster (export)	night roster (export)	day	night
1	SENTRY	guard: inner gate and gate road (INNER GATE -> GATE ROAD)	guard: inner gate (INNER GATE)	guard	guard
2	WATCH	guard: watchtower and east wood look-out (WATCHTOWER -> EAST WOOD)	guard: sandbag guard post (SANDBAG POST)	guard	guard
3	GUNNER	walk: patrol loop on the east apron	guard: machine-gun nest (MG NEST)	walk	guard
4	MECHANIC	work: wrench at the parked tank and the helicopter (MBT-2 ENGINE DECK -> HELICOPTER)	off duty: bunk room (south bunk) (BUNK ROOM)	work	off
5	LOGISTICS	work: carry crates from the depot to the ammunition bunker (FUEL DEPOT -> AMMO BUNKER)	off duty: bunk room (mess table) (MESS BENCH)	work	off
6	TECHNICIAN	work: inspect the radar and the AA gun (RADAR -> AA GUN PIT)	work: plotting table in the command room (COMMAND ROOM)	work	work
7	PATROL	walk: patrol loop inside the compound	guard: east guard tower (GUARD TOWER E)	walk	guard
8	RUNNER	run: running loop on the south apron	off duty: bunk room (north bunk) (BUNK ROOM)	run	off

24-hour grid: state at the middle of each hour = IF(day start <= h + 0.5 < day end, day kind, night kind)

#	role	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
1	SENTRY	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard
2	WATCH	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard	guard
3	GUNNER	guard	guard	guard	guard	guard	guard	walk	walk	walk	walk	walk	walk	walk	walk	walk	walk	walk	walk	guard	guard	guard	guard	guard	guard
4	MECHANIC	off	off	off	off	off	off	work	work	work	work	work	work	work	work	work	work	work	work	off	off	off	off	off	off
5	LOGISTICS	off	off	off	off	off	off	work	work	work	work	work	work	work	work	work	work	work	work	off	off	off	off	off	off
6	TECHNICIAN	work	work	work	work	work	work	work	work	work	work	work	work	work	work	work	work	work	work	work	work	work	work	work	work
7	PATROL	guard	guard	guard	guard	guard	guard	walk	walk	walk	walk	walk	walk	walk	walk	walk	walk	walk	walk	guard	guard	guard	guard	guard	guard
8	RUNNER	off	off	off	off	off	off	run	run	run	run	run	run	run	run	run	run	run	run	off	off	off	off	off	off
on duty (not off)		5	5	5	5	5	5	8	8	8	8	8	8	8	8	8	8	8	8	5	5	5	5	5	5
shift		NIGHT	NIGHT	NIGHT	NIGHT	NIGHT	NIGHT	DAY	DAY	DAY	DAY	DAY	DAY	DAY	DAY	DAY	DAY	DAY	DAY	NIGHT	NIGHT	NIGHT	NIGHT	NIGHT	NIGHT

Colours: grey off duty, red guard, blue work, green walk (patrol), yellow run.

Staggered shift change: departure delay = 1 + 2.5 rank (s, simulation time; rank = seeded order)

rank	0	1	2	3	4	5	6	7
delay (s)	1.0	3.5	6.0	8.5	11.0	13.5	16.0	18.5

The day shift follows the sun (DayBreak about 06:27, Dusk 17:36 at latitude 22.8 deg). Two shifts; the spec's 6-hour shifts are not in the code.

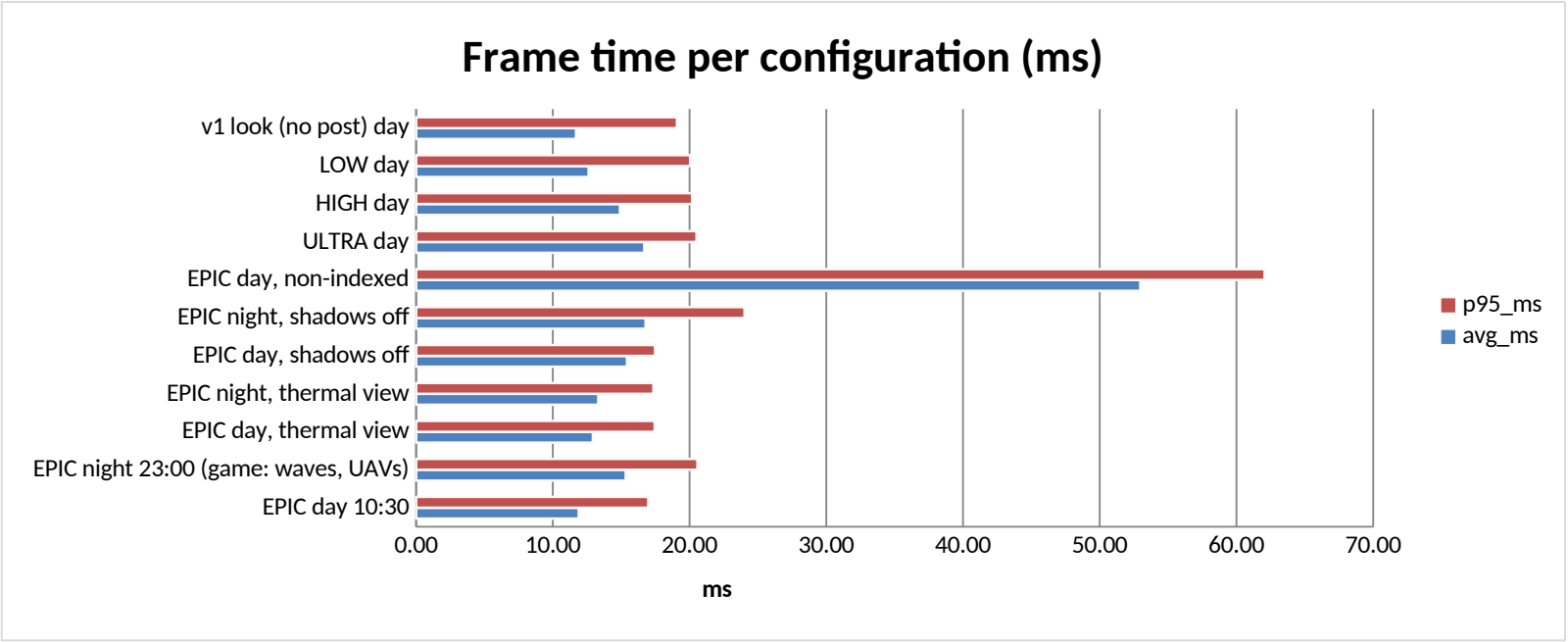
Performance: benchmark import (report/data/bench.csv)

Legend: yellow = input from the code; white = formula.

Source: report/data/bench.csv written by the benchmark run (ts --benchmark / --bench-shots); this sheet is rebuilt by tools/v10/build_cg_workbook.py.

IMPORT AREA: rows below are copied from bench.csv (yellow); derived columns are formulas.

config	avg_ms	p95_ms	avg_fps	draw_calls	triangles	1000 / avg_ms (fps)	fps check (diff)	triangles per draw
EPIC day 10:30	11.90	16.98	84.06	242	4112850	84.1	0.0	16968
EPIC night 23:00 (game: waves, UAVs)	15.33	20.58	65.24	962	4116830	65.2	0.0	4280
EPIC day, thermal view	12.93	17.45	77.34	53	1477730	77.3	0.0	27704
EPIC night, thermal view	13.32	17.36	75.07	61	1545670	75.1	0.0	25322
EPIC day, shadows off	15.43	17.48	64.82	187	4138570	64.8	0.0	22090
EPIC night, shadows off	16.78	24.02	59.58	886	4170890	59.6	0.0	4707
EPIC day, non-indexed	52.97	62.06	18.88	3828	4133000	18.9	0.0	1080
ULTRA day	16.69	20.51	59.93	246	2614220	59.9	0.0	10636
HIGH day	14.91	20.19	67.05	187	1518500	67.1	0.0	8124
LOW day	12.62	20.04	79.26	178	1385390	79.3	0.0	7794
v1 look (no post) day	11.70	19.08	85.50	154	1441730	85.5	0.0	9357



Report_Numbers: every number of the report's text, with its source in the code

Legend: yellow = input from the code; white = formula.

Source: A: the number macros of the report, read from the code and the scene export; B-D: numbers stated in words, in equations and in hand-written tables, read from the code here. tools/rep

A. Number macros of the report

file	name in the report	value	printed as	source
gen_game.tex	SquadkCount	2	2	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	accelNoise	1.5	1.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	bbB	4.5	4.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	bbBk	0.85	0.85	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	bbG	1.9	1.9	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	bbR	0.45	0.45	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	bounce	0.25	0.25	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	bounceH	0.6	0.6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	coastAfter	0.25	0.25	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	confirmHits	3	3	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	core	230	230	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	debrisFric	0.7	0.7	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	debrisRest	0.3	0.3	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	debrisRestSpeed	0.25	0.25	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	debrisSpin	0.6	0.6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	debrisSub	120	120	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	deleteAfter	3	3	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	despeckle	1	1	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	dustDrag	1.6	1.6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	emberDrag	0.3	0.3	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	emberG	0.6	0.6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	errScale	0.12	0.12	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	fireDrag	1.8	1.8	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	fireEmitA	0.5	0.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	fireEmitB	5	5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	fireLife	0.6	0.6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	fireRise	2	2	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	flashInner		1.0,0.92,0.75	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	flashOuter		1.0,0.55,0.2	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	flight_accel	130	130	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	flight_cruise_speed	60	60	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	flight_fuse_m	0.5	0.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	flight_launch_speed	8	8	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	flight_ramp_time	0.4	0.4	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	floor	204	204	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	gateChi2	9.21	9.21	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	gravity	9.81	9.81	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kBreachDamage	10	10	code constant / export value of this name (tools/v10/gen_game.py)

gen_game.tex	kBreachRadius	2	2	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kCellSize	0.5	0.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kCombinedBonus	50	50	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kCvFovYDeg	40	40	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kCvHeight	240	240	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kCvWidth	320	320	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kDamageRadiusPerPower	1.6	1.6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kDroneOwner	0	0	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kEvadeRadius	5	5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kEvadeTime	4	4	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kFirstWaveDelay	0	0	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kFriendlyRadius	4.5	4.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kGridCells	80	80	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kInfiltrateReplan	1.6	1.6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kIrSensorCount	5	5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kKd	0.25	0.25	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kKillScore	100	100	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kKp	3.2	3.2	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kLaunchInterval	4	4	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kLineHalfWidth	0.05	0.05	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kLockTime	1.5	1.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kMaxRovers	6	6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kMaxSpeed	0.75	0.75	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kMissileFuse	0.5	0.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kMissileLife	6	6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kMissileMaxSpeed	60	60	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kMissileRampTime	0.4	0.4	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kMissileTurnRate	6	6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kParticleCap	4000	4\,000	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kReplanPeriod	0.4	0.4	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kRespawnDelay	3	3	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kRoundDamage	30	30	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kRoverCount	3	3	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kRoverHealth	100	100	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kRoverTrackWidth	0.5	0.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kSensorForward	0.31	0.31	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kSensorSpacing	0.06	0.06	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kShakeDecay	0.8	0.8	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kShakeRange	25	25	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kShellSpeed	120	120	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kSpawnInterval	3.5	3.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kSpawnRadius	16.8	16.8	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kSquadEngageAlt	5	5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kSquadLeadLock	0.8	0.8	code constant / export value of this name (tools/v10/gen_game.py)

gen_game.tex	kSquadPatrolAlt	7.5	7.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kSquadReacquireTime	12	12	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kSquadSoloLock	1.3	1.3	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kSquadStandOff	6.5	6.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kSquadWingLock	0.5	0.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kSquadWingWait	2.5	2.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kTracerLife	2.5	2.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kTracerSpeed	1000	1\,000	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kUavCount	2	2	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kUavMissiles	4	4	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kUavReloadTime	6	6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kWaveBreak	6	6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kWreckBurnTime	15	15	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kWreckClearTime	25	25	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kWreckIgnoreRadius	2.5	2.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	kWreckIgnoreTime	30	30	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	maxArea	2500	2\,500	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	maxElongation	6	6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	mergeDistance	1	1	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	minArea	2	2	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	omegaMax	2.5	2.5	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	reloadMeasured	6.03	6.03	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	slowDown	0.6	0.6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	smokeGlowB	0.08	0.08	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	smokeGlowDecay	0.9	0.9	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	smokeGlowG	0.36	0.36	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	smokeGlowR	1	1	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	smokeGrowK	0.25	0.25	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	smokeHeatFloor	0.08	0.08	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	smokeHeatK	0.35	0.35	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	smokeMaxUp	3	3	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	smokeRise	0.9	0.9	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	smokeWind	0.6	0.6	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	sparkCool	0.9	0.9	code constant / export value of this name (tools/v10/gen_game.py)
gen_game.tex	sparkDrag	0.4	0.4	code constant / export value of this name (tools/v10/gen_game.py)
gen_numbers.tex	nLeaves	5474	5{,}474	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_numbers.tex	nObjects	31	31	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_numbers.tex	nTests	396	396	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_numbers.tex	nPass	396	396	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_numbers.tex	nControllables	26	26	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_numbers.tex	nInstances	5474	5{,}474	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_numbers.tex	spotInner	9	9	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_numbers.tex	spotOuter	14	14	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_numbers.tex	spotLin	0.05	0.05	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_

gen_numbers.tex	spotQuad	0.03	0.03	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_numbers.tex	beaconLin	0.35	0.35	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_numbers.tex	beaconQuad	0.45	0.45	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_numbers.tex	moonI	0.95	0.95	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_numbers.tex	daySeconds	291	291	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_numbers.tex	dayMinutes	4.8	4.8	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	bunkerParts	440	440	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	roverN	6	6	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	roverParts	26	26	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	roverAnim	8	8	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	uavN	2	2	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	uavParts	79	79	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	uavAnim	13	13	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	tankN	3	3	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	tankParts	255	255	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	tankAnim	118	118	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	soldierN	8	8	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	soldierParts	75	75	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	soldierAnim	15	15	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	partsBase	993	993	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	partsEnv	652	652	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_tables.tex	partsCamp	640	640	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_designs.tex	nDesigns	62	62	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_designs.tex	nDesignInstances	247	247	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_m3.tex	wgBaxle		(0.71, 0, -0.71)	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_m3.tex	wgCphase	0.8125	0.8125	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_m3.tex	wgDoorSwing	92	92	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_m3.tex	wgDoorHinge		(-1.3, -2.1, 4)	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_m3.tex	nSoldierParts	75	75	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_m3.tex	nSphereV	614	614	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_m3.tex	nSphereInst	810	810	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_m3.tex	nSphereTri	1224	1224	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_m3.tex	sphereTriTotal	0.99	0.99	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_m3.tex	nKindsUav	33	33	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_m3.tex	nKindsSoldier	48	48	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_m3.tex	soldierRootT		(-0.8, -0.3, 21)	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_
gen_m3.tex	soldierGroups	19	19	scene export / benchmark (tools/v10/gen_numbers.py, tools/report/gen_

B. Numbers stated in words

group	quantity	value		source
text	split distance (m)	14		src/ai/Squad.h: kSquadSplitDist
text	re-join distance (m)	9		src/ai/Squad.h: kSquadRejoinDist
text	split urgency (m to a protected point)	10		src/ai/Squad.h: kSquadSplitThreat
text	wait for a free wing (s)	8		src/ai/Squad.h: kSquadWingWaitFree

text	guards' rifle range (m)	26		src/scene/Scene.h: kRifleRange
text	guards' burst length (s)	0.6		src/scene/Scene.cpp: guards' fire
text	guards' burst period (s)	2.4		src/scene/Scene.cpp: guards' fire
text	helicopter alert idle (rotor fraction)	0.38		src/objects/Helicopter.h: kAlertIdle
text	calls of tsDot in the shaders	218		src/rendering/*.cpp (count of the calls)
text	calls of tsMix in the shaders	173		src/rendering/*.cpp (count of the calls)
text	calls of tsSmoothstep in the shaders	146		src/rendering/*.cpp (count of the calls)
text	calls of tsNormalize in the shaders	129		src/rendering/*.cpp (count of the calls)
text	calls of tsClamp in the shaders	104		src/rendering/*.cpp (count of the calls)
text	calls of tsLength in the shaders	75		src/rendering/*.cpp (count of the calls)
text	calls of tsCross in the shaders	26		src/rendering/*.cpp (count of the calls)
text	calls of tsStep in the shaders	21		src/rendering/*.cpp (count of the calls)
text	calls of tsReflect in the shaders	9		src/rendering/*.cpp (count of the calls)
text	calls of tsRefract in the shaders	2		src/rendering/*.cpp (count of the calls)
text	calls of tsTranspose in the shaders	2		src/rendering/*.cpp (count of the calls)
text	calls of the hand-written vector functions	905		src/rendering/*.cpp
text	shader source files that use them	13		src/rendering/*.cpp
text	parts shaded Flat	796		export: components[].shading
text	parts shaded Gouraud	2197		export: components[].shading
text	parts shaded Phong	2478		export: components[].shading
text	scene parts	5474		export: leaf_count
text	designs	62		export: designs
text	automated tests	396		tests/*.cpp (build/test_results.json)

C. Constants of the equations and hand-written tables

group	quantity	value		source
constant	air mass (Kasten-Young): coefficient	0.50572		src/scene/DayCycle.cpp: airMass
constant	air mass (Kasten-Young): elevation offset (deg)	6.07995		src/scene/DayCycle.cpp: airMass
constant	air mass (Kasten-Young): exponent (negative)	1.6364		src/scene/DayCycle.cpp: airMass
constant	tone curve, numerator: linear term	0.0245786		src/math/ToneMap.cpp: tone curve
constant	tone curve, numerator: constant (subtracted)	0.000090537		src/math/ToneMap.cpp: tone curve
constant	tone curve, denominator: quadratic term	0.983729		src/math/ToneMap.cpp: tone curve
constant	tone curve, denominator: linear term	0.432951		src/math/ToneMap.cpp: tone curve
constant	tone curve, denominator: constant	0.238081		src/math/ToneMap.cpp: tone curve
constant	tone curve: lowest exposure (log2, negative)	12.47393		src/math/ToneMap.h: kAgxMinEv
constant	tone curve: highest exposure (log2)	4.026069		src/math/ToneMap.h: kAgxMaxEv
constant	value noise: frequency ratio per octave	2.03		src/rendering/ShaderSources.cpp: fbm
constant	fog density at night (1/m)	0.018		src/rendering/Lighting.h: fogDensity
constant	fog density by day (1/m)	0.0042		src/scene/DayCycle.cpp: kDayFogDensity
constant	shadow depth bias, cascades	0.0004		src/rendering/ShaderSources.cpp: cascadeShadow
constant	shadow depth bias, searchlight	0.00015		src/rendering/ShaderSources.cpp: spotShadow
constant	near plane N (m)	0.05		src/scene/Camera.h: nearZ
constant	far plane F (m)	600		src/scene/Camera.h: farZ
constant	octile distance: diagonal surcharge	0.41421356		src/scene/Garrison.cpp: octile

constant	key-light shadow map (texels), LOW	1024	src/rendering/RendererFrame.cpp: qualityProfile (moonShadowRes)
constant	key-light shadow map (texels), HIGH	2048	src/rendering/RendererFrame.cpp: qualityProfile (moonShadowRes)
constant	key-light shadow map (texels), ULTRA	4096	src/rendering/RendererFrame.cpp: qualityProfile (moonShadowRes)
constant	key-light shadow map (texels), EPIC	4096	src/rendering/RendererFrame.cpp: qualityProfile (moonShadowRes)
constant	searchlight shadow map (texels; 0 = off), LOW	0	src/rendering/RendererFrame.cpp: qualityProfile (spotShadowRes)
constant	searchlight shadow map (texels; 0 = off), HIGH	1024	src/rendering/RendererFrame.cpp: qualityProfile (spotShadowRes)
constant	searchlight shadow map (texels; 0 = off), ULTR	2048	src/rendering/RendererFrame.cpp: qualityProfile (spotShadowRes)
constant	searchlight shadow map (texels; 0 = off), EPIC	2048	src/rendering/RendererFrame.cpp: qualityProfile (spotShadowRes)
constant	near cascade covers (m), LOW	10	src/rendering/RendererFrame.cpp: qualityProfile (cascadeSplit)
constant	near cascade covers (m), HIGH	12	src/rendering/RendererFrame.cpp: qualityProfile (cascadeSplit)
constant	near cascade covers (m), ULTRA	14	src/rendering/RendererFrame.cpp: qualityProfile (cascadeSplit)
constant	near cascade covers (m), EPIC	14	src/rendering/RendererFrame.cpp: qualityProfile (cascadeSplit)
constant	shadow filter taps, LOW	8	src/rendering/RendererFrame.cpp: qualityProfile (pcfTaps)
constant	shadow filter taps, HIGH	8	src/rendering/RendererFrame.cpp: qualityProfile (pcfTaps)
constant	shadow filter taps, ULTRA	16	src/rendering/RendererFrame.cpp: qualityProfile (pcfTaps)
constant	shadow filter taps, EPIC	16	src/rendering/RendererFrame.cpp: qualityProfile (pcfTaps)
constant	soft-shadow blocker taps, HIGH	8	src/rendering/RendererFrame.cpp: qualityProfile (blockerTaps)
constant	soft-shadow blocker taps, ULTRA	16	src/rendering/RendererFrame.cpp: qualityProfile (blockerTaps)
constant	soft-shadow blocker taps, EPIC	16	src/rendering/RendererFrame.cpp: qualityProfile (blockerTaps)
constant	contact-shadow steps, HIGH	10	src/rendering/RendererFrame.cpp: qualityProfile (contactSteps)
constant	contact-shadow steps, ULTRA	12	src/rendering/RendererFrame.cpp: qualityProfile (contactSteps)
constant	contact-shadow steps, EPIC	16	src/rendering/RendererFrame.cpp: qualityProfile (contactSteps)
constant	film: length (s)	218	report/video/thermalsense_final_720p.mp4 (ffprobe)
constant	film: length, minutes part	3	the length above
constant	film: length, seconds part	38	the length above
constant	film: frames per second	30	report/video/thermalsense_final_720p.mp4 (ffprobe)

D. Values derived in the text (formulas of this sheet)

group	quantity	value		source
derived	far / near ratio F/N	12000.0000		the two planes above
derived	isometric view: elevation (deg)	35.2644		Section 11: arctan(1/sqrt 2)
derived	isometric view: foreshortening of each axis	0.8165		Section 11: sqrt(2/3)
derived	focal factor f at a 50 deg field of view	2.1445		Section 11: f = 1/tan(fov/2)
derived	focal factor f at a 60 deg field of view	1.7321		Section 11: f = 1/tan(fov/2)
derived	normal of the face x/4 + y = c: x component	0.2425		Section 8: normal matrix example, S = diag(4,1,1)
derived	normal of the face x/4 + y = c: y component	0.9701		Section 8: normal matrix example
derived	angle of M n to the true normal (deg)	61.9275		Section 8: normal matrix example
derived	octile surcharge sqrt(2) - 1	0.4142		Algorithm: octile distance

macros	163
stated	80

used in (report sources)
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B08_game.tex, algorithms.tex
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B01_decomposition.tex, gen_m3.tex
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Coordinates: every component after its transformation chain, in its design's frame (cm)

Legend: yellow = input from the code; white = formula.

Source: scene export: components[].world_matrix, S_rest, R_deg. Centre p = W_root^-1 W_part (0,0,0,1); extent = box of the unit primitive under the same matrix (a root that is itself a part: its size in use; terrain disc: 2R). These are the coordinate sheets of

design	#	component	primitive	S x	S y	S z	R x (deg)	R y	R z	x	y	z	dx	dy	dz
Soldier	2	Hips	Sphere	37.8	22.0	24.0	0.0	0.0	0.0	0.0	101.5	0.0	37.8	22.0	24.0
Soldier	3	Belt	Cylinder	38.8	25.0	7.0	90.0	0.0	0.0	0.0	109.5	0.0	38.8	7.0	25.0
Soldier	4	Hip joint	Sphere	15.0	15.0	15.0	0.0	0.0	0.0	10.0	97.5	0.0	15.0	16.7	16.7
Soldier	5	Thigh	Cylinder	15.0	15.0	46.0	90.0	0.0	0.0	10.0	74.6	2.8	15.0	47.5	20.6
Soldier	6	Knee joint	Sphere	15.0	15.0	15.0	0.0	0.0	0.0	10.0	51.8	5.7	15.0	16.3	16.3
Soldier	7	Shin	Cylinder	12.0	12.0	44.0	90.0	0.0	0.0	10.0	29.9	3.7	12.0	44.9	16.0
Soldier	8	Ankle joint	Sphere	12.0	12.0	12.0	0.0	0.0	0.0	10.0	8.0	1.7	12.0	12.0	12.0
Soldier	9	Boot	Sphere	13.0	11.0	30.0	0.0	0.0	0.0	10.0	5.5	7.7	13.0	11.0	30.0
Soldier	10	Calf	Sphere	13.0	22.0	14.0	0.0	0.0	0.0	10.0	39.0	3.0	13.0	23.2	15.9
Soldier	11	Knee pad	Sphere	11.5	13.0	6.0	0.0	0.0	0.0	10.0	49.3	11.5	11.5	13.5	7.2
Soldier	12	Thigh muscle	Sphere	17.5	31.0	18.0	0.0	0.0	0.0	10.0	82.7	2.8	17.5	33.0	21.7
Soldier	13	Hip joint	Sphere	15.0	15.0	15.0	0.0	0.0	0.0	-10.0	97.5	0.0	15.0	15.4	15.4
Soldier	14	Thigh	Cylinder	15.0	15.0	46.0	90.0	0.0	0.0	-10.0	74.5	0.7	15.0	46.4	16.3
Soldier	15	Knee joint	Sphere	15.0	15.0	15.0	0.0	0.0	0.0	-10.0	51.5	1.3	15.0	17.3	17.3
Soldier	16	Shin	Cylinder	12.0	12.0	44.0	90.0	0.0	0.0	-10.0	29.8	-2.4	12.0	45.4	19.3
Soldier	17	Ankle joint	Sphere	12.0	12.0	12.0	0.0	0.0	0.0	-10.0	8.1	-6.2	12.0	12.0	12.0
Soldier	18	Boot	Sphere	13.0	11.0	30.0	0.0	0.0	0.0	-10.0	5.6	-0.2	13.0	11.0	30.0
Soldier	19	Calf	Sphere	13.0	22.0	14.0	0.0	0.0	0.0	-10.0	38.9	-2.4	13.0	24.1	17.6
Soldier	20	Knee pad	Sphere	11.5	13.0	6.0	0.0	0.0	0.0	-10.0	48.5	6.9	11.5	13.8	8.1
Soldier	21	Thigh muscle	Sphere	17.5	31.0	18.0	0.0	0.0	0.0	-10.0	82.5	1.4	17.5	31.5	18.9
Soldier	22	Abdomen	Sphere	35.7	30.0	24.0	0.0	0.0	0.0	-0.2	121.5	0.4	36.2	31.4	25.1
Soldier	23	Torso	Tapered box	39.9	33.0	23.0	0.0	0.0	180.0	-0.6	141.4	1.1	45.9	34.0	36.4
Soldier	24	Backpack	Cube	32.0	38.0	16.0	0.0	0.0	0.0	6.3	139.2	-17.8	36.2	38.7	27.3
Soldier	25	Bedroll	Cylinder	11.0	11.0	34.0	0.0	90.0	0.0	6.3	163.2	-17.9	35.9	11.6	22.4
Soldier	26	Rifle receiver	Cube	6.0	11.0	52.0	0.0	0.0	0.0	-15.9	131.6	34.5	6.6	34.2	51.1
Soldier	27	Rifle barrel	Cylinder	3.5	3.5	40.0	0.0	0.0	0.0	-15.6	112.2	75.2	3.9	21.9	37.0
Soldier	28	Rifle magazine	Cube	4.0	16.0	7.0	15.0	0.0	0.0	-15.6	117.2	38.2	4.3	16.5	16.0
Soldier	29	Pistol grip	Cube	3.5	10.0	4.5	-20.0	0.0	0.0	-15.8	129.8	18.8	3.7	10.6	5.9
Soldier	30	Handguard	Cube	5.8	6.8	24.0	0.0	0.0	0.0	-15.6	115.6	64.5	6.1	17.4	24.4
Soldier	31	Optic	Cylinder	4.2	4.2	13.0	0.0	0.0	0.0	-16.0	138.2	40.3	4.4	9.9	13.5
Soldier	32	Rifle receiver	Cube	6.0	11.0	52.0	0.0	0.0	0.0	10.4	140.7	-29.0	36.0	50.6	13.0
Soldier	33	Rifle barrel	Cylinder	3.5	3.5	40.0	0.0	0.0	0.0	30.9	180.5	-24.2	23.3	36.4	8.3
Soldier	34	Rifle magazine	Cube	4.0	16.0	7.0	15.0	0.0	0.0	24.8	143.8	-26.6	16.7	16.4	6.3
Soldier	35	Pistol grip	Cube	3.5	10.0	4.5	-20.0	0.0	0.0	11.7	124.9	-29.4	11.0	6.3	5.2
Soldier	36	Handguard	Cube	5.8	6.8	24.0	0.0	0.0	0.0	27.2	170.0	-25.2	18.6	24.3	9.2
Soldier	37	Optic	Cylinder	4.2	4.2	13.0	0.0	0.0	0.0	4.1	146.7	-29.8	10.7	13.5	6.1
Soldier	38	Carried crate	Cube	46.0	32.0	36.0	0.0	0.0	0.0	-13.9	119.9	38.0	56.1	33.6	50.7
Soldier	39	Neck	Cylinder	12.5	11.5	11.0	90.0	0.0	0.0	-1.4	157.8	3.1	16.3	12.9	16.7
Soldier	40	Head	Sphere	18.0	22.5	20.5	0.0	0.0	0.0	-2.0	168.1	5.2	19.6	27.5	25.0
Soldier	41	Helmet	Hemisphere	24.5	12.5	26.5	-6.0	0.0	0.0	-2.7	178.4	5.8	25.3	17.0	27.9
Soldier	42	Helmet brim	Cylinder	25.5	28.0	2.2	84.0	0.0	0.0	-2.4	173.8	6.0	25.6	7.0	28.2
Soldier	43	Jaw	Sphere	13.0	8.5	11.5	0.0	0.0	0.0	-1.6	161.2	7.5	13.6	11.6	13.1
Soldier	44	Nose	Sphere	3.2	5.0	4.5	-15.0	0.0	0.0	-1.8	164.2	14.4	3.6	5.4	4.7
Soldier	45	Eye	Sphere	2.4	1.6	1.2	0.0	0.0	0.0	1.3	167.5	14.7	2.5	2.0	1.5
Soldier	46	Ear	Sphere	2.2	5.8	4.2	0.0	0.0	0.0	6.9	168.3	4.7	2.6	6.7	5.4

Soldier	47	Chin strap	Cylinder	1.2	1.2	17.0	58.8	-37.1	0.0	4.3	165.3	7.1	5.9	16.5	5.3
Soldier	48	Eye	Sphere	2.4	1.6	1.2	0.0	0.0	0.0	-5.3	167.1	14.7	2.5	2.0	1.5
Soldier	49	Ear	Sphere	2.2	5.8	4.2	0.0	0.0	0.0	-10.9	167.1	4.6	2.6	6.7	5.4
Soldier	50	Chin strap	Cylinder	1.2	1.2	17.0	58.8	37.1	0.0	-8.0	164.5	7.0	7.8	16.0	5.4
Soldier	51	Goggles	Cube	15.0	3.2	3.0	-35.0	0.0	0.0	-2.8	179.2	17.1	15.2	5.1	4.1
Soldier	52	NVG mount	Cube	4.0	3.5	2.5	-10.0	0.0	0.0	-2.5	173.4	19.5	4.2	3.9	2.7
Soldier	53	Trapezius	Sphere	20.0	8.5	15.0	0.0	0.0	-20.0	7.7	152.8	3.2	25.4	17.2	21.5
Soldier	54	Trapezius	Sphere	20.0	8.5	15.0	0.0	0.0	20.0	-8.3	152.7	-2.6	25.5	17.3	21.8
Soldier	55	Shoulder joint	Sphere	13.0	13.0	13.0	0.0	0.0	0.0	20.5	150.5	9.2	22.2	21.8	19.6
Soldier	56	Upper arm	Cylinder	13.0	13.0	30.0	90.0	0.0	0.0	12.9	141.2	18.1	30.8	32.4	29.7
Soldier	57	Elbow joint	Sphere	13.0	13.0	13.0	0.0	0.0	0.0	5.3	131.8	27.0	22.3	21.0	18.7
Soldier	58	Forearm	Cylinder	11.0	11.0	26.0	90.0	0.0	0.0	-3.5	126.4	34.9	29.0	24.1	24.9
Soldier	59	Hand	Sphere	9.0	12.0	8.0	0.0	0.0	0.0	-15.7	118.8	45.8	16.9	15.0	14.7
Soldier	60	Forearm muscle	Sphere	12.0	18.0	12.0	0.0	0.0	0.0	0.5	128.9	31.3	24.6	21.9	20.9
Soldier	61	Elbow pad	Sphere	10.0	11.0	5.5	0.0	0.0	0.0	7.6	126.9	27.8	15.3	12.8	14.8
Soldier	62	Deltoid	Sphere	15.5	17.0	15.0	0.0	0.0	0.0	19.2	148.7	12.0	26.8	26.6	24.2
Soldier	63	Shoulder joint	Sphere	13.0	13.0	13.0	0.0	0.0	0.0	-21.9	150.3	-6.3	18.8	16.2	17.1
Soldier	64	Upper arm	Cylinder	13.0	13.0	30.0	90.0	0.0	0.0	-24.8	135.6	-6.8	22.1	32.9	17.7
Soldier	65	Elbow joint	Sphere	13.0	13.0	13.0	0.0	0.0	0.0	-27.6	120.9	-7.4	17.6	18.1	20.2
Soldier	66	Forearm	Cylinder	11.0	11.0	26.0	90.0	0.0	0.0	-22.7	124.6	4.1	20.6	19.6	30.3
Soldier	67	Hand	Sphere	9.0	12.0	8.0	0.0	0.0	0.0	-15.8	129.7	19.9	13.3	12.5	16.3
Soldier	68	Wrench	Cylinder	3.0	3.0	26.0	0.0	0.0	0.0	-15.0	139.5	17.5	5.2	25.9	12.2
Soldier	69	Forearm muscle	Sphere	12.0	18.0	12.0	0.0	0.0	0.0	-25.0	122.9	-1.2	18.5	18.5	23.9
Soldier	70	Elbow pad	Sphere	10.0	11.0	5.5	0.0	0.0	0.0	-27.5	116.0	-4.7	13.7	10.0	14.9
Soldier	71	Deltoid	Sphere	15.5	17.0	15.0	0.0	0.0	0.0	-23.5	147.1	-6.1	22.6	20.8	20.0
Soldier	72	Plate carrier	Cube	34.6	27.0	29.5	0.0	0.0	0.0	-0.7	138.4	1.5	43.1	28.3	40.5
Soldier	73	Magazine pouches	Cube	25.0	11.0	6.0	0.0	0.0	0.0	-6.4	130.8	17.2	25.7	11.3	14.6
Soldier	74	Admin pouch	Cube	13.0	7.0	3.5	0.0	0.0	0.0	-6.3	142.3	16.8	13.5	7.2	8.0
Soldier	75	Radio	Cube	5.0	15.0	8.0	0.0	0.0	0.0	19.1	135.7	2.7	7.7	15.3	9.7
Soldier	76	Radio antenna	Cylinder	0.8	0.8	36.9	-81.3	153.4	0.0	21.4	161.6	-0.2	4.5	36.8	3.7
Main battle tank	2	Hull core	Cube	170.0	90.0	560.0	0.0	0.0	0.0	0.0	80.0	0.0	170.0	90.0	560.0
Main battle tank	3	Hull deck	Cube	290.0	26.0	460.0	0.0	0.0	0.0	0.0	129.0	-50.0	290.0	26.0	460.0
Main battle tank	4	Glacis plate	Wedge	250.0	64.0	130.0	0.0	0.0	0.0	0.0	116.0	245.0	250.0	64.0	130.0
Main battle tank	5	Lower nose plate	Cube	220.0	10.0	52.0	67.4	180.0	0.0	0.0	54.0	300.0	220.0	51.8	29.2
Main battle tank	6	Rear plate	Cube	260.0	90.0	12.0	0.0	0.0	0.0	0.0	85.0	-290.0	260.0	90.0	12.0
Main battle tank	7	Engine deck	Cube	240.0	14.0	170.0	0.0	0.0	0.0	0.0	147.0	-195.0	240.0	14.0	170.0
Main battle tank	8	Exhaust grille	Cube	150.0	5.0	90.0	0.0	0.0	0.0	0.0	156.0	-200.0	150.0	5.0	90.0
Main battle tank	9	Louvre front	Cube	150.0	3.0	8.0	0.0	0.0	0.0	0.0	158.0	-155.0	150.0	3.0	8.0
Main battle tank	10	Louvre rear	Cube	150.0	3.0	8.0	0.0	0.0	0.0	0.0	158.0	-245.0	150.0	3.0	8.0
Main battle tank	11	Exhaust pipe	Cylinder	13.0	13.0	50.0	0.0	0.0	0.0	-90.0	100.0	-310.0	13.0	13.0	50.0
Main battle tank	12	Front fender	Cube	70.0	5.0	130.0	8.0	0.0	0.0	-110.0	118.0	255.0	70.0	23.0	129.4
Main battle tank	13	Headlight housing	Cube	30.0	22.0	20.0	4.0	0.0	0.0	-115.0	130.0	300.0	30.0	23.3	21.5
Main battle tank	14	Headlight lens	Sphere	20.0	20.0	10.0	4.0	0.0	0.0	-115.0	130.0	310.0	20.0	20.6	11.4
Main battle tank	15	Tow hook	Cube	12.0	12.0	20.0	0.0	0.0	0.0	-80.0	50.0	312.0	12.0	12.0	20.0
Main battle tank	16	Spare links	Cube	50.0	30.0	6.0	-25.0	0.0	0.0	-55.0	100.0	300.0	50.0	29.7	18.1
Main battle tank	17	Stowage box	Cube	50.0	30.0	90.0	0.0	0.0	0.0	-120.0	142.0	60.0	50.0	30.0	90.0
Main battle tank	18	Fender lamp	Cube	10.0	8.0	10.0	0.0	0.0	0.0	-135.0	125.0	300.0	10.0	8.0	10.0
Main battle tank	19	Tail light	Cube	14.0	10.0	6.0	0.0	0.0	0.0	-110.0	115.0	-298.0	14.0	10.0	6.0
Main battle tank	20	Exhaust pipe	Cylinder	13.0	13.0	50.0	0.0	0.0	0.0	90.0	100.0	-310.0	13.0	13.0	50.0
Main battle tank	21	Front fender	Cube	70.0	5.0	130.0	8.0	0.0	0.0	110.0	118.0	255.0	70.0	23.0	129.4
Main battle tank	22	Headlight housing	Cube	30.0	22.0	20.0	4.0	0.0	0.0	115.0	130.0	300.0	30.0	23.3	21.5

Main battle tank	23	Headlight lens	Sphere	20.0	20.0	10.0	4.0	0.0	0.0	115.0	130.0	310.0	20.0	20.6	11.4
Main battle tank	24	Tow hook	Cube	12.0	12.0	20.0	0.0	0.0	0.0	80.0	50.0	312.0	12.0	12.0	20.0
Main battle tank	25	Spare links	Cube	50.0	30.0	6.0	-25.0	0.0	0.0	55.0	100.0	300.0	50.0	29.7	18.1
Main battle tank	26	Stowage box	Cube	50.0	30.0	90.0	0.0	0.0	0.0	120.0	142.0	60.0	50.0	30.0	90.0
Main battle tank	27	Fender lamp	Cube	10.0	8.0	10.0	0.0	0.0	0.0	135.0	125.0	300.0	10.0	8.0	10.0
Main battle tank	28	Tail light	Cube	14.0	10.0	6.0	0.0	0.0	0.0	110.0	115.0	-298.0	14.0	10.0	6.0
Main battle tank	29	Side skirt	Cube	7.0	50.0	120.0	0.0	0.0	0.0	-147.0	95.0	-200.0	7.0	50.0	120.0
Main battle tank	30	Side skirt	Cube	7.0	50.0	120.0	0.0	0.0	0.0	-147.0	95.0	-75.0	7.0	50.0	120.0
Main battle tank	31	Side skirt	Cube	7.0	50.0	120.0	0.0	0.0	0.0	-147.0	95.0	50.0	7.0	50.0	120.0
Main battle tank	32	Side skirt	Cube	7.0	50.0	120.0	0.0	0.0	0.0	-147.0	95.0	175.0	7.0	50.0	120.0
Main battle tank	33	Side skirt	Cube	7.0	50.0	120.0	0.0	0.0	0.0	147.0	95.0	-200.0	7.0	50.0	120.0
Main battle tank	34	Side skirt	Cube	7.0	50.0	120.0	0.0	0.0	0.0	147.0	95.0	-75.0	7.0	50.0	120.0
Main battle tank	35	Side skirt	Cube	7.0	50.0	120.0	0.0	0.0	0.0	147.0	95.0	50.0	7.0	50.0	120.0
Main battle tank	36	Side skirt	Cube	7.0	50.0	120.0	0.0	0.0	0.0	147.0	95.0	175.0	7.0	50.0	120.0
Main battle tank	37	Driver hatch	Cylinder	62.0	62.0	10.0	90.0	0.0	0.0	0.0	150.0	135.0	62.0	10.0	62.0
Main battle tank	38	Driver periscope	Cube	18.0	8.0	6.0	0.0	0.0	0.0	-30.0	150.0	172.0	18.0	8.0	6.0
Main battle tank	39	Driver periscope	Cube	18.0	8.0	6.0	0.0	0.0	0.0	0.0	150.0	172.0	18.0	8.0	6.0
Main battle tank	40	Driver periscope	Cube	18.0	8.0	6.0	0.0	0.0	0.0	30.0	150.0	172.0	18.0	8.0	6.0
Main battle tank	41	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	-280.0	55.0	8.0	33.0
Main battle tank	42	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	-244.1	55.0	8.0	33.0
Main battle tank	43	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	-208.3	55.0	8.0	33.0
Main battle tank	44	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	-172.4	55.0	8.0	33.0
Main battle tank	45	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	-136.6	55.0	8.0	33.0
Main battle tank	46	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	-100.7	55.0	8.0	33.0
Main battle tank	47	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	-64.9	55.0	8.0	33.0
Main battle tank	48	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	-29.0	55.0	8.0	33.0
Main battle tank	49	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	6.8	55.0	8.0	33.0
Main battle tank	50	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	42.7	55.0	8.0	33.0
Main battle tank	51	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	78.5	55.0	8.0	33.0
Main battle tank	52	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	114.4	55.0	8.0	33.0
Main battle tank	53	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	150.2	55.0	8.0	33.0
Main battle tank	54	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	186.1	55.0	8.0	33.0
Main battle tank	55	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	222.0	55.0	8.0	33.0
Main battle tank	56	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	110.0	105.0	257.8	55.0	8.0	33.0
Main battle tank	57	Track link	Cube	55.0	8.0	33.0	15.7	0.0	0.0	110.0	103.1	293.5	55.0	16.6	33.9
Main battle tank	58	Track link	Cube	55.0	8.0	33.0	56.7	0.0	0.0	110.0	82.4	321.8	55.0	32.0	24.8
Main battle tank	59	Track link	Cube	55.0	8.0	33.0	97.8	0.0	0.0	110.0	48.2	329.5	55.0	33.8	12.4
Main battle tank	60	Track link	Cube	55.0	8.0	33.0	138.9	0.0	0.0	110.0	17.3	312.9	55.0	27.7	30.1
Main battle tank	61	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	280.0	55.0	8.0	33.0
Main battle tank	62	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	244.1	55.0	8.0	33.0
Main battle tank	63	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	208.3	55.0	8.0	33.0
Main battle tank	64	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	172.4	55.0	8.0	33.0
Main battle tank	65	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	136.6	55.0	8.0	33.0
Main battle tank	66	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	100.7	55.0	8.0	33.0
Main battle tank	67	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	64.9	55.0	8.0	33.0
Main battle tank	68	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	29.0	55.0	8.0	33.0
Main battle tank	69	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	-6.8	55.0	8.0	33.0
Main battle tank	70	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	-42.7	55.0	8.0	33.0
Main battle tank	71	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	-78.5	55.0	8.0	33.0
Main battle tank	72	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	-114.4	55.0	8.0	33.0
Main battle tank	73	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	-150.2	55.0	8.0	33.0

Main battle tank	74	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	-186.1	55.0	8.0	33.0
Main battle tank	75	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	-222.0	55.0	8.0	33.0
Main battle tank	76	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	110.0	5.0	-257.8	55.0	8.0	33.0
Main battle tank	77	Track link	Cube	55.0	8.0	33.0	195.7	0.0	0.0	110.0	6.9	-293.5	55.0	16.6	33.9
Main battle tank	78	Track link	Cube	55.0	8.0	33.0	236.7	0.0	0.0	110.0	27.6	-321.8	55.0	32.0	24.8
Main battle tank	79	Track link	Cube	55.0	8.0	33.0	277.8	0.0	0.0	110.0	61.8	-329.5	55.0	33.8	12.4
Main battle tank	80	Track link	Cube	55.0	8.0	33.0	318.9	0.0	0.0	110.0	92.7	-312.9	55.0	27.7	30.1
Main battle tank	81	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	110.0	44.0	-190.0	133.3	64.0	64.0
Main battle tank	82	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	110.0	44.0	-190.0	46.0	32.0	32.0
Main battle tank	83	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	110.0	44.0	-190.0	48.0	7.0	57.6
Main battle tank	84	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	110.0	44.0	-125.0	133.3	64.0	64.0
Main battle tank	85	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	110.0	44.0	-125.0	46.0	32.0	32.0
Main battle tank	86	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	110.0	44.0	-125.0	48.0	7.0	57.6
Main battle tank	87	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	110.0	44.0	-60.0	133.3	64.0	64.0
Main battle tank	88	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	110.0	44.0	-60.0	46.0	32.0	32.0
Main battle tank	89	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	110.0	44.0	-60.0	48.0	7.0	57.6
Main battle tank	90	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	110.0	44.0	5.0	133.3	64.0	64.0
Main battle tank	91	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	110.0	44.0	5.0	46.0	32.0	32.0
Main battle tank	92	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	110.0	44.0	5.0	48.0	7.0	57.6
Main battle tank	93	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	110.0	44.0	70.0	133.3	64.0	64.0
Main battle tank	94	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	110.0	44.0	70.0	46.0	32.0	32.0
Main battle tank	95	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	110.0	44.0	70.0	48.0	7.0	57.6
Main battle tank	96	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	110.0	44.0	135.0	133.3	64.0	64.0
Main battle tank	97	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	110.0	44.0	135.0	46.0	32.0	32.0
Main battle tank	98	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	110.0	44.0	135.0	48.0	7.0	57.6
Main battle tank	99	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	110.0	44.0	200.0	133.3	64.0	64.0
Main battle tank	100	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	110.0	44.0	200.0	46.0	32.0	32.0
Main battle tank	101	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	110.0	44.0	200.0	48.0	7.0	57.6
Main battle tank	102	Tyre	Torus	76.0	76.0	150.0	0.0	0.0	0.0	110.0	55.0	-280.0	150.0	76.0	76.0
Main battle tank	103	Hub	Cylinder	38.0	38.0	51.0	0.0	0.0	0.0	110.0	55.0	-280.0	51.0	38.0	38.0
Main battle tank	104	Spoke	Cube	68.4	7.0	53.0	0.0	0.0	0.0	110.0	55.0	-280.0	53.0	7.0	68.4
Main battle tank	105	Tyre	Torus	76.0	76.0	150.0	0.0	0.0	0.0	110.0	55.0	280.0	150.0	76.0	76.0
Main battle tank	106	Hub	Cylinder	38.0	38.0	51.0	0.0	0.0	0.0	110.0	55.0	280.0	51.0	38.0	38.0
Main battle tank	107	Spoke	Cube	68.4	7.0	53.0	0.0	0.0	0.0	110.0	55.0	280.0	53.0	7.0	68.4
Main battle tank	108	Return roller	Cylinder	18.0	18.0	20.0	0.0	90.0	0.0	110.0	90.0	-140.0	20.0	18.0	18.0
Main battle tank	109	Return roller	Cylinder	18.0	18.0	20.0	0.0	90.0	0.0	110.0	90.0	0.0	20.0	18.0	18.0
Main battle tank	110	Return roller	Cylinder	18.0	18.0	20.0	0.0	90.0	0.0	110.0	90.0	140.0	20.0	18.0	18.0
Main battle tank	111	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	-280.0	55.0	8.0	33.0
Main battle tank	112	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	-244.1	55.0	8.0	33.0
Main battle tank	113	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	-208.3	55.0	8.0	33.0
Main battle tank	114	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	-172.4	55.0	8.0	33.0
Main battle tank	115	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	-136.6	55.0	8.0	33.0
Main battle tank	116	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	-100.7	55.0	8.0	33.0
Main battle tank	117	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	-64.9	55.0	8.0	33.0
Main battle tank	118	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	-29.0	55.0	8.0	33.0
Main battle tank	119	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	6.8	55.0	8.0	33.0
Main battle tank	120	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	42.7	55.0	8.0	33.0
Main battle tank	121	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	78.5	55.0	8.0	33.0
Main battle tank	122	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	114.4	55.0	8.0	33.0
Main battle tank	123	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	150.2	55.0	8.0	33.0
Main battle tank	124	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	186.1	55.0	8.0	33.0

Main battle tank	125	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	222.0	55.0	8.0	33.0
Main battle tank	126	Track link	Cube	55.0	8.0	33.0	0.0	0.0	0.0	-110.0	105.0	257.8	55.0	8.0	33.0
Main battle tank	127	Track link	Cube	55.0	8.0	33.0	15.7	0.0	0.0	-110.0	103.1	293.5	55.0	16.6	33.9
Main battle tank	128	Track link	Cube	55.0	8.0	33.0	56.7	0.0	0.0	-110.0	82.4	321.8	55.0	32.0	24.8
Main battle tank	129	Track link	Cube	55.0	8.0	33.0	97.8	0.0	0.0	-110.0	48.2	329.5	55.0	33.8	12.4
Main battle tank	130	Track link	Cube	55.0	8.0	33.0	138.9	0.0	0.0	-110.0	17.3	312.9	55.0	27.7	30.1
Main battle tank	131	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	280.0	55.0	8.0	33.0
Main battle tank	132	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	244.1	55.0	8.0	33.0
Main battle tank	133	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	208.3	55.0	8.0	33.0
Main battle tank	134	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	172.4	55.0	8.0	33.0
Main battle tank	135	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	136.6	55.0	8.0	33.0
Main battle tank	136	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	100.7	55.0	8.0	33.0
Main battle tank	137	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	64.9	55.0	8.0	33.0
Main battle tank	138	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	29.0	55.0	8.0	33.0
Main battle tank	139	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	-6.8	55.0	8.0	33.0
Main battle tank	140	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	-42.7	55.0	8.0	33.0
Main battle tank	141	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	-78.5	55.0	8.0	33.0
Main battle tank	142	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	-114.4	55.0	8.0	33.0
Main battle tank	143	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	-150.2	55.0	8.0	33.0
Main battle tank	144	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	-186.1	55.0	8.0	33.0
Main battle tank	145	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	-222.0	55.0	8.0	33.0
Main battle tank	146	Track link	Cube	55.0	8.0	33.0	180.0	0.0	0.0	-110.0	5.0	-257.8	55.0	8.0	33.0
Main battle tank	147	Track link	Cube	55.0	8.0	33.0	195.7	0.0	0.0	-110.0	6.9	-293.5	55.0	16.6	33.9
Main battle tank	148	Track link	Cube	55.0	8.0	33.0	236.7	0.0	0.0	-110.0	27.6	-321.8	55.0	32.0	24.8
Main battle tank	149	Track link	Cube	55.0	8.0	33.0	277.8	0.0	0.0	-110.0	61.8	-329.5	55.0	33.8	12.4
Main battle tank	150	Track link	Cube	55.0	8.0	33.0	318.9	0.0	0.0	-110.0	92.7	-312.9	55.0	27.7	30.1
Main battle tank	151	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	-110.0	44.0	-190.0	133.3	64.0	64.0
Main battle tank	152	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	-110.0	44.0	-190.0	46.0	32.0	32.0
Main battle tank	153	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	-110.0	44.0	-190.0	48.0	7.0	57.6
Main battle tank	154	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	-110.0	44.0	-125.0	133.3	64.0	64.0
Main battle tank	155	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	-110.0	44.0	-125.0	46.0	32.0	32.0
Main battle tank	156	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	-110.0	44.0	-125.0	48.0	7.0	57.6
Main battle tank	157	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	-110.0	44.0	-60.0	133.3	64.0	64.0
Main battle tank	158	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	-110.0	44.0	-60.0	46.0	32.0	32.0
Main battle tank	159	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	-110.0	44.0	-60.0	48.0	7.0	57.6
Main battle tank	160	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	-110.0	44.0	5.0	133.3	64.0	64.0
Main battle tank	161	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	-110.0	44.0	5.0	46.0	32.0	32.0
Main battle tank	162	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	-110.0	44.0	5.0	48.0	7.0	57.6
Main battle tank	163	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	-110.0	44.0	70.0	133.3	64.0	64.0
Main battle tank	164	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	-110.0	44.0	70.0	46.0	32.0	32.0
Main battle tank	165	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	-110.0	44.0	70.0	48.0	7.0	57.6
Main battle tank	166	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	-110.0	44.0	135.0	133.3	64.0	64.0
Main battle tank	167	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	-110.0	44.0	135.0	46.0	32.0	32.0
Main battle tank	168	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	-110.0	44.0	135.0	48.0	7.0	57.6
Main battle tank	169	Tyre	Torus	64.0	64.0	133.3	0.0	0.0	0.0	-110.0	44.0	200.0	133.3	64.0	64.0
Main battle tank	170	Hub	Cylinder	32.0	32.0	46.0	0.0	0.0	0.0	-110.0	44.0	200.0	46.0	32.0	32.0
Main battle tank	171	Spoke	Cube	57.6	7.0	48.0	0.0	0.0	0.0	-110.0	44.0	200.0	48.0	7.0	57.6
Main battle tank	172	Tyre	Torus	76.0	76.0	150.0	0.0	0.0	0.0	-110.0	55.0	-280.0	150.0	76.0	76.0
Main battle tank	173	Hub	Cylinder	38.0	38.0	51.0	0.0	0.0	0.0	-110.0	55.0	-280.0	51.0	38.0	38.0
Main battle tank	174	Spoke	Cube	68.4	7.0	53.0	0.0	0.0	0.0	-110.0	55.0	-280.0	53.0	7.0	68.4
Main battle tank	175	Tyre	Torus	76.0	76.0	150.0	0.0	0.0	0.0	-110.0	55.0	280.0	150.0	76.0	76.0

Main battle tank	176	Hub	Cylinder	38.0	38.0	51.0	0.0	0.0	0.0	-110.0	55.0	280.0	51.0	38.0	38.0
Main battle tank	177	Spoke	Cube	68.4	7.0	53.0	0.0	0.0	0.0	-110.0	55.0	280.0	53.0	7.0	68.4
Main battle tank	178	Return roller	Cylinder	18.0	18.0	20.0	0.0	90.0	0.0	-110.0	90.0	-140.0	20.0	18.0	18.0
Main battle tank	179	Return roller	Cylinder	18.0	18.0	20.0	0.0	90.0	0.0	-110.0	90.0	0.0	20.0	18.0	18.0
Main battle tank	180	Return roller	Cylinder	18.0	18.0	20.0	0.0	90.0	0.0	-110.0	90.0	140.0	20.0	18.0	18.0
Main battle tank	181	Turret body	Tapered box	230.0	70.0	280.0	0.0	0.0	0.0	0.0	177.0	-25.0	230.0	70.0	280.0
Main battle tank	182	Turret cheek	Cube	100.0	66.0	120.0	0.0	-22.0	0.0	95.0	176.0	100.0	137.7	66.0	148.7
Main battle tank	183	Turret cheek	Cube	100.0	66.0	120.0	0.0	22.0	0.0	-95.0	176.0	100.0	137.7	66.0	148.7
Main battle tank	184	Turret front plate	Cube	100.0	62.0	40.0	0.0	0.0	0.0	0.0	176.0	135.0	100.0	62.0	40.0
Main battle tank	185	Turret roof	Cube	200.0	10.0	220.0	0.0	0.0	0.0	0.0	214.0	-25.0	200.0	10.0	220.0
Main battle tank	186	Bustle	Cube	220.0	62.0	130.0	0.0	0.0	0.0	0.0	176.0	-210.0	220.0	62.0	130.0
Main battle tank	187	Bustle roof	Cube	220.0	8.0	130.0	6.0	0.0	0.0	0.0	210.0	-210.0	220.0	21.5	130.1
Main battle tank	188	Stowage basket	Cube	230.0	30.0	8.0	0.0	0.0	0.0	0.0	192.0	-280.0	230.0	30.0	8.0
Main battle tank	189	Bedroll	Cylinder	20.0	20.0	100.0	0.0	90.0	0.0	50.0	204.0	-287.0	100.0	20.0	20.0
Main battle tank	190	Bedroll	Cylinder	20.0	20.0	100.0	0.0	90.0	0.0	-50.0	204.0	-287.0	100.0	20.0	20.0
Main battle tank	191	Gunner sight box	Cube	45.0	28.0	55.0	0.0	0.0	0.0	-55.0	232.0	55.0	45.0	28.0	55.0
Main battle tank	192	Gunner sight lens	Sphere	18.0	18.0	12.0	0.0	0.0	0.0	-55.0	234.0	85.0	18.0	18.0	12.0
Main battle tank	193	Cupola ring	Torus	85.0	85.0	66.7	90.0	0.0	0.0	65.0	228.0	-70.0	85.0	66.7	85.0
Main battle tank	194	Cupola vision block	Cube	16.0	10.0	10.0	0.0	0.0	0.0	65.0	237.0	-30.0	16.0	10.0	10.0
Main battle tank	195	Cupola vision block	Cube	16.0	10.0	10.0	0.0	60.0	0.0	99.6	237.0	-50.0	16.7	10.0	18.9
Main battle tank	196	Cupola vision block	Cube	16.0	10.0	10.0	0.0	120.0	0.0	99.6	237.0	-90.0	16.7	10.0	18.9
Main battle tank	197	Cupola vision block	Cube	16.0	10.0	10.0	0.0	180.0	0.0	65.0	237.0	-110.0	16.0	10.0	10.0
Main battle tank	198	Cupola vision block	Cube	16.0	10.0	10.0	0.0	240.0	0.0	30.4	237.0	-90.0	16.7	10.0	18.9
Main battle tank	199	Cupola vision block	Cube	16.0	10.0	10.0	0.0	300.0	0.0	30.4	237.0	-50.0	16.7	10.0	18.9
Main battle tank	200	Loader hatch	Cylinder	75.0	75.0	8.0	90.0	0.0	0.0	-65.0	218.0	-90.0	75.0	8.0	75.0
Main battle tank	201	Loader hatch handle	Cube	30.0	5.0	5.0	0.0	0.0	0.0	-65.0	224.0	-90.0	30.0	5.0	5.0
Main battle tank	202	MG pintle	Cylinder	6.0	6.0	20.0	90.0	0.0	0.0	65.0	240.0	-70.0	6.0	20.0	6.0
Main battle tank	203	MG receiver	Cube	12.0	16.0	70.0	0.0	0.0	0.0	65.0	250.0	-45.0	12.0	16.0	70.0
Main battle tank	204	MG barrel	Cylinder	4.0	4.0	70.0	0.0	0.0	0.0	65.0	250.0	15.0	4.0	4.0	70.0
Main battle tank	205	MG flash hider	Cylinder	7.0	7.0	10.0	0.0	0.0	0.0	65.0	250.0	52.0	7.0	7.0	10.0
Main battle tank	206	MG ammo box	Cube	22.0	18.0	30.0	0.0	0.0	0.0	48.0	247.0	-60.0	22.0	18.0	30.0
Main battle tank	207	Smoke launcher mount	Cube	12.0	20.0	60.0	0.0	0.0	0.0	-118.0	192.0	45.0	12.0	20.0	60.0
Main battle tank	208	Smoke grenade tube	Cylinder	9.0	9.0	22.0	-55.0	-30.0	0.0	-122.0	204.0	25.0	17.8	23.2	21.8
Main battle tank	209	Smoke grenade tube	Cylinder	9.0	9.0	22.0	-55.0	-30.0	0.0	-122.0	204.0	39.0	17.8	23.2	21.8
Main battle tank	210	Smoke grenade tube	Cylinder	9.0	9.0	22.0	-55.0	-30.0	0.0	-122.0	204.0	53.0	17.8	23.2	21.8
Main battle tank	211	Smoke grenade tube	Cylinder	9.0	9.0	22.0	-55.0	-30.0	0.0	-122.0	204.0	67.0	17.8	23.2	21.8
Main battle tank	212	Smoke launcher mount	Cube	12.0	20.0	60.0	0.0	0.0	0.0	118.0	192.0	45.0	12.0	20.0	60.0
Main battle tank	213	Smoke grenade tube	Cylinder	9.0	9.0	22.0	-55.0	30.0	0.0	122.0	204.0	25.0	17.8	23.2	21.8
Main battle tank	214	Smoke grenade tube	Cylinder	9.0	9.0	22.0	-55.0	30.0	0.0	122.0	204.0	39.0	17.8	23.2	21.8
Main battle tank	215	Smoke grenade tube	Cylinder	9.0	9.0	22.0	-55.0	30.0	0.0	122.0	204.0	53.0	17.8	23.2	21.8
Main battle tank	216	Smoke grenade tube	Cylinder	9.0	9.0	22.0	-55.0	30.0	0.0	122.0	204.0	67.0	17.8	23.2	21.8
Main battle tank	217	Antenna mount	Cylinder	16.0	16.0	12.0	90.0	0.0	0.0	-90.0	214.0	-215.0	16.0	12.0	16.0
Main battle tank	218	Whip antenna	Cylinder	3.0	3.0	200.0	-90.0	0.0	0.0	-90.0	320.0	-215.0	3.0	200.0	3.0
Main battle tank	219	Antenna mount	Cylinder	16.0	16.0	12.0	90.0	0.0	0.0	90.0	214.0	-235.0	16.0	12.0	16.0
Main battle tank	220	Whip antenna	Cylinder	3.0	3.0	140.0	-90.0	0.0	0.0	90.0	290.0	-235.0	3.0	140.0	3.0
Main battle tank	221	Gun mantlet	Cube	80.0	58.0	45.0	0.0	0.0	0.0	0.0	184.0	140.0	80.0	58.0	45.0
Main battle tank	222	Coaxial MG	Cube	9.0	9.0	50.0	0.0	0.0	0.0	38.0	172.0	170.0	9.0	9.0	50.0
Main battle tank	223	Gun barrel	Cylinder	17.0	17.0	420.0	0.0	0.0	0.0	0.0	184.0	375.0	17.0	17.0	420.0
Main battle tank	224	Thermal sleeve	Cylinder	24.0	24.0	150.0	0.0	0.0	0.0	0.0	184.0	265.0	24.0	24.0	150.0
Main battle tank	225	Fume extractor	Cylinder	27.0	27.0	50.0	0.0	0.0	0.0	0.0	184.0	435.0	27.0	27.0	50.0
Main battle tank	226	Muzzle brake	Cylinder	23.0	23.0	34.0	0.0	0.0	0.0	0.0	184.0	595.0	23.0	23.0	34.0

Main battle tank	227	Muzzle flash	Sphere	50.0	50.0	90.0	0.0	0.0	0.0	0.0	184.0	615.0	0.0	0.0	0.0
Main battle tank	228	Muzzle bore	Cylinder	12.0	12.0	2.0	0.0	0.0	0.0	0.0	184.0	612.5	12.0	12.0	2.0
Main battle tank	229	Muzzle reference sensor	Cube	10.0	7.0	14.0	0.0	0.0	0.0	0.0	197.0	565.0	10.0	7.0	14.0
Main battle tank	230	Side armour block	Cube	9.0	46.0	60.0	0.0	0.0	0.0	-120.0	178.0	-47.0	9.0	46.0	60.0
Main battle tank	231	Side armour block	Cube	9.0	46.0	60.0	0.0	0.0	0.0	-120.0	178.0	-113.0	9.0	46.0	60.0
Main battle tank	232	Turret roof edge	Cylinder	6.0	6.0	270.0	0.0	0.0	0.0	-114.0	212.0	-25.0	6.0	6.0	270.0
Main battle tank	233	Turret number	Cube	1.2	13.0	38.0	0.0	0.0	0.0	-110.6	180.0	-210.0	1.2	13.0	38.0
Main battle tank	234	Side armour block	Cube	9.0	46.0	60.0	0.0	0.0	0.0	120.0	178.0	-47.0	9.0	46.0	60.0
Main battle tank	235	Side armour block	Cube	9.0	46.0	60.0	0.0	0.0	0.0	120.0	178.0	-113.0	9.0	46.0	60.0
Main battle tank	236	Turret roof edge	Cylinder	6.0	6.0	270.0	0.0	0.0	0.0	114.0	212.0	-25.0	6.0	6.0	270.0
Main battle tank	237	Turret number	Cube	1.2	13.0	38.0	0.0	0.0	0.0	110.6	180.0	-210.0	1.2	13.0	38.0
Main battle tank	238	Bustle rear edge	Cylinder	6.0	6.0	220.0	0.0	90.0	0.0	0.0	208.0	-275.0	220.0	6.0	6.0
Main battle tank	239	Panoramic sight post	Cylinder	16.0	16.0	30.0	90.0	0.0	0.0	5.0	232.0	20.0	16.0	30.0	16.0
Main battle tank	240	Panoramic sight head	Cube	32.0	22.0	30.0	0.0	0.0	0.0	5.0	254.0	20.0	32.0	22.0	30.0
Main battle tank	241	Panoramic sight window	Cube	24.0	12.0	2.0	0.0	0.0	0.0	5.0	255.0	36.0	24.0	12.0	2.0
Main battle tank	242	Wind sensor mast	Cylinder	3.0	3.0	42.0	90.0	0.0	0.0	0.0	240.0	-160.0	3.0	42.0	3.0
Main battle tank	243	Wind sensor head	Sphere	7.0	7.0	7.0	0.0	0.0	0.0	0.0	262.0	-160.0	7.0	7.0	7.0
Main battle tank	244	Loader hatch hinge	Cylinder	6.0	6.0	40.0	0.0	90.0	0.0	-65.0	222.0	-127.0	40.0	6.0	6.0
Main battle tank	245	Deck edge	Cylinder	5.0	5.0	450.0	0.0	0.0	0.0	-145.0	142.0	-50.0	5.0	5.0	450.0
Main battle tank	246	Tow cable	Cylinder	5.0	5.0	240.0	0.0	0.0	0.0	-149.0	130.0	-120.0	5.0	5.0	240.0
Main battle tank	247	Skirt rubber flap	Cube	5.0	14.0	495.0	0.0	0.0	0.0	-147.0	64.0	-12.0	5.0	14.0	495.0
Main battle tank	248	Jerry can	Cube	34.0	46.0	16.0	0.0	0.0	0.0	-45.0	82.0	-305.0	34.0	46.0	16.0
Main battle tank	249	Headlight guard	Cube	3.0	30.0	3.0	4.0	0.0	0.0	-115.0	130.0	320.0	3.0	30.1	5.1
Main battle tank	250	Deck edge	Cylinder	5.0	5.0	450.0	0.0	0.0	0.0	145.0	142.0	-50.0	5.0	5.0	450.0
Main battle tank	251	Tow cable	Cylinder	5.0	5.0	240.0	0.0	0.0	0.0	149.0	130.0	-120.0	5.0	5.0	240.0
Main battle tank	252	Skirt rubber flap	Cube	5.0	14.0	495.0	0.0	0.0	0.0	147.0	64.0	-12.0	5.0	14.0	495.0
Main battle tank	253	Jerry can	Cube	34.0	46.0	16.0	0.0	0.0	0.0	45.0	82.0	-305.0	34.0	46.0	16.0
Main battle tank	254	Headlight guard	Cube	3.0	30.0	3.0	4.0	0.0	0.0	115.0	130.0	320.0	3.0	30.1	5.1
Main battle tank	255	Glacis top edge	Cylinder	7.0	7.0	250.0	0.0	90.0	0.0	0.0	142.0	180.0	250.0	7.0	7.0
Main battle tank	256	Convoy marker	Cube	16.0	5.0	2.0	0.0	0.0	0.0	0.0	128.0	-297.0	16.0	5.0	2.0
MLRS rocket truck	2	Chassis	Cube	110.0	32.0	800.0	0.0	0.0	0.0	0.0	95.0	0.0	110.0	32.0	800.0
MLRS rocket truck	3	Cab	Cube	245.0	160.0	190.0	0.0	0.0	0.0	0.0	195.0	300.0	245.0	160.0	190.0
MLRS rocket truck	4	Cab roof armour	Cube	250.0	10.0	195.0	0.0	0.0	0.0	0.0	280.0	300.0	250.0	10.0	195.0
MLRS rocket truck	5	Windscreen	Cube	210.0	70.0	5.0	0.0	0.0	0.0	0.0	230.0	397.0	210.0	70.0	5.0
MLRS rocket truck	6	Bumper	Cube	240.0	25.0	30.0	0.0	0.0	0.0	0.0	100.0	410.0	240.0	25.0	30.0
MLRS rocket truck	7	Headlight	Sphere	22.0	22.0	10.0	0.0	0.0	0.0	-80.0	140.0	398.0	22.0	22.0	10.0
MLRS rocket truck	8	Headlight	Sphere	22.0	22.0	10.0	0.0	0.0	0.0	80.0	140.0	398.0	22.0	22.0	10.0
MLRS rocket truck	9	Mirror	Cube	6.0	35.0	20.0	0.0	0.0	0.0	-135.0	230.0	380.0	6.0	35.0	20.0
MLRS rocket truck	10	Mirror	Cube	6.0	35.0	20.0	0.0	0.0	0.0	135.0	230.0	380.0	6.0	35.0	20.0
MLRS rocket truck	11	Cab tactical lamp	Sphere	10.0	10.0	5.0	0.0	0.0	0.0	90.0	115.0	402.0	10.0	10.0	5.0
MLRS rocket truck	12	Bed deck	Cube	250.0	20.0	500.0	0.0	0.0	0.0	0.0	125.0	-160.0	250.0	20.0	500.0
MLRS rocket truck	13	Bed side	Cube	10.0	30.0	500.0	0.0	0.0	0.0	-120.0	150.0	-160.0	10.0	30.0	500.0
MLRS rocket truck	14	Bed side	Cube	10.0	30.0	500.0	0.0	0.0	0.0	120.0	150.0	-160.0	10.0	30.0	500.0
MLRS rocket truck	15	Wheel	Cylinder	115.0	115.0	45.0	0.0	90.0	0.0	-115.0	57.5	310.0	45.0	115.0	115.0
MLRS rocket truck	16	Wheel hub	Cylinder	51.8	51.8	49.0	0.0	90.0	0.0	-115.0	57.5	310.0	49.0	51.7	51.8
MLRS rocket truck	17	Wheel	Cylinder	115.0	115.0	45.0	0.0	90.0	0.0	115.0	57.5	310.0	45.0	115.0	115.0
MLRS rocket truck	18	Wheel hub	Cylinder	51.8	51.8	49.0	0.0	90.0	0.0	115.0	57.5	310.0	49.0	51.7	51.8
MLRS rocket truck	19	Wheel	Cylinder	115.0	115.0	45.0	0.0	90.0	0.0	-115.0	57.5	-160.0	45.0	115.0	115.0
MLRS rocket truck	20	Wheel hub	Cylinder	51.8	51.8	49.0	0.0	90.0	0.0	-115.0	57.5	-160.0	49.0	51.7	51.8
MLRS rocket truck	21	Wheel	Cylinder	115.0	115.0	45.0	0.0	90.0	0.0	115.0	57.5	-160.0	45.0	115.0	115.0
MLRS rocket truck	22	Wheel hub	Cylinder	51.8	51.8	49.0	0.0	90.0	0.0	115.0	57.5	-160.0	49.0	51.7	51.8

MLRS rocket truck	23	Wheel	Cylinder	115.0	115.0	45.0	0.0	90.0	0.0	-115.0	57.5	-300.0	45.0	115.0	115.0
MLRS rocket truck	24	Wheel hub	Cylinder	51.8	51.8	49.0	0.0	90.0	0.0	-115.0	57.5	-300.0	49.0	51.7	51.8
MLRS rocket truck	25	Wheel	Cylinder	115.0	115.0	45.0	0.0	90.0	0.0	115.0	57.5	-300.0	45.0	115.0	115.0
MLRS rocket truck	26	Wheel hub	Cylinder	51.8	51.8	49.0	0.0	90.0	0.0	115.0	57.5	-300.0	49.0	51.7	51.8
MLRS rocket truck	27	Fender	Cube	50.0	8.0	130.0	0.0	0.0	0.0	-115.0	120.0	310.0	50.0	8.0	130.0
MLRS rocket truck	28	Fender	Cube	50.0	8.0	240.0	0.0	0.0	0.0	-115.0	120.0	-230.0	50.0	8.0	240.0
MLRS rocket truck	29	Outrigger	Cylinder	10.0	10.0	113.3	56.1	-108.4	0.0	-150.0	53.0	-390.0	71.0	99.6	32.1
MLRS rocket truck	30	Outrigger pad	Cube	40.0	6.0	40.0	0.0	0.0	0.0	-180.0	3.0	-400.0	40.0	6.0	40.0
MLRS rocket truck	31	Fender	Cube	50.0	8.0	130.0	0.0	0.0	0.0	115.0	120.0	310.0	50.0	8.0	130.0
MLRS rocket truck	32	Fender	Cube	50.0	8.0	240.0	0.0	0.0	0.0	115.0	120.0	-230.0	50.0	8.0	240.0
MLRS rocket truck	33	Outrigger	Cylinder	10.0	10.0	113.3	56.1	108.4	0.0	150.0	53.0	-390.0	71.0	99.6	32.1
MLRS rocket truck	34	Outrigger pad	Cube	40.0	6.0	40.0	0.0	0.0	0.0	180.0	3.0	-400.0	40.0	6.0	40.0
MLRS rocket truck	35	Cab side window	Cube	2.0	50.0	80.0	0.0	0.0	0.0	-122.8	230.0	305.0	2.0	50.0	80.0
MLRS rocket truck	36	Cab roof edge	Cylinder	5.0	5.0	194.0	0.0	0.0	0.0	-125.0	285.0	300.0	5.0	5.0	194.0
MLRS rocket truck	37	Cab side window	Cube	2.0	50.0	80.0	0.0	0.0	0.0	122.8	230.0	305.0	2.0	50.0	80.0
MLRS rocket truck	38	Cab roof edge	Cylinder	5.0	5.0	194.0	0.0	0.0	0.0	125.0	285.0	300.0	5.0	5.0	194.0
MLRS rocket truck	39	Radiator grille	Cube	110.0	42.0	2.0	0.0	0.0	0.0	0.0	150.0	396.0	110.0	42.0	2.0
MLRS rocket truck	40	Exhaust stack	Cylinder	12.0	12.0	90.0	90.0	0.0	0.0	100.0	245.0	195.0	12.0	90.0	12.0
MLRS rocket truck	41	Tyre tread	Torus	118.5	118.5	30.0	0.0	90.0	0.0	-127.6	57.5	310.0	30.0	118.4	118.4
MLRS rocket truck	42	Tyre tread	Torus	118.5	118.5	30.0	0.0	90.0	0.0	-102.4	57.5	310.0	30.0	118.4	118.4
MLRS rocket truck	43	Tyre tread	Torus	118.5	118.5	30.0	0.0	90.0	0.0	102.4	57.5	310.0	30.0	118.4	118.4
MLRS rocket truck	44	Tyre tread	Torus	118.5	118.5	30.0	0.0	90.0	0.0	127.6	57.5	310.0	30.0	118.4	118.4
MLRS rocket truck	45	Tyre tread	Torus	118.5	118.5	30.0	0.0	90.0	0.0	-127.6	57.5	-160.0	30.0	118.4	118.4
MLRS rocket truck	46	Tyre tread	Torus	118.5	118.5	30.0	0.0	90.0	0.0	-102.4	57.5	-160.0	30.0	118.4	118.4
MLRS rocket truck	47	Tyre tread	Torus	118.5	118.5	30.0	0.0	90.0	0.0	102.4	57.5	-160.0	30.0	118.4	118.4
MLRS rocket truck	48	Tyre tread	Torus	118.5	118.5	30.0	0.0	90.0	0.0	127.6	57.5	-160.0	30.0	118.4	118.4
MLRS rocket truck	49	Tyre tread	Torus	118.5	118.5	30.0	0.0	90.0	0.0	-127.6	57.5	-300.0	30.0	118.4	118.4
MLRS rocket truck	50	Tyre tread	Torus	118.5	118.5	30.0	0.0	90.0	0.0	-102.4	57.5	-300.0	30.0	118.4	118.4
MLRS rocket truck	51	Tyre tread	Torus	118.5	118.5	30.0	0.0	90.0	0.0	102.4	57.5	-300.0	30.0	118.4	118.4
MLRS rocket truck	52	Tyre tread	Torus	118.5	118.5	30.0	0.0	90.0	0.0	127.6	57.5	-300.0	30.0	118.4	118.4
MLRS rocket truck	53	Slew ring	Cylinder	170.0	170.0	20.0	90.0	0.0	0.0	0.0	145.0	-160.0	170.0	20.0	170.0
MLRS rocket truck	54	Mount base	Cube	190.0	35.0	190.0	0.0	0.0	0.0	0.0	173.0	-160.0	190.0	35.0	190.0
MLRS rocket truck	55	Pod arm	Cube	22.0	140.0	60.0	0.0	0.0	0.0	-122.0	240.0	-160.0	22.0	140.0	60.0
MLRS rocket truck	56	Arm brace	Cube	10.0	10.0	110.0	0.0	0.0	0.0	-122.0	185.0	-140.0	10.0	10.0	110.0
MLRS rocket truck	57	Pod arm	Cube	22.0	140.0	60.0	0.0	0.0	0.0	122.0	240.0	-160.0	22.0	140.0	60.0
MLRS rocket truck	58	Arm brace	Cube	10.0	10.0	110.0	0.0	0.0	0.0	122.0	185.0	-140.0	10.0	10.0	110.0
MLRS rocket truck	59	Ram foot	Cube	30.0	12.0	30.0	0.0	0.0	0.0	0.0	187.0	-250.0	30.0	12.0	30.0
MLRS rocket truck	60	Ram cylinder	Cylinder	16.0	16.0	31.6	-58.5	0.0	0.0	0.0	204.5	-241.7	16.0	35.3	30.1
MLRS rocket truck	61	Ram rod	Cylinder	9.0	9.0	28.7	-58.5	0.0	0.0	0.0	227.7	-227.5	9.0	29.2	22.7
MLRS rocket truck	62	Pod side plate	Cube	8.0	90.0	360.0	0.0	0.0	0.0	-117.0	285.0	-60.0	8.0	90.0	360.0
MLRS rocket truck	63	Trunnion	Cylinder	18.0	18.0	30.0	0.0	90.0	0.0	-120.0	285.0	-160.0	30.0	18.0	18.0
MLRS rocket truck	64	Pod side plate	Cube	8.0	90.0	360.0	0.0	0.0	0.0	117.0	285.0	-60.0	8.0	90.0	360.0
MLRS rocket truck	65	Trunnion	Cylinder	18.0	18.0	30.0	0.0	90.0	0.0	120.0	285.0	-160.0	30.0	18.0	18.0
MLRS rocket truck	66	Pod frame	Cube	230.0	8.0	360.0	0.0	0.0	0.0	0.0	241.0	-60.0	230.0	8.0	360.0
MLRS rocket truck	67	Pod frame	Cube	230.0	8.0	360.0	0.0	0.0	0.0	0.0	329.0	-60.0	230.0	8.0	360.0
MLRS rocket truck	68	Pod rear plate	Cube	230.0	90.0	8.0	0.0	0.0	0.0	0.0	285.0	-244.0	230.0	90.0	8.0
MLRS rocket truck	69	Ram lug	Cube	20.0	20.0	20.0	0.0	0.0	0.0	0.0	240.0	-220.0	20.0	20.0	20.0
MLRS rocket truck	70	Launch tube	Cylinder	30.0	30.0	360.0	0.0	0.0	0.0	-85.0	302.0	-60.0	30.0	30.0	360.0
MLRS rocket truck	71	Muzzle ring	Torus	34.0	34.0	26.7	0.0	0.0	0.0	-85.0	302.0	114.0	34.0	34.0	26.7
MLRS rocket truck	72	Tube cap	Cylinder	31.0	31.0	4.0	0.0	0.0	0.0	-85.0	302.0	121.0	31.0	31.0	4.0
MLRS rocket truck	73	Launch tube	Cylinder	30.0	30.0	360.0	0.0	0.0	0.0	-51.0	302.0	-60.0	30.0	30.0	360.0

MLRS rocket truck	74	Muzzle ring	Torus	34.0	34.0	26.7	0.0	0.0	0.0	-51.0	302.0	114.0	34.0	34.0	26.7
MLRS rocket truck	75	Tube cap	Cylinder	31.0	31.0	4.0	0.0	0.0	0.0	-51.0	302.0	121.0	31.0	31.0	4.0
MLRS rocket truck	76	Launch tube	Cylinder	30.0	30.0	360.0	0.0	0.0	0.0	-17.0	302.0	-60.0	30.0	30.0	360.0
MLRS rocket truck	77	Muzzle ring	Torus	34.0	34.0	26.7	0.0	0.0	0.0	-17.0	302.0	114.0	34.0	34.0	26.7
MLRS rocket truck	78	Tube cap	Cylinder	31.0	31.0	4.0	0.0	0.0	0.0	-17.0	302.0	121.0	31.0	31.0	4.0
MLRS rocket truck	79	Launch tube	Cylinder	30.0	30.0	360.0	0.0	0.0	0.0	17.0	302.0	-60.0	30.0	30.0	360.0
MLRS rocket truck	80	Muzzle ring	Torus	34.0	34.0	26.7	0.0	0.0	0.0	17.0	302.0	114.0	34.0	34.0	26.7
MLRS rocket truck	81	Tube cap	Cylinder	31.0	31.0	4.0	0.0	0.0	0.0	17.0	302.0	121.0	31.0	31.0	4.0
MLRS rocket truck	82	Launch tube	Cylinder	30.0	30.0	360.0	0.0	0.0	0.0	51.0	302.0	-60.0	30.0	30.0	360.0
MLRS rocket truck	83	Muzzle ring	Torus	34.0	34.0	26.7	0.0	0.0	0.0	51.0	302.0	114.0	34.0	34.0	26.7
MLRS rocket truck	84	Tube cap	Cylinder	31.0	31.0	4.0	0.0	0.0	0.0	51.0	302.0	121.0	31.0	31.0	4.0
MLRS rocket truck	85	Launch tube	Cylinder	30.0	30.0	360.0	0.0	0.0	0.0	85.0	302.0	-60.0	30.0	30.0	360.0
MLRS rocket truck	86	Muzzle ring	Torus	34.0	34.0	26.7	0.0	0.0	0.0	85.0	302.0	114.0	34.0	34.0	26.7
MLRS rocket truck	87	Tube cap	Cylinder	31.0	31.0	4.0	0.0	0.0	0.0	85.0	302.0	121.0	31.0	31.0	4.0
MLRS rocket truck	88	Launch tube	Cylinder	30.0	30.0	360.0	0.0	0.0	0.0	-85.0	268.0	-60.0	30.0	30.0	360.0
MLRS rocket truck	89	Muzzle ring	Torus	34.0	34.0	26.7	0.0	0.0	0.0	-85.0	268.0	114.0	34.0	34.0	26.7
MLRS rocket truck	90	Tube cap	Cylinder	31.0	31.0	4.0	0.0	0.0	0.0	-85.0	268.0	121.0	31.0	31.0	4.0
MLRS rocket truck	91	Launch tube	Cylinder	30.0	30.0	360.0	0.0	0.0	0.0	-51.0	268.0	-60.0	30.0	30.0	360.0
MLRS rocket truck	92	Muzzle ring	Torus	34.0	34.0	26.7	0.0	0.0	0.0	-51.0	268.0	114.0	34.0	34.0	26.7
MLRS rocket truck	93	Tube cap	Cylinder	31.0	31.0	4.0	0.0	0.0	0.0	-51.0	268.0	121.0	31.0	31.0	4.0
MLRS rocket truck	94	Launch tube	Cylinder	30.0	30.0	360.0	0.0	0.0	0.0	-17.0	268.0	-60.0	30.0	30.0	360.0
MLRS rocket truck	95	Muzzle ring	Torus	34.0	34.0	26.7	0.0	0.0	0.0	-17.0	268.0	114.0	34.0	34.0	26.7
MLRS rocket truck	96	Tube cap	Cylinder	31.0	31.0	4.0	0.0	0.0	0.0	-17.0	268.0	121.0	31.0	31.0	4.0
MLRS rocket truck	97	Launch tube	Cylinder	30.0	30.0	360.0	0.0	0.0	0.0	17.0	268.0	-60.0	30.0	30.0	360.0
MLRS rocket truck	98	Muzzle ring	Torus	34.0	34.0	26.7	0.0	0.0	0.0	17.0	268.0	114.0	34.0	34.0	26.7
MLRS rocket truck	99	Tube cap	Cylinder	31.0	31.0	4.0	0.0	0.0	0.0	17.0	268.0	121.0	31.0	31.0	4.0
MLRS rocket truck	100	Launch tube	Cylinder	30.0	30.0	360.0	0.0	0.0	0.0	51.0	268.0	-60.0	30.0	30.0	360.0
MLRS rocket truck	101	Muzzle ring	Torus	34.0	34.0	26.7	0.0	0.0	0.0	51.0	268.0	114.0	34.0	34.0	26.7
MLRS rocket truck	102	Tube cap	Cylinder	31.0	31.0	4.0	0.0	0.0	0.0	51.0	268.0	121.0	31.0	31.0	4.0
MLRS rocket truck	103	Launch tube	Cylinder	30.0	30.0	360.0	0.0	0.0	0.0	85.0	268.0	-60.0	30.0	30.0	360.0
MLRS rocket truck	104	Muzzle ring	Torus	34.0	34.0	26.7	0.0	0.0	0.0	85.0	268.0	114.0	34.0	34.0	26.7
MLRS rocket truck	105	Tube cap	Cylinder	31.0	31.0	4.0	0.0	0.0	0.0	85.0	268.0	121.0	31.0	31.0	4.0
MLRS rocket truck	106	Pod stencil	Cube	0.6	16.0	160.0	0.0	0.0	0.0	-121.3	285.0	-40.0	0.6	16.0	160.0
MLRS rocket truck	107	Pod stencil	Cube	0.6	16.0	160.0	0.0	0.0	0.0	121.3	285.0	-40.0	0.6	16.0	160.0
Scout helicopter	2	Cabin	Sphere	135.0	145.0	250.0	0.0	0.0	0.0	0.0	118.0	0.0	135.0	145.0	250.0
Scout helicopter	3	Canopy glass	Sphere	122.0	105.0	150.0	0.0	0.0	0.0	0.0	136.0	62.0	122.0	105.0	150.0
Scout helicopter	4	Chin nose	Sphere	90.0	50.0	90.0	0.0	0.0	0.0	0.0	78.0	80.0	90.0	50.0	90.0
Scout helicopter	5	Door seam left	Cube	3.0	75.0	2.0	0.0	0.0	0.0	65.5	114.0	38.0	3.0	75.0	2.0
Scout helicopter	6	Door seam right	Cube	3.0	75.0	2.0	0.0	0.0	0.0	-65.5	114.0	38.0	3.0	75.0	2.0
Scout helicopter	7	Engine cowling	Sphere	95.0	90.0	250.0	0.0	0.0	0.0	0.0	195.0	-90.0	95.0	90.0	250.0
Scout helicopter	8	Engine exhaust	Cylinder	20.0	20.0	55.0	15.0	0.0	0.0	-35.0	200.0	-210.0	20.0	33.6	58.3
Scout helicopter	9	Rotor mast	Cylinder	34.0	34.0	87.0	90.0	0.0	0.0	0.0	238.5	0.0	34.0	87.0	34.0
Scout helicopter	10	Tail boom	Cone	45.0	45.0	520.0	0.0	180.0	0.0	0.0	180.0	-390.0	45.0	45.0	520.0
Scout helicopter	11	Tail fin	Cube	7.0	105.0	95.0	0.0	0.0	0.0	0.0	220.0	-500.0	7.0	105.0	95.0
Scout helicopter	12	Horizontal stabiliser	Cube	160.0	5.0	40.0	0.0	0.0	0.0	0.0	185.0	-430.0	160.0	5.0	40.0
Scout helicopter	13	Stabiliser plate left	Cube	4.0	50.0	40.0	0.0	0.0	0.0	80.0	185.0	-430.0	4.0	50.0	40.0
Scout helicopter	14	Stabiliser plate right	Cube	4.0	50.0	40.0	0.0	0.0	0.0	-80.0	185.0	-430.0	4.0	50.0	40.0
Scout helicopter	15	Tail bumper	Cube	5.0	55.0	12.0	0.0	0.0	0.0	0.0	150.0	-505.0	5.0	55.0	12.0
Scout helicopter	16	Skid right	Cylinder	10.0	10.0	300.0	0.0	0.0	0.0	-95.0	5.0	10.0	10.0	10.0	300.0
Scout helicopter	17	Skid toe right	Cylinder	10.0	10.0	50.0	-25.0	0.0	0.0	-95.0	12.0	168.0	10.0	30.2	49.5
Scout helicopter	18	Front strut right	Cylinder	7.0	7.0	73.5	-46.1	101.3	0.0	-70.0	31.5	85.0	56.3	57.9	17.9

Scout helicopter	19	Rear strut right	Cylinder	7.0	7.0	73.5	-46.1	78.7	0.0	-70.0	31.5	-75.0	56.3	57.9	17.9
Scout helicopter	20	Skid left	Cylinder	10.0	10.0	300.0	0.0	0.0	0.0	95.0	5.0	10.0	10.0	10.0	300.0
Scout helicopter	21	Skid toe left	Cylinder	10.0	10.0	50.0	-25.0	0.0	0.0	95.0	12.0	168.0	10.0	30.2	49.5
Scout helicopter	22	Front strut left	Cylinder	7.0	7.0	73.5	-46.1	-101.3	0.0	70.0	31.5	85.0	56.3	57.9	17.9
Scout helicopter	23	Rear strut left	Cylinder	7.0	7.0	73.5	-46.1	-78.7	0.0	70.0	31.5	-75.0	56.3	57.9	17.9
Scout helicopter	24	Rotor hub	Cylinder	45.0	45.0	20.0	90.0	0.0	0.0	0.0	282.0	0.0	45.0	20.0	45.0
Scout helicopter	25	Rotor hub cap	Sphere	30.0	22.0	30.0	0.0	0.0	0.0	0.0	293.0	0.0	30.0	22.0	30.0
Scout helicopter	26	Blade	Cube	22.0	3.0	390.0	0.0	0.0	0.0	0.0	256.2	223.5	22.1	51.6	387.6
Scout helicopter	27	Blade grip	Cube	12.0	8.0	35.0	0.0	0.0	0.0	0.0	278.2	44.9	12.4	12.8	35.8
Scout helicopter	28	Blade tip cap	Cube	23.0	3.5	30.0	0.0	0.0	0.0	0.0	234.3	402.2	23.2	8.3	30.3
Scout helicopter	29	Blade erosion strip	Cube	3.5	3.6	345.0	0.0	0.0	0.0	10.0	256.1	228.6	3.7	45.8	342.9
Scout helicopter	30	Blade	Cube	22.0	3.0	390.0	0.0	0.0	0.0	212.6	256.2	69.1	375.4	51.6	140.7
Scout helicopter	31	Blade grip	Cube	12.0	8.0	35.0	0.0	0.0	0.0	42.7	278.2	13.9	37.6	12.8	22.8
Scout helicopter	32	Blade tip cap	Cube	23.0	3.5	30.0	0.0	0.0	0.0	382.5	234.3	124.3	35.9	8.3	31.3
Scout helicopter	33	Blade erosion strip	Cube	3.5	3.6	345.0	0.0	0.0	0.0	220.5	256.1	61.1	327.1	45.8	109.4
Scout helicopter	34	Blade	Cube	22.0	3.0	390.0	0.0	0.0	0.0	131.4	256.2	-180.9	245.6	51.6	326.4
Scout helicopter	35	Blade grip	Cube	12.0	8.0	35.0	0.0	0.0	0.0	26.4	278.2	-36.3	31.0	12.8	35.8
Scout helicopter	36	Blade tip cap	Cube	23.0	3.5	30.0	0.0	0.0	0.0	236.4	234.3	-325.4	36.4	8.3	37.9
Scout helicopter	37	Blade erosion strip	Cube	3.5	3.6	345.0	0.0	0.0	0.0	126.3	256.1	-190.8	204.5	45.8	279.3
Scout helicopter	38	Blade	Cube	22.0	3.0	390.0	0.0	0.0	0.0	-131.4	256.2	-180.9	245.5	51.6	326.4
Scout helicopter	39	Blade grip	Cube	12.0	8.0	35.0	0.0	0.0	0.0	-26.4	278.2	-36.3	30.4	12.8	36.1
Scout helicopter	40	Blade tip cap	Cube	23.0	3.5	30.0	0.0	0.0	0.0	-236.4	234.3	-325.4	36.3	8.3	37.9
Scout helicopter	41	Blade erosion strip	Cube	3.5	3.6	345.0	0.0	0.0	0.0	-142.4	256.1	-179.0	204.2	45.8	279.5
Scout helicopter	42	Blade	Cube	22.0	3.0	390.0	0.0	0.0	0.0	-212.6	256.2	69.1	375.2	51.6	140.6
Scout helicopter	43	Blade grip	Cube	12.0	8.0	35.0	0.0	0.0	0.0	-42.7	278.2	13.9	37.7	12.8	22.3
Scout helicopter	44	Blade tip cap	Cube	23.0	3.5	30.0	0.0	0.0	0.0	-382.5	234.3	124.3	35.7	8.3	31.1
Scout helicopter	45	Blade erosion strip	Cube	3.5	3.6	345.0	0.0	0.0	0.0	-214.3	256.1	80.1	327.2	45.8	109.2
Scout helicopter	46	Main rotor blur disc	Cylinder	840.0	840.0	2.0	90.0	0.0	0.0	0.0	292.0	0.0	0.1	0.1	0.1
Scout helicopter	47	Tail rotor hub	Cylinder	12.0	12.0	10.0	0.0	90.0	0.0	20.0	230.0	-510.0	10.0	12.0	12.0
Scout helicopter	48	Tail blade	Cube	3.0	13.0	55.0	0.0	0.0	0.0	20.0	230.0	-480.0	3.0	13.0	55.0
Scout helicopter	49	Tail blade	Cube	3.0	13.0	55.0	0.0	0.0	0.0	20.0	230.0	-540.0	3.0	13.0	55.0
Scout helicopter	50	Tail rotor blur disc	Cylinder	120.0	120.0	1.0	0.0	90.0	0.0	23.0	230.0	-510.0	0.1	0.1	0.1
Scout helicopter	51	Nav light left (red)	Sphere	14.0	14.0	14.0	0.0	0.0	0.0	66.0	90.0	55.0	14.0	14.0	14.0
Scout helicopter	52	Nav light right (green)	Sphere	14.0	14.0	14.0	0.0	0.0	0.0	-66.0	90.0	55.0	14.0	14.0	14.0
Scout helicopter	53	Nav light tail (white)	Sphere	12.0	12.0	12.0	0.0	0.0	0.0	0.0	195.0	-552.0	12.0	12.0	12.0
Scout helicopter	54	Anti-collision strobe	Sphere	16.0	12.0	16.0	0.0	0.0	0.0	0.0	278.0	-500.0	16.0	12.0	16.0
Scout helicopter	55	Searchlight yoke	Cube	30.0	8.0	30.0	0.0	0.0	0.0	0.0	47.0	90.0	30.0	8.0	30.0
Scout helicopter	56	Searchlight housing	Cylinder	26.0	26.0	30.0	0.0	0.0	0.0	0.0	30.0	90.0	26.0	39.5	38.5
Scout helicopter	57	Searchlight lens	Sphere	22.0	22.0	4.0	0.0	0.0	0.0	0.0	16.9	99.2	22.0	15.9	20.3
Scout helicopter	58	Searchlight cone	Cylinder (frustum)	100.0	100.0	100.0	0.0	0.0	0.0	0.0	13.6	101.5	0.1	0.1	0.1
Scout helicopter	59	Swashplate	Cylinder	50.0	50.0	8.0	90.0	0.0	0.0	0.0	242.0	0.0	50.0	8.0	50.0
Scout helicopter	60	Pitch link	Cylinder	3.0	3.0	48.7	-80.5	20.0	0.0	8.9	266.0	24.4	6.6	48.5	11.3
Scout helicopter	61	Pitch link	Cylinder	3.0	3.0	48.7	-80.5	92.0	0.0	26.0	266.0	-0.9	11.1	48.5	3.4
Scout helicopter	62	Pitch link	Cylinder	3.0	3.0	48.7	-80.5	164.0	0.0	7.2	266.0	-25.0	5.9	48.5	11.4
Scout helicopter	63	Pitch link	Cylinder	3.0	3.0	48.7	-80.5	-124.0	0.0	-21.6	266.0	-14.5	10.8	48.5	8.6
Scout helicopter	64	Pitch link	Cylinder	3.0	3.0	48.7	-80.5	-52.0	0.0	-20.5	266.0	16.0	10.5	48.5	9.1
Scout helicopter	65	Bench seat right	Cube	35.0	5.0	130.0	0.0	0.0	0.0	-98.0	75.0	5.0	35.0	5.0	130.0
Scout helicopter	66	Bench back right	Cube	5.0	45.0	130.0	0.0	0.0	0.0	-80.0	100.0	5.0	5.0	45.0	130.0
Scout helicopter	67	Bench strut right	Cylinder	5.0	5.0	67.0	-90.0	0.0	0.0	-95.0	38.5	5.0	5.0	67.0	5.0
Scout helicopter	68	Bench seat left	Cube	35.0	5.0	130.0	0.0	0.0	0.0	98.0	75.0	5.0	35.0	5.0	130.0
Scout helicopter	69	Bench back left	Cube	5.0	45.0	130.0	0.0	0.0	0.0	80.0	100.0	5.0	5.0	45.0	130.0

Scout helicopter	70	Bench strut left	Cylinder	5.0	5.0	67.0	-90.0	0.0	0.0	95.0	38.5	5.0	5.0	67.0	5.0
Scout helicopter	71	Wire cutter	Cylinder	4.0	4.0	85.0	-61.9	180.0	0.0	0.0	172.5	110.0	4.0	76.9	43.5
Scout helicopter	72	Whip antenna	Cylinder	2.0	2.0	121.2	-65.1	168.7	0.0	30.0	235.0	-265.0	12.3	110.8	52.2
Scout helicopter	73	Tail rotor gearbox	Cylinder	20.0	20.0	30.0	0.0	90.0	0.0	10.0	230.0	-510.0	30.0	20.0	20.0
Scout helicopter	74	Pitot tube	Cylinder	3.0	3.0	40.0	0.0	0.0	0.0	-50.0	110.0	145.0	3.0	3.0	40.0
Scout helicopter	75	Canopy centre frame	Sphere	4.0	107.0	152.0	0.0	0.0	0.0	0.0	136.0	62.0	4.0	107.0	152.0
Scout helicopter	76	Canopy sill frame	Sphere	124.5	4.0	152.5	0.0	0.0	0.0	0.0	136.0	62.0	124.5	4.0	152.5
Scout helicopter	77	Cabin panel seam	Sphere	125.5	134.5	3.0	0.0	0.0	0.0	0.0	118.0	-50.0	125.5	134.5	3.0
Scout helicopter	78	Door window right	Sphere	10.0	50.0	70.0	0.0	0.0	0.0	-63.0	136.0	-8.0	10.0	50.0	70.0
Scout helicopter	79	Engine intake right	Sphere	10.0	22.0	42.0	0.0	0.0	0.0	-44.0	205.0	-80.0	10.0	22.0	42.0
Scout helicopter	80	Tail number right	Cube	1.0	11.0	42.0	0.0	-2.4	0.0	-17.8	180.0	-245.0	2.8	11.0	42.0
Scout helicopter	81	Formation light right	Cube	1.2	3.0	50.0	0.0	-2.4	0.0	-16.0	190.0	-300.0	3.3	3.0	50.0
Scout helicopter	82	Skid end cap right	Sphere	10.0	10.0	10.0	0.0	0.0	0.0	-95.0	5.0	-140.0	10.0	10.0	10.0
Scout helicopter	83	Door window left	Sphere	10.0	50.0	70.0	0.0	0.0	0.0	63.0	136.0	-8.0	10.0	50.0	70.0
Scout helicopter	84	Engine intake left	Sphere	10.0	22.0	42.0	0.0	0.0	0.0	44.0	205.0	-80.0	10.0	22.0	42.0
Scout helicopter	85	Tail number left	Cube	1.0	11.0	42.0	0.0	2.4	0.0	17.8	180.0	-245.0	2.8	11.0	42.0
Scout helicopter	86	Formation light left	Cube	1.2	3.0	50.0	0.0	2.4	0.0	16.0	190.0	-300.0	3.3	3.0	50.0
Scout helicopter	87	Skid end cap left	Sphere	10.0	10.0	10.0	0.0	0.0	0.0	95.0	5.0	-140.0	10.0	10.0	10.0
Scout helicopter	88	FLIR ball	Sphere	24.0	24.0	24.0	0.0	0.0	0.0	0.0	80.0	128.0	24.0	24.0	24.0
Scout helicopter	89	FLIR window	Sphere	13.0	13.0	5.0	0.0	0.0	0.0	0.0	80.0	139.0	13.0	13.0	5.0
Scout helicopter	90	Drive shaft cover	Cylinder	7.0	7.0	330.0	2.5	180.0	0.0	0.0	194.5	-315.0	7.0	21.3	330.0
Scout helicopter	91	Belly antenna	Cube	2.0	18.0	25.0	0.0	0.0	0.0	0.0	38.0	-45.0	2.0	18.0	25.0
Scout helicopter	92	VHF antenna	Cube	2.0	20.0	20.0	0.0	0.0	0.0	0.0	235.0	-170.0	2.0	20.0	20.0
Autonomous UAV	2	Fuselage	Cube	16.0	7.0	26.0	0.0	0.0	0.0	0.0	0.0	0.0	16.0	7.0	26.0
Autonomous UAV	3	Canopy	Sphere	15.0	8.0	22.0	0.0	0.0	0.0	0.0	4.0	1.0	15.0	8.0	22.0
Autonomous UAV	4	Battery	Cube	10.0	4.0	16.0	0.0	0.0	0.0	0.0	-5.5	-2.0	10.0	4.0	16.0
Autonomous UAV	5	GPS mast	Cylinder	0.8	0.8	5.0	90.0	0.0	0.0	0.0	7.0	-7.0	0.8	5.0	0.8
Autonomous UAV	6	GPS puck	Cylinder	5.0	5.0	1.2	90.0	0.0	0.0	0.0	9.8	-7.0	5.0	1.2	5.0
Autonomous UAV	7	Arm Front-left	Cube	2.5	1.8	22.0	0.0	45.0	0.0	10.6	0.0	10.6	17.3	1.8	17.3
Autonomous UAV	8	Motor	Cylinder	4.5	4.5	3.5	90.0	0.0	0.0	18.4	3.2	18.4	4.5	3.5	4.5
Autonomous UAV	9	Prop hub	Sphere	2.5	2.5	2.5	0.0	0.0	0.0	18.4	5.7	18.4	2.5	2.5	2.5
Autonomous UAV	10	Prop blade	Cube	20.0	0.4	1.8	0.0	0.0	0.0	18.4	5.7	18.4	20.0	0.4	1.8
Autonomous UAV	11	Rotor blur disc	Cylinder	20.5	20.5	0.2	90.0	0.0	0.0	18.4	5.7	18.4	20.5	0.2	20.5
Autonomous UAV	12	Nav LED	Sphere	1.8	1.8	1.8	0.0	0.0	0.0	18.4	0.0	18.4	1.8	1.8	1.8
Autonomous UAV	13	Motor cooling ring	Torus	5.0	5.0	3.3	90.0	0.0	0.0	18.4	1.6	18.4	5.0	3.3	5.0
Autonomous UAV	14	Arm Front-right	Cube	2.5	1.8	22.0	0.0	-45.0	0.0	-10.6	0.0	10.6	17.3	1.8	17.3
Autonomous UAV	15	Motor	Cylinder	4.5	4.5	3.5	90.0	0.0	0.0	-18.4	3.2	18.4	4.5	3.5	4.5
Autonomous UAV	16	Prop hub	Sphere	2.5	2.5	2.5	0.0	0.0	0.0	-18.4	5.7	18.4	3.5	2.5	3.5
Autonomous UAV	17	Prop blade	Cube	20.0	0.4	1.8	0.0	0.0	0.0	-18.4	5.7	18.4	17.1	0.4	13.5
Autonomous UAV	18	Rotor blur disc	Cylinder	20.5	20.5	0.2	90.0	0.0	0.0	-18.4	5.7	18.4	20.5	0.2	20.5
Autonomous UAV	19	Nav LED	Sphere	1.8	1.8	1.8	0.0	0.0	0.0	-18.4	0.0	18.4	1.8	1.8	1.8
Autonomous UAV	20	Motor cooling ring	Torus	5.0	5.0	3.3	90.0	0.0	0.0	-18.4	1.6	18.4	5.0	3.3	5.0
Autonomous UAV	21	Arm Rear-left	Cube	2.5	1.8	22.0	0.0	135.0	0.0	10.6	0.0	-10.6	17.3	1.8	17.3
Autonomous UAV	22	Motor	Cylinder	4.5	4.5	3.5	90.0	0.0	0.0	18.4	3.2	-18.4	4.5	3.5	4.5
Autonomous UAV	23	Prop hub	Sphere	2.5	2.5	2.5	0.0	0.0	0.0	18.4	5.7	-18.4	3.1	2.5	3.1
Autonomous UAV	24	Prop blade	Cube	20.0	0.4	1.8	0.0	0.0	0.0	18.4	5.7	-18.4	7.2	0.4	19.7
Autonomous UAV	25	Rotor blur disc	Cylinder	20.5	20.5	0.2	90.0	0.0	0.0	18.4	5.7	-18.4	20.5	0.2	20.5
Autonomous UAV	26	Motor cooling ring	Torus	5.0	5.0	3.3	90.0	0.0	0.0	18.4	1.6	-18.4	5.0	3.3	5.0
Autonomous UAV	27	Status LED	Sphere	1.6	1.6	1.6	0.0	0.0	0.0	18.4	0.0	-18.4	1.6	1.6	1.6
Autonomous UAV	28	Arm Rear-right	Cube	2.5	1.8	22.0	0.0	-135.0	0.0	-10.6	0.0	-10.6	17.3	1.8	17.3
Autonomous UAV	29	Motor	Cylinder	4.5	4.5	3.5	90.0	0.0	0.0	-18.4	3.2	-18.4	4.5	3.5	4.5

Autonomous UAV	30	Prop hub	Sphere	2.5	2.5	2.5	0.0	0.0	0.0	-18.4	5.7	-18.4	3.2	2.5	3.2
Autonomous UAV	31	Prop blade	Cube	20.0	0.4	1.8	0.0	0.0	0.0	-18.4	5.7	-18.4	8.8	0.4	19.3
Autonomous UAV	32	Rotor blur disc	Cylinder	20.5	20.5	0.2	90.0	0.0	0.0	-18.4	5.7	-18.4	20.5	0.2	20.5
Autonomous UAV	33	Motor cooling ring	Torus	5.0	5.0	3.3	90.0	0.0	0.0	-18.4	1.6	-18.4	5.0	3.3	5.0
Autonomous UAV	34	Status LED	Sphere	1.6	1.6	1.6	0.0	0.0	0.0	-18.4	0.0	-18.4	1.6	1.6	1.6
Autonomous UAV	35	Strobe	Sphere	2.2	1.6	2.2	0.0	0.0	0.0	0.0	2.0	-13.5	2.2	1.6	2.2
Autonomous UAV	36	Landing strut	Cylinder	1.2	1.2	9.0	90.0	0.0	0.0	-6.0	-8.0	-7.0	1.2	9.0	1.2
Autonomous UAV	37	Landing strut	Cylinder	1.2	1.2	9.0	90.0	0.0	0.0	-6.0	-8.0	7.0	1.2	9.0	1.2
Autonomous UAV	38	Landing skid	Cylinder	1.4	1.4	24.0	0.0	0.0	0.0	-6.0	-12.5	0.0	1.4	1.4	24.0
Autonomous UAV	39	Landing strut	Cylinder	1.2	1.2	9.0	90.0	0.0	0.0	6.0	-8.0	-7.0	1.2	9.0	1.2
Autonomous UAV	40	Landing strut	Cylinder	1.2	1.2	9.0	90.0	0.0	0.0	6.0	-8.0	7.0	1.2	9.0	1.2
Autonomous UAV	41	Landing skid	Cylinder	1.4	1.4	24.0	0.0	0.0	0.0	6.0	-12.5	0.0	1.4	1.4	24.0
Autonomous UAV	42	Gimbal yoke	Cube	7.0	3.0	5.0	0.0	0.0	0.0	0.0	-7.5	10.0	7.0	3.0	5.0
Autonomous UAV	43	Lamp housing	Cylinder	6.0	6.0	8.0	0.0	0.0	0.0	0.0	-11.0	10.0	6.0	10.0	9.7
Autonomous UAV	44	Lens	Sphere	5.0	5.0	1.5	0.0	0.0	0.0	0.0	-14.4	12.9	5.0	4.4	4.8
Autonomous UAV	45	Thermal camera	Cube	3.0	2.5	3.5	0.0	0.0	0.0	0.0	-8.3	13.2	3.0	4.3	4.2
Autonomous UAV	46	Light cone	Cylinder (frustum)	485.9	485.9	974.4	0.0	0.0	0.0	0.0	-387.7	326.1	485.9	1058.7	998.5
Autonomous UAV	47	Camera window	Sphere	1.8	1.8	0.6	0.0	0.0	0.0	0.0	-9.7	14.4	1.8	1.6	1.8
Autonomous UAV	48	Rail	Cube	1.2	1.2	22.0	0.0	0.0	0.0	10.5	-6.3	0.0	1.2	1.2	22.0
Autonomous UAV	49	Rail strut	Cube	0.8	2.0	0.8	0.0	0.0	0.0	10.5	-5.0	5.0	0.8	2.0	0.8
Autonomous UAV	50	Rail	Cube	1.2	1.2	22.0	0.0	0.0	0.0	-10.5	-6.3	0.0	1.2	1.2	22.0
Autonomous UAV	51	Rail strut	Cube	0.8	2.0	0.8	0.0	0.0	0.0	-10.5	-5.0	5.0	0.8	2.0	0.8
Autonomous UAV	52	Telemetry antenna	Cylinder	0.5	0.5	9.0	-35.0	0.0	0.0	-5.0	-4.0	-13.0	0.5	5.6	7.7
Autonomous UAV	53	Telemetry antenna	Cylinder	0.5	0.5	9.0	-35.0	0.0	0.0	5.0	-4.0	-13.0	0.5	5.6	7.7
Autonomous UAV	54	Battery strap	Cube	10.6	4.4	1.2	0.0	0.0	0.0	0.0	-5.5	-7.0	10.6	4.4	1.2
Autonomous UAV	55	Battery strap	Cube	10.6	4.4	1.2	0.0	0.0	0.0	0.0	-5.5	3.0	10.6	4.4	1.2
Autonomous UAV	56	Arming LED	Sphere	1.2	0.8	1.2	0.0	0.0	0.0	4.0	7.5	-3.0	1.2	0.8	1.2
Cargo truck	2	Chassis	Cube	100.0	30.0	700.0	0.0	0.0	0.0	0.0	75.0	-30.0	100.0	30.0	700.0
Cargo truck	3	Hood	Cube	200.0	95.0	160.0	0.0	0.0	0.0	0.0	145.0	290.0	200.0	95.0	160.0
Cargo truck	4	Cab	Cube	235.0	155.0	150.0	0.0	0.0	0.0	0.0	185.0	135.0	235.0	155.0	150.0
Cargo truck	5	Windscreen	Cube	200.0	70.0	4.0	0.0	0.0	0.0	0.0	210.0	212.0	200.0	70.0	4.0
Cargo truck	6	Bumper	Cube	230.0	25.0	30.0	0.0	0.0	0.0	0.0	85.0	380.0	230.0	25.0	30.0
Cargo truck	7	Cargo bed	Cube	240.0	20.0	420.0	0.0	0.0	0.0	0.0	125.0	-165.0	240.0	20.0	420.0
Cargo truck	8	Bed side	Cube	8.0	55.0	420.0	0.0	0.0	0.0	-120.0	160.0	-165.0	8.0	55.0	420.0
Cargo truck	9	Bed side	Cube	8.0	55.0	420.0	0.0	0.0	0.0	120.0	160.0	-165.0	8.0	55.0	420.0
Cargo truck	10	Canvas cover	Cylinder	230.0	190.0	400.0	0.0	0.0	0.0	0.0	195.0	-165.0	230.0	190.0	400.0
Cargo truck	11	Wheel	Cylinder	105.0	105.0	40.0	0.0	90.0	0.0	-110.0	53.0	275.0	40.0	105.0	105.0
Cargo truck	12	Wheel	Cylinder	105.0	105.0	40.0	0.0	90.0	0.0	110.0	53.0	275.0	40.0	105.0	105.0
Cargo truck	13	Wheel	Cylinder	105.0	105.0	40.0	0.0	90.0	0.0	-110.0	53.0	-120.0	40.0	105.0	105.0
Cargo truck	14	Wheel	Cylinder	105.0	105.0	40.0	0.0	90.0	0.0	-110.0	53.0	-270.0	40.0	105.0	105.0
Cargo truck	15	Fender	Cube	25.0	10.0	130.0	0.0	0.0	0.0	-110.0	112.0	275.0	25.0	10.0	130.0
Cargo truck	16	Wheel	Cylinder	105.0	105.0	40.0	0.0	90.0	0.0	110.0	53.0	-120.0	40.0	105.0	105.0
Cargo truck	17	Wheel	Cylinder	105.0	105.0	40.0	0.0	90.0	0.0	110.0	53.0	-270.0	40.0	105.0	105.0
Cargo truck	18	Fender	Cube	25.0	10.0	130.0	0.0	0.0	0.0	110.0	112.0	275.0	25.0	10.0	130.0
Cargo truck	19	Headlight	Sphere	22.0	22.0	10.0	0.0	0.0	0.0	-70.0	155.0	372.0	22.0	22.0	10.0
Cargo truck	20	Headlight	Sphere	22.0	22.0	10.0	0.0	0.0	0.0	70.0	155.0	372.0	22.0	22.0	10.0
Cargo truck	21	Wheel hub	Cylinder	45.0	45.0	42.0	0.0	90.0	0.0	-110.0	53.0	275.0	42.0	45.0	45.0
Cargo truck	22	Wheel hub	Cylinder	45.0	45.0	42.0	0.0	90.0	0.0	110.0	53.0	275.0	42.0	45.0	45.0
Cargo truck	23	Wheel hub	Cylinder	45.0	45.0	42.0	0.0	90.0	0.0	-110.0	53.0	-120.0	42.0	45.0	45.0
Cargo truck	24	Wheel hub	Cylinder	45.0	45.0	42.0	0.0	90.0	0.0	110.0	53.0	-120.0	42.0	45.0	45.0
Cargo truck	25	Wheel hub	Cylinder	45.0	45.0	42.0	0.0	90.0	0.0	-110.0	53.0	-270.0	42.0	45.0	45.0

Cargo truck	26	Wheel hub	Cylinder	45.0	45.0	42.0	0.0	90.0	0.0	110.0	53.0	-270.0	42.0	45.0	45.0
Cargo truck	27	Cab side window	Cube	2.0	50.0	70.0	0.0	0.0	0.0	-118.0	220.0	140.0	2.0	50.0	70.0
Cargo truck	28	Mirror	Cube	5.0	30.0	18.0	0.0	0.0	0.0	-132.0	220.0	195.0	5.0	30.0	18.0
Cargo truck	29	Cab side window	Cube	2.0	50.0	70.0	0.0	0.0	0.0	118.0	220.0	140.0	2.0	50.0	70.0
Cargo truck	30	Mirror	Cube	5.0	30.0	18.0	0.0	0.0	0.0	132.0	220.0	195.0	5.0	30.0	18.0
Cargo truck	31	Radiator grille	Cube	100.0	60.0	2.0	0.0	0.0	0.0	0.0	145.0	371.0	100.0	60.0	2.0
Cargo truck	32	Exhaust stack	Cylinder	12.0	12.0	100.0	90.0	0.0	0.0	105.0	250.0	50.0	12.0	100.0	12.0
Target rover	2	Chassis	Cube	42.0	10.0	62.0	0.0	0.0	0.0	0.0	14.0	0.0	42.0	10.0	62.0
Target rover	3	Upper hull	Cube	34.0	10.0	40.0	0.0	0.0	0.0	0.0	24.0	-4.0	34.0	10.0	40.0
Target rover	4	Motor + battery box	Cube	22.0	7.0	16.0	0.0	0.0	0.0	0.0	32.5	-14.0	22.0	7.0	16.0
Target rover	5	Front bumper	Cube	44.0	5.0	4.0	0.0	0.0	0.0	0.0	12.0	33.0	44.0	5.0	4.0
Target rover	6	Wheel	Cylinder	16.0	16.0	7.0	0.0	90.0	0.0	25.0	8.0	-22.0	7.0	16.0	16.0
Target rover	7	Wheel	Cylinder	16.0	16.0	7.0	0.0	90.0	0.0	-25.0	8.0	-22.0	7.0	16.0	16.0
Target rover	8	Hub	Sphere	7.0	7.0	3.0	0.0	90.0	0.0	29.0	8.0	-22.0	3.0	7.0	7.0
Target rover	9	Hub	Sphere	7.0	7.0	3.0	0.0	90.0	0.0	-29.0	8.0	-22.0	3.0	7.0	7.0
Target rover	10	Wheel	Cylinder	16.0	16.0	7.0	0.0	90.0	0.0	25.0	8.0	0.0	7.0	16.0	16.0
Target rover	11	Wheel	Cylinder	16.0	16.0	7.0	0.0	90.0	0.0	-25.0	8.0	0.0	7.0	16.0	16.0
Target rover	12	Hub	Sphere	7.0	7.0	3.0	0.0	90.0	0.0	29.0	8.0	0.0	3.0	7.0	7.0
Target rover	13	Hub	Sphere	7.0	7.0	3.0	0.0	90.0	0.0	-29.0	8.0	0.0	3.0	7.0	7.0
Target rover	14	Wheel	Cylinder	16.0	16.0	7.0	0.0	90.0	0.0	25.0	8.0	22.0	7.0	16.0	16.0
Target rover	15	Wheel	Cylinder	16.0	16.0	7.0	0.0	90.0	0.0	-25.0	8.0	22.0	7.0	16.0	16.0
Target rover	16	Hub	Sphere	7.0	7.0	3.0	0.0	90.0	0.0	29.0	8.0	22.0	3.0	7.0	7.0
Target rover	17	Hub	Sphere	7.0	7.0	3.0	0.0	90.0	0.0	-29.0	8.0	22.0	3.0	7.0	7.0
Target rover	18	IR sensor bar	Cube	30.0	2.0	4.0	0.0	0.0	0.0	0.0	7.0	30.0	30.0	2.0	4.0
Target rover	19	IR sensor	Sphere	2.5	2.5	2.5	0.0	0.0	0.0	-12.0	5.5	31.0	2.5	2.5	2.5
Target rover	20	IR sensor	Sphere	2.5	2.5	2.5	0.0	0.0	0.0	-6.0	5.5	31.0	2.5	2.5	2.5
Target rover	21	IR sensor	Sphere	2.5	2.5	2.5	0.0	0.0	0.0	0.0	5.5	31.0	2.5	2.5	2.5
Target rover	22	IR sensor	Sphere	2.5	2.5	2.5	0.0	0.0	0.0	6.0	5.5	31.0	2.5	2.5	2.5
Target rover	23	IR sensor	Sphere	2.5	2.5	2.5	0.0	0.0	0.0	12.0	5.5	31.0	2.5	2.5	2.5
Target rover	24	Sensor mast	Cylinder	2.5	2.5	16.0	90.0	0.0	0.0	8.0	37.0	10.0	2.5	16.0	2.5
Target rover	25	Scanner head	Sphere	7.0	7.0	7.0	0.0	0.0	0.0	8.0	46.0	10.0	7.0	7.0	7.0
Target rover	26	Scanner window	Cube	5.0	3.0	2.0	0.0	0.0	0.0	8.0	46.0	13.5	5.0	3.0	2.0
Target rover	27	Whip antenna	Cylinder	1.2	1.2	30.0	90.0	0.0	0.0	-13.0	44.0	-20.0	1.2	30.0	1.2
SAM launcher	2	Trailer frame	Cube	130.0	28.0	680.0	0.0	0.0	0.0	0.0	86.0	0.0	130.0	28.0	680.0
SAM launcher	3	Trailer side rail	Cube	12.0	30.0	680.0	0.0	0.0	0.0	-110.0	90.0	0.0	12.0	30.0	680.0
SAM launcher	4	Trailer side rail	Cube	12.0	30.0	680.0	0.0	0.0	0.0	110.0	90.0	0.0	12.0	30.0	680.0
SAM launcher	5	Trailer deck	Cube	260.0	10.0	620.0	0.0	0.0	0.0	0.0	109.0	0.0	260.0	10.0	620.0
SAM launcher	6	Wheel	Cylinder	100.0	100.0	40.0	0.0	90.0	0.0	-125.0	50.0	-170.0	40.0	100.0	100.0
SAM launcher	7	Wheel hub	Cylinder	45.0	45.0	44.0	0.0	90.0	0.0	-125.0	50.0	-170.0	44.0	45.0	45.0
SAM launcher	8	Wheel	Cylinder	100.0	100.0	40.0	0.0	90.0	0.0	125.0	50.0	-170.0	40.0	100.0	100.0
SAM launcher	9	Wheel hub	Cylinder	45.0	45.0	44.0	0.0	90.0	0.0	125.0	50.0	-170.0	44.0	45.0	45.0
SAM launcher	10	Wheel	Cylinder	100.0	100.0	40.0	0.0	90.0	0.0	-125.0	50.0	-290.0	40.0	100.0	100.0
SAM launcher	11	Wheel hub	Cylinder	45.0	45.0	44.0	0.0	90.0	0.0	-125.0	50.0	-290.0	44.0	45.0	45.0
SAM launcher	12	Wheel	Cylinder	100.0	100.0	40.0	0.0	90.0	0.0	125.0	50.0	-290.0	40.0	100.0	100.0
SAM launcher	13	Wheel hub	Cylinder	45.0	45.0	44.0	0.0	90.0	0.0	125.0	50.0	-290.0	44.0	45.0	45.0
SAM launcher	14	Fender	Cube	50.0	8.0	210.0	0.0	0.0	0.0	-125.0	105.0	-230.0	50.0	8.0	210.0
SAM launcher	15	Fender	Cube	50.0	8.0	210.0	0.0	0.0	0.0	125.0	105.0	-230.0	50.0	8.0	210.0
SAM launcher	16	Tow bar	Cylinder	12.0	12.0	191.6	7.5	180.0	0.0	0.0	77.5	-435.0	12.0	36.9	191.6
SAM launcher	17	Tow eye	Sphere	22.0	22.0	22.0	0.0	0.0	0.0	0.0	65.0	-535.0	22.0	22.0	22.0
SAM launcher	18	Landing leg	Cylinder	10.0	10.0	85.0	90.0	0.0	0.0	0.0	47.5	-320.0	10.0	85.0	10.0
SAM launcher	19	Outrigger	Cylinder	10.0	10.0	105.6	52.7	-128.7	0.0	-155.0	48.0	-280.0	62.5	90.1	52.8

SAM launcher	20	Outrigger pad	Cube	40.0	6.0	40.0	0.0	0.0	0.0	-180.0	3.0	-300.0	40.0	6.0	40.0
SAM launcher	21	Outrigger	Cylinder	10.0	10.0	105.6	52.7	-51.3	0.0	-155.0	48.0	280.0	62.5	90.1	52.8
SAM launcher	22	Outrigger pad	Cube	40.0	6.0	40.0	0.0	0.0	0.0	-180.0	3.0	300.0	40.0	6.0	40.0
SAM launcher	23	Outrigger	Cylinder	10.0	10.0	105.6	52.7	128.7	0.0	155.0	48.0	-280.0	62.5	90.1	52.8
SAM launcher	24	Outrigger pad	Cube	40.0	6.0	40.0	0.0	0.0	0.0	180.0	3.0	-300.0	40.0	6.0	40.0
SAM launcher	25	Outrigger	Cylinder	10.0	10.0	105.6	52.7	51.3	0.0	155.0	48.0	280.0	62.5	90.1	52.8
SAM launcher	26	Outrigger pad	Cube	40.0	6.0	40.0	0.0	0.0	0.0	180.0	3.0	300.0	40.0	6.0	40.0
SAM launcher	27	Control cabin	Cube	180.0	90.0	140.0	0.0	0.0	0.0	0.0	159.0	250.0	180.0	90.0	140.0
SAM launcher	28	Cabin door	Cube	60.0	70.0	4.0	0.0	0.0	0.0	40.0	155.0	321.0	60.0	70.0	4.0
SAM launcher	29	Cabin louvre	Cube	40.0	4.0	3.0	0.0	0.0	0.0	-55.0	140.0	321.0	40.0	4.0	3.0
SAM launcher	30	Cabin louvre	Cube	40.0	4.0	3.0	0.0	0.0	0.0	-55.0	152.0	321.0	40.0	4.0	3.0
SAM launcher	31	Cabin louvre	Cube	40.0	4.0	3.0	0.0	0.0	0.0	-55.0	164.0	321.0	40.0	4.0	3.0
SAM launcher	32	Cabin exhaust	Cylinder	10.0	10.0	50.0	90.0	0.0	0.0	70.0	225.0	210.0	10.0	50.0	10.0
SAM launcher	33	Cabin tactical lamp	Sphere	10.0	10.0	5.0	0.0	0.0	0.0	-40.0	190.0	322.0	10.0	10.0	5.0
SAM launcher	34	Power cable	Cylinder	7.0	7.0	90.0	-1.3	180.0	0.0	0.0	121.0	135.0	7.0	9.0	90.2
SAM launcher	35	Status panel	Cube	1.2	26.0	42.0	0.0	0.0	0.0	-90.6	172.0	250.0	1.2	26.0	42.0
SAM launcher	36	Status LED	Sphere	4.0	4.0	4.0	0.0	0.0	0.0	-91.2	192.0	238.0	4.0	4.0	4.0
SAM launcher	37	Status LED	Sphere	4.0	4.0	4.0	0.0	0.0	0.0	-91.2	192.0	262.0	4.0	4.0	4.0
SAM launcher	38	Generator grille	Cube	1.2	42.0	60.0	0.0	0.0	0.0	90.6	150.0	250.0	1.2	42.0	60.0
SAM launcher	39	Cabin roof edge	Cylinder	5.0	5.0	140.0	0.0	0.0	0.0	-90.0	204.0	250.0	5.0	5.0	140.0
SAM launcher	40	Cabin roof edge	Cylinder	5.0	5.0	140.0	0.0	0.0	0.0	90.0	204.0	250.0	5.0	5.0	140.0
SAM launcher	41	Tyre tread	Torus	103.0	103.0	26.7	0.0	90.0	0.0	-136.2	50.0	-170.0	26.7	103.0	103.0
SAM launcher	42	Tyre tread	Torus	103.0	103.0	26.7	0.0	90.0	0.0	-113.8	50.0	-170.0	26.7	103.0	103.0
SAM launcher	43	Tyre tread	Torus	103.0	103.0	26.7	0.0	90.0	0.0	113.8	50.0	-170.0	26.7	103.0	103.0
SAM launcher	44	Tyre tread	Torus	103.0	103.0	26.7	0.0	90.0	0.0	136.2	50.0	-170.0	26.7	103.0	103.0
SAM launcher	45	Tyre tread	Torus	103.0	103.0	26.7	0.0	90.0	0.0	-136.2	50.0	-290.0	26.7	103.0	103.0
SAM launcher	46	Tyre tread	Torus	103.0	103.0	26.7	0.0	90.0	0.0	-113.8	50.0	-290.0	26.7	103.0	103.0
SAM launcher	47	Tyre tread	Torus	103.0	103.0	26.7	0.0	90.0	0.0	113.8	50.0	-290.0	26.7	103.0	103.0
SAM launcher	48	Tyre tread	Torus	103.0	103.0	26.7	0.0	90.0	0.0	136.2	50.0	-290.0	26.7	103.0	103.0
SAM launcher	49	Slew ring	Cylinder	190.0	190.0	22.0	90.0	0.0	0.0	0.0	125.0	40.0	190.0	22.0	190.0
SAM launcher	50	Slew gear tooth	Cube	10.0	12.0	10.0	0.0	0.0	0.0	0.0	125.0	138.0	10.0	12.0	10.0
SAM launcher	51	Slew gear tooth	Cube	10.0	12.0	10.0	0.0	-30.0	0.0	49.0	125.0	124.9	13.7	12.0	13.7
SAM launcher	52	Slew gear tooth	Cube	10.0	12.0	10.0	0.0	-60.0	0.0	84.9	125.0	89.0	13.7	12.0	13.7
SAM launcher	53	Slew gear tooth	Cube	10.0	12.0	10.0	0.0	-90.0	0.0	98.0	125.0	40.0	10.0	12.0	10.0
SAM launcher	54	Slew gear tooth	Cube	10.0	12.0	10.0	0.0	-120.0	0.0	84.9	125.0	-9.0	13.7	12.0	13.7
SAM launcher	55	Slew gear tooth	Cube	10.0	12.0	10.0	0.0	-150.0	0.0	49.0	125.0	-44.9	13.7	12.0	13.7
SAM launcher	56	Slew gear tooth	Cube	10.0	12.0	10.0	0.0	-180.0	0.0	0.0	125.0	-58.0	10.0	12.0	10.0
SAM launcher	57	Slew gear tooth	Cube	10.0	12.0	10.0	0.0	-210.0	0.0	-49.0	125.0	-44.9	13.7	12.0	13.7
SAM launcher	58	Slew gear tooth	Cube	10.0	12.0	10.0	0.0	-240.0	0.0	-84.9	125.0	-9.0	13.7	12.0	13.7
SAM launcher	59	Slew gear tooth	Cube	10.0	12.0	10.0	0.0	-270.0	0.0	-98.0	125.0	40.0	10.0	12.0	10.0
SAM launcher	60	Slew gear tooth	Cube	10.0	12.0	10.0	0.0	-300.0	0.0	-84.9	125.0	89.0	13.7	12.0	13.7
SAM launcher	61	Slew gear tooth	Cube	10.0	12.0	10.0	0.0	-330.0	0.0	-49.0	125.0	124.9	13.7	12.0	13.7
SAM launcher	62	Turntable deck	Cube	190.0	12.0	300.0	0.0	0.0	0.0	0.0	142.0	30.0	190.0	12.0	300.0
SAM launcher	63	Cradle upright	Cube	14.0	140.0	50.0	0.0	0.0	0.0	-80.0	219.0	40.0	14.0	140.0	50.0
SAM launcher	64	Upright brace	Cube	10.0	10.0	90.0	0.0	0.0	0.0	-80.0	164.0	60.0	10.0	10.0	90.0
SAM launcher	65	Cradle upright	Cube	14.0	140.0	50.0	0.0	0.0	0.0	80.0	219.0	40.0	14.0	140.0	50.0
SAM launcher	66	Upright brace	Cube	10.0	10.0	90.0	0.0	0.0	0.0	80.0	164.0	60.0	10.0	10.0	90.0
SAM launcher	67	Ram foot	Cube	30.0	16.0	30.0	0.0	0.0	0.0	0.0	148.0	-90.0	30.0	16.0	30.0
SAM launcher	68	Ram cylinder	Cylinder	16.0	16.0	47.5	-45.9	0.0	0.0	0.0	169.1	-73.5	16.0	45.2	44.5
SAM launcher	69	Ram rod	Cylinder	9.0	9.0	43.1	-45.9	0.0	0.0	0.0	198.5	-45.0	9.0	37.3	36.5
SAM launcher	70	Cradle rail	Cube	10.0	10.0	340.0	0.0	0.0	0.0	-62.0	274.0	110.0	10.0	10.0	340.0

SAM launcher	71	Trunnion	Cylinder	16.0	16.0	34.0	0.0	90.0	0.0	-72.0	274.0	40.0	34.0	16.0	16.0
SAM launcher	72	Cradle rail	Cube	10.0	10.0	340.0	0.0	0.0	0.0	62.0	274.0	110.0	10.0	10.0	340.0
SAM launcher	73	Trunnion	Cylinder	16.0	16.0	34.0	0.0	90.0	0.0	72.0	274.0	40.0	34.0	16.0	16.0
SAM launcher	74	Cradle cross beam	Cube	140.0	10.0	12.0	0.0	0.0	0.0	0.0	219.0	-40.0	140.0	10.0	12.0
SAM launcher	75	Canister strap	Cube	135.0	6.0	10.0	0.0	0.0	0.0	0.0	326.0	-40.0	135.0	6.0	10.0
SAM launcher	76	Cradle cross beam	Cube	140.0	10.0	12.0	0.0	0.0	0.0	0.0	219.0	130.0	140.0	10.0	12.0
SAM launcher	77	Canister strap	Cube	135.0	6.0	10.0	0.0	0.0	0.0	0.0	326.0	130.0	135.0	6.0	10.0
SAM launcher	78	Cradle cross beam	Cube	140.0	10.0	12.0	0.0	0.0	0.0	0.0	219.0	250.0	140.0	10.0	12.0
SAM launcher	79	Canister strap	Cube	135.0	6.0	10.0	0.0	0.0	0.0	0.0	326.0	250.0	135.0	6.0	10.0
SAM launcher	80	Ram lug	Cube	20.0	20.0	20.0	0.0	0.0	0.0	0.0	214.0	-30.0	20.0	20.0	20.0
SAM launcher	81	Canister tube	Cube	44.0	44.0	340.0	0.0	0.0	0.0	-28.0	300.0	110.0	44.0	44.0	340.0
SAM launcher	82	Hazard band	Cube	46.0	46.0	18.0	0.0	0.0	0.0	-28.0	300.0	230.0	46.0	46.0	18.0
SAM launcher	83	Muzzle collar	Cube	50.0	50.0	6.0	0.0	0.0	0.0	-28.0	300.0	275.0	50.0	50.0	6.0
SAM launcher	84	Canister cap	Cube	40.0	40.0	4.0	0.0	0.0	0.0	-28.0	300.0	281.0	40.0	40.0	4.0
SAM launcher	85	Rear hatch	Cube	46.0	46.0	6.0	0.0	0.0	0.0	-28.0	300.0	-62.0	46.0	46.0	6.0
SAM launcher	86	Canister rib	Cube	47.0	47.0	7.0	0.0	0.0	0.0	-28.0	300.0	20.0	47.0	47.0	7.0
SAM launcher	87	Canister stencil	Cube	0.6	12.0	90.0	0.0	0.0	0.0	-50.3	300.0	130.0	0.6	12.0	90.0
SAM launcher	88	Canister tube	Cube	44.0	44.0	340.0	0.0	0.0	0.0	28.0	300.0	110.0	44.0	44.0	340.0
SAM launcher	89	Hazard band	Cube	46.0	46.0	18.0	0.0	0.0	0.0	28.0	300.0	230.0	46.0	46.0	18.0
SAM launcher	90	Muzzle collar	Cube	50.0	50.0	6.0	0.0	0.0	0.0	28.0	300.0	275.0	50.0	50.0	6.0
SAM launcher	91	Canister cap	Cube	40.0	40.0	4.0	0.0	0.0	0.0	28.0	300.0	281.0	40.0	40.0	4.0
SAM launcher	92	Rear hatch	Cube	46.0	46.0	6.0	0.0	0.0	0.0	28.0	300.0	-62.0	46.0	46.0	6.0
SAM launcher	93	Canister rib	Cube	47.0	47.0	7.0	0.0	0.0	0.0	28.0	300.0	20.0	47.0	47.0	7.0
SAM launcher	94	Canister stencil	Cube	0.6	12.0	90.0	0.0	0.0	0.0	50.3	300.0	130.0	0.6	12.0	90.0
SAM launcher	95	Canister tube	Cube	44.0	44.0	340.0	0.0	0.0	0.0	-28.0	248.0	110.0	44.0	44.0	340.0
SAM launcher	96	Hazard band	Cube	46.0	46.0	18.0	0.0	0.0	0.0	-28.0	248.0	230.0	46.0	46.0	18.0
SAM launcher	97	Muzzle collar	Cube	50.0	50.0	6.0	0.0	0.0	0.0	-28.0	248.0	275.0	50.0	50.0	6.0
SAM launcher	98	Canister cap	Cube	40.0	40.0	4.0	0.0	0.0	0.0	-28.0	248.0	281.0	40.0	40.0	4.0
SAM launcher	99	Rear hatch	Cube	46.0	46.0	6.0	0.0	0.0	0.0	-28.0	248.0	-62.0	46.0	46.0	6.0
SAM launcher	100	Canister rib	Cube	47.0	47.0	7.0	0.0	0.0	0.0	-28.0	248.0	20.0	47.0	47.0	7.0
SAM launcher	101	Canister stencil	Cube	0.6	12.0	90.0	0.0	0.0	0.0	-50.3	248.0	130.0	0.6	12.0	90.0
SAM launcher	102	Canister tube	Cube	44.0	44.0	340.0	0.0	0.0	0.0	28.0	248.0	110.0	44.0	44.0	340.0
SAM launcher	103	Hazard band	Cube	46.0	46.0	18.0	0.0	0.0	0.0	28.0	248.0	230.0	46.0	46.0	18.0
SAM launcher	104	Muzzle collar	Cube	50.0	50.0	6.0	0.0	0.0	0.0	28.0	248.0	275.0	50.0	50.0	6.0
SAM launcher	105	Canister cap	Cube	40.0	40.0	4.0	0.0	0.0	0.0	28.0	248.0	281.0	40.0	40.0	4.0
SAM launcher	106	Rear hatch	Cube	46.0	46.0	6.0	0.0	0.0	0.0	28.0	248.0	-62.0	46.0	46.0	6.0
SAM launcher	107	Canister rib	Cube	47.0	47.0	7.0	0.0	0.0	0.0	28.0	248.0	20.0	47.0	47.0	7.0
SAM launcher	108	Canister stencil	Cube	0.6	12.0	90.0	0.0	0.0	0.0	50.3	248.0	130.0	0.6	12.0	90.0
AA Gatling gun	2	Base plate	Cylinder	220.0	220.0	15.0	90.0	0.0	0.0	0.0	7.5	0.0	220.0	15.0	220.0
AA Gatling gun	3	Pedestal column	Cylinder	90.0	90.0	85.0	90.0	0.0	0.0	0.0	57.5	0.0	90.0	85.0	90.0
AA Gatling gun	4	Pedestal collar	Cylinder	120.0	120.0	12.0	90.0	0.0	0.0	0.0	95.0	0.0	120.0	12.0	120.0
AA Gatling gun	5	Pedestal brace	Cube	12.0	60.0	12.0	-14.0	45.0	0.0	43.8	40.0	43.8	27.0	61.1	27.0
AA Gatling gun	6	Pedestal brace	Cube	12.0	60.0	12.0	-14.0	135.0	0.0	43.8	40.0	-43.8	27.0	61.1	27.0
AA Gatling gun	7	Pedestal brace	Cube	12.0	60.0	12.0	-14.0	225.0	0.0	-43.8	40.0	-43.8	27.0	61.1	27.0
AA Gatling gun	8	Pedestal brace	Cube	12.0	60.0	12.0	-14.0	315.0	0.0	-43.8	40.0	43.8	27.0	61.1	27.0
AA Gatling gun	9	Turntable	Cylinder	150.0	150.0	20.0	90.0	0.0	0.0	0.0	115.0	0.0	211.9	20.0	211.9
AA Gatling gun	10	Mount body	Cube	140.0	50.0	140.0	0.0	0.0	0.0	0.0	150.0	0.0	197.7	50.0	197.7
AA Gatling gun	11	Cheek	Cube	15.0	90.0	110.0	0.0	0.0	0.0	-29.5	200.0	-47.5	91.6	90.0	84.9
AA Gatling gun	12	Cheek	Cube	15.0	90.0	110.0	0.0	0.0	0.0	44.3	200.0	34.1	91.6	90.0	84.9
AA Gatling gun	13	Ammo drum	Cylinder	90.0	90.0	70.0	90.0	0.0	0.0	-63.0	200.0	57.0	127.1	70.0	127.1
AA Gatling gun	14	Ammo drum lid	Cylinder	95.0	95.0	8.0	90.0	0.0	0.0	-63.0	235.0	57.0	134.2	8.0	134.2

AA Gatling gun	15	Radome mast	Cylinder	40.0	40.0	50.0	90.0	0.0	0.0	-63.0	265.0	57.0	56.5	50.0	56.5
AA Gatling gun	16	Radome	Hemisphere	80.0	60.0	80.0	0.0	0.0	0.0	-63.0	320.0	57.0	113.0	60.0	113.0
AA Gatling gun	17	Antenna mast	Cylinder	5.0	5.0	30.0	90.0	0.0	0.0	-63.0	360.0	57.0	7.1	30.0	7.1
AA Gatling gun	18	Search antenna plate	Cube	55.0	25.0	3.0	0.0	0.0	0.0	-69.7	377.0	49.6	39.1	25.0	42.8
AA Gatling gun	19	Gun housing	Cube	60.0	55.0	130.0	0.0	0.0	0.0	28.8	236.8	-26.0	144.8	96.1	139.1
AA Gatling gun	20	Trunnion	Cylinder	25.0	25.0	20.0	0.0	90.0	0.0	-18.7	230.0	-50.5	37.2	32.0	36.3
AA Gatling gun	21	Trunnion	Cylinder	25.0	25.0	20.0	0.0	90.0	0.0	48.4	230.0	23.7	37.2	32.0	36.3
AA Gatling gun	22	Front shroud	Cylinder	50.0	50.0	40.0	0.0	0.0	0.0	81.0	262.5	-73.3	74.1	60.7	73.8
AA Gatling gun	23	Feed chute	Cube	25.0	12.0	90.0	0.0	0.0	0.0	-12.4	184.7	11.2	82.5	42.1	78.0
AA Gatling gun	24	Ejection port	Cube	10.0	12.0	20.0	0.0	0.0	0.0	-2.7	244.7	-40.7	23.7	18.1	22.8
AA Gatling gun	25	EO sensor pod	Cylinder	20.0	20.0	18.0	0.0	0.0	0.0	82.9	286.8	-18.4	31.0	25.0	30.8
AA Gatling gun	26	EO sensor lens	Sphere	14.0	14.0	8.0	0.0	0.0	0.0	89.9	290.3	-24.7	18.5	15.9	18.6
AA Gatling gun	27	Gatling barrel	Cylinder	7.0	7.0	200.0	0.0	0.0	0.0	161.1	316.7	-145.8	145.8	75.0	132.9
AA Gatling gun	28	Gatling barrel	Cylinder	7.0	7.0	200.0	0.0	0.0	0.0	154.7	310.1	-156.3	145.8	75.0	132.9
AA Gatling gun	29	Gatling barrel	Cylinder	7.0	7.0	200.0	0.0	0.0	0.0	158.3	297.0	-159.6	145.8	75.0	132.9
AA Gatling gun	30	Gatling barrel	Cylinder	7.0	7.0	200.0	0.0	0.0	0.0	168.2	290.4	-152.2	145.8	75.0	132.9
AA Gatling gun	31	Gatling barrel	Cylinder	7.0	7.0	200.0	0.0	0.0	0.0	174.6	297.0	-141.6	145.8	75.0	132.9
AA Gatling gun	32	Gatling barrel	Cylinder	7.0	7.0	200.0	0.0	0.0	0.0	171.0	310.1	-138.4	145.8	75.0	132.9
AA Gatling gun	33	Cluster shaft	Cylinder	10.0	10.0	200.0	0.0	0.0	0.0	164.6	303.5	-149.0	148.6	77.8	135.8
AA Gatling gun	34	Barrel clamp rear	Cylinder	40.0	40.0	6.0	0.0	0.0	0.0	133.3	288.1	-120.6	41.2	39.6	42.6
AA Gatling gun	35	Barrel clamp mid	Cylinder	40.0	40.0	6.0	0.0	0.0	0.0	196.0	318.9	-177.3	41.2	39.6	42.6
AA Gatling gun	36	Barrel clamp muzzle	Cylinder	42.0	42.0	6.0	0.0	0.0	0.0	232.2	336.7	-210.1	43.0	41.5	44.6
AA Gatling gun	37	Rotation marker	Cube	5.0	5.0	25.0	0.0	0.0	0.0	119.2	288.1	-136.2	22.0	13.2	20.6
AA Gatling gun	38	Muzzle flash	Sphere	50.0	50.0	90.0	0.0	0.0	0.0	251.7	346.3	-227.8	0.0	0.0	0.0
AA Gatling gun	39	Housing vent	Cube	42.0	2.5	4.5	0.0	0.0	0.0	4.2	254.6	-3.8	31.9	3.9	34.6
AA Gatling gun	40	Housing vent	Cube	42.0	2.5	4.5	0.0	0.0	0.0	14.7	259.7	-13.3	31.9	3.9	34.6
AA Gatling gun	41	Housing vent	Cube	42.0	2.5	4.5	0.0	0.0	0.0	25.1	264.9	-22.8	31.9	3.9	34.6
AA Gatling gun	42	Radome seam	Cylinder	82.0	82.0	3.0	90.0	0.0	0.0	-63.0	310.0	57.0	115.8	3.0	115.8
AA Gatling gun	43	Fire-control console	Cube	7.0	34.0	46.0	0.0	0.0	0.0	-27.1	155.0	-74.6	38.8	34.0	36.1
AA Gatling gun	44	Fire-control display	Cube	1.0	22.0	32.0	0.0	0.0	0.0	-29.5	157.0	-77.4	24.4	22.0	22.2
AA Gatling gun	45	Ready LED	Sphere	3.5	3.5	3.5	0.0	0.0	0.0	-38.4	175.0	-69.3	4.9	3.5	4.9
AA Gatling gun	46	Armed LED	Sphere	3.5	3.5	3.5	0.0	0.0	0.0	-20.6	175.0	-85.4	4.9	3.5	4.9
AA Gatling gun	47	Danger stencil	Cube	1.0	12.0	45.0	0.0	0.0	0.0	53.5	220.0	36.8	34.0	12.0	30.9
AA Gatling gun	48	Mount body edge	Cylinder	5.0	5.0	140.0	0.0	0.0	0.0	47.0	175.0	51.9	107.2	5.0	97.6
AA Gatling gun	49	Mount body edge	Cylinder	5.0	5.0	140.0	0.0	90.0	0.0	-51.9	175.0	47.0	97.6	5.0	107.2
AA Gatling gun	50	Danger stencil	Cube	1.0	12.0	45.0	0.0	0.0	0.0	-31.3	220.0	-56.9	34.0	12.0	30.9
AA Gatling gun	51	Mount body edge	Cylinder	5.0	5.0	140.0	0.0	0.0	0.0	-47.0	175.0	-51.9	107.2	5.0	97.6
AA Gatling gun	52	Mount body edge	Cylinder	5.0	5.0	140.0	0.0	90.0	0.0	51.9	175.0	-47.0	97.6	5.0	107.2
AA Gatling gun	53	Grab handle	Cylinder	4.0	4.0	50.0	0.0	90.0	0.0	54.9	165.0	-49.6	36.5	4.0	39.8
AA Gatling gun	54	Anchor bolt	Cylinder	9.0	9.0	6.0	90.0	0.0	0.0	0.0	17.0	95.0	9.0	6.0	9.0
AA Gatling gun	55	Anchor bolt	Cylinder	9.0	9.0	6.0	90.0	0.0	0.0	82.3	17.0	47.5	9.0	6.0	9.0
AA Gatling gun	56	Anchor bolt	Cylinder	9.0	9.0	6.0	90.0	0.0	0.0	82.3	17.0	-47.5	9.0	6.0	9.0
AA Gatling gun	57	Anchor bolt	Cylinder	9.0	9.0	6.0	90.0	0.0	0.0	0.0	17.0	-95.0	9.0	6.0	9.0
AA Gatling gun	58	Anchor bolt	Cylinder	9.0	9.0	6.0	90.0	0.0	0.0	-82.3	17.0	-47.5	9.0	6.0	9.0
AA Gatling gun	59	Anchor bolt	Cylinder	9.0	9.0	6.0	90.0	0.0	0.0	-82.3	17.0	47.5	9.0	6.0	9.0
SAM missile	2	Round body	Cylinder	28.0	28.0	260.0	0.0	0.0	0.0	0.0	0.0	130.0	28.0	28.0	260.0
SAM missile	3	Round nose	Cone	28.0	28.0	72.8	0.0	0.0	0.0	0.0	0.0	296.4	28.0	28.0	72.8
SAM missile	4	Round fins	Cube	84.0	2.0	31.2	0.0	0.0	0.0	0.0	0.0	31.2	84.0	2.0	31.2
SAM missile	5	Round fins	Cube	2.0	84.0	31.2	0.0	0.0	0.0	0.0	0.0	31.2	2.0	84.0	31.2
SAM missile	6	Round flame	Sphere	36.4	36.4	156.0	0.0	0.0	0.0	0.0	0.0	-65.0	36.4	36.4	156.0
MLRS rocket	2	Round body	Cylinder	22.0	22.0	220.0	0.0	0.0	0.0	0.0	0.0	110.0	22.0	22.0	220.0

MLRS rocket	3	Round nose	Cone	22.0	22.0	61.6	0.0	0.0	0.0	0.0	0.0	250.8	22.0	22.0	61.6
MLRS rocket	4	Round fins	Cube	66.0	2.0	26.4	0.0	0.0	0.0	0.0	0.0	26.4	66.0	2.0	26.4
MLRS rocket	5	Round fins	Cube	2.0	66.0	26.4	0.0	0.0	0.0	0.0	0.0	26.4	2.0	66.0	26.4
MLRS rocket	6	Round flame	Sphere	28.6	28.6	132.0	0.0	0.0	0.0	0.0	0.0	-55.0	28.6	28.6	132.0
UAV round	2	Round body	Cylinder	2.2	2.2	17.0	0.0	0.0	0.0	0.0	0.0	0.0	2.2	2.2	17.0
UAV round	3	Round nose	Cone	2.2	2.2	4.0	0.0	0.0	0.0	0.0	0.0	10.5	2.2	2.2	4.0
UAV round	4	Round fin	Cube	0.2	3.0	3.0	0.0	0.0	45.0	0.0	0.0	-7.0	2.3	2.3	3.0
UAV round	5	Round fin	Cube	0.2	3.0	3.0	0.0	0.0	135.0	0.0	0.0	-7.0	2.3	2.3	3.0
UAV round	6	Round fin	Cube	0.2	3.0	3.0	0.0	0.0	225.0	0.0	0.0	-7.0	2.3	2.3	3.0
UAV round	7	Round fin	Cube	0.2	3.0	3.0	0.0	0.0	315.0	0.0	0.0	-7.0	2.3	2.3	3.0
AA shell casing	2	Shell casing	Cylinder	3.0	3.0	10.0	0.0	0.0	0.0	0.0	0.0	0.0	3.0	3.0	10.0
Command bunker	2	Ramp floor	Wedge	320.0	326.6	700.0	0.0	180.0	0.0	0.0	-176.7	950.0	320.0	326.6	700.0
Command bunker	3	Ramp grip strip	Cube	290.0	3.0	10.0	-25.0	0.0	0.0	0.0	-305.3	670.0	290.0	6.9	10.3
Command bunker	4	Ramp grip strip	Cube	290.0	3.0	10.0	-25.0	0.0	0.0	0.0	-258.7	770.0	290.0	6.9	10.3
Command bunker	5	Ramp grip strip	Cube	290.0	3.0	10.0	-25.0	0.0	0.0	0.0	-212.0	870.0	290.0	6.9	10.3
Command bunker	6	Ramp grip strip	Cube	290.0	3.0	10.0	-25.0	0.0	0.0	0.0	-165.4	970.0	290.0	6.9	10.3
Command bunker	7	Ramp grip strip	Cube	290.0	3.0	10.0	-25.0	0.0	0.0	0.0	-118.7	1070.0	290.0	6.9	10.3
Command bunker	8	Ramp grip strip	Cube	290.0	3.0	10.0	-25.0	0.0	0.0	0.0	-72.1	1170.0	290.0	6.9	10.3
Command bunker	9	Ramp grip strip	Cube	290.0	3.0	10.0	-25.0	0.0	0.0	0.0	-25.4	1270.0	290.0	6.9	10.3
Command bunker	10	Landing slab	Cube	320.0	30.0	235.0	0.0	0.0	0.0	0.0	-355.0	482.5	320.0	30.0	235.0
Command bunker	11	Retaining wall W	Cube	35.0	385.7	124.5	0.0	0.0	0.0	-167.5	-182.1	426.2	35.0	385.7	124.5
Command bunker	12	Coping slab W	Cube	85.0	16.0	126.5	0.0	0.0	0.0	-182.0	16.7	426.2	85.0	16.0	126.5
Command bunker	13	Retaining wall W	Cube	35.0	391.4	124.5	0.0	0.0	0.0	-167.5	-179.3	548.8	35.0	391.4	124.5
Command bunker	14	Coping slab W	Cube	85.0	16.0	126.5	0.0	0.0	0.0	-182.0	22.4	548.8	85.0	16.0	126.5
Command bunker	15	Retaining wall W	Cube	35.0	386.0	124.5	0.0	0.0	0.0	-167.5	-177.3	671.2	35.0	386.0	124.5
Command bunker	16	Coping slab W	Cube	85.0	16.0	126.5	0.0	0.0	0.0	-182.0	21.7	671.2	85.0	16.0	126.5
Command bunker	17	Retaining wall W	Cube	35.0	321.5	124.5	0.0	0.0	0.0	-167.5	-152.4	793.7	35.0	321.5	124.5
Command bunker	18	Coping slab W	Cube	85.0	16.0	126.5	0.0	0.0	0.0	-182.0	14.3	793.7	85.0	16.0	126.5
Command bunker	19	Retaining wall W	Cube	35.0	260.8	124.5	0.0	0.0	0.0	-167.5	-125.6	916.2	35.0	260.8	124.5
Command bunker	20	Coping slab W	Cube	85.0	16.0	126.5	0.0	0.0	0.0	-182.0	10.8	916.2	85.0	16.0	126.5
Command bunker	21	Retaining wall W	Cube	35.0	194.2	124.5	0.0	0.0	0.0	-167.5	-101.7	1038.7	35.0	194.2	124.5
Command bunker	22	Coping slab W	Cube	85.0	16.0	126.5	0.0	0.0	0.0	-182.0	1.4	1038.7	85.0	16.0	126.5
Command bunker	23	Retaining wall W	Cube	35.0	142.2	124.5	0.0	0.0	0.0	-167.5	-70.6	1161.3	35.0	142.2	124.5
Command bunker	24	Coping slab W	Cube	85.0	16.0	126.5	0.0	0.0	0.0	-182.0	6.5	1161.3	85.0	16.0	126.5
Command bunker	25	Retaining wall W	Cube	35.0	91.6	124.5	0.0	0.0	0.0	-167.5	-38.8	1283.7	35.0	91.6	124.5
Command bunker	26	Coping slab W	Cube	85.0	16.0	126.5	0.0	0.0	0.0	-182.0	13.0	1283.7	85.0	16.0	126.5
Command bunker	27	Counterfort rib	Cube	18.0	361.3	30.0	0.0	0.0	0.0	-141.0	-164.3	540.0	18.0	361.3	30.0
Command bunker	28	Counterfort rib	Cube	18.0	263.5	30.0	0.0	0.0	0.0	-141.0	-124.6	790.0	18.0	263.5	30.0
Command bunker	29	Counterfort rib	Cube	18.0	130.8	30.0	0.0	0.0	0.0	-141.0	-74.3	1040.0	18.0	130.8	30.0
Command bunker	30	Counterfort rib	Cube	18.0	39.3	30.0	0.0	0.0	0.0	-141.0	-17.4	1260.0	18.0	39.3	30.0
Command bunker	31	Retaining wall E	Cube	35.0	385.7	124.5	0.0	0.0	0.0	167.5	-182.1	426.2	35.0	385.7	124.5
Command bunker	32	Coping slab E	Cube	85.0	16.0	126.5	0.0	0.0	0.0	182.0	16.7	426.2	85.0	16.0	126.5
Command bunker	33	Retaining wall E	Cube	35.0	391.4	124.5	0.0	0.0	0.0	167.5	-179.3	548.8	35.0	391.4	124.5
Command bunker	34	Coping slab E	Cube	85.0	16.0	126.5	0.0	0.0	0.0	182.0	22.4	548.8	85.0	16.0	126.5
Command bunker	35	Retaining wall E	Cube	35.0	386.0	124.5	0.0	0.0	0.0	167.5	-177.3	671.2	35.0	386.0	124.5
Command bunker	36	Coping slab E	Cube	85.0	16.0	126.5	0.0	0.0	0.0	182.0	21.7	671.2	85.0	16.0	126.5
Command bunker	37	Retaining wall E	Cube	35.0	321.5	124.5	0.0	0.0	0.0	167.5	-152.4	793.7	35.0	321.5	124.5
Command bunker	38	Coping slab E	Cube	85.0	16.0	126.5	0.0	0.0	0.0	182.0	14.3	793.7	85.0	16.0	126.5
Command bunker	39	Retaining wall E	Cube	35.0	260.8	124.5	0.0	0.0	0.0	167.5	-125.6	916.2	35.0	260.8	124.5
Command bunker	40	Coping slab E	Cube	85.0	16.0	126.5	0.0	0.0	0.0	182.0	10.8	916.2	85.0	16.0	126.5
Command bunker	41	Retaining wall E	Cube	35.0	194.2	124.5	0.0	0.0	0.0	167.5	-101.7	1038.7	35.0	194.2	124.5

Command bunker	42	Coping slab E	Cube	85.0	16.0	126.5	0.0	0.0	0.0	182.0	1.4	1038.7	85.0	16.0	126.5
Command bunker	43	Retaining wall E	Cube	35.0	142.2	124.5	0.0	0.0	0.0	167.5	-70.6	1161.3	35.0	142.2	124.5
Command bunker	44	Coping slab E	Cube	85.0	16.0	126.5	0.0	0.0	0.0	182.0	6.5	1161.3	85.0	16.0	126.5
Command bunker	45	Retaining wall E	Cube	35.0	91.6	124.5	0.0	0.0	0.0	167.5	-38.8	1283.7	35.0	91.6	124.5
Command bunker	46	Coping slab E	Cube	85.0	16.0	126.5	0.0	0.0	0.0	182.0	13.0	1283.7	85.0	16.0	126.5
Command bunker	47	Counterfort rib	Cube	18.0	361.3	30.0	0.0	0.0	0.0	141.0	-164.3	540.0	18.0	361.3	30.0
Command bunker	48	Counterfort rib	Cube	18.0	263.5	30.0	0.0	0.0	0.0	141.0	-124.6	790.0	18.0	263.5	30.0
Command bunker	49	Counterfort rib	Cube	18.0	130.8	30.0	0.0	0.0	0.0	141.0	-74.3	1040.0	18.0	130.8	30.0
Command bunker	50	Counterfort rib	Cube	18.0	39.3	30.0	0.0	0.0	0.0	141.0	-17.4	1260.0	18.0	39.3	30.0
Command bunker	51	Guard rail post	Cylinder	6.0	6.0	100.0	90.0	0.0	0.0	-195.0	77.6	490.0	6.0	100.0	6.0
Command bunker	52	Guard rail post	Cylinder	6.0	6.0	100.0	90.0	0.0	0.0	-195.0	74.7	720.0	6.0	100.0	6.0
Command bunker	53	Guard rail post	Cylinder	6.0	6.0	100.0	90.0	0.0	0.0	-195.0	65.3	950.0	6.0	100.0	6.0
Command bunker	54	Guard rail post	Cylinder	6.0	6.0	100.0	90.0	0.0	0.0	-195.0	64.0	1180.0	6.0	100.0	6.0
Command bunker	55	Guard rail	Cylinder	4.0	4.0	230.0	0.7	0.0	0.0	-195.0	126.2	605.0	4.0	6.9	230.1
Command bunker	56	Guard rail	Cylinder	4.0	4.0	230.0	0.7	0.0	0.0	-195.0	76.2	605.0	4.0	6.9	230.1
Command bunker	57	Guard rail	Cylinder	4.0	4.0	230.2	2.4	0.0	0.0	-195.0	120.0	835.0	4.0	13.4	230.2
Command bunker	58	Guard rail	Cylinder	4.0	4.0	230.2	2.4	0.0	0.0	-195.0	70.0	835.0	4.0	13.4	230.2
Command bunker	59	Guard rail	Cylinder	4.0	4.0	230.0	0.3	0.0	0.0	-195.0	114.7	1065.0	4.0	5.2	230.0
Command bunker	60	Guard rail	Cylinder	4.0	4.0	230.0	0.3	0.0	0.0	-195.0	64.7	1065.0	4.0	5.2	230.0
Command bunker	61	Guard rail post	Cylinder	6.0	6.0	100.0	90.0	0.0	0.0	195.0	77.6	490.0	6.0	100.0	6.0
Command bunker	62	Guard rail post	Cylinder	6.0	6.0	100.0	90.0	0.0	0.0	195.0	74.7	720.0	6.0	100.0	6.0
Command bunker	63	Guard rail post	Cylinder	6.0	6.0	100.0	90.0	0.0	0.0	195.0	65.3	950.0	6.0	100.0	6.0
Command bunker	64	Guard rail post	Cylinder	6.0	6.0	100.0	90.0	0.0	0.0	195.0	64.0	1180.0	6.0	100.0	6.0
Command bunker	65	Guard rail	Cylinder	4.0	4.0	230.0	0.7	0.0	0.0	195.0	126.2	605.0	4.0	6.9	230.1
Command bunker	66	Guard rail	Cylinder	4.0	4.0	230.0	0.7	0.0	0.0	195.0	76.2	605.0	4.0	6.9	230.1
Command bunker	67	Guard rail	Cylinder	4.0	4.0	230.2	2.4	0.0	0.0	195.0	120.0	835.0	4.0	13.4	230.2
Command bunker	68	Guard rail	Cylinder	4.0	4.0	230.2	2.4	0.0	0.0	195.0	70.0	835.0	4.0	13.4	230.2
Command bunker	69	Guard rail	Cylinder	4.0	4.0	230.0	0.3	0.0	0.0	195.0	114.7	1065.0	4.0	5.2	230.0
Command bunker	70	Guard rail	Cylinder	4.0	4.0	230.0	0.3	0.0	0.0	195.0	64.7	1065.0	4.0	5.2	230.0
Command bunker	71	Drain gutter	Cube	290.0	2.0	45.0	0.0	0.0	0.0	0.0	-339.5	650.0	290.0	2.0	45.0
Command bunker	72	Drain frame	Cube	300.0	4.0	55.0	0.0	0.0	0.0	0.0	-338.0	650.0	300.0	4.0	55.0
Command bunker	73	Drain grate bar	Cube	3.0	4.0	50.0	0.0	0.0	0.0	-120.0	-335.5	650.0	3.0	4.0	50.0
Command bunker	74	Drain grate bar	Cube	3.0	4.0	50.0	0.0	0.0	0.0	-72.0	-335.5	650.0	3.0	4.0	50.0
Command bunker	75	Drain grate bar	Cube	3.0	4.0	50.0	0.0	0.0	0.0	-24.0	-335.5	650.0	3.0	4.0	50.0
Command bunker	76	Drain grate bar	Cube	3.0	4.0	50.0	0.0	0.0	0.0	24.0	-335.5	650.0	3.0	4.0	50.0
Command bunker	77	Drain grate bar	Cube	3.0	4.0	50.0	0.0	0.0	0.0	72.0	-335.5	650.0	3.0	4.0	50.0
Command bunker	78	Drain grate bar	Cube	3.0	4.0	50.0	0.0	0.0	0.0	120.0	-335.5	650.0	3.0	4.0	50.0
Command bunker	79	Landing lamp housing	Cube	30.0	12.0	20.0	0.0	0.0	0.0	135.0	-80.0	560.0	30.0	12.0	20.0
Command bunker	80	Landing floodlight	Sphere	18.0	18.0	6.0	90.0	0.0	0.0	115.0	-85.0	560.0	18.0	6.0	18.0
Command bunker	81	Wall lamp housing	Cube	18.0	12.0	22.0	0.0	0.0	0.0	-141.0	-120.0	500.0	18.0	12.0	22.0
Command bunker	82	Wall lamp	Sphere	12.0	12.0	12.0	0.0	0.0	0.0	-130.0	-122.0	500.0	12.0	12.0	12.0
Command bunker	83	Wall lamp housing	Cube	18.0	12.0	22.0	0.0	0.0	0.0	141.0	-45.3	760.0	18.0	12.0	22.0
Command bunker	84	Wall lamp	Sphere	12.0	12.0	12.0	0.0	0.0	0.0	130.0	-47.3	760.0	12.0	12.0	12.0
Command bunker	85	Wall lamp housing	Cube	18.0	12.0	22.0	0.0	0.0	0.0	-141.0	-51.2	1020.0	18.0	12.0	22.0
Command bunker	86	Wall lamp	Sphere	12.0	12.0	12.0	0.0	0.0	0.0	-130.0	-53.2	1020.0	12.0	12.0	12.0
Command bunker	87	Wall lamp housing	Cube	18.0	12.0	22.0	0.0	0.0	0.0	141.0	-120.0	500.0	18.0	12.0	22.0
Command bunker	88	Wall lamp	Sphere	12.0	12.0	12.0	0.0	0.0	0.0	130.0	-122.0	500.0	12.0	12.0	12.0
Command bunker	89	Portal head wall	Cube	480.0	96.4	55.0	0.0	0.0	0.0	0.0	-31.8	390.0	480.0	96.4	55.0
Command bunker	90	Earth berm	Hemisphere	540.0	50.0	200.0	0.0	0.0	0.0	0.0	41.4	280.0	540.0	50.0	200.0
Command bunker	91	Hazard jamb	Cube	20.0	260.0	5.0	0.0	0.0	0.0	-140.0	-210.0	402.5	20.0	260.0	5.0
Command bunker	92	Hazard jamb	Cube	20.0	260.0	5.0	0.0	0.0	0.0	140.0	-210.0	402.5	20.0	260.0	5.0

Command bunker	93	Hazard header	Cube	300.0	30.0	5.0	0.0	0.0	0.0	0.0	-65.0	402.5	300.0	30.0	5.0
Command bunker	94	Warning sign	Cube	90.0	35.0	3.0	0.0	0.0	0.0	0.0	-20.0	402.0	90.0	35.0	3.0
Command bunker	95	Gate lintel	Cube	320.0	34.0	46.0	0.0	0.0	0.0	0.0	-43.0	423.0	320.0	34.0	46.0
Command bunker	96	Gate threshold	Cube	260.0	3.0	30.0	0.0	0.0	0.0	0.0	-338.5	430.0	260.0	3.0	30.0
Command bunker	97	Entry guide stripe	Cube	14.0	2.0	70.0	0.0	0.0	0.0	0.0	-338.8	490.0	14.0	2.0	70.0
Command bunker	98	Entry guide stripe	Cube	14.0	2.0	70.0	0.0	0.0	0.0	0.0	-338.8	545.0	14.0	2.0	70.0
Command bunker	99	Entry guide stripe	Cube	14.0	2.0	70.0	0.0	0.0	0.0	0.0	-338.8	600.0	14.0	2.0	70.0
Command bunker	100	Entry guide stripe	Cube	14.0	2.0	70.0	0.0	0.0	0.0	0.0	-338.8	655.0	14.0	2.0	70.0
Command bunker	101	Entry guide stripe	Cube	14.0	2.0	70.0	0.0	0.0	0.0	0.0	-338.8	710.0	14.0	2.0	70.0
Command bunker	102	Gate lamp	Sphere	16.0	16.0	10.0	90.0	0.0	0.0	-75.0	-65.0	452.0	16.0	10.0	16.0
Command bunker	103	Gate lamp	Sphere	16.0	16.0	10.0	90.0	0.0	0.0	75.0	-65.0	452.0	16.0	10.0	16.0
Command bunker	104	Leaf plate	Cube	130.0	260.0	20.0	0.0	0.0	0.0	-65.0	-210.0	410.0	130.0	260.0	20.0
Command bunker	105	Leaf stiffener	Cube	10.0	230.0	5.0	0.0	0.0	0.0	-105.0	-210.0	422.5	10.0	230.0	5.0
Command bunker	106	Leaf stiffener	Cube	10.0	230.0	5.0	0.0	0.0	0.0	-67.0	-210.0	422.5	10.0	230.0	5.0
Command bunker	107	Leaf stiffener	Cube	10.0	230.0	5.0	0.0	0.0	0.0	-29.0	-210.0	422.5	10.0	230.0	5.0
Command bunker	108	Leaf edge stripe	Cube	12.0	260.0	4.0	0.0	0.0	0.0	-6.0	-210.0	422.0	12.0	260.0	4.0
Command bunker	109	Leaf hinge barrel	Cylinder	14.0	14.0	260.0	90.0	0.0	0.0	-130.0	-210.0	404.0	14.0	260.0	14.0
Command bunker	110	Locking bolt	Cylinder	12.0	12.0	12.0	0.0	0.0	0.0	-20.0	-155.0	426.0	12.0	12.0	12.0
Command bunker	111	Leaf hold-open stop	Cube	10.0	10.0	40.0	0.0	0.0	0.0	-15.0	-335.0	440.0	10.0	10.0	40.0
Command bunker	112	Leaf plate	Cube	130.0	260.0	20.0	0.0	0.0	0.0	65.0	-210.0	410.0	130.0	260.0	20.0
Command bunker	113	Leaf stiffener	Cube	10.0	230.0	5.0	0.0	0.0	0.0	105.0	-210.0	422.5	10.0	230.0	5.0
Command bunker	114	Leaf stiffener	Cube	10.0	230.0	5.0	0.0	0.0	0.0	67.0	-210.0	422.5	10.0	230.0	5.0
Command bunker	115	Leaf stiffener	Cube	10.0	230.0	5.0	0.0	0.0	0.0	29.0	-210.0	422.5	10.0	230.0	5.0
Command bunker	116	Leaf edge stripe	Cube	12.0	260.0	4.0	0.0	0.0	0.0	6.0	-210.0	422.0	12.0	260.0	4.0
Command bunker	117	Leaf hinge barrel	Cylinder	14.0	14.0	260.0	90.0	0.0	0.0	130.0	-210.0	404.0	14.0	260.0	14.0
Command bunker	118	Locking bolt	Cylinder	12.0	12.0	12.0	0.0	0.0	0.0	20.0	-155.0	426.0	12.0	12.0	12.0
Command bunker	119	Leaf hold-open stop	Cube	10.0	10.0	40.0	0.0	0.0	0.0	15.0	-335.0	440.0	10.0	10.0	40.0
Command bunker	120	Door top rail	Cube	640.0	14.0	20.0	0.0	0.0	0.0	0.0	-87.0	355.0	640.0	14.0	20.0
Command bunker	121	Door floor rail	Cube	640.0	4.0	20.0	0.0	0.0	0.0	0.0	-338.0	382.5	640.0	4.0	20.0
Command bunker	122	Door drive motor	Cube	60.0	45.0	35.0	0.0	0.0	0.0	-240.0	-135.0	347.0	60.0	45.0	35.0
Command bunker	123	Door drive motor	Cube	60.0	45.0	35.0	0.0	0.0	0.0	240.0	-135.0	347.0	60.0	45.0	35.0
Command bunker	124	Beacon base	Cylinder	22.0	22.0	10.0	90.0	0.0	0.0	-110.0	-73.0	410.0	22.0	10.0	22.0
Command bunker	125	Beacon lamp	Sphere	16.0	16.0	16.0	0.0	0.0	0.0	-100.0	-54.0	410.0	16.0	16.0	16.0
Command bunker	126	Beacon lamp	Sphere	16.0	16.0	16.0	0.0	0.0	0.0	-120.0	-54.0	410.0	16.0	16.0	16.0
Command bunker	127	Beacon reflector	Cube	2.0	14.0	18.0	0.0	0.0	0.0	-110.0	-54.0	410.0	2.0	14.0	18.0
Command bunker	128	Beacon base	Cylinder	22.0	22.0	10.0	90.0	0.0	0.0	110.0	-73.0	410.0	22.0	10.0	22.0
Command bunker	129	Beacon lamp	Sphere	16.0	16.0	16.0	0.0	0.0	0.0	110.0	-54.0	400.0	16.0	16.0	16.0
Command bunker	130	Beacon lamp	Sphere	16.0	16.0	16.0	0.0	0.0	0.0	110.0	-54.0	420.0	16.0	16.0	16.0
Command bunker	131	Beacon reflector	Cube	2.0	14.0	18.0	0.0	0.0	0.0	110.0	-54.0	410.0	18.0	14.0	2.0
Command bunker	132	Status lamp housing	Cube	18.0	28.0	8.0	0.0	0.0	0.0	-140.0	-170.0	404.0	18.0	28.0	8.0
Command bunker	133	Status lamp	Sphere	12.0	12.0	8.0	0.0	0.0	0.0	-140.0	-170.0	409.0	12.0	12.0	8.0
Command bunker	134	Front wall W	Cube	420.0	260.0	35.0	0.0	0.0	0.0	-340.0	-210.0	382.5	420.0	260.0	35.0
Command bunker	135	Front wall E	Cube	420.0	260.0	35.0	0.0	0.0	0.0	340.0	-210.0	382.5	420.0	260.0	35.0
Command bunker	136	Floor slab	Cube	1100.0	30.0	1250.0	0.0	0.0	0.0	0.0	-355.0	-225.0	1100.0	30.0	1250.0
Command bunker	137	West wall	Cube	30.0	260.0	1250.0	0.0	0.0	0.0	-535.0	-210.0	-225.0	30.0	260.0	1250.0
Command bunker	138	East wall	Cube	30.0	260.0	1060.0	0.0	0.0	0.0	535.0	-210.0	-130.0	30.0	260.0	1060.0
Command bunker	139	South wall	Cube	910.0	260.0	30.0	0.0	0.0	0.0	-95.0	-210.0	-835.0	910.0	260.0	30.0
Command bunker	140	Roof slab	Cube	1100.0	35.0	1060.0	0.0	0.0	0.0	0.0	-62.5	-130.0	1100.0	35.0	1060.0
Command bunker	141	Roof slab	Cube	910.0	35.0	190.0	0.0	0.0	0.0	-95.0	-62.5	-755.0	910.0	35.0	190.0
Command bunker	142	Tunnel wall W1	Cube	30.0	260.0	270.0	0.0	0.0	0.0	-165.0	-210.0	-45.0	30.0	260.0	270.0
Command bunker	143	Tunnel wall W2	Cube	30.0	260.0	175.0	0.0	0.0	0.0	-165.0	-210.0	277.5	30.0	260.0	175.0

Command bunker	144	Doorway lintel W	Cube	30.0	50.0	100.0	0.0	0.0	0.0	-165.0	-105.0	140.0	30.0	50.0	100.0
Command bunker	145	Tunnel wall E1	Cube	30.0	260.0	240.0	0.0	0.0	0.0	165.0	-210.0	-60.0	30.0	260.0	240.0
Command bunker	146	Tunnel wall E2	Cube	30.0	260.0	185.0	0.0	0.0	0.0	165.0	-210.0	272.5	30.0	260.0	185.0
Command bunker	147	Doorway lintel E	Cube	30.0	50.0	120.0	0.0	0.0	0.0	165.0	-105.0	120.0	30.0	50.0	120.0
Command bunker	148	Cross wall W	Cube	460.0	260.0	30.0	0.0	0.0	0.0	-320.0	-210.0	-195.0	460.0	260.0	30.0
Command bunker	149	Cross wall E	Cube	460.0	260.0	30.0	0.0	0.0	0.0	320.0	-210.0	-195.0	460.0	260.0	30.0
Command bunker	150	Doorway lintel CIC	Cube	180.0	40.0	30.0	0.0	0.0	0.0	0.0	-100.0	-195.0	180.0	40.0	30.0
Command bunker	151	Shaft wall W	Cube	30.0	348.3	190.0	0.0	0.0	0.0	375.0	-165.9	-755.0	30.0	348.3	190.0
Command bunker	152	Shaft wall E	Cube	30.0	348.3	190.0	0.0	0.0	0.0	535.0	-165.9	-755.0	30.0	348.3	190.0
Command bunker	153	Shaft wall S	Cube	130.0	348.3	30.0	0.0	0.0	0.0	455.0	-165.9	-835.0	130.0	348.3	30.0
Command bunker	154	Shaft wall N1	Cube	10.0	348.3	30.0	0.0	0.0	0.0	395.0	-165.9	-675.0	10.0	348.3	30.0
Command bunker	155	Shaft wall N2	Cube	30.0	348.3	30.0	0.0	0.0	0.0	505.0	-165.9	-675.0	30.0	348.3	30.0
Command bunker	156	Shaft door lintel	Cube	90.0	148.3	30.0	0.0	0.0	0.0	445.0	-65.9	-675.0	90.0	148.3	30.0
Command bunker	157	Fixture housing	Cube	130.0	5.0	30.0	0.0	0.0	0.0	0.0	-83.0	300.0	130.0	5.0	30.0
Command bunker	158	Tunnel tube	Cube	120.0	5.0	22.0	0.0	0.0	0.0	0.0	-87.0	300.0	120.0	5.0	22.0
Command bunker	159	Fixture housing	Cube	130.0	5.0	30.0	0.0	0.0	0.0	0.0	-83.0	100.0	130.0	5.0	30.0
Command bunker	160	Tunnel tube (light)	Cube	120.0	5.0	22.0	0.0	0.0	0.0	0.0	-87.0	100.0	120.0	5.0	22.0
Command bunker	161	Fixture housing	Cube	130.0	5.0	30.0	0.0	0.0	0.0	0.0	-83.0	-90.0	130.0	5.0	30.0
Command bunker	162	Tunnel tube (light)	Cube	120.0	5.0	22.0	0.0	0.0	0.0	0.0	-87.0	-90.0	120.0	5.0	22.0
Command bunker	163	Wall rib W	Cube	12.0	260.0	30.0	0.0	0.0	0.0	-144.0	-210.0	310.0	12.0	260.0	30.0
Command bunker	164	Wall rib W	Cube	12.0	260.0	30.0	0.0	0.0	0.0	-144.0	-210.0	20.0	12.0	260.0	30.0
Command bunker	165	Wall rib W	Cube	12.0	260.0	30.0	0.0	0.0	0.0	-144.0	-210.0	-130.0	12.0	260.0	30.0
Command bunker	166	Wall rib E	Cube	12.0	260.0	30.0	0.0	0.0	0.0	144.0	-210.0	310.0	12.0	260.0	30.0
Command bunker	167	Wall rib E	Cube	12.0	260.0	30.0	0.0	0.0	0.0	144.0	-210.0	20.0	12.0	260.0	30.0
Command bunker	168	Wall rib E	Cube	12.0	260.0	30.0	0.0	0.0	0.0	144.0	-210.0	-130.0	12.0	260.0	30.0
Command bunker	169	Ceiling beam	Cube	300.0	20.0	24.0	0.0	0.0	0.0	0.0	-90.0	230.0	300.0	20.0	24.0
Command bunker	170	Ceiling beam	Cube	300.0	20.0	24.0	0.0	0.0	0.0	0.0	-90.0	0.0	300.0	20.0	24.0
Command bunker	171	Ceiling beam	Cube	300.0	20.0	24.0	0.0	0.0	0.0	0.0	-90.0	-150.0	300.0	20.0	24.0
Command bunker	172	Cable tray	Cube	40.0	6.0	520.0	0.0	0.0	0.0	-125.0	-109.0	90.0	40.0	6.0	520.0
Command bunker	173	Cable bundle	Cube	20.0	10.0	520.0	0.0	0.0	0.0	-125.0	-101.0	90.0	20.0	10.0	520.0
Command bunker	174	Water pipe	Cylinder	14.0	14.0	520.0	0.0	0.0	0.0	115.0	-95.0	90.0	14.0	14.0	520.0
Command bunker	175	Air duct pipe	Cylinder	10.0	10.0	520.0	0.0	0.0	0.0	70.0	-95.0	90.0	10.0	10.0	520.0
Command bunker	176	Floor line	Cube	8.0	1.2	500.0	0.0	0.0	0.0	-112.0	-339.4	90.0	8.0	1.2	500.0
Command bunker	177	Floor line	Cube	8.0	1.2	500.0	0.0	0.0	0.0	112.0	-339.4	90.0	8.0	1.2	500.0
Command bunker	178	Door sill hazard	Cube	260.0	1.2	40.0	0.0	0.0	0.0	0.0	-339.4	340.0	260.0	1.2	40.0
Command bunker	179	Floor drain	Cube	60.0	1.4	40.0	0.0	0.0	0.0	0.0	-339.3	-20.0	60.0	1.4	40.0
Command bunker	180	Emergency exit sign	Cube	70.0	22.0	4.0	0.0	0.0	0.0	0.0	-111.0	342.0	70.0	22.0	4.0
Command bunker	181	Emergency lamp	Sphere	12.0	10.0	12.0	0.0	0.0	0.0	135.0	-125.0	100.0	12.0	10.0	12.0
Command bunker	182	Fire extinguisher	Cylinder	16.0	16.0	45.0	90.0	0.0	0.0	140.0	-270.0	260.0	16.0	45.0	16.0
Command bunker	183	Extinguisher bracket	Cube	10.0	10.0	20.0	0.0	0.0	0.0	145.0	-245.0	260.0	10.0	10.0	20.0
Command bunker	184	Sign board	Cube	4.0	50.0	100.0	0.0	0.0	0.0	148.0	-165.0	260.0	4.0	50.0	100.0
Command bunker	185	Fixture housing	Cube	130.0	5.0	30.0	0.0	0.0	0.0	-320.0	-83.0	-320.0	130.0	5.0	30.0
Command bunker	186	CIC tube (light)	Cube	120.0	5.0	22.0	0.0	0.0	0.0	-320.0	-87.0	-320.0	120.0	5.0	22.0
Command bunker	187	Fixture housing	Cube	130.0	5.0	30.0	0.0	0.0	0.0	320.0	-83.0	-320.0	130.0	5.0	30.0
Command bunker	188	CIC tube (light)	Cube	120.0	5.0	22.0	0.0	0.0	0.0	320.0	-87.0	-320.0	120.0	5.0	22.0
Command bunker	189	Fixture housing	Cube	130.0	5.0	30.0	0.0	0.0	0.0	-320.0	-83.0	-740.0	130.0	5.0	30.0
Command bunker	190	CIC tube (light)	Cube	120.0	5.0	22.0	0.0	0.0	0.0	-320.0	-87.0	-740.0	120.0	5.0	22.0
Command bunker	191	Fixture housing	Cube	130.0	5.0	30.0	0.0	0.0	0.0	320.0	-83.0	-740.0	130.0	5.0	30.0
Command bunker	192	CIC tube (light)	Cube	120.0	5.0	22.0	0.0	0.0	0.0	320.0	-87.0	-740.0	120.0	5.0	22.0
Command bunker	193	Fixture housing	Cube	130.0	5.0	30.0	0.0	0.0	0.0	0.0	-83.0	-540.0	130.0	5.0	30.0
Command bunker	194	CIC tube (light)	Cube	120.0	5.0	22.0	0.0	0.0	0.0	0.0	-87.0	-540.0	120.0	5.0	22.0

Command bunker	195	Emergency lamp	Sphere	14.0	12.0	14.0	0.0	0.0	0.0	-505.0	-125.0	-240.0	14.0	12.0	14.0
Command bunker	196	Emergency lamp	Sphere	14.0	12.0	14.0	0.0	0.0	0.0	505.0	-125.0	-790.0	14.0	12.0	14.0
Command bunker	197	Wall rib W	Cube	12.0	260.0	30.0	0.0	0.0	0.0	-514.0	-210.0	-240.0	12.0	260.0	30.0
Command bunker	198	Wall rib W	Cube	12.0	260.0	30.0	0.0	0.0	0.0	-514.0	-210.0	-650.0	12.0	260.0	30.0
Command bunker	199	Wall rib W	Cube	12.0	260.0	30.0	0.0	0.0	0.0	-514.0	-210.0	-790.0	12.0	260.0	30.0
Command bunker	200	Wall rib E	Cube	12.0	260.0	30.0	0.0	0.0	0.0	514.0	-210.0	-260.0	12.0	260.0	30.0
Command bunker	201	Wall rib E	Cube	12.0	260.0	30.0	0.0	0.0	0.0	514.0	-210.0	-470.0	12.0	260.0	30.0
Command bunker	202	Wall rib E	Cube	12.0	260.0	30.0	0.0	0.0	0.0	514.0	-210.0	-640.0	12.0	260.0	30.0
Command bunker	203	Wall rib S	Cube	30.0	260.0	12.0	0.0	0.0	0.0	-460.0	-210.0	-814.0	30.0	260.0	12.0
Command bunker	204	Wall rib S	Cube	30.0	260.0	12.0	0.0	0.0	0.0	-280.0	-210.0	-814.0	30.0	260.0	12.0
Command bunker	205	Wall rib S	Cube	30.0	260.0	12.0	0.0	0.0	0.0	220.0	-210.0	-814.0	30.0	260.0	12.0
Command bunker	206	Ceiling beam	Cube	24.0	22.0	610.0	0.0	0.0	0.0	-360.0	-91.0	-515.0	24.0	22.0	610.0
Command bunker	207	Ceiling beam	Cube	24.0	22.0	610.0	0.0	0.0	0.0	-120.0	-91.0	-515.0	24.0	22.0	610.0
Command bunker	208	Ceiling beam	Cube	24.0	22.0	610.0	0.0	0.0	0.0	120.0	-91.0	-515.0	24.0	22.0	610.0
Command bunker	209	Ceiling beam	Cube	24.0	22.0	610.0	0.0	0.0	0.0	360.0	-91.0	-515.0	24.0	22.0	610.0
Command bunker	210	Cable tray	Cube	1020.0	6.0	40.0	0.0	0.0	0.0	0.0	-111.0	-680.0	1020.0	6.0	40.0
Command bunker	211	Cable bundle	Cube	1000.0	10.0	20.0	0.0	0.0	0.0	0.0	-103.0	-680.0	1000.0	10.0	20.0
Command bunker	212	Ventilation duct	Cylinder	30.0	30.0	1000.0	0.0	90.0	0.0	0.0	-110.0	-270.0	1000.0	30.0	30.0
Command bunker	213	Pipe	Cylinder	9.0	9.0	1000.0	0.0	90.0	0.0	0.0	-92.0	-235.0	1000.0	9.0	9.0
Command bunker	214	Floor line	Cube	8.0	1.2	190.0	0.0	0.0	0.0	-95.0	-339.4	-315.0	8.0	1.2	190.0
Command bunker	215	Floor line	Cube	8.0	1.2	190.0	0.0	0.0	0.0	95.0	-339.4	-315.0	8.0	1.2	190.0
Command bunker	216	Floor line ring N	Cube	360.0	1.2	8.0	0.0	0.0	0.0	0.0	-339.4	-496.0	360.0	1.2	8.0
Command bunker	217	Floor drain	Cube	60.0	1.4	40.0	0.0	0.0	0.0	-170.0	-339.3	-740.0	60.0	1.4	40.0
Command bunker	218	Cable cover	Cube	660.0	3.0	10.0	0.0	0.0	0.0	-10.0	-338.5	-455.0	660.0	3.0	10.0
Command bunker	219	Plotting table base	Cube	240.0	85.0	120.0	0.0	0.0	0.0	0.0	-297.5	-640.0	240.0	85.0	120.0
Command bunker	220	Plotting table top	Cube	250.0	5.0	130.0	0.0	0.0	0.0	0.0	-252.5	-640.0	250.0	5.0	130.0
Command bunker	221	Wall map frame	Cube	440.0	140.0	7.0	0.0	0.0	0.0	-50.0	-165.0	-816.5	440.0	140.0	7.0
Command bunker	222	Wall map screen	Cube	420.0	126.0	3.0	0.0	0.0	0.0	-50.0	-165.0	-811.5	420.0	126.0	3.0
Command bunker	223	Console desk	Cube	160.0	75.0	70.0	0.0	0.0	0.0	-320.0	-302.5	-340.0	160.0	75.0	70.0
Command bunker	224	Console keyboard screen	Cube	130.0	4.0	45.0	-18.0	0.0	0.0	-320.0	-262.0	-335.0	130.0	17.7	44.0
Command bunker	225	Console monitor	Cube	100.0	55.0	5.0	0.0	0.0	0.0	-320.0	-217.5	-367.5	100.0	55.0	5.0
Command bunker	226	Chair seat	Cube	50.0	6.0	50.0	0.0	0.0	0.0	-320.0	-295.0	-265.0	50.0	6.0	50.0
Command bunker	227	Chair back	Cube	50.0	47.0	5.0	0.0	0.0	0.0	-320.0	-268.5	-240.5	50.0	47.0	5.0
Command bunker	228	Chair post	Cylinder	7.0	7.0	42.0	90.0	0.0	0.0	-320.0	-319.0	-265.0	7.0	42.0	7.0
Command bunker	229	Console desk	Cube	160.0	75.0	70.0	0.0	0.0	0.0	320.0	-302.5	-340.0	160.0	75.0	70.0
Command bunker	230	Console keyboard screen	Cube	130.0	4.0	45.0	-18.0	0.0	0.0	320.0	-262.0	-335.0	130.0	17.7	44.0
Command bunker	231	Console monitor	Cube	100.0	55.0	5.0	0.0	0.0	0.0	320.0	-217.5	-367.5	100.0	55.0	5.0
Command bunker	232	Task lamp arm	Cylinder	3.0	3.0	35.0	90.0	0.0	0.0	385.0	-247.0	-360.0	3.0	35.0	3.0
Command bunker	233	Task lamp	Sphere	13.0	10.0	13.0	0.0	0.0	0.0	385.0	-226.0	-360.0	13.0	10.0	13.0
Command bunker	234	Chair seat	Cube	50.0	6.0	50.0	0.0	0.0	0.0	320.0	-295.0	-265.0	50.0	6.0	50.0
Command bunker	235	Chair back	Cube	50.0	47.0	5.0	0.0	0.0	0.0	320.0	-268.5	-240.5	50.0	47.0	5.0
Command bunker	236	Chair post	Cylinder	7.0	7.0	42.0	90.0	0.0	0.0	320.0	-319.0	-265.0	7.0	42.0	7.0
Command bunker	237	Console desk	Cube	160.0	75.0	70.0	0.0	0.0	0.0	-320.0	-302.5	-560.0	160.0	75.0	70.0
Command bunker	238	Console keyboard screen	Cube	130.0	4.0	45.0	-18.0	0.0	0.0	-320.0	-262.0	-555.0	130.0	17.7	44.0
Command bunker	239	Console monitor	Cube	100.0	55.0	5.0	0.0	0.0	0.0	-320.0	-217.5	-587.5	100.0	55.0	5.0
Command bunker	240	Task lamp arm	Cylinder	3.0	3.0	35.0	90.0	0.0	0.0	-255.0	-247.0	-580.0	3.0	35.0	3.0
Command bunker	241	Task lamp	Sphere	13.0	10.0	13.0	0.0	0.0	0.0	-255.0	-226.0	-580.0	13.0	10.0	13.0
Command bunker	242	Chair seat	Cube	50.0	6.0	50.0	0.0	0.0	0.0	-320.0	-295.0	-485.0	50.0	6.0	50.0
Command bunker	243	Chair back	Cube	50.0	47.0	5.0	0.0	0.0	0.0	-320.0	-268.5	-460.5	50.0	47.0	5.0
Command bunker	244	Chair post	Cylinder	7.0	7.0	42.0	90.0	0.0	0.0	-320.0	-319.0	-485.0	7.0	42.0	7.0
Command bunker	245	Console desk	Cube	160.0	75.0	70.0	0.0	0.0	0.0	300.0	-302.5	-540.0	160.0	75.0	70.0

Command bunker	246	Console keyboard screen	Cube	130.0	4.0	45.0	-18.0	0.0	0.0	300.0	-262.0	-535.0	130.0	17.7	44.0
Command bunker	247	Console monitor	Cube	100.0	55.0	5.0	0.0	0.0	0.0	300.0	-217.5	-567.5	100.0	55.0	5.0
Command bunker	248	Chair seat	Cube	50.0	6.0	50.0	0.0	0.0	0.0	300.0	-295.0	-465.0	50.0	6.0	50.0
Command bunker	249	Chair back	Cube	50.0	47.0	5.0	0.0	0.0	0.0	300.0	-268.5	-440.5	50.0	47.0	5.0
Command bunker	250	Chair post	Cylinder	7.0	7.0	42.0	90.0	0.0	0.0	300.0	-319.0	-465.0	7.0	42.0	7.0
Command bunker	251	Server rack	Cube	60.0	200.0	80.0	0.0	0.0	0.0	-490.0	-240.0	-360.0	60.0	200.0	80.0
Command bunker	252	Rack LED row	Cube	3.0	6.0	60.0	0.0	0.0	0.0	-458.5	-277.0	-360.0	3.0	6.0	60.0
Command bunker	253	Rack LED row	Cube	3.0	6.0	60.0	0.0	0.0	0.0	-458.5	-232.0	-360.0	3.0	6.0	60.0
Command bunker	254	Rack LED row	Cube	3.0	6.0	60.0	0.0	0.0	0.0	-458.5	-187.0	-360.0	3.0	6.0	60.0
Command bunker	255	Server rack	Cube	60.0	200.0	80.0	0.0	0.0	0.0	-490.0	-240.0	-460.0	60.0	200.0	80.0
Command bunker	256	Rack LED row	Cube	3.0	6.0	60.0	0.0	0.0	0.0	-458.5	-277.0	-460.0	3.0	6.0	60.0
Command bunker	257	Rack LED row	Cube	3.0	6.0	60.0	0.0	0.0	0.0	-458.5	-232.0	-460.0	3.0	6.0	60.0
Command bunker	258	Rack LED row	Cube	3.0	6.0	60.0	0.0	0.0	0.0	-458.5	-187.0	-460.0	3.0	6.0	60.0
Command bunker	259	Server rack	Cube	60.0	200.0	80.0	0.0	0.0	0.0	-490.0	-240.0	-560.0	60.0	200.0	80.0
Command bunker	260	Rack LED row	Cube	3.0	6.0	60.0	0.0	0.0	0.0	-458.5	-277.0	-560.0	3.0	6.0	60.0
Command bunker	261	Rack LED row	Cube	3.0	6.0	60.0	0.0	0.0	0.0	-458.5	-232.0	-560.0	3.0	6.0	60.0
Command bunker	262	Rack LED row	Cube	3.0	6.0	60.0	0.0	0.0	0.0	-458.5	-187.0	-560.0	3.0	6.0	60.0
Command bunker	263	Notice board	Cube	10.0	70.0	80.0	0.0	0.0	0.0	515.0	-185.0	-320.0	10.0	70.0	80.0
Command bunker	264	Room sign	Cube	120.0	25.0	4.0	0.0	0.0	0.0	0.0	-112.5	-180.0	120.0	25.0	4.0
Command bunker	265	Fire extinguisher	Cylinder	16.0	16.0	45.0	90.0	0.0	0.0	-500.0	-300.0	-195.0	16.0	45.0	16.0
Command bunker	266	Fixture housing	Cube	130.0	5.0	30.0	0.0	0.0	0.0	-350.0	-83.0	100.0	130.0	5.0	30.0
Command bunker	267	Bunk room tube (light)	Cube	120.0	5.0	22.0	0.0	0.0	0.0	-350.0	-87.0	100.0	120.0	5.0	22.0
Command bunker	268	Ceiling beam	Cube	340.0	20.0	24.0	0.0	0.0	0.0	-350.0	-90.0	-70.0	340.0	20.0	24.0
Command bunker	269	Ceiling beam	Cube	340.0	20.0	24.0	0.0	0.0	0.0	-350.0	-90.0	200.0	340.0	20.0	24.0
Command bunker	270	Wall rib	Cube	12.0	260.0	30.0	0.0	0.0	0.0	-514.0	-210.0	-100.0	12.0	260.0	30.0
Command bunker	271	Wall rib	Cube	12.0	260.0	30.0	0.0	0.0	0.0	-514.0	-210.0	120.0	12.0	260.0	30.0
Command bunker	272	Wall rib	Cube	12.0	260.0	30.0	0.0	0.0	0.0	-514.0	-210.0	310.0	12.0	260.0	30.0
Command bunker	273	Bunk frame	Cube	90.0	170.0	100.0	0.0	0.0	0.0	-470.0	-255.0	30.0	90.0	170.0	100.0
Command bunker	274	Lower mattress	Cube	84.0	17.0	96.0	0.0	0.0	0.0	-470.0	-296.5	30.0	84.0	17.0	96.0
Command bunker	275	Upper mattress	Cube	84.0	17.0	96.0	0.0	0.0	0.0	-470.0	-206.5	30.0	84.0	17.0	96.0
Command bunker	276	Pillow	Cube	50.0	8.0	30.0	0.0	0.0	0.0	-475.0	-194.0	0.0	50.0	8.0	30.0
Command bunker	277	Bunk frame	Cube	90.0	170.0	100.0	0.0	0.0	0.0	-470.0	-255.0	250.0	90.0	170.0	100.0
Command bunker	278	Lower mattress	Cube	84.0	17.0	96.0	0.0	0.0	0.0	-470.0	-296.5	250.0	84.0	17.0	96.0
Command bunker	279	Upper mattress	Cube	84.0	17.0	96.0	0.0	0.0	0.0	-470.0	-206.5	250.0	84.0	17.0	96.0
Command bunker	280	Pillow	Cube	50.0	8.0	30.0	0.0	0.0	0.0	-475.0	-194.0	220.0	50.0	8.0	30.0
Command bunker	281	Weapon rack	Cube	180.0	130.0	25.0	0.0	0.0	0.0	-310.0	-275.0	-167.5	180.0	130.0	25.0
Command bunker	282	Rifle	Cylinder	6.0	6.0	80.0	-90.0	0.0	0.0	-380.0	-265.0	-150.0	6.0	80.0	6.0
Command bunker	283	Rifle	Cylinder	6.0	6.0	80.0	-90.0	0.0	0.0	-325.0	-265.0	-150.0	6.0	80.0	6.0
Command bunker	284	Rifle	Cylinder	6.0	6.0	80.0	-90.0	0.0	0.0	-270.0	-265.0	-150.0	6.0	80.0	6.0
Command bunker	285	Lockers	Cube	190.0	190.0	50.0	0.0	0.0	0.0	-395.0	-245.0	340.0	190.0	190.0	50.0
Command bunker	286	Locker door line	Cube	2.0	170.0	1.0	0.0	0.0	0.0	-450.0	-245.0	314.5	2.0	170.0	1.0
Command bunker	287	Locker door line	Cube	2.0	170.0	1.0	0.0	0.0	0.0	-387.0	-245.0	314.5	2.0	170.0	1.0
Command bunker	288	Locker door line	Cube	2.0	170.0	1.0	0.0	0.0	0.0	-324.0	-245.0	314.5	2.0	170.0	1.0
Command bunker	289	Table	Cube	100.0	75.0	90.0	0.0	0.0	0.0	-300.0	-302.5	95.0	100.0	75.0	90.0
Command bunker	290	Bench	Cube	120.0	42.0	30.0	0.0	0.0	0.0	-300.0	-319.0	165.0	120.0	42.0	30.0
Command bunker	291	Room sign	Cube	4.0	25.0	80.0	0.0	0.0	0.0	-176.0	-112.5	140.0	4.0	25.0	80.0
Command bunker	292	Fire extinguisher	Cylinder	16.0	16.0	45.0	90.0	0.0	0.0	-200.0	-300.0	340.0	16.0	45.0	16.0
Command bunker	293	Floor line	Cube	200.0	1.2	10.0	0.0	0.0	0.0	-300.0	-339.4	-135.0	200.0	1.2	10.0
Command bunker	294	Fixture housing	Cube	130.0	5.0	30.0	0.0	0.0	0.0	350.0	-83.0	60.0	130.0	5.0	30.0
Command bunker	295	Plant room tube (light)	Cube	120.0	5.0	22.0	0.0	0.0	0.0	350.0	-87.0	60.0	120.0	5.0	22.0
Command bunker	296	Ceiling beam	Cube	340.0	20.0	24.0	0.0	0.0	0.0	350.0	-90.0	-90.0	340.0	20.0	24.0

Command bunker	297	Ceiling beam	Cube	340.0	20.0	24.0	0.0	0.0	0.0	350.0	-90.0	200.0	340.0	20.0	24.0
Command bunker	298	Wall rib	Cube	12.0	260.0	30.0	0.0	0.0	0.0	514.0	-210.0	-120.0	12.0	260.0	30.0
Command bunker	299	Wall rib	Cube	12.0	260.0	30.0	0.0	0.0	0.0	514.0	-210.0	0.0	12.0	260.0	30.0
Command bunker	300	Wall rib	Cube	12.0	260.0	30.0	0.0	0.0	0.0	514.0	-210.0	320.0	12.0	260.0	30.0
Command bunker	301	Generator skid	Cube	270.0	18.0	130.0	0.0	0.0	0.0	365.0	-331.0	265.0	270.0	18.0	130.0
Command bunker	302	Generator engine block	Cube	170.0	82.0	90.0	0.0	0.0	0.0	315.0	-281.0	265.0	170.0	82.0	90.0
Command bunker	303	Generator alternator	Cylinder	70.0	70.0	90.0	0.0	90.0	0.0	445.0	-275.0	265.0	90.0	70.0	70.0
Command bunker	304	Generator radiator	Cube	30.0	80.0	90.0	0.0	0.0	0.0	215.0	-275.0	265.0	30.0	80.0	90.0
Command bunker	305	Generator cylinder head	Cylinder	25.0	25.0	30.0	90.0	0.0	0.0	310.0	-225.0	260.0	25.0	30.0	25.0
Command bunker	306	Generator cylinder head	Cylinder	25.0	25.0	30.0	90.0	0.0	0.0	360.0	-225.0	260.0	25.0	30.0	25.0
Command bunker	307	Exhaust pipe	Cylinder	12.0	12.0	105.0	-49.6	107.1	0.0	422.5	-190.0	250.0	77.3	87.8	34.2
Command bunker	308	Exhaust silencer	Cylinder	30.0	30.0	60.0	90.0	0.0	0.0	455.0	-130.0	240.0	30.0	60.0	30.0
Command bunker	309	Exhaust riser	Cylinder	12.0	12.0	115.0	90.0	0.0	0.0	455.0	-42.5	240.0	12.0	115.0	12.0
Command bunker	310	Control cabinet	Cube	50.0	190.0	80.0	0.0	0.0	0.0	495.0	-245.0	140.0	50.0	190.0	80.0
Command bunker	311	Switchgear cabinet	Cube	50.0	190.0	80.0	0.0	0.0	0.0	495.0	-245.0	40.0	50.0	190.0	80.0
Command bunker	312	Cabinet display	Cube	4.0	30.0	60.0	0.0	0.0	0.0	468.0	-195.0	140.0	4.0	30.0	60.0
Command bunker	313	Fuel tank	Cylinder	100.0	100.0	220.0	0.0	90.0	0.0	340.0	-255.0	-100.0	220.0	100.0	100.0
Command bunker	314	Tank saddle	Cube	30.0	45.0	80.0	0.0	0.0	0.0	275.0	-317.5	-100.0	30.0	45.0	80.0
Command bunker	315	Tank saddle	Cube	30.0	45.0	80.0	0.0	0.0	0.0	405.0	-317.5	-100.0	30.0	45.0	80.0
Command bunker	316	Fuel line	Cylinder	5.0	5.0	320.0	0.0	0.0	0.0	390.0	-250.0	60.0	5.0	5.0	320.0
Command bunker	317	Ventilation unit	Cube	80.0	40.0	70.0	0.0	0.0	0.0	460.0	-115.0	-35.0	80.0	40.0	70.0
Command bunker	318	Ventilation riser	Cylinder	45.0	45.0	110.0	90.0	0.0	0.0	460.0	-40.0	-40.0	45.0	110.0	45.0
Command bunker	319	Floor line	Cube	10.0	1.2	200.0	0.0	0.0	0.0	195.0	-339.4	90.0	10.0	1.2	200.0
Command bunker	320	Floor drain	Cube	50.0	1.4	40.0	0.0	0.0	0.0	325.0	-339.3	80.0	50.0	1.4	40.0
Command bunker	321	Hazard marking	Cube	270.0	1.2	40.0	0.0	0.0	0.0	365.0	-339.4	170.0	270.0	1.2	40.0
Command bunker	322	Room sign	Cube	4.0	25.0	120.0	0.0	0.0	0.0	176.0	-112.5	120.0	4.0	25.0	120.0
Command bunker	323	Door leaf	Cube	5.0	206.0	117.0	0.0	0.0	0.0	176.5	-235.0	120.5	5.0	206.0	117.0
Command bunker	324	Hinge stile	Cube	1.2	206.0	11.0	0.0	0.0	0.0	173.4	-235.0	67.5	1.2	206.0	11.0
Command bunker	325	Lock stile	Cube	1.2	206.0	11.0	0.0	0.0	0.0	173.4	-235.0	173.5	1.2	206.0	11.0
Command bunker	326	Top rail	Cube	1.2	12.0	95.0	0.0	0.0	0.0	173.4	-138.0	120.5	1.2	12.0	95.0
Command bunker	327	Lock rail	Cube	1.2	14.0	95.0	0.0	0.0	0.0	173.4	-238.0	120.5	1.2	14.0	95.0
Command bunker	328	Bottom rail	Cube	1.2	22.0	95.0	0.0	0.0	0.0	173.4	-327.0	120.5	1.2	22.0	95.0
Command bunker	329	Kick plate	Cube	0.4	20.0	111.0	0.0	0.0	0.0	172.6	-327.0	120.5	0.4	20.0	111.0
Command bunker	330	Handle rose	Cylinder	6.5	6.5	1.2	0.0	90.0	0.0	172.2	-238.0	171.5	1.2	6.5	6.5
Command bunker	331	Handle spindle	Cylinder	2.0	2.0	5.0	0.0	90.0	0.0	169.5	-238.0	171.5	5.0	2.0	2.0
Command bunker	332	Handle lever	Cube	2.2	2.4	13.0	0.0	0.0	0.0	167.2	-238.0	166.0	2.2	2.4	13.0
Command bunker	333	Key cylinder	Cylinder	3.0	3.0	1.6	0.0	90.0	0.0	172.0	-251.0	171.5	1.6	3.0	3.0
Command bunker	334	Hinge stile	Cube	1.2	206.0	11.0	0.0	0.0	0.0	179.6	-235.0	67.5	1.2	206.0	11.0
Command bunker	335	Lock stile	Cube	1.2	206.0	11.0	0.0	0.0	0.0	179.6	-235.0	173.5	1.2	206.0	11.0
Command bunker	336	Top rail	Cube	1.2	12.0	95.0	0.0	0.0	0.0	179.6	-138.0	120.5	1.2	12.0	95.0
Command bunker	337	Lock rail	Cube	1.2	14.0	95.0	0.0	0.0	0.0	179.6	-238.0	120.5	1.2	14.0	95.0
Command bunker	338	Bottom rail	Cube	1.2	22.0	95.0	0.0	0.0	0.0	179.6	-327.0	120.5	1.2	22.0	95.0
Command bunker	339	Kick plate	Cube	0.4	20.0	111.0	0.0	0.0	0.0	180.4	-327.0	120.5	0.4	20.0	111.0
Command bunker	340	Handle rose	Cylinder	6.5	6.5	1.2	0.0	90.0	0.0	180.8	-238.0	171.5	1.2	6.5	6.5
Command bunker	341	Handle spindle	Cylinder	2.0	2.0	5.0	0.0	90.0	0.0	183.5	-238.0	171.5	5.0	2.0	2.0
Command bunker	342	Handle lever	Cube	2.2	2.4	13.0	0.0	0.0	0.0	185.8	-238.0	166.0	2.2	2.4	13.0
Command bunker	343	Key cylinder	Cylinder	3.0	3.0	1.6	0.0	90.0	0.0	181.0	-251.0	171.5	1.6	3.0	3.0
Command bunker	344	Hinge knuckle	Cylinder	2.6	2.6	11.0	90.0	0.0	0.0	179.0	-316.0	61.0	2.6	11.0	2.6
Command bunker	345	Hinge leaf	Cube	0.4	10.0	6.0	0.0	0.0	0.0	179.0	-316.0	64.0	0.4	10.0	6.0
Command bunker	346	Hinge knuckle	Cylinder	2.6	2.6	11.0	90.0	0.0	0.0	179.0	-225.0	61.0	2.6	11.0	2.6
Command bunker	347	Hinge leaf	Cube	0.4	10.0	6.0	0.0	0.0	0.0	179.0	-225.0	64.0	0.4	10.0	6.0

Command bunker	348	Hinge knuckle	Cylinder	2.6	2.6	11.0	90.0	0.0	0.0	179.0	-154.0	61.0	2.6	11.0	2.6
Command bunker	349	Hinge leaf	Cube	0.4	10.0	6.0	0.0	0.0	0.0	179.0	-154.0	64.0	0.4	10.0	6.0
Command bunker	350	Fire extinguisher	Cylinder	16.0	16.0	45.0	90.0	0.0	0.0	210.0	-300.0	340.0	16.0	45.0	16.0
Command bunker	351	Ceiling pipe	Cylinder	9.0	9.0	310.0	0.0	90.0	0.0	345.0	-92.0	320.0	310.0	9.0	9.0
Command bunker	352	Shaft lamp	Sphere	14.0	14.0	14.0	0.0	0.0	0.0	400.0	-110.0	-760.0	14.0	14.0	14.0
Command bunker	353	Ladder rail	Cylinder	5.0	5.0	333.3	90.0	0.0	0.0	430.0	-173.4	-810.0	5.0	333.3	5.0
Command bunker	354	Ladder rail	Cylinder	5.0	5.0	333.3	90.0	0.0	0.0	480.0	-173.4	-810.0	5.0	333.3	5.0
Command bunker	355	Ladder rung	Cylinder	3.0	3.0	50.0	0.0	90.0	0.0	455.0	-305.0	-810.0	50.0	3.0	3.0
Command bunker	356	Ladder rung	Cylinder	3.0	3.0	50.0	0.0	90.0	0.0	455.0	-275.0	-810.0	50.0	3.0	3.0
Command bunker	357	Ladder rung	Cylinder	3.0	3.0	50.0	0.0	90.0	0.0	455.0	-245.0	-810.0	50.0	3.0	3.0
Command bunker	358	Ladder rung	Cylinder	3.0	3.0	50.0	0.0	90.0	0.0	455.0	-215.0	-810.0	50.0	3.0	3.0
Command bunker	359	Ladder rung	Cylinder	3.0	3.0	50.0	0.0	90.0	0.0	455.0	-185.0	-810.0	50.0	3.0	3.0
Command bunker	360	Ladder rung	Cylinder	3.0	3.0	50.0	0.0	90.0	0.0	455.0	-155.0	-810.0	50.0	3.0	3.0
Command bunker	361	Ladder rung	Cylinder	3.0	3.0	50.0	0.0	90.0	0.0	455.0	-125.0	-810.0	50.0	3.0	3.0
Command bunker	362	Ladder rung	Cylinder	3.0	3.0	50.0	0.0	90.0	0.0	455.0	-95.0	-810.0	50.0	3.0	3.0
Command bunker	363	Ladder rung	Cylinder	3.0	3.0	50.0	0.0	90.0	0.0	455.0	-65.0	-810.0	50.0	3.0	3.0
Command bunker	364	Ladder rung	Cylinder	3.0	3.0	50.0	0.0	90.0	0.0	455.0	-35.0	-810.0	50.0	3.0	3.0
Command bunker	365	Exit sign	Cube	50.0	20.0	4.0	0.0	0.0	0.0	455.0	-100.0	-693.0	50.0	20.0	4.0
Command bunker	366	Vent stack A (exhaust)	Cube	75.0	120.0	75.0	0.0	0.0	0.0	455.0	59.5	240.0	75.0	120.0	75.0
Command bunker	367	Stack louvre	Cube	78.0	5.0	78.0	0.0	0.0	0.0	455.0	34.5	240.0	78.0	5.0	78.0
Command bunker	368	Stack louvre	Cube	78.0	5.0	78.0	0.0	0.0	0.0	455.0	59.5	240.0	78.0	5.0	78.0
Command bunker	369	Stack rain cap rim	Cube	85.0	6.0	85.0	0.0	0.0	0.0	455.0	122.5	240.0	85.0	6.0	85.0
Command bunker	370	Fan disc	Cylinder	60.0	60.0	3.0	0.0	0.0	0.0	455.0	129.5	240.0	60.0	3.0	60.0
Command bunker	371	Fan blade	Cube	58.0	7.0	2.0	0.0	0.0	20.0	455.0	126.5	240.0	56.9	2.0	26.4
Command bunker	372	Fan blade	Cube	58.0	7.0	2.0	0.0	0.0	110.0	455.0	126.5	240.0	26.4	2.0	56.9
Command bunker	373	Vent stack B (intake)	Cube	75.0	120.0	75.0	0.0	0.0	0.0	460.0	49.5	-40.0	75.0	120.0	75.0
Command bunker	374	Stack louvre	Cube	78.0	5.0	78.0	0.0	0.0	0.0	460.0	24.5	-40.0	78.0	5.0	78.0
Command bunker	375	Stack louvre	Cube	78.0	5.0	78.0	0.0	0.0	0.0	460.0	49.5	-40.0	78.0	5.0	78.0
Command bunker	376	Stack rain cap rim	Cube	85.0	6.0	85.0	0.0	0.0	0.0	460.0	112.5	-40.0	85.0	6.0	85.0
Command bunker	377	Fan disc	Cylinder	60.0	60.0	3.0	0.0	0.0	0.0	460.0	119.5	-40.0	60.0	3.0	60.0
Command bunker	378	Fan blade	Cube	58.0	7.0	2.0	0.0	0.0	20.0	460.0	116.5	-40.0	56.9	2.0	26.4
Command bunker	379	Fan blade	Cube	58.0	7.0	2.0	0.0	0.0	110.0	460.0	116.5	-40.0	26.4	2.0	56.9
Command bunker	380	Beacon mast base	Cube	40.0	30.0	40.0	0.0	0.0	0.0	-360.0	9.9	160.0	40.0	30.0	40.0
Command bunker	381	Beacon mast	Cylinder	9.0	9.0	400.0	90.0	0.0	0.0	-360.0	209.9	160.0	9.0	400.0	9.0
Command bunker	382	Beacon housing	Cylinder	22.0	22.0	10.0	90.0	0.0	0.0	-360.0	414.9	160.0	22.0	10.0	22.0
Command bunker	383	Beacon bulb	Sphere	20.0	20.0	20.0	0.0	0.0	0.0	-360.0	429.9	160.0	20.0	20.0	20.0
Command bunker	384	Periscope tube	Cylinder	16.0	16.0	90.0	90.0	0.0	0.0	80.0	41.9	-460.0	16.0	90.0	16.0
Command bunker	385	Periscope head	Cube	30.0	18.0	30.0	0.0	0.0	0.0	80.0	93.9	-460.0	30.0	18.0	30.0
Command bunker	386	Periscope lens	Cylinder	10.0	10.0	4.0	0.0	0.0	0.0	80.0	93.9	-444.0	10.0	10.0	4.0
Command bunker	387	Antenna base	Cube	40.0	25.0	40.0	0.0	0.0	0.0	-400.0	13.0	-600.0	40.0	25.0	40.0
Command bunker	388	Antenna mast	Cylinder	5.0	5.0	320.0	90.0	0.0	0.0	-400.0	178.0	-600.0	5.0	320.0	5.0
Command bunker	389	Antenna dish	Sphere	70.0	70.0	12.0	0.0	0.0	0.0	-400.0	268.0	-590.0	70.0	70.0	12.0
Command bunker	390	Hatch collar	Cube	180.0	30.0	180.0	0.0	0.0	0.0	455.0	-6.7	-755.0	180.0	30.0	180.0
Command bunker	391	Hatch lid	Cube	145.0	10.0	145.0	0.0	0.0	0.0	455.0	13.3	-755.0	145.0	10.0	145.0
Command bunker	392	Hatch wheel	Cylinder	40.0	40.0	4.0	90.0	0.0	0.0	455.0	22.3	-755.0	40.0	4.0	40.0
Command bunker	393	Hatch wheel spoke	Cube	44.0	5.0	5.0	0.0	0.0	0.0	455.0	26.3	-755.0	44.0	5.0	5.0
Command bunker	394	Hatch wheel spoke	Cube	5.0	5.0	44.0	0.0	0.0	0.0	455.0	26.3	-755.0	5.0	5.0	44.0
Command bunker	395	Hatch hinge	Cube	100.0	6.0	10.0	0.0	0.0	0.0	455.0	19.3	-830.0	100.0	6.0	10.0
Command bunker	396	Sandbag	Capsule	24.0	18.0	50.0	0.0	-127.7	0.0	336.4	-14.4	-840.7	54.2	18.0	49.6
Command bunker	397	Sandbag	Capsule	24.0	18.0	50.0	0.0	-149.5	0.0	378.9	-13.5	-875.6	46.1	18.0	55.3
Command bunker	398	Sandbag	Capsule	24.0	18.0	50.0	0.0	-171.3	0.0	432.2	-15.4	-893.4	31.3	18.0	53.1

Command bunker	399	Sandbag	Capsule	24.0	18.0	50.0	0.0	-193.0	0.0	488.8	-14.6	-891.4	34.7	18.0	54.1
Command bunker	400	Sandbag	Capsule	24.0	18.0	50.0	0.0	-214.8	0.0	540.6	-12.8	-869.9	48.2	18.0	54.8
Command bunker	401	Sandbag	Capsule	24.0	18.0	50.0	0.0	-236.6	0.0	580.2	-14.0	-832.1	55.0	18.0	47.6
Command bunker	402	Sandbag	Capsule	24.0	18.0	50.0	0.0	-258.4	0.0	601.9	-18.5	-783.3	53.8	18.0	33.6
Command bunker	403	Sandbag	Capsule	24.0	18.0	50.0	0.0	-280.1	0.0	602.7	-18.7	-730.4	53.4	18.0	32.4
Command bunker	404	Sandbag	Capsule	24.0	18.0	50.0	0.0	-301.9	0.0	582.3	-14.8	-681.0	55.1	18.0	46.8
Barracks hall	2	Floor slab	Cube	920.0	17.0	660.0	0.0	0.0	0.0	2310.0	-27.5	-1180.0	920.0	17.0	660.0
Barracks hall	3	Plank floor	Cube	856.0	1.2	596.0	0.0	0.0	0.0	2310.0	-18.6	-1180.0	856.0	1.2	596.0
Barracks hall	4	North wall	Cube	95.0	282.0	22.0	0.0	0.0	0.0	1907.5	121.0	-871.0	95.0	282.0	22.0
Barracks hall	5	North wall	Cube	90.0	197.0	22.0	0.0	0.0	0.0	2000.0	78.5	-871.0	90.0	197.0	22.0
Barracks hall	6	North wall	Cube	90.0	35.0	22.0	0.0	0.0	0.0	2000.0	244.5	-871.0	90.0	35.0	22.0
Barracks hall	7	Opening frame	Cube	6.0	50.0	26.0	0.0	0.0	0.0	1952.0	202.0	-871.0	6.0	50.0	26.0
Barracks hall	8	Opening frame	Cube	6.0	50.0	26.0	0.0	0.0	0.0	2048.0	202.0	-871.0	6.0	50.0	26.0
Barracks hall	9	Opening frame	Cube	102.0	6.0	26.0	0.0	0.0	0.0	2000.0	230.0	-871.0	102.0	6.0	26.0
Barracks hall	10	Opening frame	Cube	102.0	5.0	26.0	0.0	0.0	0.0	2000.0	174.5	-871.0	102.0	5.0	26.0
Barracks hall	11	North wall	Cube	240.0	282.0	22.0	0.0	0.0	0.0	2165.0	121.0	-871.0	240.0	282.0	22.0
Barracks hall	12	North wall	Cube	130.0	2.0	22.0	0.0	0.0	0.0	2350.0	-19.0	-871.0	130.0	2.0	22.0
Barracks hall	13	North wall	Cube	130.0	60.0	22.0	0.0	0.0	0.0	2350.0	232.0	-871.0	130.0	60.0	22.0
Barracks hall	14	Opening frame	Cube	6.0	220.0	26.0	0.0	0.0	0.0	2282.0	92.0	-871.0	6.0	220.0	26.0
Barracks hall	15	Opening frame	Cube	6.0	220.0	26.0	0.0	0.0	0.0	2418.0	92.0	-871.0	6.0	220.0	26.0
Barracks hall	16	Opening frame	Cube	142.0	6.0	26.0	0.0	0.0	0.0	2350.0	205.0	-871.0	142.0	6.0	26.0
Barracks hall	17	North wall	Cube	165.0	282.0	22.0	0.0	0.0	0.0	2497.5	121.0	-871.0	165.0	282.0	22.0
Barracks hall	18	North wall	Cube	120.0	102.0	22.0	0.0	0.0	0.0	2640.0	31.0	-871.0	120.0	102.0	22.0
Barracks hall	19	North wall	Cube	120.0	90.0	22.0	0.0	0.0	0.0	2640.0	217.0	-871.0	120.0	90.0	22.0
Barracks hall	20	Opening frame	Cube	6.0	90.0	26.0	0.0	0.0	0.0	2577.0	127.0	-871.0	6.0	90.0	26.0
Barracks hall	21	Opening frame	Cube	6.0	90.0	26.0	0.0	0.0	0.0	2703.0	127.0	-871.0	6.0	90.0	26.0
Barracks hall	22	Opening frame	Cube	132.0	6.0	26.0	0.0	0.0	0.0	2640.0	175.0	-871.0	132.0	6.0	26.0
Barracks hall	23	Opening frame	Cube	132.0	5.0	26.0	0.0	0.0	0.0	2640.0	79.5	-871.0	132.0	5.0	26.0
Barracks hall	24	North wall	Cube	60.0	282.0	22.0	0.0	0.0	0.0	2730.0	121.0	-871.0	60.0	282.0	22.0
Barracks hall	25	South wall	Cube	330.0	282.0	22.0	0.0	0.0	0.0	2025.0	121.0	-1489.0	330.0	282.0	22.0
Barracks hall	26	South wall	Cube	100.0	102.0	22.0	0.0	0.0	0.0	2240.0	31.0	-1489.0	100.0	102.0	22.0
Barracks hall	27	South wall	Cube	100.0	90.0	22.0	0.0	0.0	0.0	2240.0	217.0	-1489.0	100.0	90.0	22.0
Barracks hall	28	Opening frame	Cube	6.0	90.0	26.0	0.0	0.0	0.0	2187.0	127.0	-1489.0	6.0	90.0	26.0
Barracks hall	29	Opening frame	Cube	6.0	90.0	26.0	0.0	0.0	0.0	2293.0	127.0	-1489.0	6.0	90.0	26.0
Barracks hall	30	Opening frame	Cube	112.0	6.0	26.0	0.0	0.0	0.0	2240.0	175.0	-1489.0	112.0	6.0	26.0
Barracks hall	31	Opening frame	Cube	112.0	5.0	26.0	0.0	0.0	0.0	2240.0	79.5	-1489.0	112.0	5.0	26.0
Barracks hall	32	South wall	Cube	240.0	282.0	22.0	0.0	0.0	0.0	2410.0	121.0	-1489.0	240.0	282.0	22.0
Barracks hall	33	South wall	Cube	120.0	102.0	22.0	0.0	0.0	0.0	2590.0	31.0	-1489.0	120.0	102.0	22.0
Barracks hall	34	South wall	Cube	120.0	90.0	22.0	0.0	0.0	0.0	2590.0	217.0	-1489.0	120.0	90.0	22.0
Barracks hall	35	Opening frame	Cube	6.0	90.0	26.0	0.0	0.0	0.0	2527.0	127.0	-1489.0	6.0	90.0	26.0
Barracks hall	36	Opening frame	Cube	6.0	90.0	26.0	0.0	0.0	0.0	2653.0	127.0	-1489.0	6.0	90.0	26.0
Barracks hall	37	Opening frame	Cube	132.0	6.0	26.0	0.0	0.0	0.0	2590.0	175.0	-1489.0	132.0	6.0	26.0
Barracks hall	38	Opening frame	Cube	132.0	5.0	26.0	0.0	0.0	0.0	2590.0	79.5	-1489.0	132.0	5.0	26.0
Barracks hall	39	South wall	Cube	110.0	282.0	22.0	0.0	0.0	0.0	2705.0	121.0	-1489.0	110.0	282.0	22.0
Barracks hall	40	East wall	Cube	22.0	282.0	118.0	0.0	0.0	0.0	2749.0	121.0	-1419.0	22.0	282.0	118.0
Barracks hall	41	East wall	Cube	22.0	102.0	120.0	0.0	0.0	0.0	2749.0	31.0	-1300.0	22.0	102.0	120.0
Barracks hall	42	East wall	Cube	22.0	90.0	120.0	0.0	0.0	0.0	2749.0	217.0	-1300.0	22.0	90.0	120.0
Barracks hall	43	Opening frame	Cube	26.0	90.0	6.0	0.0	0.0	0.0	2749.0	127.0	-1363.0	26.0	90.0	6.0
Barracks hall	44	Opening frame	Cube	26.0	90.0	6.0	0.0	0.0	0.0	2749.0	127.0	-1237.0	26.0	90.0	6.0
Barracks hall	45	Opening frame	Cube	26.0	6.0	132.0	0.0	0.0	0.0	2749.0	175.0	-1300.0	26.0	6.0	132.0
Barracks hall	46	Opening frame	Cube	26.0	5.0	132.0	0.0	0.0	0.0	2749.0	79.5	-1300.0	26.0	5.0	132.0

Barracks hall	47	East wall	Cube	22.0	282.0	140.0	0.0	0.0	0.0	2749.0	121.0	-1170.0	22.0	282.0	140.0
Barracks hall	48	East wall	Cube	22.0	102.0	120.0	0.0	0.0	0.0	2749.0	31.0	-1040.0	22.0	102.0	120.0
Barracks hall	49	East wall	Cube	22.0	90.0	120.0	0.0	0.0	0.0	2749.0	217.0	-1040.0	22.0	90.0	120.0
Barracks hall	50	Opening frame	Cube	26.0	90.0	6.0	0.0	0.0	0.0	2749.0	127.0	-1103.0	26.0	90.0	6.0
Barracks hall	51	Opening frame	Cube	26.0	90.0	6.0	0.0	0.0	0.0	2749.0	127.0	-977.0	26.0	90.0	6.0
Barracks hall	52	Opening frame	Cube	26.0	6.0	132.0	0.0	0.0	0.0	2749.0	175.0	-1040.0	26.0	6.0	132.0
Barracks hall	53	Opening frame	Cube	26.0	5.0	132.0	0.0	0.0	0.0	2749.0	79.5	-1040.0	26.0	5.0	132.0
Barracks hall	54	East wall	Cube	22.0	282.0	98.0	0.0	0.0	0.0	2749.0	121.0	-931.0	22.0	282.0	98.0
Barracks hall	55	West wall	Cube	22.0	282.0	118.0	0.0	0.0	0.0	1871.0	121.0	-1419.0	22.0	282.0	118.0
Barracks hall	56	West wall	Cube	22.0	202.0	70.0	0.0	0.0	0.0	1871.0	81.0	-1325.0	22.0	202.0	70.0
Barracks hall	57	West wall	Cube	22.0	35.0	70.0	0.0	0.0	0.0	1871.0	244.5	-1325.0	22.0	35.0	70.0
Barracks hall	58	Opening frame	Cube	26.0	45.0	6.0	0.0	0.0	0.0	1871.0	204.5	-1363.0	26.0	45.0	6.0
Barracks hall	59	Opening frame	Cube	26.0	45.0	6.0	0.0	0.0	0.0	1871.0	204.5	-1287.0	26.0	45.0	6.0
Barracks hall	60	Opening frame	Cube	26.0	6.0	82.0	0.0	0.0	0.0	1871.0	230.0	-1325.0	26.0	6.0	82.0
Barracks hall	61	Opening frame	Cube	26.0	5.0	82.0	0.0	0.0	0.0	1871.0	179.5	-1325.0	26.0	5.0	82.0
Barracks hall	62	West wall	Cube	22.0	282.0	80.0	0.0	0.0	0.0	1871.0	121.0	-1250.0	22.0	282.0	80.0
Barracks hall	63	West wall	Cube	22.0	202.0	70.0	0.0	0.0	0.0	1871.0	81.0	-1175.0	22.0	202.0	70.0
Barracks hall	64	West wall	Cube	22.0	35.0	70.0	0.0	0.0	0.0	1871.0	244.5	-1175.0	22.0	35.0	70.0
Barracks hall	65	Opening frame	Cube	26.0	45.0	6.0	0.0	0.0	0.0	1871.0	204.5	-1213.0	26.0	45.0	6.0
Barracks hall	66	Opening frame	Cube	26.0	45.0	6.0	0.0	0.0	0.0	1871.0	204.5	-1137.0	26.0	45.0	6.0
Barracks hall	67	Opening frame	Cube	26.0	6.0	82.0	0.0	0.0	0.0	1871.0	230.0	-1175.0	26.0	6.0	82.0
Barracks hall	68	Opening frame	Cube	26.0	5.0	82.0	0.0	0.0	0.0	1871.0	179.5	-1175.0	26.0	5.0	82.0
Barracks hall	69	West wall	Cube	22.0	282.0	258.0	0.0	0.0	0.0	1871.0	121.0	-1011.0	22.0	282.0	258.0
Barracks hall	70	Wall lining	Cube	1.2	60.0	596.0	0.0	0.0	0.0	1882.6	12.0	-1180.0	1.2	60.0	596.0
Barracks hall	71	Wall lining	Cube	1.2	60.0	596.0	0.0	0.0	0.0	2737.4	12.0	-1180.0	1.2	60.0	596.0
Barracks hall	72	Ceiling	Cube	856.0	8.0	596.0	0.0	0.0	0.0	2310.0	256.0	-1180.0	856.0	8.0	596.0
Barracks hall	73	Roof sheet	Cube	960.0	5.0	371.9	-17.4	0.0	0.0	2310.0	310.5	-1357.5	960.0	115.7	356.5
Barracks hall	74	Roof sheet	Cube	960.0	5.0	371.9	17.4	0.0	0.0	2310.0	310.5	-1002.5	960.0	115.7	356.5
Barracks hall	75	Ridge cap	Cube	960.0	8.0	24.0	0.0	0.0	0.0	2310.0	366.0	-1180.0	960.0	8.0	24.0
Barracks hall	76	Gable	Cube	22.0	20.0	576.0	0.0	0.0	0.0	1871.0	272.0	-1180.0	22.0	20.0	576.0
Barracks hall	77	Gable	Cube	22.0	20.0	448.0	0.0	0.0	0.0	1871.0	292.0	-1180.0	22.0	20.0	448.0
Barracks hall	78	Gable	Cube	22.0	20.0	320.0	0.0	0.0	0.0	1871.0	312.0	-1180.0	22.0	20.0	320.0
Barracks hall	79	Gable	Cube	22.0	20.0	192.0	0.0	0.0	0.0	1871.0	332.0	-1180.0	22.0	20.0	192.0
Barracks hall	80	Gable	Cube	22.0	20.0	64.0	0.0	0.0	0.0	1871.0	352.0	-1180.0	22.0	20.0	64.0
Barracks hall	81	Gable	Cube	22.0	20.0	576.0	0.0	0.0	0.0	2749.0	272.0	-1180.0	22.0	20.0	576.0
Barracks hall	82	Gable	Cube	22.0	20.0	448.0	0.0	0.0	0.0	2749.0	292.0	-1180.0	22.0	20.0	448.0
Barracks hall	83	Gable	Cube	22.0	20.0	320.0	0.0	0.0	0.0	2749.0	312.0	-1180.0	22.0	20.0	320.0
Barracks hall	84	Gable	Cube	22.0	20.0	192.0	0.0	0.0	0.0	2749.0	332.0	-1180.0	22.0	20.0	192.0
Barracks hall	85	Gable	Cube	22.0	20.0	64.0	0.0	0.0	0.0	2749.0	352.0	-1180.0	22.0	20.0	64.0
Barracks hall	86	Door step	Cube	180.0	14.0	70.0	0.0	0.0	0.0	2350.0	-27.0	-825.0	180.0	14.0	70.0
Barracks hall	87	Door leaf	Cube	5.0	218.0	127.8	0.0	0.0	0.0	2349.9	92.2	-863.5	127.8	218.0	5.0
Barracks hall	88	Hinge stile	Cube	1.2	218.0	11.0	0.0	0.0	0.0	2408.3	92.2	-866.6	11.0	218.0	1.2
Barracks hall	89	Lock stile	Cube	1.2	218.0	11.0	0.0	0.0	0.0	2291.5	92.2	-866.6	11.0	218.0	1.2
Barracks hall	90	Top rail	Cube	1.2	12.0	105.8	0.0	0.0	0.0	2349.9	195.2	-866.6	105.8	12.0	1.2
Barracks hall	91	Lock rail	Cube	1.2	14.0	105.8	0.0	0.0	0.0	2349.9	83.2	-866.6	105.8	14.0	1.2
Barracks hall	92	Bottom rail	Cube	1.2	22.0	105.8	0.0	0.0	0.0	2349.9	-5.8	-866.6	105.8	22.0	1.2
Barracks hall	93	Kick plate	Cube	0.4	20.0	121.8	0.0	0.0	0.0	2349.9	-5.8	-867.4	121.8	20.0	0.4
Barracks hall	94	Handle rose	Cylinder	6.5	6.5	1.2	0.0	90.0	0.0	2293.5	83.2	-867.8	6.5	6.5	1.2
Barracks hall	95	Handle spindle	Cylinder	2.0	2.0	5.0	0.0	90.0	0.0	2293.5	83.2	-870.5	2.0	2.0	5.0
Barracks hall	96	Handle lever	Cube	2.2	2.4	13.0	0.0	0.0	0.0	2299.0	83.2	-872.8	13.0	2.4	2.2
Barracks hall	97	Key cylinder	Cylinder	3.0	3.0	1.6	0.0	90.0	0.0	2293.5	70.2	-868.0	3.0	3.0	1.6

Barracks hall	98	Hinge stile	Cube	1.2	218.0	11.0	0.0	0.0	0.0	2408.3	92.2	-860.4	11.0	218.0	1.2
Barracks hall	99	Lock stile	Cube	1.2	218.0	11.0	0.0	0.0	0.0	2291.5	92.2	-860.4	11.0	218.0	1.2
Barracks hall	100	Top rail	Cube	1.2	12.0	105.8	0.0	0.0	0.0	2349.9	195.2	-860.4	105.8	12.0	1.2
Barracks hall	101	Lock rail	Cube	1.2	14.0	105.8	0.0	0.0	0.0	2349.9	83.2	-860.4	105.8	14.0	1.2
Barracks hall	102	Bottom rail	Cube	1.2	22.0	105.8	0.0	0.0	0.0	2349.9	-5.8	-860.4	105.8	22.0	1.2
Barracks hall	103	Kick plate	Cube	0.4	20.0	121.8	0.0	0.0	0.0	2349.9	-5.8	-859.6	121.8	20.0	0.4
Barracks hall	104	Handle rose	Cylinder	6.5	6.5	1.2	0.0	90.0	0.0	2293.5	83.2	-859.2	6.5	6.5	1.2
Barracks hall	105	Handle spindle	Cylinder	2.0	2.0	5.0	0.0	90.0	0.0	2293.5	83.2	-856.5	2.0	2.0	5.0
Barracks hall	106	Handle lever	Cube	2.2	2.4	13.0	0.0	0.0	0.0	2299.0	83.2	-854.2	13.0	2.4	2.2
Barracks hall	107	Key cylinder	Cylinder	3.0	3.0	1.6	0.0	90.0	0.0	2293.5	70.2	-859.0	3.0	3.0	1.6
Barracks hall	108	Hinge knuckle	Cylinder	2.6	2.6	11.0	90.0	0.0	0.0	2415.0	5.2	-858.5	2.6	11.0	2.6
Barracks hall	109	Hinge leaf	Cube	2.5	10.0	6.2	0.0	0.0	0.0	2411.9	5.2	-859.7	6.2	10.0	2.5
Barracks hall	110	Hinge knuckle	Cylinder	2.6	2.6	11.0	90.0	0.0	0.0	2415.0	102.2	-858.5	2.6	11.0	2.6
Barracks hall	111	Hinge leaf	Cube	2.5	10.0	6.2	0.0	0.0	0.0	2411.9	102.2	-859.7	6.2	10.0	2.5
Barracks hall	112	Hinge knuckle	Cylinder	2.6	2.6	11.0	90.0	0.0	0.0	2415.0	179.2	-858.5	2.6	11.0	2.6
Barracks hall	113	Hinge leaf	Cube	2.5	10.0	6.2	0.0	0.0	0.0	2411.9	179.2	-859.7	6.2	10.0	2.5
Barracks hall	114	Door lamp hood	Cube	30.0	4.0	24.0	0.0	0.0	0.0	2350.0	237.0	-846.0	30.0	4.0	24.0
Barracks hall	115	Hall door lamp	Sphere	13.0	8.0	13.0	0.0	0.0	0.0	2350.0	230.0	-846.0	13.0	8.0	13.0
Barracks hall	116	Sign board	Cube	100.0	32.0	3.0	0.0	0.0	0.0	2225.0	202.0	-857.0	100.0	32.0	3.0
Barracks hall	117	Sign letters	Cube	80.0	10.0	1.0	0.0	0.0	0.0	2225.0	202.0	-855.0	80.0	10.0	1.0
Barracks hall	118	Hall AC unit	Cube	45.0	60.0	90.0	0.0	0.0	0.0	2782.5	202.0	-1175.0	45.0	60.0	90.0
Barracks hall	119	Hall AC fan	Cylinder	45.0	45.0	4.0	0.0	90.0	0.0	2807.0	202.0	-1175.0	4.0	45.0	45.0
Barracks hall	120	Bunk post	Cube	5.0	192.0	5.0	0.0	0.0	0.0	1888.0	78.0	-1444.5	5.0	192.0	5.0
Barracks hall	121	Bunk post	Cube	5.0	192.0	5.0	0.0	0.0	0.0	1888.0	78.0	-1355.5	5.0	192.0	5.0
Barracks hall	122	Bunk post	Cube	5.0	192.0	5.0	0.0	0.0	0.0	2082.0	78.0	-1444.5	5.0	192.0	5.0
Barracks hall	123	Bunk post	Cube	5.0	192.0	5.0	0.0	0.0	0.0	2082.0	78.0	-1355.5	5.0	192.0	5.0
Barracks hall	124	Bunk rail	Cube	200.0	6.0	4.0	0.0	0.0	0.0	1985.0	10.0	-1444.5	200.0	6.0	4.0
Barracks hall	125	Bunk rail	Cube	200.0	6.0	4.0	0.0	0.0	0.0	1985.0	10.0	-1355.5	200.0	6.0	4.0
Barracks hall	126	Bunk springs	Cube	192.0	3.0	83.0	0.0	0.0	0.0	1985.0	11.5	-1400.0	192.0	3.0	83.0
Barracks hall	127	Mattress	Cube	190.0	14.0	81.0	0.0	0.0	0.0	1985.0	20.0	-1400.0	190.0	14.0	81.0
Barracks hall	128	Pillow	Cube	34.0	9.0	62.0	0.0	0.0	0.0	1910.0	31.5	-1400.0	34.0	9.0	62.0
Barracks hall	129	Blanket	Cube	125.0	3.5	85.0	0.0	0.0	0.0	2017.0	28.7	-1400.0	125.0	3.5	85.0
Barracks hall	130	Head rail	Cube	4.0	30.0	89.0	0.0	0.0	0.0	1887.0	37.0	-1400.0	4.0	30.0	89.0
Barracks hall	131	Bunk rail	Cube	200.0	6.0	4.0	0.0	0.0	0.0	1985.0	115.0	-1444.5	200.0	6.0	4.0
Barracks hall	132	Bunk rail	Cube	200.0	6.0	4.0	0.0	0.0	0.0	1985.0	115.0	-1355.5	200.0	6.0	4.0
Barracks hall	133	Bunk springs	Cube	192.0	3.0	83.0	0.0	0.0	0.0	1985.0	116.5	-1400.0	192.0	3.0	83.0
Barracks hall	134	Mattress	Cube	190.0	14.0	81.0	0.0	0.0	0.0	1985.0	125.0	-1400.0	190.0	14.0	81.0
Barracks hall	135	Pillow	Cube	34.0	9.0	62.0	0.0	0.0	0.0	1910.0	136.5	-1400.0	34.0	9.0	62.0
Barracks hall	136	Blanket	Cube	125.0	3.5	85.0	0.0	0.0	0.0	2017.0	133.7	-1400.0	125.0	3.5	85.0
Barracks hall	137	Head rail	Cube	4.0	30.0	89.0	0.0	0.0	0.0	1887.0	142.0	-1400.0	4.0	30.0	89.0
Barracks hall	138	Ladder rail	Cube	3.0	155.0	3.0	0.0	0.0	0.0	2087.0	60.0	-1381.5	3.0	155.0	3.0
Barracks hall	139	Ladder rail	Cube	3.0	155.0	3.0	0.0	0.0	0.0	2087.0	60.0	-1347.5	3.0	155.0	3.0
Barracks hall	140	Ladder rung	Cylinder	2.5	2.5	34.0	0.0	0.0	0.0	2087.0	17.0	-1364.5	2.5	2.5	34.0
Barracks hall	141	Ladder rung	Cylinder	2.5	2.5	34.0	0.0	0.0	0.0	2087.0	52.0	-1364.5	2.5	2.5	34.0
Barracks hall	142	Ladder rung	Cylinder	2.5	2.5	34.0	0.0	0.0	0.0	2087.0	87.0	-1364.5	2.5	2.5	34.0
Barracks hall	143	Ladder rung	Cylinder	2.5	2.5	34.0	0.0	0.0	0.0	2087.0	122.0	-1364.5	2.5	2.5	34.0
Barracks hall	144	Bunk post	Cube	5.0	192.0	5.0	0.0	0.0	0.0	1888.0	78.0	-1294.5	5.0	192.0	5.0
Barracks hall	145	Bunk post	Cube	5.0	192.0	5.0	0.0	0.0	0.0	1888.0	78.0	-1205.5	5.0	192.0	5.0
Barracks hall	146	Bunk post	Cube	5.0	192.0	5.0	0.0	0.0	0.0	2082.0	78.0	-1294.5	5.0	192.0	5.0
Barracks hall	147	Bunk post	Cube	5.0	192.0	5.0	0.0	0.0	0.0	2082.0	78.0	-1205.5	5.0	192.0	5.0
Barracks hall	148	Bunk rail	Cube	200.0	6.0	4.0	0.0	0.0	0.0	1985.0	10.0	-1294.5	200.0	6.0	4.0

Barracks hall	149	Bunk rail	Cube	200.0	6.0	4.0	0.0	0.0	0.0	1985.0	10.0	-1205.5	200.0	6.0	4.0
Barracks hall	150	Bunk springs	Cube	192.0	3.0	83.0	0.0	0.0	0.0	1985.0	11.5	-1250.0	192.0	3.0	83.0
Barracks hall	151	Mattress	Cube	190.0	14.0	81.0	0.0	0.0	0.0	1985.0	20.0	-1250.0	190.0	14.0	81.0
Barracks hall	152	Pillow	Cube	34.0	9.0	62.0	0.0	0.0	0.0	1910.0	31.5	-1250.0	34.0	9.0	62.0
Barracks hall	153	Blanket	Cube	125.0	3.5	85.0	0.0	0.0	0.0	2017.0	28.7	-1250.0	125.0	3.5	85.0
Barracks hall	154	Head rail	Cube	4.0	30.0	89.0	0.0	0.0	0.0	1887.0	37.0	-1250.0	4.0	30.0	89.0
Barracks hall	155	Bunk rail	Cube	200.0	6.0	4.0	0.0	0.0	0.0	1985.0	115.0	-1294.5	200.0	6.0	4.0
Barracks hall	156	Bunk rail	Cube	200.0	6.0	4.0	0.0	0.0	0.0	1985.0	115.0	-1205.5	200.0	6.0	4.0
Barracks hall	157	Bunk springs	Cube	192.0	3.0	83.0	0.0	0.0	0.0	1985.0	116.5	-1250.0	192.0	3.0	83.0
Barracks hall	158	Mattress	Cube	190.0	14.0	81.0	0.0	0.0	0.0	1985.0	125.0	-1250.0	190.0	14.0	81.0
Barracks hall	159	Pillow	Cube	34.0	9.0	62.0	0.0	0.0	0.0	1910.0	136.5	-1250.0	34.0	9.0	62.0
Barracks hall	160	Blanket	Cube	125.0	3.5	85.0	0.0	0.0	0.0	2017.0	133.7	-1250.0	125.0	3.5	85.0
Barracks hall	161	Head rail	Cube	4.0	30.0	89.0	0.0	0.0	0.0	1887.0	142.0	-1250.0	4.0	30.0	89.0
Barracks hall	162	Ladder rail	Cube	3.0	155.0	3.0	0.0	0.0	0.0	2087.0	60.0	-1231.5	3.0	155.0	3.0
Barracks hall	163	Ladder rail	Cube	3.0	155.0	3.0	0.0	0.0	0.0	2087.0	60.0	-1197.5	3.0	155.0	3.0
Barracks hall	164	Ladder rung	Cylinder	2.5	2.5	34.0	0.0	0.0	0.0	2087.0	17.0	-1214.5	2.5	2.5	34.0
Barracks hall	165	Ladder rung	Cylinder	2.5	2.5	34.0	0.0	0.0	0.0	2087.0	52.0	-1214.5	2.5	2.5	34.0
Barracks hall	166	Ladder rung	Cylinder	2.5	2.5	34.0	0.0	0.0	0.0	2087.0	87.0	-1214.5	2.5	2.5	34.0
Barracks hall	167	Ladder rung	Cylinder	2.5	2.5	34.0	0.0	0.0	0.0	2087.0	122.0	-1214.5	2.5	2.5	34.0
Barracks hall	168	Bunk post	Cube	5.0	192.0	5.0	0.0	0.0	0.0	1888.0	78.0	-1144.5	5.0	192.0	5.0
Barracks hall	169	Bunk post	Cube	5.0	192.0	5.0	0.0	0.0	0.0	1888.0	78.0	-1055.5	5.0	192.0	5.0
Barracks hall	170	Bunk post	Cube	5.0	192.0	5.0	0.0	0.0	0.0	2082.0	78.0	-1144.5	5.0	192.0	5.0
Barracks hall	171	Bunk post	Cube	5.0	192.0	5.0	0.0	0.0	0.0	2082.0	78.0	-1055.5	5.0	192.0	5.0
Barracks hall	172	Bunk rail	Cube	200.0	6.0	4.0	0.0	0.0	0.0	1985.0	10.0	-1144.5	200.0	6.0	4.0
Barracks hall	173	Bunk rail	Cube	200.0	6.0	4.0	0.0	0.0	0.0	1985.0	10.0	-1055.5	200.0	6.0	4.0
Barracks hall	174	Bunk springs	Cube	192.0	3.0	83.0	0.0	0.0	0.0	1985.0	11.5	-1100.0	192.0	3.0	83.0
Barracks hall	175	Mattress	Cube	190.0	14.0	81.0	0.0	0.0	0.0	1985.0	20.0	-1100.0	190.0	14.0	81.0
Barracks hall	176	Pillow	Cube	34.0	9.0	62.0	0.0	0.0	0.0	1910.0	31.5	-1100.0	34.0	9.0	62.0
Barracks hall	177	Blanket	Cube	125.0	3.5	85.0	0.0	0.0	0.0	2017.0	28.7	-1100.0	125.0	3.5	85.0
Barracks hall	178	Head rail	Cube	4.0	30.0	89.0	0.0	0.0	0.0	1887.0	37.0	-1100.0	4.0	30.0	89.0
Barracks hall	179	Bunk rail	Cube	200.0	6.0	4.0	0.0	0.0	0.0	1985.0	115.0	-1144.5	200.0	6.0	4.0
Barracks hall	180	Bunk rail	Cube	200.0	6.0	4.0	0.0	0.0	0.0	1985.0	115.0	-1055.5	200.0	6.0	4.0
Barracks hall	181	Bunk springs	Cube	192.0	3.0	83.0	0.0	0.0	0.0	1985.0	116.5	-1100.0	192.0	3.0	83.0
Barracks hall	182	Mattress	Cube	190.0	14.0	81.0	0.0	0.0	0.0	1985.0	125.0	-1100.0	190.0	14.0	81.0
Barracks hall	183	Pillow	Cube	34.0	9.0	62.0	0.0	0.0	0.0	1910.0	136.5	-1100.0	34.0	9.0	62.0
Barracks hall	184	Blanket	Cube	125.0	3.5	85.0	0.0	0.0	0.0	2017.0	133.7	-1100.0	125.0	3.5	85.0
Barracks hall	185	Head rail	Cube	4.0	30.0	89.0	0.0	0.0	0.0	1887.0	142.0	-1100.0	4.0	30.0	89.0
Barracks hall	186	Ladder rail	Cube	3.0	155.0	3.0	0.0	0.0	0.0	2087.0	60.0	-1081.5	3.0	155.0	3.0
Barracks hall	187	Ladder rail	Cube	3.0	155.0	3.0	0.0	0.0	0.0	2087.0	60.0	-1047.5	3.0	155.0	3.0
Barracks hall	188	Ladder rung	Cylinder	2.5	2.5	34.0	0.0	0.0	0.0	2087.0	17.0	-1064.5	2.5	2.5	34.0
Barracks hall	189	Ladder rung	Cylinder	2.5	2.5	34.0	0.0	0.0	0.0	2087.0	52.0	-1064.5	2.5	2.5	34.0
Barracks hall	190	Ladder rung	Cylinder	2.5	2.5	34.0	0.0	0.0	0.0	2087.0	87.0	-1064.5	2.5	2.5	34.0
Barracks hall	191	Ladder rung	Cylinder	2.5	2.5	34.0	0.0	0.0	0.0	2087.0	122.0	-1064.5	2.5	2.5	34.0
Barracks hall	192	Locker	Cube	54.7	185.0	48.0	0.0	0.0	0.0	1928.3	74.5	-906.0	54.7	185.0	48.0
Barracks hall	193	Locker vents	Cube	28.3	12.0	1.0	0.0	0.0	0.0	1928.3	142.0	-930.5	28.3	12.0	1.0
Barracks hall	194	Locker handle	Cube	3.0	12.0	3.0	0.0	0.0	0.0	1948.7	87.0	-931.5	3.0	12.0	3.0
Barracks hall	195	Name tag	Cube	14.0	5.0	0.5	0.0	0.0	0.0	1928.3	122.0	-930.3	14.0	5.0	0.5
Barracks hall	196	Locker	Cube	54.7	185.0	48.0	0.0	0.0	0.0	1985.0	74.5	-906.0	54.7	185.0	48.0
Barracks hall	197	Locker vents	Cube	28.3	12.0	1.0	0.0	0.0	0.0	1985.0	142.0	-930.5	28.3	12.0	1.0
Barracks hall	198	Locker handle	Cube	3.0	12.0	3.0	0.0	0.0	0.0	2005.3	87.0	-931.5	3.0	12.0	3.0
Barracks hall	199	Name tag	Cube	14.0	5.0	0.5	0.0	0.0	0.0	1985.0	122.0	-930.3	14.0	5.0	0.5

Barracks hall	200	Locker	Cube	54.7	185.0	48.0	0.0	0.0	0.0	2041.7	74.5	-906.0	54.7	185.0	48.0
Barracks hall	201	Locker vents	Cube	28.3	12.0	1.0	0.0	0.0	0.0	2041.7	142.0	-930.5	28.3	12.0	1.0
Barracks hall	202	Locker handle	Cube	3.0	12.0	3.0	0.0	0.0	0.0	2062.0	87.0	-931.5	3.0	12.0	3.0
Barracks hall	203	Name tag	Cube	14.0	5.0	0.5	0.0	0.0	0.0	2041.7	122.0	-930.3	14.0	5.0	0.5
Barracks hall	204	Locker	Cube	54.7	185.0	48.0	0.0	0.0	0.0	2098.3	74.5	-906.0	54.7	185.0	48.0
Barracks hall	205	Locker vents	Cube	28.3	12.0	1.0	0.0	0.0	0.0	2098.3	142.0	-930.5	28.3	12.0	1.0
Barracks hall	206	Locker handle	Cube	3.0	12.0	3.0	0.0	0.0	0.0	2118.7	87.0	-931.5	3.0	12.0	3.0
Barracks hall	207	Name tag	Cube	14.0	5.0	0.5	0.0	0.0	0.0	2098.3	122.0	-930.3	14.0	5.0	0.5
Barracks hall	208	Locker	Cube	54.7	185.0	48.0	0.0	0.0	0.0	2155.0	74.5	-906.0	54.7	185.0	48.0
Barracks hall	209	Locker vents	Cube	28.3	12.0	1.0	0.0	0.0	0.0	2155.0	142.0	-930.5	28.3	12.0	1.0
Barracks hall	210	Locker handle	Cube	3.0	12.0	3.0	0.0	0.0	0.0	2175.3	87.0	-931.5	3.0	12.0	3.0
Barracks hall	211	Name tag	Cube	14.0	5.0	0.5	0.0	0.0	0.0	2155.0	122.0	-930.3	14.0	5.0	0.5
Barracks hall	212	Locker	Cube	54.7	185.0	48.0	0.0	0.0	0.0	2211.7	74.5	-906.0	54.7	185.0	48.0
Barracks hall	213	Locker vents	Cube	28.3	12.0	1.0	0.0	0.0	0.0	2211.7	142.0	-930.5	28.3	12.0	1.0
Barracks hall	214	Locker handle	Cube	3.0	12.0	3.0	0.0	0.0	0.0	2232.0	87.0	-931.5	3.0	12.0	3.0
Barracks hall	215	Name tag	Cube	14.0	5.0	0.5	0.0	0.0	0.0	2211.7	122.0	-930.3	14.0	5.0	0.5
Barracks hall	216	Duffel bag	Cylinder	32.0	32.0	70.0	0.0	90.0	0.0	1960.0	182.0	-907.0	70.0	32.0	32.0
Barracks hall	217	Rifle rack	Cube	80.0	12.0	25.0	0.0	0.0	0.0	2485.0	-12.0	-894.5	80.0	12.0	25.0
Barracks hall	218	Racked rifle	Cube	5.0	95.0	6.0	-8.0	0.0	0.0	2460.0	42.0	-896.0	5.0	94.9	19.2
Barracks hall	219	Racked rifle	Cube	5.0	95.0	6.0	-8.0	0.0	0.0	2485.0	42.0	-896.0	5.0	94.9	19.2
Barracks hall	220	Racked rifle	Cube	5.0	95.0	6.0	-8.0	0.0	0.0	2510.0	42.0	-896.0	5.0	94.9	19.2
Barracks hall	221	Notice board	Cube	90.0	60.0	2.0	0.0	0.0	0.0	2485.0	147.0	-883.2	90.0	60.0	2.0
Barracks hall	222	Notice	Cube	18.0	24.0	0.4	0.0	0.0	-6.0	2455.0	150.0	-884.4	20.4	25.8	0.4
Barracks hall	223	Notice	Cube	18.0	24.0	0.4	0.0	0.0	-2.0	2475.0	156.0	-884.4	18.8	24.6	0.4
Barracks hall	224	Notice	Cube	18.0	24.0	0.4	0.0	0.0	2.0	2495.0	150.0	-884.4	18.8	24.6	0.4
Barracks hall	225	Notice	Cube	18.0	24.0	0.4	0.0	0.0	6.0	2515.0	156.0	-884.4	20.4	25.8	0.4
Barracks hall	226	Mess table top	Cube	220.0	5.0	80.0	0.0	0.0	0.0	2570.0	56.5	-1100.0	220.0	5.0	80.0
Barracks hall	227	Table leg	Cube	8.0	72.0	8.0	0.0	0.0	0.0	2468.0	18.0	-1130.0	8.0	72.0	8.0
Barracks hall	228	Table leg	Cube	8.0	72.0	8.0	0.0	0.0	0.0	2468.0	18.0	-1070.0	8.0	72.0	8.0
Barracks hall	229	Table leg	Cube	8.0	72.0	8.0	0.0	0.0	0.0	2672.0	18.0	-1130.0	8.0	72.0	8.0
Barracks hall	230	Table leg	Cube	8.0	72.0	8.0	0.0	0.0	0.0	2672.0	18.0	-1070.0	8.0	72.0	8.0
Barracks hall	231	Bench seat	Cube	220.0	4.0	30.0	0.0	0.0	0.0	2570.0	26.0	-1010.0	220.0	4.0	30.0
Barracks hall	232	Bench leg	Cube	8.0	42.0	24.0	0.0	0.0	0.0	2470.0	3.0	-1010.0	8.0	42.0	24.0
Barracks hall	233	Bench leg	Cube	8.0	42.0	24.0	0.0	0.0	0.0	2670.0	3.0	-1010.0	8.0	42.0	24.0
Barracks hall	234	Bench seat	Cube	220.0	4.0	30.0	0.0	0.0	0.0	2570.0	26.0	-1190.0	220.0	4.0	30.0
Barracks hall	235	Bench leg	Cube	8.0	42.0	24.0	0.0	0.0	0.0	2470.0	3.0	-1190.0	8.0	42.0	24.0
Barracks hall	236	Bench leg	Cube	8.0	42.0	24.0	0.0	0.0	0.0	2670.0	3.0	-1190.0	8.0	42.0	24.0
Barracks hall	237	Mug	Cylinder	8.0	8.0	10.0	90.0	0.0	0.0	2510.0	64.0	-1078.0	8.0	10.0	8.0
Barracks hall	238	Mug	Cylinder	8.0	8.0	10.0	90.0	0.0	0.0	2630.0	64.0	-1078.0	8.0	10.0	8.0
Barracks hall	239	Mug	Cylinder	8.0	8.0	10.0	90.0	0.0	0.0	2525.0	64.0	-1124.0	8.0	10.0	8.0
Barracks hall	240	Mug	Cylinder	8.0	8.0	10.0	90.0	0.0	0.0	2645.0	64.0	-1124.0	8.0	10.0	8.0
Barracks hall	241	Map sheet	Cube	60.0	0.4	45.0	0.0	12.0	0.0	2560.0	59.2	-1100.0	68.0	0.4	56.5
Barracks hall	242	Field radio	Cube	30.0	16.0	22.0	0.0	-10.0	0.0	2655.0	67.0	-1095.0	33.4	16.0	26.9
Barracks hall	243	Radio antenna	Cylinder	1.0	1.0	45.0	90.0	0.0	0.0	2665.0	97.0	-1090.0	1.0	45.0	1.0
Barracks hall	244	Playing cards	Cube	9.0	1.2	6.0	0.0	25.0	0.0	2615.0	59.6	-1110.0	10.7	1.2	9.2
Barracks hall	245	Pull-up post	Cylinder	6.0	6.0	235.0	90.0	0.0	0.0	2530.0	99.5	-1415.0	6.0	235.0	6.0
Barracks hall	246	Post foot	Cube	30.0	2.0	30.0	0.0	0.0	0.0	2530.0	-17.0	-1415.0	30.0	2.0	30.0
Barracks hall	247	Pull-up post	Cylinder	6.0	6.0	235.0	90.0	0.0	0.0	2650.0	99.5	-1415.0	6.0	235.0	6.0
Barracks hall	248	Post foot	Cube	30.0	2.0	30.0	0.0	0.0	0.0	2650.0	-17.0	-1415.0	30.0	2.0	30.0
Barracks hall	249	Pull-up bar	Cylinder	3.5	3.5	130.0	0.0	90.0	0.0	2590.0	207.0	-1415.0	130.0	3.5	3.5
Barracks hall	250	Exercise mat	Cube	220.0	2.0	80.0	0.0	0.0	0.0	2320.0	-17.0	-1410.0	220.0	2.0	80.0

Barracks hall	251	Exercise mat	Cube	220.0	2.0	80.0	0.0	0.0	0.0	2320.0	-17.0	-1315.0	220.0	2.0	80.0
Barracks hall	252	Dumbbell rack	Cube	36.0	75.0	100.0	0.0	0.0	0.0	2718.0	19.5	-1325.0	36.0	75.0	100.0
Barracks hall	253	Dumbbell	Cylinder	11.0	11.0	30.0	0.0	90.0	0.0	2718.0	24.0	-1360.0	30.0	11.0	11.0
Barracks hall	254	Dumbbell	Cylinder	11.0	11.0	30.0	0.0	90.0	0.0	2718.0	54.0	-1360.0	30.0	11.0	11.0
Barracks hall	255	Dumbbell	Cylinder	13.0	13.0	30.0	0.0	90.0	0.0	2718.0	24.0	-1335.0	30.0	13.0	13.0
Barracks hall	256	Dumbbell	Cylinder	13.0	13.0	30.0	0.0	90.0	0.0	2718.0	54.0	-1335.0	30.0	13.0	13.0
Barracks hall	257	Dumbbell	Cylinder	15.0	15.0	30.0	0.0	90.0	0.0	2718.0	24.0	-1310.0	30.0	15.0	15.0
Barracks hall	258	Dumbbell	Cylinder	15.0	15.0	30.0	0.0	90.0	0.0	2718.0	54.0	-1310.0	30.0	15.0	15.0
Barracks hall	259	Dumbbell	Cylinder	17.0	17.0	30.0	0.0	90.0	0.0	2718.0	24.0	-1285.0	30.0	17.0	17.0
Barracks hall	260	Dumbbell	Cylinder	17.0	17.0	30.0	0.0	90.0	0.0	2718.0	54.0	-1285.0	30.0	17.0	17.0
Barracks hall	261	Punching bag	Cylinder	36.0	36.0	100.0	90.0	0.0	0.0	2693.0	97.0	-1433.0	36.0	100.0	36.0
Barracks hall	262	Bag chain	Cylinder	2.0	2.0	90.0	90.0	0.0	0.0	2693.0	202.0	-1433.0	2.0	90.0	2.0
Barracks hall	263	Kettlebell	Sphere	22.0	22.0	22.0	0.0	0.0	0.0	2703.0	-7.0	-1245.0	22.0	22.0	22.0
Barracks hall	264	Map board	Cube	2.0	80.0	120.0	0.0	0.0	0.0	2736.8	137.0	-1175.0	2.0	80.0	120.0
Barracks hall	265	Map pins	Cube	1.0	40.0	60.0	0.0	0.0	0.0	2735.5	142.0	-1170.0	1.0	40.0	60.0
Barracks hall	266	Wall flag	Cube	2.0	60.0	90.0	0.0	0.0	0.0	1883.2	207.0	-1250.0	2.0	60.0	90.0
Barracks hall	267	Wall clock	Cylinder	28.0	28.0	3.0	0.0	0.0	0.0	2570.0	202.0	-884.0	28.0	28.0	3.0
Barracks hall	268	Light fitting	Cube	124.0	6.0	20.0	0.0	0.0	0.0	2060.0	249.0	-1180.0	124.0	6.0	20.0
Barracks hall	269	Ceiling tube (light)	Cylinder	5.0	5.0	115.0	0.0	90.0	0.0	2060.0	244.0	-1180.0	115.0	5.0	5.0
Barracks hall	270	Light fitting	Cube	124.0	6.0	20.0	0.0	0.0	0.0	2330.0	249.0	-1180.0	124.0	6.0	20.0
Barracks hall	271	Ceiling tube (light)	Cylinder	5.0	5.0	115.0	0.0	90.0	0.0	2330.0	244.0	-1180.0	115.0	5.0	5.0
Barracks hall	272	Light fitting	Cube	124.0	6.0	20.0	0.0	0.0	0.0	2600.0	249.0	-1180.0	124.0	6.0	20.0
Barracks hall	273	Ceiling tube (light)	Cylinder	5.0	5.0	115.0	0.0	90.0	0.0	2600.0	244.0	-1180.0	115.0	5.0	5.0
Barracks hall	274	PT frame post	Cylinder	8.0	8.0	250.0	90.0	0.0	0.0	2490.0	95.0	-630.0	8.0	250.0	8.0
Barracks hall	275	PT frame post	Cylinder	8.0	8.0	250.0	90.0	0.0	0.0	2630.0	95.0	-630.0	8.0	250.0	8.0
Barracks hall	276	PT bar	Cylinder	4.0	4.0	150.0	0.0	90.0	0.0	2560.0	205.0	-630.0	150.0	4.0	4.0
Barracks hall	277	PT tyre	Cylinder	90.0	90.0	28.0	90.0	0.0	0.0	2760.0	-16.0	-560.0	90.0	28.0	90.0
Barracks hall	278	PT tyre	Cylinder	90.0	90.0	28.0	90.0	0.0	0.0	2775.0	12.0	-560.0	90.0	28.0	90.0
Air-defence radar	2	Concrete plinth	Cube	460.0	60.0	460.0	0.0	0.0	0.0	0.0	30.0	0.0	460.0	60.0	460.0
Air-defence radar	3	Mast leg	Cylinder	14.0	14.0	408.0	-83.0	-135.0	0.0	62.5	262.5	62.5	54.7	406.7	54.7
Air-defence radar	4	Mast leg	Cylinder	14.0	14.0	408.0	-83.0	-45.0	0.0	62.5	262.5	-62.5	54.7	406.7	54.7
Air-defence radar	5	Mast leg	Cylinder	14.0	14.0	408.0	-83.0	45.0	0.0	-62.5	262.5	-62.5	54.7	406.7	54.7
Air-defence radar	6	Mast leg	Cylinder	14.0	14.0	408.0	-83.0	135.0	0.0	-62.5	262.5	62.5	54.7	406.7	54.7
Air-defence radar	7	Mast ring bar	Cylinder	7.0	7.0	136.7	0.0	180.0	0.0	68.3	195.0	0.0	7.0	7.0	136.7
Air-defence radar	8	Mast ring bar	Cylinder	7.0	7.0	136.7	0.0	-90.0	0.0	0.0	195.0	-68.3	136.7	7.0	7.0
Air-defence radar	9	Mast ring bar	Cylinder	7.0	7.0	136.7	0.0	0.0	0.0	-68.3	195.0	0.0	7.0	7.0	136.7
Air-defence radar	10	Mast ring bar	Cylinder	7.0	7.0	136.7	0.0	90.0	0.0	0.0	195.0	68.3	136.7	7.0	7.0
Air-defence radar	11	Mast ring bar	Cylinder	7.0	7.0	113.3	0.0	180.0	0.0	56.7	330.0	0.0	7.0	7.0	113.3
Air-defence radar	12	Mast ring bar	Cylinder	7.0	7.0	113.3	0.0	-90.0	0.0	0.0	330.0	-56.7	113.3	7.0	7.0
Air-defence radar	13	Mast ring bar	Cylinder	7.0	7.0	113.3	0.0	0.0	0.0	-56.7	330.0	0.0	7.0	7.0	113.3
Air-defence radar	14	Mast ring bar	Cylinder	7.0	7.0	113.3	0.0	90.0	0.0	0.0	330.0	56.7	113.3	7.0	7.0
Air-defence radar	15	Mast ring bar	Cylinder	7.0	7.0	90.0	0.0	180.0	0.0	45.0	465.0	0.0	7.0	7.0	90.0
Air-defence radar	16	Mast ring bar	Cylinder	7.0	7.0	90.0	0.0	-90.0	0.0	0.0	465.0	-45.0	90.0	7.0	7.0
Air-defence radar	17	Mast ring bar	Cylinder	7.0	7.0	90.0	0.0	0.0	0.0	-45.0	465.0	0.0	7.0	7.0	90.0
Air-defence radar	18	Mast ring bar	Cylinder	7.0	7.0	90.0	0.0	90.0	0.0	0.0	465.0	45.0	90.0	7.0	7.0
Air-defence radar	19	Mast brace	Cylinder	5.0	5.0	200.9	-42.2	-175.5	0.0	74.2	127.5	5.8	16.9	138.7	152.1
Air-defence radar	20	Mast brace	Cylinder	5.0	5.0	200.9	42.2	-94.5	0.0	-5.8	127.5	-74.2	152.1	138.7	16.9
Air-defence radar	21	Mast brace	Cylinder	5.0	5.0	200.9	-42.2	4.5	0.0	-74.2	127.5	-5.8	16.9	138.7	152.1
Air-defence radar	22	Mast brace	Cylinder	5.0	5.0	200.9	42.2	85.5	0.0	5.8	127.5	74.2	152.1	138.7	16.9
Air-defence radar	23	Mast brace	Cylinder	5.0	5.0	184.4	47.1	174.7	0.0	62.5	262.5	-5.8	17.0	138.4	129.1
Air-defence radar	24	Mast brace	Cylinder	5.0	5.0	184.4	-47.1	-84.7	0.0	5.8	262.5	-62.5	129.1	138.4	17.0

Air-defence radar	25	Mast brace	Cylinder	5.0	5.0	184.4	47.1	-5.3	0.0	-62.5	262.5	5.8	17.0	138.4	129.1
Air-defence radar	26	Mast brace	Cylinder	5.0	5.0	184.4	-47.1	95.3	0.0	-5.8	262.5	62.5	129.1	138.4	17.0
Air-defence radar	27	Mast brace	Cylinder	5.0	5.0	169.4	-52.8	-173.5	0.0	50.8	397.5	5.8	17.1	138.0	106.2
Air-defence radar	28	Mast brace	Cylinder	5.0	5.0	169.4	52.8	-96.5	0.0	-5.8	397.5	-50.8	106.2	138.0	17.1
Air-defence radar	29	Mast brace	Cylinder	5.0	5.0	169.4	-52.8	6.5	0.0	-50.8	397.5	-5.8	17.1	138.0	106.2
Air-defence radar	30	Mast brace	Cylinder	5.0	5.0	169.4	52.8	83.5	0.0	5.8	397.5	50.8	106.2	138.0	17.1
Air-defence radar	31	Mast platform	Cube	150.0	12.0	150.0	0.0	0.0	0.0	0.0	470.0	0.0	150.0	12.0	150.0
Air-defence radar	32	Bearing housing	Cylinder	100.0	100.0	30.0	90.0	0.0	0.0	0.0	490.0	0.0	100.0	30.0	100.0
Air-defence radar	33	On-air lamp	Sphere	14.0	14.0	14.0	0.0	0.0	0.0	65.0	482.0	65.0	14.0	14.0	14.0
Air-defence radar	34	Equipment shelter	Cube	240.0	230.0	300.0	0.0	0.0	0.0	-460.0	130.0	50.0	240.0	230.0	300.0
Air-defence radar	35	Shelter slab	Cube	300.0	15.0	360.0	0.0	0.0	0.0	-460.0	7.5	50.0	300.0	15.0	360.0
Air-defence radar	36	Shelter door	Cube	6.0	190.0	90.0	0.0	0.0	0.0	-337.0	110.0	0.0	6.0	190.0	90.0
Air-defence radar	37	Door handle	Cube	8.0	6.0	20.0	0.0	0.0	0.0	-332.0	110.0	30.0	8.0	6.0	20.0
Air-defence radar	38	Door step	Cube	50.0	20.0	120.0	0.0	0.0	0.0	-310.0	25.0	0.0	50.0	20.0	120.0
Air-defence radar	39	Lit vent	Cube	8.0	30.0	90.0	0.0	0.0	0.0	-583.0	190.0	60.0	8.0	30.0	90.0
Air-defence radar	40	Roof air conditioner	Cube	90.0	40.0	90.0	0.0	0.0	0.0	-460.0	265.0	120.0	90.0	40.0	90.0
Air-defence radar	41	AC fan	Cylinder	70.0	70.0	5.0	90.0	0.0	0.0	-460.0	287.0	120.0	70.0	5.0	70.0
Air-defence radar	42	Power cable	Cylinder	9.0	9.0	111.8	0.0	100.3	0.0	-285.0	50.0	130.0	111.6	9.0	28.9
Air-defence radar	43	Mast cable duct	Cylinder	7.0	7.0	445.0	-64.0	112.6	0.0	-140.0	260.0	82.5	188.5	403.1	83.9
Air-defence radar	44	Pedestal	Cube	130.0	40.0	130.0	0.0	0.0	0.0	0.0	520.0	0.0	130.0	40.0	130.0
Air-defence radar	45	Yoke cheek left	Cube	18.0	110.0	80.0	0.0	0.0	0.0	70.0	585.0	0.0	18.0	110.0	80.0
Air-defence radar	46	Yoke cheek right	Cube	18.0	110.0	80.0	0.0	0.0	0.0	-70.0	585.0	0.0	18.0	110.0	80.0
Air-defence radar	47	Beacon pole	Cylinder	7.0	7.0	360.0	90.0	0.0	0.0	0.0	720.0	0.0	7.0	360.0	7.0
Air-defence radar	48	Aviation light	Sphere	20.0	20.0	20.0	0.0	0.0	0.0	0.0	905.0	0.0	20.0	20.0	20.0
Air-defence radar	49	Trunnion	Cylinder	14.0	14.0	150.0	0.0	90.0	0.0	0.0	590.0	0.0	150.0	14.9	14.9
Air-defence radar	50	Array tile c1 r1	Cube	64.0	78.0	5.0	0.0	22.5	0.0	-191.3	519.8	143.3	61.0	79.8	34.5
Air-defence radar	51	Array tile c1 r2	Cube	64.0	78.0	5.0	0.0	22.5	0.0	-191.3	599.6	137.7	61.0	79.8	34.5
Air-defence radar	52	Array tile c1 r3	Cube	64.0	78.0	5.0	0.0	22.5	0.0	-191.3	679.4	132.1	61.0	79.8	34.5
Air-defence radar	53	Antenna rib	Cube	66.0	8.0	8.0	0.0	22.5	0.0	-188.3	559.2	133.1	64.0	10.3	33.1
Air-defence radar	54	Antenna rib	Cube	66.0	8.0	8.0	0.0	22.5	0.0	-188.3	639.0	127.6	64.0	10.3	33.1
Air-defence radar	55	Array tile c2 r1	Cube	64.0	78.0	5.0	0.0	15.0	0.0	-129.4	518.4	122.3	63.1	79.3	26.8
Air-defence radar	56	Array tile c2 r2	Cube	64.0	78.0	5.0	0.0	15.0	0.0	-129.4	598.2	116.8	63.1	79.3	26.8
Air-defence radar	57	Array tile c2 r3	Cube	64.0	78.0	5.0	0.0	15.0	0.0	-129.4	678.0	111.2	63.1	79.3	26.8
Air-defence radar	58	Antenna rib	Cube	66.0	8.0	8.0	0.0	15.0	0.0	-127.3	557.7	111.8	65.8	9.7	25.3
Air-defence radar	59	Antenna rib	Cube	66.0	8.0	8.0	0.0	15.0	0.0	-127.3	637.5	106.3	65.8	9.7	25.3
Air-defence radar	60	Array tile c3 r1	Cube	64.0	78.0	5.0	0.0	7.5	0.0	-65.3	517.5	109.6	64.1	78.7	18.7
Air-defence radar	61	Array tile c3 r2	Cube	64.0	78.0	5.0	0.0	7.5	0.0	-65.3	597.3	104.0	64.1	78.7	18.7
Air-defence radar	62	Array tile c3 r3	Cube	64.0	78.0	5.0	0.0	7.5	0.0	-65.3	677.1	98.4	64.1	78.7	18.7
Air-defence radar	63	Antenna rib	Cube	66.0	8.0	8.0	0.0	7.5	0.0	-64.2	556.8	98.9	66.5	9.1	17.1
Air-defence radar	64	Antenna rib	Cube	66.0	8.0	8.0	0.0	7.5	0.0	-64.2	636.6	93.3	66.5	9.1	17.1
Air-defence radar	65	Array tile c4 r1	Cube	64.0	78.0	5.0	0.0	0.0	0.0	0.0	517.2	105.3	64.0	78.2	10.4
Air-defence radar	66	Array tile c4 r2	Cube	64.0	78.0	5.0	0.0	0.0	0.0	0.0	597.0	99.8	64.0	78.2	10.4
Air-defence radar	67	Array tile c4 r3	Cube	64.0	78.0	5.0	0.0	0.0	0.0	0.0	676.8	94.2	64.0	78.2	10.4
Air-defence radar	68	Antenna rib	Cube	66.0	8.0	8.0	0.0	0.0	0.0	0.0	556.5	94.6	66.0	8.5	8.5
Air-defence radar	69	Antenna rib	Cube	66.0	8.0	8.0	0.0	0.0	0.0	0.0	636.3	89.0	66.0	8.5	8.5
Air-defence radar	70	Array tile c5 r1	Cube	64.0	78.0	5.0	0.0	-7.5	0.0	65.3	517.5	109.6	64.1	78.7	18.7
Air-defence radar	71	Array tile c5 r2	Cube	64.0	78.0	5.0	0.0	-7.5	0.0	65.3	597.3	104.0	64.1	78.7	18.7
Air-defence radar	72	Array tile c5 r3	Cube	64.0	78.0	5.0	0.0	-7.5	0.0	65.3	677.1	98.4	64.1	78.7	18.7
Air-defence radar	73	Antenna rib	Cube	66.0	8.0	8.0	0.0	-7.5	0.0	64.2	556.8	98.9	66.5	9.1	17.1
Air-defence radar	74	Antenna rib	Cube	66.0	8.0	8.0	0.0	-7.5	0.0	64.2	636.6	93.3	66.5	9.1	17.1
Air-defence radar	75	Array tile c6 r1	Cube	64.0	78.0	5.0	0.0	-15.0	0.0	129.4	518.4	122.3	63.1	79.3	26.8

Air-defence radar	76	Array tile c6 r2	Cube	64.0	78.0	5.0	0.0	-15.0	0.0	129.4	598.2	116.8	63.1	79.3	26.8
Air-defence radar	77	Array tile c6 r3	Cube	64.0	78.0	5.0	0.0	-15.0	0.0	129.4	678.0	111.2	63.1	79.3	26.8
Air-defence radar	78	Antenna rib	Cube	66.0	8.0	8.0	0.0	-15.0	0.0	127.3	557.7	111.8	65.8	9.7	25.3
Air-defence radar	79	Antenna rib	Cube	66.0	8.0	8.0	0.0	-15.0	0.0	127.3	637.5	106.3	65.8	9.7	25.3
Air-defence radar	80	Array tile c7 r1	Cube	64.0	78.0	5.0	0.0	-22.5	0.0	191.3	519.8	143.3	61.0	79.8	34.5
Air-defence radar	81	Array tile c7 r2	Cube	64.0	78.0	5.0	0.0	-22.5	0.0	191.3	599.6	137.7	61.0	79.8	34.5
Air-defence radar	82	Array tile c7 r3	Cube	64.0	78.0	5.0	0.0	-22.5	0.0	191.3	679.4	132.1	61.0	79.8	34.5
Air-defence radar	83	Antenna rib	Cube	66.0	8.0	8.0	0.0	-22.5	0.0	188.3	559.2	133.1	64.0	10.3	33.1
Air-defence radar	84	Antenna rib	Cube	66.0	8.0	8.0	0.0	-22.5	0.0	188.3	639.0	127.6	64.0	10.3	33.1
Air-defence radar	85	Antenna spine	Cube	14.0	260.0	14.0	0.0	0.0	0.0	0.0	596.3	89.8	14.0	260.3	32.1
Air-defence radar	86	Arm left	Cube	12.0	12.0	91.2	0.0	-9.5	0.0	37.5	593.1	44.9	26.8	18.4	92.6
Air-defence radar	87	Arm right	Cube	12.0	12.0	91.2	0.0	9.5	0.0	-37.5	593.1	44.9	26.8	18.4	92.6
Air-defence radar	88	Feed horn	Cone	40.0	40.0	55.0	0.0	0.0	0.0	0.0	613.7	339.2	40.0	43.7	57.7
Air-defence radar	89	Feed waveguide	Cylinder	8.0	8.0	215.0	0.0	0.0	0.0	0.0	604.5	207.0	8.0	23.0	215.0
Air-defence radar	90	Feed strut left	Cylinder	6.0	6.0	251.0	0.0	-39.0	0.0	91.0	606.2	231.9	162.7	19.9	198.7
Air-defence radar	91	Feed strut right	Cylinder	6.0	6.0	251.0	0.0	39.0	0.0	-91.0	606.2	231.9	162.7	19.9	198.7
Air-defence radar	92	Feed strut top	Cylinder	6.0	6.0	249.6	25.6	0.0	0.0	0.0	671.0	212.4	6.0	97.6	234.2
Air-defence radar	93	Counterweight arm	Cylinder	12.0	12.0	130.0	0.0	180.0	0.0	0.0	584.8	-74.8	12.0	21.0	130.5
Air-defence radar	94	Counterweight	Cube	90.0	70.0	60.0	0.0	0.0	0.0	0.0	578.1	-169.6	90.0	74.0	64.7
Air-defence radar	95	IFF bar	Cube	240.0	10.0	14.0	0.0	0.0	0.0	0.0	758.6	88.5	240.0	11.0	14.7
Air-defence radar	96	IFF post left	Cube	6.0	30.0	6.0	0.0	0.0	0.0	80.0	741.3	84.7	6.0	30.3	8.1
Air-defence radar	97	IFF post right	Cube	6.0	30.0	6.0	0.0	0.0	0.0	-80.0	741.3	84.7	6.0	30.3	8.1
Air-defence radar	98	IFF dipole	Cube	4.0	20.0	4.0	0.0	0.0	0.0	-105.0	774.5	87.3	4.0	20.2	5.4
Air-defence radar	99	IFF dipole	Cube	4.0	20.0	4.0	0.0	0.0	0.0	-63.0	774.5	87.3	4.0	20.2	5.4
Air-defence radar	100	IFF dipole	Cube	4.0	20.0	4.0	0.0	0.0	0.0	-21.0	774.5	87.3	4.0	20.2	5.4
Air-defence radar	101	IFF dipole	Cube	4.0	20.0	4.0	0.0	0.0	0.0	21.0	774.5	87.3	4.0	20.2	5.4
Air-defence radar	102	IFF dipole	Cube	4.0	20.0	4.0	0.0	0.0	0.0	63.0	774.5	87.3	4.0	20.2	5.4
Air-defence radar	103	IFF dipole	Cube	4.0	20.0	4.0	0.0	0.0	0.0	105.0	774.5	87.3	4.0	20.2	5.4
Air-defence radar	104	Shelter status display	Cube	1.2	30.0	26.0	0.0	0.0	0.0	-339.3	150.0	80.0	1.2	30.0	26.0
Air-defence radar	105	Shelter status LED	Sphere	5.0	5.0	5.0	0.0	0.0	0.0	-339.5	176.0	80.0	5.0	5.0	5.0
Air-defence radar	106	Door placard	Cube	1.2	26.0	36.0	0.0	0.0	0.0	-339.3	155.0	-75.0	1.2	26.0	36.0
Air-defence radar	107	Shelter roof edge	Cylinder	5.0	5.0	300.0	0.0	0.0	0.0	-580.0	245.0	50.0	5.0	5.0	300.0
Air-defence radar	108	Shelter roof edge	Cylinder	5.0	5.0	300.0	0.0	0.0	0.0	-340.0	245.0	50.0	5.0	5.0	300.0
Air-defence radar	109	Bearing seal	Cylinder	104.0	104.0	4.0	90.0	0.0	0.0	0.0	506.0	0.0	104.0	4.0	104.0
Motor pool	2	Floodlight pole	Cylinder	16.0	16.0	520.0	90.0	0.0	0.0	0.0	260.0	792.0	16.0	520.0	16.0
Motor pool	3	Floodlight arm	Cube	10.0	10.0	50.0	0.0	45.0	0.0	0.0	530.0	792.0	42.4	10.0	42.4
Motor pool	4	Floodlight housing	Cylinder	42.0	42.0	36.0	29.1	180.0	0.0	0.0	540.0	792.0	42.0	54.2	51.9
Motor pool	5	Floodlight lens	Sphere	38.0	38.0	6.0	29.1	180.0	0.0	0.0	530.3	774.5	38.0	36.1	23.7
Helipad	2	Helipad disc	Cylinder	1000.0	1000.0	8.0	90.0	0.0	0.0	0.0	4.0	0.0	1000.0	8.0	1000.0
Helipad	3	H bar	Cube	50.0	1.2	320.0	0.0	0.0	0.0	-95.0	8.6	0.0	50.0	1.2	320.0
Helipad	4	H bar	Cube	50.0	1.2	320.0	0.0	0.0	0.0	95.0	8.6	0.0	50.0	1.2	320.0
Helipad	5	H crossbar	Cube	150.0	1.2	50.0	0.0	0.0	0.0	0.0	8.6	0.0	150.0	1.2	50.0
Helipad	6	Circle mark	Cube	70.0	1.2	20.0	0.0	0.0	0.0	390.0	8.6	0.0	70.0	1.2	20.0
Helipad	7	Circle mark	Cube	70.0	1.2	20.0	0.0	-15.0	0.0	376.7	8.6	100.9	72.8	1.2	37.4
Helipad	8	Circle mark	Cube	70.0	1.2	20.0	0.0	-30.0	0.0	337.7	8.6	195.0	70.6	1.2	52.3
Helipad	9	Circle mark	Cube	70.0	1.2	20.0	0.0	-45.0	0.0	275.8	8.6	275.8	63.6	1.2	63.6
Helipad	10	Circle mark	Cube	70.0	1.2	20.0	0.0	-60.0	0.0	195.0	8.6	337.7	52.3	1.2	70.6
Helipad	11	Circle mark	Cube	70.0	1.2	20.0	0.0	-75.0	0.0	100.9	8.6	376.7	37.4	1.2	72.8
Helipad	12	Circle mark	Cube	70.0	1.2	20.0	0.0	-90.0	0.0	0.0	8.6	390.0	20.0	1.2	70.0
Helipad	13	Circle mark	Cube	70.0	1.2	20.0	0.0	-105.0	0.0	-100.9	8.6	376.7	37.4	1.2	72.8
Helipad	14	Circle mark	Cube	70.0	1.2	20.0	0.0	-120.0	0.0	-195.0	8.6	337.7	52.3	1.2	70.6

Helipad	15	Circle mark	Cube	70.0	1.2	20.0	0.0	-135.0	0.0	-275.8	8.6	275.8	63.6	1.2	63.6
Helipad	16	Circle mark	Cube	70.0	1.2	20.0	0.0	-150.0	0.0	-337.7	8.6	195.0	70.6	1.2	52.3
Helipad	17	Circle mark	Cube	70.0	1.2	20.0	0.0	-165.0	0.0	-376.7	8.6	100.9	72.8	1.2	37.4
Helipad	18	Circle mark	Cube	70.0	1.2	20.0	0.0	180.0	0.0	-390.0	8.6	0.0	70.0	1.2	20.0
Helipad	19	Circle mark	Cube	70.0	1.2	20.0	0.0	165.0	0.0	-376.7	8.6	-100.9	72.8	1.2	37.4
Helipad	20	Circle mark	Cube	70.0	1.2	20.0	0.0	150.0	0.0	-337.7	8.6	-195.0	70.6	1.2	52.3
Helipad	21	Circle mark	Cube	70.0	1.2	20.0	0.0	135.0	0.0	-275.8	8.6	-275.8	63.6	1.2	63.6
Helipad	22	Circle mark	Cube	70.0	1.2	20.0	0.0	120.0	0.0	-195.0	8.6	-337.7	52.3	1.2	70.6
Helipad	23	Circle mark	Cube	70.0	1.2	20.0	0.0	105.0	0.0	-100.9	8.6	-376.7	37.4	1.2	72.8
Helipad	24	Circle mark	Cube	70.0	1.2	20.0	0.0	90.0	0.0	0.0	8.6	-390.0	20.0	1.2	70.0
Helipad	25	Circle mark	Cube	70.0	1.2	20.0	0.0	75.0	0.0	100.9	8.6	-376.7	37.4	1.2	72.8
Helipad	26	Circle mark	Cube	70.0	1.2	20.0	0.0	60.0	0.0	195.0	8.6	-337.7	52.3	1.2	70.6
Helipad	27	Circle mark	Cube	70.0	1.2	20.0	0.0	45.0	0.0	275.8	8.6	-275.8	63.6	1.2	63.6
Helipad	28	Circle mark	Cube	70.0	1.2	20.0	0.0	30.0	0.0	337.8	8.6	-195.0	70.6	1.2	52.3
Helipad	29	Circle mark	Cube	70.0	1.2	20.0	0.0	15.0	0.0	376.7	8.6	-100.9	72.8	1.2	37.4
Helipad	30	Edge mark	Cube	35.0	1.2	100.0	0.0	90.0	0.0	460.0	8.6	0.0	100.0	1.2	35.0
Helipad	31	Edge mark	Cube	35.0	1.2	100.0	0.0	180.0	0.0	0.0	8.6	460.0	35.0	1.2	100.0
Helipad	32	Edge mark	Cube	35.0	1.2	100.0	0.0	270.0	0.0	-460.0	8.6	0.0	100.0	1.2	35.0
Helipad	33	Edge mark	Cube	35.0	1.2	100.0	0.0	360.0	0.0	0.0	8.6	-460.0	35.0	1.2	100.0
Helipad	34	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	475.7	7.0	94.6	20.0	7.0	20.0
Helipad	35	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	403.3	7.0	269.5	20.0	7.0	20.0
Helipad	36	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	269.5	7.0	403.3	20.0	7.0	20.0
Helipad	37	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	94.6	7.0	475.7	20.0	7.0	20.0
Helipad	38	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	-94.6	7.0	475.7	20.0	7.0	20.0
Helipad	39	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	-269.5	7.0	403.3	20.0	7.0	20.0
Helipad	40	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	-403.3	7.0	269.5	20.0	7.0	20.0
Helipad	41	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	-475.7	7.0	94.6	20.0	7.0	20.0
Helipad	42	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	-475.7	7.0	-94.6	20.0	7.0	20.0
Helipad	43	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	-403.3	7.0	-269.5	20.0	7.0	20.0
Helipad	44	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	-269.5	7.0	-403.3	20.0	7.0	20.0
Helipad	45	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	-94.6	7.0	-475.7	20.0	7.0	20.0
Helipad	46	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	94.6	7.0	-475.7	20.0	7.0	20.0
Helipad	47	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	269.5	7.0	-403.3	20.0	7.0	20.0
Helipad	48	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	403.3	7.0	-269.5	20.0	7.0	20.0
Helipad	49	Perimeter light	Sphere	20.0	7.0	20.0	0.0	0.0	0.0	475.7	7.0	-94.6	20.0	7.0	20.0
Helipad	50	Pad floodlight pole	Cylinder	14.0	14.0	400.0	90.0	0.0	0.0	-620.0	200.0	400.0	14.0	400.0	14.0
Helipad	51	Pad floodlight housing	Cylinder	36.0	36.0	30.0	23.4	122.8	0.0	-620.0	420.0	400.0	54.7	45.0	52.9
Helipad	52	Pad floodlight lens	Sphere	32.0	32.0	5.0	23.4	122.8	0.0	-606.9	413.2	391.5	31.9	31.3	36.3
Ammunition bunker	2	Concrete chamber	Cube	560.0	130.0	520.0	0.0	0.0	0.0	0.0	65.0	-40.0	560.0	130.0	520.0
Ammunition bunker	3	Earth mound	Sphere	820.0	440.0	740.0	0.0	0.0	0.0	0.0	0.0	-40.0	820.0	440.0	740.0
Ammunition bunker	4	Earth mound lobe	Sphere	420.0	240.0	340.0	0.0	20.0	0.0	-220.0	10.0	-180.0	511.0	240.0	463.1
Ammunition bunker	5	Earth mound lobe	Sphere	380.0	220.0	300.0	0.0	-25.0	0.0	230.0	10.0	-120.0	471.2	220.0	432.5
Ammunition bunker	6	Turf clump	Sphere	100.0	40.0	90.0	0.0	0.0	0.0	-180.0	150.0	-120.0	100.0	40.0	90.0
Ammunition bunker	7	Turf clump	Sphere	100.0	40.0	90.0	0.0	37.0	0.0	-90.0	160.0	-90.0	134.0	40.0	132.1
Ammunition bunker	8	Turf clump	Sphere	100.0	40.0	90.0	0.0	74.0	0.0	0.0	170.0	-60.0	114.1	40.0	120.9
Ammunition bunker	9	Turf clump	Sphere	100.0	40.0	90.0	0.0	111.0	0.0	90.0	160.0	-30.0	119.9	40.0	125.6
Ammunition bunker	10	Turf clump	Sphere	100.0	40.0	90.0	0.0	148.0	0.0	180.0	150.0	0.0	132.5	40.0	129.3
Ammunition bunker	11	Headwall block	Cube	130.0	190.0	50.0	0.0	0.0	0.0	-115.0	95.0	240.0	130.0	190.0	50.0
Ammunition bunker	12	Headwall block	Cube	130.0	190.0	50.0	0.0	0.0	0.0	115.0	95.0	240.0	130.0	190.0	50.0
Ammunition bunker	13	Headwall lintel	Cube	360.0	40.0	50.0	0.0	0.0	0.0	0.0	170.0	240.0	360.0	40.0	50.0
Ammunition bunker	14	Blast door	Cube	100.0	150.0	12.0	0.0	0.0	0.0	0.0	75.0	220.0	100.0	150.0	12.0

Ammunition bunker	15	Door frame stripe	Cube	120.0	10.0	14.0	0.0	0.0	0.0	0.0	155.0	220.0	120.0	10.0	14.0
Ammunition bunker	16	Door step	Cube	140.0	14.0	50.0	0.0	0.0	0.0	0.0	7.0	285.0	140.0	14.0	50.0
Ammunition bunker	17	Door step marker	Sphere	9.0	4.0	9.0	0.0	0.0	0.0	50.0	16.0	295.0	9.0	4.0	9.0
Ammunition bunker	18	Door lamp hood	Cube	34.0	4.0	30.0	0.0	0.0	0.0	0.0	200.0	275.0	34.0	4.0	30.0
Ammunition bunker	19	Door lamp	Sphere	16.0	16.0	6.0	90.0	0.0	0.0	0.0	193.0	275.0	16.0	6.0	16.0
Ammunition bunker	20	Warning sign	Cube	50.0	35.0	3.0	0.0	0.0	0.0	-115.0	130.0	267.0	50.0	35.0	3.0
Ammunition bunker	21	Wing wall core	Cube	60.0	130.0	240.0	0.0	0.0	0.0	-200.0	65.0	350.0	60.0	130.0	240.0
Ammunition bunker	22	Wing wall slope	Wedge	240.0	130.0	65.0	0.0	-90.0	0.0	-262.5	65.0	350.0	65.0	130.0	240.0
Ammunition bunker	23	Wing sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-200.0	138.0	250.0	50.0	17.0	28.0
Ammunition bunker	24	Wing sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-200.0	138.0	310.0	50.0	17.0	28.0
Ammunition bunker	25	Wing sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-200.0	138.0	370.0	50.0	17.0	28.0
Ammunition bunker	26	Wing sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-200.0	138.0	430.0	50.0	17.0	28.0
Ammunition bunker	27	Wing wall core	Cube	60.0	130.0	240.0	0.0	0.0	0.0	200.0	65.0	350.0	60.0	130.0	240.0
Ammunition bunker	28	Wing wall slope	Wedge	240.0	130.0	65.0	0.0	90.0	0.0	262.5	65.0	350.0	65.0	130.0	240.0
Ammunition bunker	29	Wing sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	200.0	138.0	250.0	50.0	17.0	28.0
Ammunition bunker	30	Wing sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	200.0	138.0	310.0	50.0	17.0	28.0
Ammunition bunker	31	Wing sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	200.0	138.0	370.0	50.0	17.0	28.0
Ammunition bunker	32	Wing sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	200.0	138.0	430.0	50.0	17.0	28.0
Ammunition bunker	33	Blast baffle	Cube	260.0	100.0	50.0	0.0	0.0	0.0	0.0	50.0	540.0	260.0	100.0	50.0
Ammunition bunker	34	Vent pipe	Cylinder	20.0	20.0	100.0	90.0	0.0	0.0	-130.0	200.0	-100.0	20.0	100.0	20.0
Ammunition bunker	35	Vent cap	Cylinder	34.0	34.0	6.0	90.0	0.0	0.0	-130.0	255.0	-100.0	34.0	6.0	34.0
Ammunition bunker	36	Vent pipe	Cylinder	20.0	20.0	100.0	90.0	0.0	0.0	130.0	200.0	-100.0	20.0	100.0	20.0
Ammunition bunker	37	Vent cap	Cylinder	34.0	34.0	6.0	90.0	0.0	0.0	130.0	255.0	-100.0	34.0	6.0	34.0
Guard tower	2	Tower leg	Cylinder	16.0	16.0	420.4	-87.6	-135.0	0.0	109.3	190.0	109.3	35.2	420.7	35.2
Guard tower	3	Tower leg	Cylinder	16.0	16.0	420.4	-87.6	-45.0	0.0	109.3	190.0	-109.3	35.2	420.7	35.2
Guard tower	4	Tower leg	Cylinder	16.0	16.0	420.4	-87.6	45.0	0.0	-109.3	190.0	-109.3	35.2	420.7	35.2
Guard tower	5	Tower leg	Cylinder	16.0	16.0	420.4	-87.6	135.0	0.0	-109.3	190.0	109.3	35.2	420.7	35.2
Guard tower	6	Tower brace	Cylinder	7.0	7.0	379.6	-54.7	-177.6	0.0	109.4	185.0	4.7	16.5	314.0	224.9
Guard tower	7	Tower brace	Cylinder	7.0	7.0	379.6	-54.7	-87.6	0.0	4.7	185.0	-109.4	224.9	314.0	16.5
Guard tower	8	Tower brace	Cylinder	7.0	7.0	379.6	-54.7	2.4	0.0	-109.4	185.0	-4.7	16.5	314.0	224.9
Guard tower	9	Tower brace	Cylinder	7.0	7.0	379.6	-54.7	92.4	0.0	-4.6	185.0	109.5	224.9	314.0	16.5
Guard tower	10	Platform	Cube	290.0	14.0	290.0	0.0	0.0	0.0	0.0	407.0	0.0	290.0	14.0	290.0
Guard tower	11	Parapet front	Cube	290.0	95.0	50.0	0.0	0.0	0.0	0.0	462.0	120.0	290.0	95.0	50.0
Guard tower	12	Parapet side	Cube	50.0	95.0	200.0	0.0	0.0	0.0	120.0	462.0	0.0	50.0	95.0	200.0
Guard tower	13	Parapet side	Cube	50.0	95.0	200.0	0.0	0.0	0.0	-120.0	462.0	0.0	50.0	95.0	200.0
Guard tower	14	Parapet cap	Cube	290.0	14.0	50.0	0.0	0.0	0.0	0.0	517.0	120.0	290.0	14.0	50.0
Guard tower	15	Roof post	Cylinder	10.0	10.0	210.0	90.0	0.0	0.0	135.0	519.0	135.0	10.0	210.0	10.0
Guard tower	16	Roof post	Cylinder	10.0	10.0	210.0	90.0	0.0	0.0	135.0	519.0	-135.0	10.0	210.0	10.0
Guard tower	17	Roof post	Cylinder	10.0	10.0	210.0	90.0	0.0	0.0	-135.0	519.0	-135.0	10.0	210.0	10.0
Guard tower	18	Roof post	Cylinder	10.0	10.0	210.0	90.0	0.0	0.0	-135.0	519.0	135.0	10.0	210.0	10.0
Guard tower	19	Roof slab	Cube	350.0	10.0	350.0	0.0	0.0	0.0	0.0	630.0	0.0	350.0	10.0	350.0
Guard tower	20	Roof lip	Cube	350.0	12.0	8.0	0.0	0.0	0.0	0.0	638.0	170.0	350.0	12.0	8.0
Guard tower	21	Ladder rail	Cylinder	5.0	5.0	422.4	-83.9	0.0	0.0	-25.0	190.0	-167.5	5.0	420.5	50.0
Guard tower	22	Ladder rail	Cylinder	5.0	5.0	422.4	-83.9	0.0	0.0	25.0	190.0	-167.5	5.0	420.5	50.0
Guard tower	23	Ladder rung	Cylinder	3.5	3.5	50.0	0.0	90.0	0.0	0.0	33.5	-184.3	50.0	3.5	3.5
Guard tower	24	Ladder rung	Cylinder	3.5	3.5	50.0	0.0	90.0	0.0	0.0	109.8	-176.1	50.0	3.5	3.5
Guard tower	25	Ladder rung	Cylinder	3.5	3.5	50.0	0.0	90.0	0.0	0.0	186.2	-167.9	50.0	3.5	3.5
Guard tower	26	Ladder rung	Cylinder	3.5	3.5	50.0	0.0	90.0	0.0	0.0	262.5	-159.7	50.0	3.5	3.5
Guard tower	27	Ladder rung	Cylinder	3.5	3.5	50.0	0.0	90.0	0.0	0.0	338.9	-151.5	50.0	3.5	3.5
Guard tower	28	MG mount	Cube	12.0	30.0	12.0	0.0	0.0	0.0	50.0	535.0	115.0	12.0	30.0	12.0
Guard tower	29	MG receiver	Cube	14.0	16.0	50.0	0.0	0.0	0.0	50.0	558.0	120.0	14.0	16.0	50.0

Guard tower	30	MG barrel	Cylinder	4.0	4.0	70.0	0.0	0.0	0.0	50.0	558.0	172.0	4.0	4.0	70.0
Guard tower	31	Floodlight bracket	Cylinder	6.0	6.0	25.0	90.0	0.0	0.0	-80.0	617.5	162.0	6.0	25.0	6.0
Guard tower	32	Floodlight housing	Cylinder	36.0	36.0	34.0	26.7	1.7	0.0	-80.0	605.0	162.0	37.3	47.4	47.6
Guard tower	33	Floodlight lens	Sphere	32.0	32.0	6.0	26.7	1.7	0.0	-79.5	596.9	178.1	32.6	31.3	20.7
Guard tower	34	Sandbag course	Cylinder	10.0	10.0	280.0	0.0	90.0	0.0	0.0	436.0	145.0	280.0	10.0	10.0
Guard tower	35	Sandbag course	Cylinder	10.0	10.0	280.0	0.0	90.0	0.0	0.0	476.0	145.0	280.0	10.0	10.0
Guard tower	36	MG ammo box	Cube	26.0	18.0	12.0	0.0	0.0	0.0	80.0	533.0	115.0	26.0	18.0	12.0
Guard tower	37	Field radio	Cube	30.0	35.0	15.0	0.0	0.0	0.0	-85.0	432.0	85.0	30.0	35.0	15.0
Guard tower	38	Radio LED	Sphere	3.5	3.5	2.0	0.0	0.0	0.0	-85.0	442.0	77.2	3.5	3.5	2.0
Watchtower	2	Tower leg	Cylinder	14.0	14.0	530.8	-86.9	-135.0	0.0	95.0	235.0	95.0	39.8	530.7	39.8
Watchtower	3	Tower leg	Cylinder	14.0	14.0	530.8	-86.9	-45.0	0.0	95.0	235.0	-95.0	39.8	530.7	39.8
Watchtower	4	Tower leg	Cylinder	14.0	14.0	530.8	-86.9	45.0	0.0	-95.0	235.0	-95.0	39.8	530.7	39.8
Watchtower	5	Tower leg	Cylinder	14.0	14.0	530.8	-86.9	135.0	0.0	-95.0	235.0	95.0	39.8	530.7	39.8
Watchtower	6	Tower brace	Cylinder	8.0	8.0	341.8	-55.0	-176.9	0.0	97.8	160.0	5.3	18.9	284.6	202.6
Watchtower	7	Tower brace	Cylinder	8.0	8.0	341.8	-55.0	-86.9	0.0	5.3	160.0	-97.8	202.6	284.6	18.9
Watchtower	8	Tower brace	Cylinder	8.0	8.0	341.8	-55.0	3.1	0.0	-97.8	160.0	-5.3	18.9	284.6	202.6
Watchtower	9	Tower brace	Cylinder	8.0	8.0	341.8	-55.0	93.1	0.0	-5.3	160.0	97.8	202.6	284.6	18.9
Watchtower	10	Platform	Cube	240.0	12.0	240.0	0.0	0.0	0.0	0.0	310.0	0.0	240.0	12.0	240.0
Watchtower	11	Parapet front	Cube	230.0	90.0	6.0	0.0	0.0	0.0	0.0	361.0	-112.0	230.0	90.0	6.0
Watchtower	12	Parapet side	Cube	6.0	90.0	230.0	0.0	0.0	0.0	112.0	361.0	0.0	6.0	90.0	230.0
Watchtower	13	Parapet side	Cube	6.0	90.0	230.0	0.0	0.0	0.0	-112.0	361.0	0.0	6.0	90.0	230.0
Watchtower	14	Hand rail	Cylinder	5.0	5.0	220.0	0.0	90.0	0.0	0.0	400.0	112.0	220.0	5.0	5.0
Watchtower	15	Roof slab	Cube	250.0	8.0	250.0	0.0	0.0	0.0	0.0	502.0	0.0	250.0	8.0	250.0
Watchtower	16	Roof cone	Cone	340.0	340.0	90.0	-90.0	45.0	0.0	0.0	551.0	0.0	480.8	90.0	480.8
Watchtower	17	Ladder rail	Cylinder	5.0	5.0	337.0	-83.7	180.0	0.0	-25.0	147.5	126.5	5.0	335.5	42.0
Watchtower	18	Ladder rail	Cylinder	5.0	5.0	337.0	-83.7	180.0	0.0	25.0	147.5	126.5	5.0	335.5	42.0
Watchtower	19	Ladder rung	Cylinder	3.5	3.5	50.0	0.0	90.0	0.0	0.0	16.1	141.0	50.0	3.5	3.5
Watchtower	20	Ladder rung	Cylinder	3.5	3.5	50.0	0.0	90.0	0.0	0.0	67.6	135.3	50.0	3.5	3.5
Watchtower	21	Ladder rung	Cylinder	3.5	3.5	50.0	0.0	90.0	0.0	0.0	119.2	129.6	50.0	3.5	3.5
Watchtower	22	Ladder rung	Cylinder	3.5	3.5	50.0	0.0	90.0	0.0	0.0	170.7	123.9	50.0	3.5	3.5
Watchtower	23	Ladder rung	Cylinder	3.5	3.5	50.0	0.0	90.0	0.0	0.0	222.2	118.2	50.0	3.5	3.5
Watchtower	24	Ladder rung	Cylinder	3.5	3.5	50.0	0.0	90.0	0.0	0.0	273.8	112.6	50.0	3.5	3.5
Watchtower	25	Floodlight bracket	Cylinder	6.0	6.0	44.0	90.0	0.0	0.0	-95.0	428.0	-95.0	6.0	44.0	6.0
Watchtower	26	Floodlight housing	Cylinder	34.0	34.0	32.0	25.8	-133.6	0.0	-95.0	466.0	-95.0	55.0	44.5	54.7
Watchtower	27	Floodlight lens	Sphere	30.0	30.0	6.0	25.8	-133.6	0.0	-106.1	458.6	-105.6	34.0	29.6	34.4
Fuel depot	2	Dike wall	Cube	840.0	55.0	50.0	0.0	0.0	0.0	0.0	27.5	250.0	840.0	55.0	50.0
Fuel depot	3	Dike wall	Cube	840.0	55.0	50.0	0.0	0.0	0.0	0.0	27.5	-250.0	840.0	55.0	50.0
Fuel depot	4	Dike wall	Cube	50.0	55.0	500.0	0.0	0.0	0.0	400.0	27.5	0.0	50.0	55.0	500.0
Fuel depot	5	Dike wall	Cube	50.0	55.0	500.0	0.0	0.0	0.0	-400.0	27.5	0.0	50.0	55.0	500.0
Fuel depot	6	Fuel bladder	Sphere	340.0	110.0	210.0	0.0	0.0	0.0	-200.0	50.0	30.0	340.0	110.0	210.0
Fuel depot	7	Bladder fill cap	Cylinder	25.0	25.0	10.0	90.0	0.0	0.0	-200.0	102.0	30.0	25.0	10.0	25.0
Fuel depot	8	Fuel bladder	Sphere	340.0	110.0	210.0	0.0	0.0	0.0	200.0	50.0	30.0	340.0	110.0	210.0
Fuel depot	9	Bladder fill cap	Cylinder	25.0	25.0	10.0	90.0	0.0	0.0	200.0	102.0	30.0	25.0	10.0	25.0
Fuel depot	10	Pump skid	Cube	90.0	40.0	60.0	0.0	0.0	0.0	0.0	20.0	-170.0	90.0	40.0	60.0
Fuel depot	11	Fuel hose	Cylinder	8.0	8.0	188.9	-3.0	-58.0	0.0	-120.0	45.0	-120.0	164.6	18.0	107.0
Fuel depot	12	Fuel hose	Cylinder	8.0	8.0	188.9	-3.0	58.0	0.0	120.0	45.0	-120.0	164.6	18.0	107.0
Fuel depot	13	Fuel drum	Cylinder	55.0	55.0	85.0	90.0	0.0	0.0	-320.0	43.0	-360.0	55.0	85.0	55.0
Fuel depot	14	Fuel drum	Cylinder	55.0	55.0	85.0	90.0	0.0	0.0	-260.0	43.0	-360.0	55.0	85.0	55.0
Fuel depot	15	Fuel drum	Cylinder	55.0	55.0	85.0	90.0	0.0	0.0	-200.0	43.0	-360.0	55.0	85.0	55.0
Fuel depot	16	Fuel drum	Cylinder	55.0	55.0	85.0	90.0	0.0	0.0	-140.0	43.0	-360.0	55.0	85.0	55.0
Fuel depot	17	Generator housing	Cube	100.0	110.0	200.0	0.0	0.0	0.0	720.0	70.0	-60.0	200.0	110.0	100.0

Fuel depot	18	Generator skid	Cube	120.0	15.0	220.0	0.0	0.0	0.0	720.0	7.0	-60.0	220.0	15.0	120.0
Fuel depot	19	Exhaust stack	Cylinder	10.0	10.0	70.0	90.0	0.0	0.0	800.0	160.0	-90.0	10.0	70.0	10.0
Fuel depot	20	Control panel	Cube	50.0	40.0	4.0	0.0	0.0	0.0	823.0	85.0	-60.0	4.0	40.0	50.0
Fuel depot	21	Power cable	Cylinder	6.0	6.0	230.5	3.7	0.0	0.0	945.0	12.5	-60.0	230.4	21.0	6.0
Quonset hut	2	Hut shell	Cylinder	440.0	440.0	850.0	0.0	0.0	0.0	0.0	15.0	0.0	440.0	440.0	850.0
Quonset hut	3	Hut door	Cube	95.0	190.0	8.0	0.0	0.0	0.0	-50.0	95.0	428.0	95.0	190.0	8.0
Quonset hut	4	Hut door step	Cube	120.0	18.0	60.0	0.0	0.0	0.0	-50.0	9.0	458.0	120.0	18.0	60.0
Quonset hut	5	Hut window	Cube	70.0	50.0	8.0	0.0	0.0	0.0	95.0	135.0	428.0	70.0	50.0	8.0
Quonset hut	6	Hut AC unit	Cube	90.0	60.0	40.0	0.0	0.0	0.0	90.0	75.0	-448.0	90.0	60.0	40.0
Quonset hut	7	Hut AC fan	Cylinder	45.0	45.0	5.0	0.0	0.0	0.0	90.0	75.0	-471.0	45.0	45.0	5.0
Quonset hut	8	Hut flank window	Cube	70.0	3.0	50.0	0.0	0.0	58.0	-187.4	132.1	-240.0	39.6	61.0	50.0
Quonset hut	9	Hut flank window	Cube	70.0	3.0	50.0	0.0	0.0	58.0	-187.4	132.1	0.0	39.6	61.0	50.0
Quonset hut	10	Hut flank window	Cube	70.0	3.0	50.0	0.0	0.0	58.0	-187.4	132.1	240.0	39.6	61.0	50.0
Quonset hut	11	Hut flank window	Cube	70.0	3.0	50.0	0.0	0.0	-58.0	187.4	132.1	-240.0	39.6	61.0	50.0
Quonset hut	12	Hut flank window	Cube	70.0	3.0	50.0	0.0	0.0	-58.0	187.4	132.1	0.0	39.6	61.0	50.0
Quonset hut	13	Hut flank window	Cube	70.0	3.0	50.0	0.0	0.0	-58.0	187.4	132.1	240.0	39.6	61.0	50.0
Quonset hut	14	Hut door lamp hood	Cube	30.0	4.0	26.0	0.0	0.0	0.0	-50.0	220.0	444.0	30.0	4.0	26.0
Quonset hut	15	Hut door lamp	Sphere	14.0	14.0	6.0	90.0	0.0	0.0	-50.0	213.0	444.0	14.0	6.0	14.0
Quonset hut	16	Door step marker	Sphere	9.0	4.0	9.0	0.0	0.0	0.0	-10.0	20.0	483.0	9.0	4.0	9.0
Quonset hut	17	Hut stove pipe	Cylinder	12.0	12.0	80.0	90.0	0.0	0.0	-60.0	240.0	-150.0	12.0	80.0	12.0
Quonset hut	18	Blast wall course	Cube	300.0	40.0	60.0	0.0	0.0	0.0	0.0	20.0	628.0	300.0	40.0	60.0
Quonset hut	19	Blast wall course	Cube	270.0	40.0	60.0	0.0	0.0	0.0	0.0	60.0	628.0	270.0	40.0	60.0
Quonset hut	20	Blast wall course	Cube	240.0	40.0	60.0	0.0	0.0	0.0	0.0	100.0	628.0	240.0	40.0	60.0
Radio mast	2	Mast footing	Cube	130.0	20.0	130.0	0.0	0.0	0.0	0.0	2.0	0.0	130.0	20.0	130.0
Radio mast	3	Equipment shelter	Cube	110.0	100.0	90.0	0.0	0.0	0.0	150.0	45.0	20.0	110.0	100.0	90.0
Radio mast	4	Mast leg	Cylinder	7.0	7.0	900.4	-88.3	180.0	0.0	0.0	450.0	27.0	7.0	900.2	33.0
Radio mast	5	Mast leg	Cylinder	7.0	7.0	900.4	-88.3	60.0	0.0	-23.4	450.0	-13.5	32.1	900.2	22.6
Radio mast	6	Mast leg	Cylinder	7.0	7.0	900.4	-88.3	-60.0	0.0	23.4	450.0	-13.5	32.1	900.2	22.6
Radio mast	7	Mast brace	Cylinder	4.0	4.0	153.9	-65.5	-151.8	0.0	-15.1	110.0	10.7	35.4	141.7	61.3
Radio mast	8	Mast brace	Cylinder	4.0	4.0	151.1	-67.9	88.0	0.0	-1.8	250.0	-16.4	60.6	141.5	6.2
Radio mast	9	Mast brace	Cylinder	4.0	4.0	148.6	-70.4	-32.3	0.0	13.3	390.0	5.7	32.0	141.3	47.4
Radio mast	10	Mast brace	Cylinder	4.0	4.0	146.4	-73.0	-152.7	0.0	-9.8	530.0	7.7	24.9	141.2	43.3
Radio mast	11	Mast brace	Cylinder	4.0	4.0	144.5	-75.7	86.8	0.0	-1.8	670.0	-10.3	39.9	141.0	6.2
Radio mast	12	Mast brace	Cylinder	4.0	4.0	142.9	-78.4	-34.0	0.0	8.1	810.0	2.6	21.6	140.8	29.4
Radio mast	13	Whip antenna	Cylinder	3.0	3.0	140.0	90.0	0.0	0.0	0.0	970.0	0.0	3.0	140.0	3.0
Radio mast	14	Aviation light	Sphere	18.0	18.0	18.0	0.0	0.0	0.0	0.0	905.0	0.0	18.0	18.0	18.0
Radio mast	15	Aviation light	Sphere	15.0	15.0	15.0	0.0	0.0	0.0	0.0	460.0	0.0	15.0	15.0	15.0
Radio mast	16	Guy wire	Cylinder	1.5	1.5	775.7	62.3	60.0	0.0	155.9	356.4	90.0	313.7	687.8	182.0
Radio mast	17	Guy anchor	Cube	30.0	25.0	30.0	0.0	0.0	0.0	311.8	7.9	180.0	30.0	25.0	30.0
Radio mast	18	Guy wire	Cylinder	1.5	1.5	746.0	61.1	-60.0	0.0	-155.9	373.3	90.0	313.7	654.2	182.0
Radio mast	19	Guy anchor	Cube	30.0	25.0	30.0	0.0	0.0	0.0	-311.8	41.6	180.0	30.0	25.0	30.0
Radio mast	20	Guy wire	Cylinder	1.5	1.5	780.4	62.5	180.0	0.0	0.0	353.8	-180.0	1.5	693.1	361.3
Radio mast	21	Guy anchor	Cube	30.0	25.0	30.0	0.0	0.0	0.0	0.0	2.6	-360.0	30.0	25.0	30.0
Camouflaged tent	2	Ridge pole	Cylinder	7.0	7.0	320.0	0.0	90.0	0.0	0.0	170.0	0.0	320.0	7.0	7.0
Camouflaged tent	3	Tent pole	Cylinder	7.0	7.0	190.0	90.0	0.0	0.0	-150.0	75.0	0.0	7.0	190.0	7.0
Camouflaged tent	4	Tent pole	Cylinder	7.0	7.0	190.0	90.0	0.0	0.0	150.0	75.0	0.0	7.0	190.0	7.0
Camouflaged tent	5	Tent roof	Cube	310.0	4.0	185.1	51.6	180.0	0.0	0.0	97.5	-57.5	310.0	147.5	118.1
Camouflaged tent	6	Tent roof	Cube	310.0	4.0	185.1	51.6	0.0	0.0	0.0	97.5	57.5	310.0	147.5	118.1
Camouflaged tent	7	Tent back wall	Cube	4.0	120.0	120.0	45.0	0.0	0.0	-155.0	85.0	0.0	4.0	169.7	169.7
Camouflaged tent	8	Camo net	Sphere	460.0	100.0	380.0	0.0	0.0	0.0	0.0	155.0	0.0	460.0	100.0	380.0
Camouflaged tent	9	Camo net	Sphere	240.0	70.0	220.0	0.0	25.0	0.0	130.0	162.0	60.0	310.5	70.0	300.8

Camouflaged tent	10	Camo net	Sphere	220.0	60.0	200.0	0.0	-20.0	0.0	-120.0	158.0	-70.0	275.1	60.0	263.2
Camouflaged tent	11	Net pole	Cylinder	5.0	5.0	160.0	90.0	0.0	0.0	-210.0	60.0	-170.0	5.0	160.0	5.0
Camouflaged tent	12	Net pole	Cylinder	5.0	5.0	160.0	90.0	0.0	0.0	-210.0	60.0	170.0	5.0	160.0	5.0
Camouflaged tent	13	Net pole	Cylinder	5.0	5.0	160.0	90.0	0.0	0.0	210.0	60.0	-170.0	5.0	160.0	5.0
Camouflaged tent	14	Net pole	Cylinder	5.0	5.0	160.0	90.0	0.0	0.0	210.0	60.0	170.0	5.0	160.0	5.0
Camouflaged tent	15	Lantern	Sphere	12.0	16.0	12.0	0.0	0.0	0.0	90.0	142.0	0.0	12.0	16.0	12.0
Container	2	Container body	Cube	244.0	259.0	606.0	0.0	0.0	0.0	0.0	129.5	0.0	244.0	259.0	606.0
Container	3	Door bar	Cylinder	4.5	4.5	240.0	90.0	0.0	0.0	-55.0	130.0	306.0	4.5	240.0	4.5
Container	4	Door bar	Cylinder	4.5	4.5	240.0	90.0	0.0	0.0	55.0	130.0	306.0	4.5	240.0	4.5
Container	5	Door seam	Cube	3.0	240.0	3.0	0.0	0.0	0.0	0.0	130.0	306.0	3.0	240.0	3.0
Container	6	Corner casting	Cube	18.0	18.0	18.0	0.0	0.0	0.0	115.0	255.0	295.0	18.0	18.0	18.0
Container	7	Corner casting	Cube	18.0	18.0	18.0	0.0	0.0	0.0	-115.0	255.0	295.0	18.0	18.0	18.0
Container	8	Corner casting	Cube	18.0	18.0	18.0	0.0	0.0	0.0	115.0	255.0	-295.0	18.0	18.0	18.0
Container	9	Corner casting	Cube	18.0	18.0	18.0	0.0	0.0	0.0	-115.0	255.0	-295.0	18.0	18.0	18.0
Container	10	Corner casting	Cube	18.0	18.0	18.0	0.0	0.0	0.0	115.0	9.0	295.0	18.0	18.0	18.0
Container	11	Corner casting	Cube	18.0	18.0	18.0	0.0	0.0	0.0	-115.0	9.0	295.0	18.0	18.0	18.0
Container	12	Container ID	Cube	1.0	30.0	140.0	0.0	0.0	0.0	122.6	220.0	160.0	1.0	30.0	140.0
Guard booth	2	Booth floor	Cube	240.0	20.0	240.0	0.0	0.0	0.0	0.0	10.0	0.0	240.0	20.0	240.0
Guard booth	3	Booth body	Cube	220.0	230.0	220.0	0.0	0.0	0.0	0.0	135.0	0.0	220.0	230.0	220.0
Guard booth	4	Booth roof	Cube	270.0	12.0	270.0	0.0	0.0	0.0	0.0	256.0	0.0	270.0	12.0	270.0
Guard booth	5	Booth window	Cube	4.0	70.0	130.0	0.0	0.0	0.0	-112.0	165.0	0.0	4.0	70.0	130.0
Guard booth	6	Booth door	Cube	90.0	180.0	4.0	0.0	0.0	0.0	30.0	110.0	112.0	90.0	180.0	4.0
Guard booth	7	Booth AC unit	Cube	70.0	50.0	35.0	0.0	90.0	0.0	125.0	110.0	-50.0	35.0	50.0	70.0
Guard booth	8	Booth antenna	Cylinder	3.0	3.0	140.0	90.0	0.0	0.0	80.0	330.0	-80.0	3.0	140.0	3.0
Guard booth	9	Door handle	Cube	15.0	4.0	5.0	0.0	0.0	0.0	65.0	110.0	116.0	15.0	4.0	5.0
Guard booth	10	Booth roof edge	Cylinder	12.0	12.0	270.0	0.0	90.0	0.0	0.0	256.0	-135.0	270.0	12.0	12.0
Guard booth	11	Booth roof edge	Cylinder	12.0	12.0	270.0	0.0	90.0	0.0	0.0	256.0	135.0	270.0	12.0	12.0
Guard booth	12	Checkpoint placard	Cube	1.0	30.0	80.0	0.0	0.0	0.0	-111.5	85.0	0.0	1.0	30.0	80.0
UAV pad	2	Pad slab	Cylinder	130.0	130.0	4.0	90.0	0.0	0.0	0.0	2.0	0.0	130.0	4.0	130.0
UAV pad	3	Pad ring	Annulus	118.0	118.0	0.4	90.0	0.0	0.0	0.0	4.2	0.0	118.0	0.4	118.0
UAV pad	4	Pad centre	Cylinder	102.0	102.0	0.4	90.0	0.0	0.0	0.0	4.4	0.0	102.0	0.4	102.0
UAV pad	5	Landing rail	Cube	3.5	2.0	32.0	0.0	0.0	0.0	-6.0	5.0	0.0	3.5	2.0	32.0
UAV pad	6	Landing rail	Cube	3.5	2.0	32.0	0.0	0.0	0.0	6.0	5.0	0.0	3.5	2.0	32.0
UAV pad	7	Pad light	Sphere	5.0	3.0	5.0	0.0	0.0	0.0	42.4	5.2	42.4	5.0	3.0	5.0
UAV pad	8	Pad light	Sphere	5.0	3.0	5.0	0.0	0.0	0.0	-42.4	5.2	42.4	5.0	3.0	5.0
UAV pad	9	Pad light	Sphere	5.0	3.0	5.0	0.0	0.0	0.0	-42.4	5.2	-42.4	5.0	3.0	5.0
UAV pad	10	Pad light	Sphere	5.0	3.0	5.0	0.0	0.0	0.0	42.4	5.2	-42.4	5.0	3.0	5.0
HESCO perimeter	2	HESCO cell	Cube	103.0	140.0	106.0	0.0	-90.8	0.0	3399.6	70.0	49.9	107.5	140.0	104.5
HESCO perimeter	3	HESCO cell	Cube	103.0	140.0	106.0	0.0	-92.5	0.0	3396.7	70.0	149.7	110.4	140.0	107.6
HESCO perimeter	4	HESCO cell	Cube	103.0	140.0	106.0	0.0	-94.2	0.0	3390.8	70.0	249.3	113.3	140.0	110.5
HESCO perimeter	5	HESCO cell	Cube	103.0	140.0	106.0	0.0	-95.9	0.0	3382.1	70.0	348.8	116.0	140.0	113.3
HESCO perimeter	6	HESCO cell	Cube	103.0	140.0	106.0	0.0	-97.6	0.0	3370.4	70.0	447.9	118.6	140.0	116.1
HESCO perimeter	7	HESCO cell	Cube	103.0	140.0	106.0	0.0	-99.3	0.0	3355.8	70.0	546.7	121.2	140.0	118.7
HESCO perimeter	8	HESCO cell	Cube	103.0	140.0	106.0	0.0	-100.9	0.0	3338.3	70.0	644.9	123.6	140.0	121.2
HESCO perimeter	9	HESCO cell	Cube	103.0	140.0	106.0	0.0	-102.6	0.0	3317.9	70.0	742.7	125.9	140.0	123.7
HESCO perimeter	10	HESCO cell	Cube	103.0	140.0	106.0	0.0	-104.3	0.0	3294.7	70.0	839.7	128.2	140.0	126.0
HESCO perimeter	11	HESCO cell	Cube	103.0	140.0	106.0	0.0	-106.0	0.0	3268.6	70.0	936.1	130.3	140.0	128.2
HESCO perimeter	12	HESCO cell	Cube	103.0	140.0	106.0	0.0	-107.7	0.0	3239.7	70.0	1031.7	132.3	140.0	130.3
HESCO perimeter	13	HESCO cell	Cube	103.0	140.0	106.0	0.0	-109.3	0.0	3208.0	70.0	1126.3	134.1	140.0	132.3
HESCO perimeter	14	HESCO cell	Cube	103.0	140.0	106.0	0.0	-111.0	0.0	3173.6	70.0	1220.0	135.9	140.0	134.2
HESCO perimeter	15	HESCO cell	Cube	103.0	140.0	106.0	0.0	-112.7	0.0	3136.4	70.0	1312.6	137.5	140.0	135.9

HESCO perimeter	16	HESCO cell	Cube	103.0	140.0	106.0	0.0	-114.4	0.0	3096.5	70.0	1404.2	139.1	140.0	137.6
HESCO perimeter	17	HESCO cell	Cube	103.0	140.0	106.0	0.0	-116.1	0.0	3054.0	70.0	1494.4	140.5	140.0	139.1
HESCO perimeter	18	HESCO cell	Cube	103.0	140.0	106.0	0.0	-117.8	0.0	3008.8	70.0	1583.5	141.8	140.0	140.5
HESCO perimeter	19	HESCO cell	Cube	103.0	140.0	106.0	0.0	-119.4	0.0	2961.0	70.0	1671.1	142.9	140.0	141.8
HESCO perimeter	20	HESCO cell	Cube	103.0	140.0	106.0	0.0	-121.1	0.0	2910.6	70.0	1757.3	144.0	140.0	143.0
HESCO perimeter	21	HESCO cell	Cube	103.0	140.0	106.0	0.0	-122.8	0.0	2857.8	70.0	1842.0	144.9	140.0	144.0
HESCO perimeter	22	HESCO cell	Cube	103.0	140.0	106.0	0.0	-124.5	0.0	2802.5	70.0	1925.1	145.7	140.0	144.9
HESCO perimeter	23	HESCO cell	Cube	103.0	140.0	106.0	0.0	-126.2	0.0	2744.8	70.0	2006.5	146.4	140.0	145.7
HESCO perimeter	24	HESCO cell	Cube	103.0	140.0	106.0	0.0	-127.9	0.0	2684.7	70.0	2086.2	146.9	140.0	146.4
HESCO perimeter	25	HESCO cell	Cube	103.0	140.0	106.0	0.0	-129.5	0.0	2622.3	70.0	2164.2	147.3	140.0	146.9
HESCO perimeter	26	HESCO cell	Cube	103.0	140.0	106.0	0.0	-131.2	0.0	2557.6	70.0	2240.2	147.6	140.0	147.3
HESCO perimeter	27	HESCO cell	Cube	103.0	140.0	106.0	0.0	-132.9	0.0	2490.8	70.0	2314.3	147.8	140.0	147.6
HESCO perimeter	28	HESCO cell	Cube	103.0	140.0	106.0	0.0	-134.6	0.0	2421.7	70.0	2386.5	147.8	140.0	147.8
HESCO perimeter	29	HESCO cell	Cube	103.0	140.0	106.0	0.0	-136.3	0.0	2350.6	70.0	2456.5	147.7	140.0	147.8
HESCO perimeter	30	HESCO cell	Cube	103.0	140.0	106.0	0.0	-137.9	0.0	2277.5	70.0	2524.5	147.5	140.0	147.7
HESCO perimeter	31	HESCO cell	Cube	103.0	140.0	106.0	0.0	-139.6	0.0	2202.4	70.0	2590.2	147.1	140.0	147.5
HESCO perimeter	32	HESCO cell	Cube	103.0	140.0	106.0	0.0	-141.3	0.0	2125.4	70.0	2653.8	146.7	140.0	147.1
HESCO perimeter	33	HESCO cell	Cube	103.0	140.0	106.0	0.0	-143.0	0.0	2046.6	70.0	2715.0	146.1	140.0	146.6
HESCO perimeter	34	HESCO cell	Cube	103.0	140.0	106.0	0.0	-144.7	0.0	1966.0	70.0	2773.9	145.3	140.0	146.0
HESCO perimeter	35	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-144.7	0.0	1966.0	210.0	2773.9	145.3	140.0	146.0
HESCO perimeter	36	HESCO cell	Cube	103.0	140.0	106.0	0.0	-146.4	0.0	1883.7	70.0	2830.5	144.5	140.0	145.3
HESCO perimeter	37	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-146.4	0.0	1883.7	210.0	2830.5	144.5	140.0	145.3
HESCO perimeter	38	HESCO cell	Cube	103.0	140.0	106.0	0.0	-148.0	0.0	1799.8	70.0	2884.5	143.5	140.0	144.5
HESCO perimeter	39	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-148.0	0.0	1799.8	210.0	2884.5	143.5	140.0	144.5
HESCO perimeter	40	HESCO cell	Cube	103.0	140.0	106.0	0.0	-149.7	0.0	1714.4	70.0	2936.1	142.4	140.0	143.5
HESCO perimeter	41	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-149.7	0.0	1714.4	210.0	2936.1	142.4	140.0	143.5
HESCO perimeter	42	HESCO cell	Cube	103.0	140.0	106.0	0.0	-151.4	0.0	1627.5	70.0	2985.2	141.2	140.0	142.4
HESCO perimeter	43	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-151.4	0.0	1627.5	210.0	2985.2	141.2	140.0	142.4
HESCO perimeter	44	HESCO cell	Cube	103.0	140.0	106.0	0.0	-153.1	0.0	1539.1	70.0	3031.7	139.8	140.0	141.1
HESCO perimeter	45	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-153.1	0.0	1539.1	210.0	3031.7	139.8	140.0	141.1
HESCO perimeter	46	HESCO cell	Cube	103.0	140.0	106.0	0.0	-154.8	0.0	1449.5	70.0	3075.6	138.4	140.0	139.8
HESCO perimeter	47	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-154.8	0.0	1449.5	210.0	3075.6	138.4	140.0	139.8
HESCO perimeter	48	HESCO cell	Cube	103.0	140.0	106.0	0.0	-156.4	0.0	1358.5	70.0	3116.8	136.8	140.0	138.3
HESCO perimeter	49	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-156.4	0.0	1358.5	210.0	3116.8	136.8	140.0	138.3
HESCO perimeter	50	HESCO cell	Cube	103.0	140.0	106.0	0.0	-158.1	0.0	1266.5	70.0	3155.3	135.1	140.0	136.7
HESCO perimeter	51	HESCO cell	Cube	103.0	140.0	106.0	0.0	-159.8	0.0	1173.3	70.0	3191.1	133.3	140.0	135.0
HESCO perimeter	52	HESCO cell	Cube	103.0	140.0	106.0	0.0	-161.5	0.0	1079.1	70.0	3224.2	131.3	140.0	133.2
HESCO perimeter	53	HESCO cell	Cube	103.0	140.0	106.0	0.0	-163.2	0.0	984.0	70.0	3254.5	129.3	140.0	131.3
HESCO perimeter	54	HESCO cell	Cube	103.0	140.0	106.0	0.0	-164.9	0.0	888.0	70.0	3282.0	127.1	140.0	129.2
HESCO perimeter	55	HESCO cell	Cube	103.0	140.0	106.0	0.0	-166.5	0.0	791.3	70.0	3306.6	124.8	140.0	127.1
HESCO perimeter	56	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-166.5	0.0	791.3	210.0	3306.6	124.8	140.0	127.1
HESCO perimeter	57	HESCO cell	Cube	103.0	140.0	106.0	0.0	-168.2	0.0	693.9	70.0	3328.4	122.5	140.0	124.8
HESCO perimeter	58	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-168.2	0.0	693.9	210.0	3328.4	122.5	140.0	124.8
HESCO perimeter	59	HESCO cell	Cube	103.0	140.0	106.0	0.0	-169.9	0.0	595.9	70.0	3347.4	120.0	140.0	122.4
HESCO perimeter	60	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-169.9	0.0	595.9	210.0	3347.4	120.0	140.0	122.4
HESCO perimeter	61	HESCO cell	Cube	103.0	140.0	106.0	0.0	-171.6	0.0	497.3	70.0	3363.4	117.4	140.0	119.9
HESCO perimeter	62	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-171.6	0.0	497.3	210.0	3363.4	117.4	140.0	119.9
HESCO perimeter	63	HESCO cell	Cube	103.0	140.0	106.0	0.0	169.9	0.0	-595.9	70.0	3347.4	120.0	140.0	122.4
HESCO perimeter	64	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	169.9	0.0	-595.9	210.0	3347.4	120.0	140.0	122.4
HESCO perimeter	65	HESCO cell	Cube	103.0	140.0	106.0	0.0	168.2	0.0	-693.9	70.0	3328.4	122.5	140.0	124.8
HESCO perimeter	66	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	168.2	0.0	-693.9	210.0	3328.4	122.5	140.0	124.8

HESCO perimeter	67	HESCO cell	Cube	103.0	140.0	106.0	0.0	166.5	0.0	-791.3	70.0	3306.6	124.8	140.0	127.1
HESCO perimeter	68	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	166.5	0.0	-791.3	210.0	3306.6	124.8	140.0	127.1
HESCO perimeter	69	HESCO cell	Cube	103.0	140.0	106.0	0.0	164.9	0.0	-888.0	70.0	3282.0	127.1	140.0	129.2
HESCO perimeter	70	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	164.9	0.0	-888.0	210.0	3282.0	127.1	140.0	129.2
HESCO perimeter	71	HESCO cell	Cube	103.0	140.0	106.0	0.0	163.2	0.0	-984.0	70.0	3254.5	129.3	140.0	131.3
HESCO perimeter	72	HESCO cell	Cube	103.0	140.0	106.0	0.0	161.5	0.0	-1079.1	70.0	3224.2	131.3	140.0	133.2
HESCO perimeter	73	HESCO cell	Cube	103.0	140.0	106.0	0.0	159.8	0.0	-1173.3	70.0	3191.1	133.3	140.0	135.0
HESCO perimeter	74	HESCO cell	Cube	103.0	140.0	106.0	0.0	158.1	0.0	-1266.5	70.0	3155.3	135.1	140.0	136.7
HESCO perimeter	75	HESCO cell	Cube	103.0	140.0	106.0	0.0	156.4	0.0	-1358.5	70.0	3116.8	136.8	140.0	138.3
HESCO perimeter	76	HESCO cell	Cube	103.0	140.0	106.0	0.0	154.8	0.0	-1449.5	70.0	3075.6	138.4	140.0	139.8
HESCO perimeter	77	HESCO cell	Cube	103.0	140.0	106.0	0.0	153.1	0.0	-1539.1	70.0	3031.7	139.8	140.0	141.1
HESCO perimeter	78	HESCO cell	Cube	103.0	140.0	106.0	0.0	151.4	0.0	-1627.5	70.0	2985.2	141.2	140.0	142.4
HESCO perimeter	79	HESCO cell	Cube	103.0	140.0	106.0	0.0	149.7	0.0	-1714.4	70.0	2936.1	142.4	140.0	143.5
HESCO perimeter	80	HESCO cell	Cube	103.0	140.0	106.0	0.0	148.0	0.0	-1799.8	70.0	2884.5	143.5	140.0	144.5
HESCO perimeter	81	HESCO cell	Cube	103.0	140.0	106.0	0.0	146.4	0.0	-1883.7	70.0	2830.5	144.5	140.0	145.3
HESCO perimeter	82	HESCO cell	Cube	103.0	140.0	106.0	0.0	144.7	0.0	-1966.0	70.0	2773.9	145.3	140.0	146.0
HESCO perimeter	83	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	144.7	0.0	-1966.0	210.0	2773.9	145.3	140.0	146.0
HESCO perimeter	84	HESCO cell	Cube	103.0	140.0	106.0	0.0	143.0	0.0	-2046.6	70.0	2715.0	146.1	140.0	146.6
HESCO perimeter	85	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	143.0	0.0	-2046.6	210.0	2715.0	146.1	140.0	146.6
HESCO perimeter	86	HESCO cell	Cube	103.0	140.0	106.0	0.0	141.3	0.0	-2125.4	70.0	2653.8	146.7	140.0	147.1
HESCO perimeter	87	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	141.3	0.0	-2125.4	210.0	2653.8	146.7	140.0	147.1
HESCO perimeter	88	HESCO cell	Cube	103.0	140.0	106.0	0.0	139.6	0.0	-2202.4	70.0	2590.2	147.1	140.0	147.5
HESCO perimeter	89	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	139.6	0.0	-2202.4	210.0	2590.2	147.1	140.0	147.5
HESCO perimeter	90	HESCO cell	Cube	103.0	140.0	106.0	0.0	137.9	0.0	-2277.5	70.0	2524.5	147.5	140.0	147.7
HESCO perimeter	91	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	137.9	0.0	-2277.5	210.0	2524.5	147.5	140.0	147.7
HESCO perimeter	92	HESCO cell	Cube	103.0	140.0	106.0	0.0	136.3	0.0	-2350.6	70.0	2456.5	147.7	140.0	147.8
HESCO perimeter	93	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	136.3	0.0	-2350.6	210.0	2456.5	147.7	140.0	147.8
HESCO perimeter	94	HESCO cell	Cube	103.0	140.0	106.0	0.0	134.6	0.0	-2421.7	70.0	2386.5	147.8	140.0	147.8
HESCO perimeter	95	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	134.6	0.0	-2421.7	210.0	2386.5	147.8	140.0	147.8
HESCO perimeter	96	HESCO cell	Cube	103.0	140.0	106.0	0.0	132.9	0.0	-2490.8	70.0	2314.3	147.8	140.0	147.6
HESCO perimeter	97	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	132.9	0.0	-2490.8	210.0	2314.3	147.8	140.0	147.6
HESCO perimeter	98	HESCO cell	Cube	103.0	140.0	106.0	0.0	131.2	0.0	-2557.6	70.0	2240.2	147.6	140.0	147.3
HESCO perimeter	99	HESCO cell	Cube	103.0	140.0	106.0	0.0	129.5	0.0	-2622.3	70.0	2164.2	147.3	140.0	146.9
HESCO perimeter	100	HESCO cell	Cube	103.0	140.0	106.0	0.0	127.9	0.0	-2684.7	70.0	2086.2	146.9	140.0	146.4
HESCO perimeter	101	HESCO cell	Cube	103.0	140.0	106.0	0.0	126.2	0.0	-2744.8	70.0	2006.5	146.4	140.0	145.7
HESCO perimeter	102	HESCO cell	Cube	103.0	140.0	106.0	0.0	124.5	0.0	-2802.5	70.0	1925.1	145.7	140.0	144.9
HESCO perimeter	103	HESCO cell	Cube	103.0	140.0	106.0	0.0	122.8	0.0	-2857.8	70.0	1842.0	144.9	140.0	144.0
HESCO perimeter	104	HESCO cell	Cube	103.0	140.0	106.0	0.0	121.1	0.0	-2910.6	70.0	1757.3	144.0	140.0	143.0
HESCO perimeter	105	HESCO cell	Cube	103.0	140.0	106.0	0.0	119.4	0.0	-2961.0	70.0	1671.1	142.9	140.0	141.8
HESCO perimeter	106	HESCO cell	Cube	103.0	140.0	106.0	0.0	117.8	0.0	-3008.8	70.0	1583.5	141.8	140.0	140.5
HESCO perimeter	107	HESCO cell	Cube	103.0	140.0	106.0	0.0	116.1	0.0	-3054.0	70.0	1494.4	140.5	140.0	139.1
HESCO perimeter	108	HESCO cell	Cube	103.0	140.0	106.0	0.0	114.4	0.0	-3096.5	70.0	1404.2	139.1	140.0	137.6
HESCO perimeter	109	HESCO cell	Cube	103.0	140.0	106.0	0.0	112.7	0.0	-3136.4	70.0	1312.6	137.5	140.0	135.9
HESCO perimeter	110	HESCO cell	Cube	103.0	140.0	106.0	0.0	111.0	0.0	-3173.6	70.0	1220.0	135.9	140.0	134.2
HESCO perimeter	111	HESCO cell	Cube	103.0	140.0	106.0	0.0	109.3	0.0	-3208.0	70.0	1126.3	134.1	140.0	132.3
HESCO perimeter	112	HESCO cell	Cube	103.0	140.0	106.0	0.0	107.7	0.0	-3239.7	70.0	1031.7	132.3	140.0	130.3
HESCO perimeter	113	HESCO cell	Cube	103.0	140.0	106.0	0.0	106.0	0.0	-3268.6	70.0	936.1	130.3	140.0	128.2
HESCO perimeter	114	HESCO cell	Cube	103.0	140.0	106.0	0.0	104.3	0.0	-3294.7	70.0	839.7	128.2	140.0	126.0
HESCO perimeter	115	HESCO cell	Cube	103.0	140.0	106.0	0.0	102.6	0.0	-3317.9	70.0	742.7	125.9	140.0	123.7
HESCO perimeter	116	HESCO cell	Cube	103.0	140.0	106.0	0.0	100.9	0.0	-3338.3	70.0	644.9	123.6	140.0	121.2
HESCO perimeter	117	HESCO cell	Cube	103.0	140.0	106.0	0.0	99.3	0.0	-3355.8	70.0	546.7	121.2	140.0	118.7

HESCO perimeter	118	HESCO cell	Cube	103.0	140.0	106.0	0.0	97.6	0.0	-3370.4	70.0	447.9	118.6	140.0	116.1
HESCO perimeter	119	HESCO cell	Cube	103.0	140.0	106.0	0.0	95.9	0.0	-3382.1	70.0	348.8	116.0	140.0	113.3
HESCO perimeter	120	HESCO cell	Cube	103.0	140.0	106.0	0.0	94.2	0.0	-3390.8	70.0	249.3	113.3	140.0	110.5
HESCO perimeter	121	HESCO cell	Cube	103.0	140.0	106.0	0.0	92.5	0.0	-3396.7	70.0	149.7	110.4	140.0	107.6
HESCO perimeter	122	HESCO cell	Cube	103.0	140.0	106.0	0.0	90.8	0.0	-3399.6	70.0	49.9	107.5	140.0	104.5
HESCO perimeter	123	HESCO cell	Cube	103.0	140.0	106.0	0.0	89.2	0.0	-3399.6	70.0	-49.9	107.5	140.0	104.5
HESCO perimeter	124	HESCO cell	Cube	103.0	140.0	106.0	0.0	87.5	0.0	-3396.7	70.0	-149.7	110.4	140.0	107.6
HESCO perimeter	125	HESCO cell	Cube	103.0	140.0	106.0	0.0	85.8	0.0	-3390.8	70.0	-249.3	113.3	140.0	110.5
HESCO perimeter	126	HESCO cell	Cube	103.0	140.0	106.0	0.0	84.1	0.0	-3382.1	70.0	-348.8	116.0	140.0	113.3
HESCO perimeter	127	HESCO cell	Cube	103.0	140.0	106.0	0.0	82.4	0.0	-3370.4	70.0	-447.9	118.6	140.0	116.1
HESCO perimeter	128	HESCO cell	Cube	103.0	140.0	106.0	0.0	80.7	0.0	-3355.8	70.0	-546.7	121.2	140.0	118.7
HESCO perimeter	129	HESCO cell	Cube	103.0	140.0	106.0	0.0	79.1	0.0	-3338.3	70.0	-644.9	123.6	140.0	121.2
HESCO perimeter	130	HESCO cell	Cube	103.0	140.0	106.0	0.0	77.4	0.0	-3317.9	70.0	-742.7	125.9	140.0	123.7
HESCO perimeter	131	HESCO cell	Cube	103.0	140.0	106.0	0.0	75.7	0.0	-3294.7	70.0	-839.7	128.2	140.0	126.0
HESCO perimeter	132	HESCO cell	Cube	103.0	140.0	106.0	0.0	74.0	0.0	-3268.6	70.0	-936.1	130.3	140.0	128.2
HESCO perimeter	133	HESCO cell	Cube	103.0	140.0	106.0	0.0	72.3	0.0	-3239.7	70.0	-1031.7	132.3	140.0	130.3
HESCO perimeter	134	HESCO cell	Cube	103.0	140.0	106.0	0.0	70.7	0.0	-3208.0	70.0	-1126.3	134.1	140.0	132.3
HESCO perimeter	135	HESCO cell	Cube	103.0	140.0	106.0	0.0	69.0	0.0	-3173.6	70.0	-1220.0	135.9	140.0	134.2
HESCO perimeter	136	HESCO cell	Cube	103.0	140.0	106.0	0.0	67.3	0.0	-3136.4	70.0	-1312.6	137.5	140.0	135.9
HESCO perimeter	137	HESCO cell	Cube	103.0	140.0	106.0	0.0	65.6	0.0	-3096.5	70.0	-1404.2	139.1	140.0	137.6
HESCO perimeter	138	HESCO cell	Cube	103.0	140.0	106.0	0.0	63.9	0.0	-3054.0	70.0	-1494.4	140.5	140.0	139.1
HESCO perimeter	139	HESCO cell	Cube	103.0	140.0	106.0	0.0	62.2	0.0	-3008.8	70.0	-1583.5	141.8	140.0	140.5
HESCO perimeter	140	HESCO cell	Cube	103.0	140.0	106.0	0.0	60.6	0.0	-2961.0	70.0	-1671.1	142.9	140.0	141.8
HESCO perimeter	141	HESCO cell	Cube	103.0	140.0	106.0	0.0	58.9	0.0	-2910.6	70.0	-1757.3	144.0	140.0	143.0
HESCO perimeter	142	HESCO cell	Cube	103.0	140.0	106.0	0.0	57.2	0.0	-2857.8	70.0	-1842.0	144.9	140.0	144.0
HESCO perimeter	143	HESCO cell	Cube	103.0	140.0	106.0	0.0	55.5	0.0	-2802.5	70.0	-1925.1	145.7	140.0	144.9
HESCO perimeter	144	HESCO cell	Cube	103.0	140.0	106.0	0.0	53.8	0.0	-2744.8	70.0	-2006.5	146.4	140.0	145.7
HESCO perimeter	145	HESCO cell	Cube	103.0	140.0	106.0	0.0	52.2	0.0	-2684.7	70.0	-2086.3	146.9	140.0	146.4
HESCO perimeter	146	HESCO cell	Cube	103.0	140.0	106.0	0.0	50.5	0.0	-2622.3	70.0	-2164.2	147.3	140.0	146.9
HESCO perimeter	147	HESCO cell	Cube	103.0	140.0	106.0	0.0	48.8	0.0	-2557.6	70.0	-2240.2	147.6	140.0	147.3
HESCO perimeter	148	HESCO cell	Cube	103.0	140.0	106.0	0.0	47.1	0.0	-2490.8	70.0	-2314.3	147.8	140.0	147.6
HESCO perimeter	149	HESCO cell	Cube	103.0	140.0	106.0	0.0	45.4	0.0	-2421.7	70.0	-2386.5	147.8	140.0	147.8
HESCO perimeter	150	HESCO cell	Cube	103.0	140.0	106.0	0.0	43.7	0.0	-2350.6	70.0	-2456.5	147.7	140.0	147.8
HESCO perimeter	151	HESCO cell	Cube	103.0	140.0	106.0	0.0	42.1	0.0	-2277.5	70.0	-2524.5	147.5	140.0	147.7
HESCO perimeter	152	HESCO cell	Cube	103.0	140.0	106.0	0.0	40.4	0.0	-2202.4	70.0	-2590.2	147.1	140.0	147.5
HESCO perimeter	153	HESCO cell	Cube	103.0	140.0	106.0	0.0	38.7	0.0	-2125.4	70.0	-2653.8	146.7	140.0	147.1
HESCO perimeter	154	HESCO cell	Cube	103.0	140.0	106.0	0.0	37.0	0.0	-2046.6	70.0	-2715.0	146.1	140.0	146.6
HESCO perimeter	155	HESCO cell	Cube	103.0	140.0	106.0	0.0	35.3	0.0	-1966.0	70.0	-2773.9	145.3	140.0	146.0
HESCO perimeter	156	HESCO cell	Cube	103.0	140.0	106.0	0.0	33.6	0.0	-1883.7	70.0	-2830.5	144.5	140.0	145.3
HESCO perimeter	157	HESCO cell	Cube	103.0	140.0	106.0	0.0	32.0	0.0	-1799.8	70.0	-2884.5	143.5	140.0	144.5
HESCO perimeter	158	HESCO cell	Cube	103.0	140.0	106.0	0.0	30.3	0.0	-1714.4	70.0	-2936.1	142.4	140.0	143.5
HESCO perimeter	159	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	30.3	0.0	-1714.4	210.0	-2936.1	142.4	140.0	143.5
HESCO perimeter	160	HESCO cell	Cube	103.0	140.0	106.0	0.0	28.6	0.0	-1627.5	70.0	-2985.2	141.2	140.0	142.4
HESCO perimeter	161	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	28.6	0.0	-1627.5	210.0	-2985.2	141.2	140.0	142.4
HESCO perimeter	162	HESCO cell	Cube	103.0	140.0	106.0	0.0	26.9	0.0	-1539.1	70.0	-3031.7	139.8	140.0	141.1
HESCO perimeter	163	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	26.9	0.0	-1539.1	210.0	-3031.7	139.8	140.0	141.1
HESCO perimeter	164	HESCO cell	Cube	103.0	140.0	106.0	0.0	25.2	0.0	-1449.5	70.0	-3075.6	138.4	140.0	139.8
HESCO perimeter	165	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	25.2	0.0	-1449.5	210.0	-3075.6	138.4	140.0	139.8
HESCO perimeter	166	HESCO cell	Cube	103.0	140.0	106.0	0.0	23.6	0.0	-1358.5	70.0	-3116.8	136.8	140.0	138.3
HESCO perimeter	167	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	23.6	0.0	-1358.5	210.0	-3116.8	136.8	140.0	138.3
HESCO perimeter	168	HESCO cell	Cube	103.0	140.0	106.0	0.0	21.9	0.0	-1266.5	70.0	-3155.3	135.1	140.0	136.7

HESCO perimeter	169	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	21.9	0.0	-1266.5	210.0	-3155.3	135.1	140.0	136.7
HESCO perimeter	170	HESCO cell	Cube	103.0	140.0	106.0	0.0	20.2	0.0	-1173.3	70.0	-3191.1	133.3	140.0	135.0
HESCO perimeter	171	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	20.2	0.0	-1173.3	210.0	-3191.1	133.3	140.0	135.0
HESCO perimeter	172	HESCO cell	Cube	103.0	140.0	106.0	0.0	18.5	0.0	-1079.1	70.0	-3224.2	131.3	140.0	133.2
HESCO perimeter	173	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	18.5	0.0	-1079.1	210.0	-3224.2	131.3	140.0	133.2
HESCO perimeter	174	HESCO cell	Cube	103.0	140.0	106.0	0.0	16.8	0.0	-984.0	70.0	-3254.5	129.3	140.0	131.3
HESCO perimeter	175	HESCO cell	Cube	103.0	140.0	106.0	0.0	15.1	0.0	-888.0	70.0	-3282.0	127.1	140.0	129.2
HESCO perimeter	176	HESCO cell	Cube	103.0	140.0	106.0	0.0	13.5	0.0	-791.3	70.0	-3306.6	124.8	140.0	127.1
HESCO perimeter	177	HESCO cell	Cube	103.0	140.0	106.0	0.0	11.8	0.0	-693.9	70.0	-3328.4	122.5	140.0	124.8
HESCO perimeter	178	HESCO cell	Cube	103.0	140.0	106.0	0.0	10.1	0.0	-595.9	70.0	-3347.4	120.0	140.0	122.4
HESCO perimeter	179	HESCO cell	Cube	103.0	140.0	106.0	0.0	8.4	0.0	-497.3	70.0	-3363.4	117.4	140.0	119.9
HESCO perimeter	180	HESCO cell	Cube	103.0	140.0	106.0	0.0	6.7	0.0	-398.4	70.0	-3376.6	114.7	140.0	117.3
HESCO perimeter	181	HESCO cell	Cube	103.0	140.0	106.0	0.0	5.0	0.0	-299.1	70.0	-3386.8	111.9	140.0	114.6
HESCO perimeter	182	HESCO cell	Cube	103.0	140.0	106.0	0.0	3.4	0.0	-199.5	70.0	-3394.1	109.0	140.0	111.9
HESCO perimeter	183	HESCO cell	Cube	103.0	140.0	106.0	0.0	1.7	0.0	-99.8	70.0	-3398.5	106.1	140.0	109.0
HESCO perimeter	184	HESCO cell	Cube	103.0	140.0	106.0	0.0	0.0	0.0	0.0	70.0	-3400.0	103.0	140.0	106.0
HESCO perimeter	185	HESCO cell	Cube	103.0	140.0	106.0	0.0	-1.7	0.0	99.8	70.0	-3398.5	106.1	140.0	109.0
HESCO perimeter	186	HESCO cell	Cube	103.0	140.0	106.0	0.0	-3.4	0.0	199.5	70.0	-3394.1	109.0	140.0	111.9
HESCO perimeter	187	HESCO cell	Cube	103.0	140.0	106.0	0.0	-5.0	0.0	299.1	70.0	-3386.8	111.9	140.0	114.6
HESCO perimeter	188	HESCO cell	Cube	103.0	140.0	106.0	0.0	-6.7	0.0	398.4	70.0	-3376.6	114.7	140.0	117.3
HESCO perimeter	189	HESCO cell	Cube	103.0	140.0	106.0	0.0	-8.4	0.0	497.3	70.0	-3363.4	117.4	140.0	119.9
HESCO perimeter	190	HESCO cell	Cube	103.0	140.0	106.0	0.0	-10.1	0.0	595.9	70.0	-3347.4	120.0	140.0	122.4
HESCO perimeter	191	HESCO cell	Cube	103.0	140.0	106.0	0.0	-11.8	0.0	693.9	70.0	-3328.4	122.5	140.0	124.8
HESCO perimeter	192	HESCO cell	Cube	103.0	140.0	106.0	0.0	-13.5	0.0	791.3	70.0	-3306.6	124.8	140.0	127.1
HESCO perimeter	193	HESCO cell	Cube	103.0	140.0	106.0	0.0	-15.1	0.0	888.0	70.0	-3282.0	127.1	140.0	129.2
HESCO perimeter	194	HESCO cell	Cube	103.0	140.0	106.0	0.0	-16.8	0.0	984.0	70.0	-3254.5	129.3	140.0	131.3
HESCO perimeter	195	HESCO cell	Cube	103.0	140.0	106.0	0.0	-18.5	0.0	1079.1	70.0	-3224.2	131.3	140.0	133.2
HESCO perimeter	196	HESCO cell	Cube	103.0	140.0	106.0	0.0	-20.2	0.0	1173.3	70.0	-3191.1	133.3	140.0	135.0
HESCO perimeter	197	HESCO cell	Cube	103.0	140.0	106.0	0.0	-21.9	0.0	1266.5	70.0	-3155.3	135.1	140.0	136.7
HESCO perimeter	198	HESCO cell	Cube	103.0	140.0	106.0	0.0	-23.6	0.0	1358.5	70.0	-3116.8	136.8	140.0	138.3
HESCO perimeter	199	HESCO cell	Cube	103.0	140.0	106.0	0.0	-25.2	0.0	1449.5	70.0	-3075.6	138.4	140.0	139.8
HESCO perimeter	200	HESCO cell	Cube	103.0	140.0	106.0	0.0	-26.9	0.0	1539.1	70.0	-3031.7	139.8	140.0	141.1
HESCO perimeter	201	HESCO cell	Cube	103.0	140.0	106.0	0.0	-28.6	0.0	1627.5	70.0	-2985.2	141.2	140.0	142.4
HESCO perimeter	202	HESCO cell	Cube	103.0	140.0	106.0	0.0	-30.3	0.0	1714.4	70.0	-2936.1	142.4	140.0	143.5
HESCO perimeter	203	HESCO cell	Cube	103.0	140.0	106.0	0.0	-32.0	0.0	1799.8	70.0	-2884.5	143.5	140.0	144.5
HESCO perimeter	204	HESCO cell	Cube	103.0	140.0	106.0	0.0	-33.6	0.0	1883.7	70.0	-2830.5	144.5	140.0	145.3
HESCO perimeter	205	HESCO cell	Cube	103.0	140.0	106.0	0.0	-35.3	0.0	1966.0	70.0	-2773.9	145.3	140.0	146.0
HESCO perimeter	206	HESCO cell	Cube	103.0	140.0	106.0	0.0	-37.0	0.0	2046.6	70.0	-2715.0	146.1	140.0	146.6
HESCO perimeter	207	HESCO cell	Cube	103.0	140.0	106.0	0.0	-38.7	0.0	2125.4	70.0	-2653.8	146.7	140.0	147.1
HESCO perimeter	208	HESCO cell	Cube	103.0	140.0	106.0	0.0	-40.4	0.0	2202.4	70.0	-2590.2	147.1	140.0	147.5
HESCO perimeter	209	HESCO cell	Cube	103.0	140.0	106.0	0.0	-42.1	0.0	2277.5	70.0	-2524.5	147.5	140.0	147.7
HESCO perimeter	210	HESCO cell	Cube	103.0	140.0	106.0	0.0	-43.7	0.0	2350.6	70.0	-2456.5	147.7	140.0	147.8
HESCO perimeter	211	HESCO cell	Cube	103.0	140.0	106.0	0.0	-45.4	0.0	2421.7	70.0	-2386.5	147.8	140.0	147.8
HESCO perimeter	212	HESCO cell	Cube	103.0	140.0	106.0	0.0	-47.1	0.0	2490.8	70.0	-2314.3	147.8	140.0	147.6
HESCO perimeter	213	HESCO cell	Cube	103.0	140.0	106.0	0.0	-48.8	0.0	2557.6	70.0	-2240.2	147.6	140.0	147.3
HESCO perimeter	214	HESCO cell	Cube	103.0	140.0	106.0	0.0	-50.5	0.0	2622.3	70.0	-2164.2	147.3	140.0	146.9
HESCO perimeter	215	HESCO cell	Cube	103.0	140.0	106.0	0.0	-52.2	0.0	2684.7	70.0	-2086.2	146.9	140.0	146.4
HESCO perimeter	216	HESCO cell	Cube	103.0	140.0	106.0	0.0	-53.8	0.0	2744.8	70.0	-2006.5	146.4	140.0	145.7
HESCO perimeter	217	HESCO cell	Cube	103.0	140.0	106.0	0.0	-55.5	0.0	2802.5	70.0	-1925.1	145.7	140.0	144.9
HESCO perimeter	218	HESCO cell	Cube	103.0	140.0	106.0	0.0	-57.2	0.0	2857.8	70.0	-1842.0	144.9	140.0	144.0
HESCO perimeter	219	HESCO cell	Cube	103.0	140.0	106.0	0.0	-58.9	0.0	2910.6	70.0	-1757.3	144.0	140.0	143.0

HESCO perimeter	220	HESCO cell	Cube	103.0	140.0	106.0	0.0	-60.6	0.0	2961.0	70.0	-1671.1	142.9	140.0	141.8
HESCO perimeter	221	HESCO cell	Cube	103.0	140.0	106.0	0.0	-62.2	0.0	3008.8	70.0	-1583.5	141.8	140.0	140.5
HESCO perimeter	222	HESCO cell	Cube	103.0	140.0	106.0	0.0	-63.9	0.0	3054.0	70.0	-1494.4	140.5	140.0	139.1
HESCO perimeter	223	HESCO cell	Cube	103.0	140.0	106.0	0.0	-65.6	0.0	3096.5	70.0	-1404.2	139.1	140.0	137.6
HESCO perimeter	224	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-65.6	0.0	3096.5	210.0	-1404.2	139.1	140.0	137.6
HESCO perimeter	225	HESCO cell	Cube	103.0	140.0	106.0	0.0	-67.3	0.0	3136.4	70.0	-1312.6	137.5	140.0	135.9
HESCO perimeter	226	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-67.3	0.0	3136.4	210.0	-1312.6	137.5	140.0	135.9
HESCO perimeter	227	HESCO cell	Cube	103.0	140.0	106.0	0.0	-69.0	0.0	3173.6	70.0	-1220.0	135.9	140.0	134.2
HESCO perimeter	228	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-69.0	0.0	3173.6	210.0	-1220.0	135.9	140.0	134.2
HESCO perimeter	229	HESCO cell	Cube	103.0	140.0	106.0	0.0	-70.7	0.0	3208.0	70.0	-1126.3	134.1	140.0	132.3
HESCO perimeter	230	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-70.7	0.0	3208.0	210.0	-1126.3	134.1	140.0	132.3
HESCO perimeter	231	HESCO cell	Cube	103.0	140.0	106.0	0.0	-72.3	0.0	3239.7	70.0	-1031.6	132.3	140.0	130.3
HESCO perimeter	232	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-72.3	0.0	3239.7	210.0	-1031.6	132.3	140.0	130.3
HESCO perimeter	233	HESCO cell	Cube	103.0	140.0	106.0	0.0	-74.0	0.0	3268.6	70.0	-936.1	130.3	140.0	128.2
HESCO perimeter	234	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-74.0	0.0	3268.6	210.0	-936.1	130.3	140.0	128.2
HESCO perimeter	235	HESCO cell	Cube	103.0	140.0	106.0	0.0	-75.7	0.0	3294.7	70.0	-839.7	128.2	140.0	126.0
HESCO perimeter	236	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-75.7	0.0	3294.7	210.0	-839.7	128.2	140.0	126.0
HESCO perimeter	237	HESCO cell	Cube	103.0	140.0	106.0	0.0	-77.4	0.0	3317.9	70.0	-742.7	125.9	140.0	123.7
HESCO perimeter	238	HESCO cell (upper)	Cube	103.0	140.0	106.0	0.0	-77.4	0.0	3317.9	210.0	-742.7	125.9	140.0	123.7
HESCO perimeter	239	HESCO cell	Cube	103.0	140.0	106.0	0.0	-79.1	0.0	3338.3	70.0	-644.9	123.6	140.0	121.2
HESCO perimeter	240	HESCO cell	Cube	103.0	140.0	106.0	0.0	-80.7	0.0	3355.8	70.0	-546.7	121.2	140.0	118.7
HESCO perimeter	241	HESCO cell	Cube	103.0	140.0	106.0	0.0	-82.4	0.0	3370.4	70.0	-447.9	118.6	140.0	116.1
HESCO perimeter	242	HESCO cell	Cube	103.0	140.0	106.0	0.0	-84.1	0.0	3382.1	70.0	-348.8	116.0	140.0	113.3
HESCO perimeter	243	HESCO cell	Cube	103.0	140.0	106.0	0.0	-85.8	0.0	3390.8	70.0	-249.3	113.3	140.0	110.5
HESCO perimeter	244	HESCO cell	Cube	103.0	140.0	106.0	0.0	-87.5	0.0	3396.7	70.0	-149.7	110.4	140.0	107.6
HESCO perimeter	245	HESCO cell	Cube	103.0	140.0	106.0	0.0	-89.2	0.0	3399.6	70.0	-49.9	107.5	140.0	104.5
Concertina wire	2	Wire coil	Helix	80.0	80.0	88.4	0.0	-0.8	0.0	3399.6	180.0	49.9	81.3	80.0	89.6
Concertina wire	3	Wire coil	Helix	80.0	80.0	88.4	0.0	-7.6	0.0	3370.4	180.0	447.9	91.0	80.0	98.2
Concertina wire	4	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-4.2	0.0	3385.0	210.0	248.9	32.3	3.0	398.2
Concertina wire	5	Wire coil	Helix	80.0	80.0	88.4	0.0	-14.3	0.0	3294.7	180.0	839.7	99.4	80.0	105.4
Concertina wire	6	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-10.9	0.0	3332.5	210.0	643.8	78.6	3.0	392.4
Concertina wire	7	Wire coil	Helix	80.0	80.0	88.4	0.0	-21.0	0.0	3173.6	180.0	1220.0	106.4	80.0	111.2
Concertina wire	8	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-17.7	0.0	3234.1	210.0	1029.9	123.9	3.0	381.2
Concertina wire	9	Wire coil	Helix	80.0	80.0	88.4	0.0	-27.8	0.0	3008.8	180.0	1583.5	112.0	80.0	115.5
Concertina wire	10	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-24.4	0.0	3091.2	210.0	1401.7	167.5	3.0	364.7
Concertina wire	11	Wire coil	Helix	80.0	80.0	88.4	0.0	-34.5	0.0	2802.5	180.0	1925.1	116.0	80.0	118.2
Concertina wire	12	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-31.1	0.0	2905.6	210.0	1754.3	208.8	3.0	343.2
Concertina wire	13	Wire coil	Helix	80.0	80.0	88.4	0.0	-41.2	0.0	2557.6	180.0	2240.2	118.4	80.0	119.2
Concertina wire	14	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-37.9	0.0	2680.1	210.0	2082.7	247.2	3.0	317.0
Concertina wire	15	Wire coil	Helix	80.0	80.0	88.4	0.0	-47.9	0.0	2277.5	180.0	2524.5	119.2	80.0	118.6
Concertina wire	16	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-44.6	0.0	2417.6	210.0	2382.3	282.2	3.0	286.4
Concertina wire	17	Wire coil	Helix	80.0	80.0	88.4	0.0	-54.7	0.0	1966.0	320.0	2773.9	118.4	80.0	116.4
Concertina wire	18	Razor strand	Cylinder	3.0	3.0	422.9	-19.3	-51.3	0.0	2121.8	280.0	2649.2	314.1	142.8	252.4
Concertina wire	19	Wire coil	Helix	80.0	80.0	88.4	0.0	-61.4	0.0	1627.5	320.0	2985.2	115.9	80.0	112.6
Concertina wire	20	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-58.0	0.0	1796.7	350.0	2879.6	340.2	3.0	213.8
Concertina wire	21	Wire coil	Helix	80.0	80.0	88.4	0.0	-68.1	0.0	1266.5	180.0	3155.3	111.9	80.0	107.2
Concertina wire	22	Razor strand	Cylinder	3.0	3.0	422.9	19.3	-64.8	0.0	1447.0	280.0	3070.3	363.2	142.8	173.3
Concertina wire	23	Wire coil	Helix	80.0	80.0	88.4	0.0	-74.9	0.0	888.0	180.0	3282.0	106.2	80.0	100.3
Concertina wire	24	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-71.5	0.0	1077.2	210.0	3218.7	379.4	3.0	129.5
Concertina wire	25	Wire coil	Helix	80.0	80.0	88.4	0.0	-81.6	0.0	497.3	320.0	3363.4	99.2	80.0	92.1
Concertina wire	26	Razor strand	Cylinder	3.0	3.0	422.9	-19.3	-78.2	0.0	692.7	280.0	3322.7	392.3	142.8	84.6

Concertina wire	27	Wire coil	Helix	80.0	80.0	88.4	0.0	-101.8	0.0	-693.9	320.0	3328.4	102.9	80.0	96.4
Concertina wire	28	Wire coil	Helix	80.0	80.0	88.4	0.0	-108.5	0.0	-1079.1	180.0	3224.2	109.2	80.0	103.9
Concertina wire	29	Razor strand	Cylinder	3.0	3.0	422.9	19.3	-105.1	0.0	-886.5	280.0	3276.3	387.0	142.8	107.4
Concertina wire	30	Wire coil	Helix	80.0	80.0	88.4	0.0	-115.2	0.0	-1449.5	180.0	3075.6	114.1	80.0	110.1
Concertina wire	31	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-111.9	0.0	-1264.3	210.0	3149.9	371.5	3.0	151.4
Concertina wire	32	Wire coil	Helix	80.0	80.0	88.4	0.0	-122.0	0.0	-1799.8	180.0	2884.5	117.4	80.0	114.7
Concertina wire	33	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-118.6	0.0	-1624.7	210.0	2980.0	351.8	3.0	193.7
Concertina wire	34	Wire coil	Helix	80.0	80.0	88.4	0.0	-128.7	0.0	-2125.4	320.0	2653.8	119.0	80.0	117.7
Concertina wire	35	Razor strand	Cylinder	3.0	3.0	422.9	-19.3	-125.3	0.0	-1962.6	280.0	2769.2	328.1	142.8	233.8
Concertina wire	36	Wire coil	Helix	80.0	80.0	88.4	0.0	-135.4	0.0	-2421.7	320.0	2386.5	119.0	80.0	119.1
Concertina wire	37	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-132.1	0.0	-2273.6	350.0	2520.1	298.3	3.0	269.6
Concertina wire	38	Wire coil	Helix	80.0	80.0	88.4	0.0	-142.2	0.0	-2684.7	180.0	2086.2	117.4	80.0	118.9
Concertina wire	39	Razor strand	Cylinder	3.0	3.0	422.9	19.3	-138.8	0.0	-2553.2	280.0	2236.4	265.9	142.8	302.9
Concertina wire	40	Wire coil	Helix	80.0	80.0	88.4	0.0	-148.9	0.0	-2910.6	180.0	1757.3	114.2	80.0	117.0
Concertina wire	41	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-145.5	0.0	-2797.7	210.0	1921.8	228.4	3.0	330.6
Concertina wire	42	Wire coil	Helix	80.0	80.0	88.4	0.0	-155.6	0.0	-3096.5	180.0	1404.2	109.4	80.0	113.6
Concertina wire	43	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-152.2	0.0	-3003.6	210.0	1580.7	188.5	3.0	354.6
Concertina wire	44	Wire coil	Helix	80.0	80.0	88.4	0.0	-162.3	0.0	-3239.7	180.0	1031.7	103.1	80.0	108.5
Concertina wire	45	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-159.0	0.0	-3168.1	210.0	1217.9	146.0	3.0	373.6
Concertina wire	46	Wire coil	Helix	80.0	80.0	88.4	0.0	-169.1	0.0	-3338.3	180.0	644.9	95.3	80.0	102.0
Concertina wire	47	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-165.7	0.0	-3289.0	210.0	838.3	101.5	3.0	387.5
Concertina wire	48	Wire coil	Helix	80.0	80.0	88.4	0.0	-175.8	0.0	-3390.8	180.0	249.3	86.3	80.0	94.0
Concertina wire	49	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-172.4	0.0	-3364.6	210.0	447.1	55.5	3.0	396.0
Concertina wire	50	Wire coil	Helix	80.0	80.0	88.4	0.0	177.5	0.0	-3396.7	180.0	-149.7	83.8	80.0	91.9
Concertina wire	51	Razor strand	Cylinder	3.0	3.0	399.1	0.0	-179.2	0.0	-3393.8	210.0	49.8	8.9	3.0	399.1
Concertina wire	52	Wire coil	Helix	80.0	80.0	88.4	0.0	170.7	0.0	-3355.8	180.0	-546.7	93.2	80.0	100.1
Concertina wire	53	Razor strand	Cylinder	3.0	3.0	399.1	0.0	174.1	0.0	-3376.2	210.0	-348.2	43.9	3.0	397.3
Concertina wire	54	Wire coil	Helix	80.0	80.0	88.4	0.0	164.0	0.0	-3268.6	180.0	-936.1	101.3	80.0	107.0
Concertina wire	55	Razor strand	Cylinder	3.0	3.0	399.1	0.0	167.4	0.0	-3312.2	210.0	-741.4	90.1	3.0	390.1
Concertina wire	56	Wire coil	Helix	80.0	80.0	88.4	0.0	157.3	0.0	-3136.4	180.0	-1312.6	107.9	80.0	112.5
Concertina wire	57	Razor strand	Cylinder	3.0	3.0	399.1	0.0	160.7	0.0	-3202.5	210.0	-1124.4	135.0	3.0	377.5
Concertina wire	58	Wire coil	Helix	80.0	80.0	88.4	0.0	150.6	0.0	-2961.0	180.0	-1671.1	113.1	80.0	116.3
Concertina wire	59	Razor strand	Cylinder	3.0	3.0	399.1	0.0	153.9	0.0	-3048.7	210.0	-1491.9	178.1	3.0	359.8
Concertina wire	60	Wire coil	Helix	80.0	80.0	88.4	0.0	143.8	0.0	-2744.8	180.0	-2006.5	116.8	80.0	118.6
Concertina wire	61	Razor strand	Cylinder	3.0	3.0	399.1	0.0	147.2	0.0	-2852.9	210.0	-1838.8	218.7	3.0	337.1
Concertina wire	62	Wire coil	Helix	80.0	80.0	88.4	0.0	137.1	0.0	-2490.8	180.0	-2314.3	118.8	80.0	119.2
Concertina wire	63	Razor strand	Cylinder	3.0	3.0	399.1	0.0	140.5	0.0	-2617.8	210.0	-2160.4	256.3	3.0	309.7
Concertina wire	64	Wire coil	Helix	80.0	80.0	88.4	0.0	130.4	0.0	-2202.4	180.0	-2590.2	119.2	80.0	118.2
Concertina wire	65	Razor strand	Cylinder	3.0	3.0	399.1	0.0	133.7	0.0	-2346.6	210.0	-2452.3	290.4	3.0	278.1
Concertina wire	66	Wire coil	Helix	80.0	80.0	88.4	0.0	123.6	0.0	-1883.7	180.0	-2830.5	117.9	80.0	115.6
Concertina wire	67	Razor strand	Cylinder	3.0	3.0	399.1	0.0	127.0	0.0	-2043.1	210.0	-2710.3	320.5	3.0	242.6
Concertina wire	68	Wire coil	Helix	80.0	80.0	88.4	0.0	116.9	0.0	-1539.1	320.0	-3031.7	115.1	80.0	111.4
Concertina wire	69	Razor strand	Cylinder	3.0	3.0	422.9	-19.3	120.3	0.0	-1711.4	280.0	-2931.1	347.0	142.8	204.3
Concertina wire	70	Wire coil	Helix	80.0	80.0	88.4	0.0	110.2	0.0	-1173.3	320.0	-3191.1	110.6	80.0	105.6
Concertina wire	71	Razor strand	Cylinder	3.0	3.0	399.1	0.0	113.6	0.0	-1356.2	350.0	-3111.4	367.0	3.0	162.2
Concertina wire	72	Wire coil	Helix	80.0	80.0	88.4	0.0	103.5	0.0	-791.3	180.0	-3306.6	104.6	80.0	98.4
Concertina wire	73	Razor strand	Cylinder	3.0	3.0	422.9	19.3	106.8	0.0	-982.3	280.0	-3248.9	383.8	142.8	118.7
Concertina wire	74	Wire coil	Helix	80.0	80.0	88.4	0.0	96.7	0.0	-398.4	180.0	-3376.6	97.2	80.0	89.8
Concertina wire	75	Razor strand	Cylinder	3.0	3.0	399.1	0.0	100.1	0.0	-594.8	210.0	-3341.6	393.4	3.0	72.9
Concertina wire	76	Wire coil	Helix	80.0	80.0	88.4	0.0	90.0	0.0	0.0	180.0	-3400.0	88.4	80.0	80.0
Concertina wire	77	Razor strand	Cylinder	3.0	3.0	399.1	0.0	93.4	0.0	-199.2	210.0	-3388.3	398.6	3.0	26.4

Concertina wire	78	Wire coil	Helix	80.0	80.0	88.4	0.0	83.3	0.0	398.4	180.0	-3376.6	97.2	80.0	89.8
Concertina wire	79	Razor strand	Cylinder	3.0	3.0	399.1	0.0	86.6	0.0	199.2	210.0	-3388.3	398.6	3.0	26.4
Concertina wire	80	Wire coil	Helix	80.0	80.0	88.4	0.0	76.5	0.0	791.3	180.0	-3306.6	104.6	80.0	98.4
Concertina wire	81	Razor strand	Cylinder	3.0	3.0	399.1	0.0	79.9	0.0	594.8	210.0	-3341.6	393.4	3.0	72.9
Concertina wire	82	Wire coil	Helix	80.0	80.0	88.4	0.0	69.8	0.0	1173.3	180.0	-3191.1	110.6	80.0	105.6
Concertina wire	83	Razor strand	Cylinder	3.0	3.0	399.1	0.0	73.2	0.0	982.3	210.0	-3248.9	382.9	3.0	118.4
Concertina wire	84	Wire coil	Helix	80.0	80.0	88.4	0.0	63.1	0.0	1539.1	180.0	-3031.7	115.1	80.0	111.4
Concertina wire	85	Razor strand	Cylinder	3.0	3.0	399.1	0.0	66.4	0.0	1356.2	210.0	-3111.4	367.0	3.0	162.2
Concertina wire	86	Wire coil	Helix	80.0	80.0	88.4	0.0	56.4	0.0	1883.7	180.0	-2830.5	117.9	80.0	115.6
Concertina wire	87	Razor strand	Cylinder	3.0	3.0	399.1	0.0	59.7	0.0	1711.4	210.0	-2931.1	346.1	3.0	203.8
Concertina wire	88	Wire coil	Helix	80.0	80.0	88.4	0.0	49.6	0.0	2202.4	180.0	-2590.2	119.2	80.0	118.2
Concertina wire	89	Razor strand	Cylinder	3.0	3.0	399.1	0.0	53.0	0.0	2043.1	210.0	-2710.3	320.5	3.0	242.6
Concertina wire	90	Wire coil	Helix	80.0	80.0	88.4	0.0	42.9	0.0	2490.8	180.0	-2314.3	118.8	80.0	119.2
Concertina wire	91	Razor strand	Cylinder	3.0	3.0	399.1	0.0	46.3	0.0	2346.6	210.0	-2452.3	290.4	3.0	278.1
Concertina wire	92	Wire coil	Helix	80.0	80.0	88.4	0.0	36.2	0.0	2744.8	180.0	-2006.5	116.8	80.0	118.6
Concertina wire	93	Razor strand	Cylinder	3.0	3.0	399.1	0.0	39.5	0.0	2617.8	210.0	-2160.4	256.3	3.0	309.7
Concertina wire	94	Wire coil	Helix	80.0	80.0	88.4	0.0	29.4	0.0	2961.0	180.0	-1671.1	113.1	80.0	116.3
Concertina wire	95	Razor strand	Cylinder	3.0	3.0	399.1	0.0	32.8	0.0	2852.9	210.0	-1838.8	218.7	3.0	337.1
Concertina wire	96	Wire coil	Helix	80.0	80.0	88.4	0.0	22.7	0.0	3136.4	320.0	-1312.6	107.9	80.0	112.5
Concertina wire	97	Razor strand	Cylinder	3.0	3.0	422.9	-19.3	26.1	0.0	3048.7	280.0	-1491.9	178.5	142.8	360.7
Concertina wire	98	Wire coil	Helix	80.0	80.0	88.4	0.0	16.0	0.0	3268.6	320.0	-936.1	101.3	80.0	107.0
Concertina wire	99	Razor strand	Cylinder	3.0	3.0	399.1	0.0	19.3	0.0	3202.5	350.0	-1124.4	135.0	3.0	377.5
Concertina wire	100	Wire coil	Helix	80.0	80.0	88.4	0.0	9.3	0.0	3355.8	180.0	-546.7	93.2	80.0	100.1
Concertina wire	101	Razor strand	Cylinder	3.0	3.0	422.9	19.3	12.6	0.0	3312.2	280.0	-741.4	90.3	142.8	391.1
Concertina wire	102	Wire coil	Helix	80.0	80.0	88.4	0.0	2.5	0.0	3396.7	180.0	-149.7	83.8	80.0	91.9
Concertina wire	103	Razor strand	Cylinder	3.0	3.0	399.1	0.0	5.9	0.0	3376.2	210.0	-348.2	43.9	3.0	397.3
Perimeter fence	2	Gate post	Cylinder	18.0	18.0	175.0	90.0	0.0	0.0	-297.2	59.1	1876.6	18.0	175.0	18.0
Perimeter fence	3	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	-862.6	50.4	1692.9	10.0	155.0	10.0
Perimeter fence	4	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	-1343.5	45.7	1343.5	10.0	155.0	10.0
Perimeter fence	5	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	-1692.9	46.0	862.6	10.0	155.0	10.0
Perimeter fence	6	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	-1876.6	50.5	297.2	10.0	155.0	10.0
Perimeter fence	7	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	-1876.6	50.5	-297.2	10.0	155.0	10.0
Perimeter fence	8	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	-1692.9	41.5	-862.6	10.0	155.0	10.0
Perimeter fence	9	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	-1343.5	45.0	-1343.5	10.0	155.0	10.0
Perimeter fence	10	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	-862.6	50.0	-1692.9	10.0	155.0	10.0
Perimeter fence	11	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	-297.2	45.4	-1876.6	10.0	155.0	10.0
Perimeter fence	12	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	297.2	46.7	-1876.6	10.0	155.0	10.0
Perimeter fence	13	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	862.6	47.1	-1692.9	10.0	155.0	10.0
Perimeter fence	14	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	1343.5	50.5	-1343.5	10.0	155.0	10.0
Perimeter fence	15	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	1692.9	28.0	-862.6	10.0	155.0	10.0
Perimeter fence	16	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	1876.6	47.5	-297.2	10.0	155.0	10.0
Perimeter fence	17	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	1876.6	41.5	297.2	10.0	155.0	10.0
Perimeter fence	18	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	1692.9	44.6	862.6	10.0	155.0	10.0
Perimeter fence	19	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	1343.5	50.6	1343.5	10.0	155.0	10.0
Perimeter fence	20	Fence post	Cylinder	10.0	10.0	155.0	90.0	0.0	0.0	862.6	42.5	1692.9	10.0	155.0	10.0
Perimeter fence	21	Gate post	Cylinder	18.0	18.0	175.0	90.0	0.0	0.0	297.2	51.3	1876.6	18.0	175.0	18.0
Perimeter fence	22	Fence wire	Cylinder	1.2	1.2	594.5	-0.1	-108.0	0.0	-579.9	37.3	1784.8	565.7	2.5	184.8
Perimeter fence	23	Fence wire	Cylinder	1.2	1.2	594.5	-0.1	-108.0	0.0	-579.9	77.3	1784.8	565.7	2.5	184.8
Perimeter fence	24	Fence wire	Cylinder	1.2	1.2	594.5	-0.1	-108.0	0.0	-579.9	117.3	1784.8	565.7	2.5	184.8
Perimeter fence	25	Fence wire	Cylinder	1.2	1.2	594.5	0.5	-126.0	0.0	-1103.0	35.6	1518.2	481.6	6.0	350.4
Perimeter fence	26	Fence wire	Cylinder	1.2	1.2	594.5	0.5	-126.0	0.0	-1103.0	75.6	1518.2	481.6	6.0	350.4

Perimeter fence	27	Fence wire	Cylinder	1.2	1.2	594.5	0.5	-126.0	0.0	-1103.0	115.6	1518.2	481.6	6.0	350.4
Perimeter fence	28	Fence wire	Cylinder	1.2	1.2	594.5	0.0	-144.0	0.0	-1518.2	33.3	1103.0	350.4	1.5	481.6
Perimeter fence	29	Fence wire	Cylinder	1.2	1.2	594.5	0.0	-144.0	0.0	-1518.2	73.3	1103.0	350.4	1.5	481.6
Perimeter fence	30	Fence wire	Cylinder	1.2	1.2	594.5	0.0	-144.0	0.0	-1518.2	113.3	1103.0	350.4	1.5	481.6
Perimeter fence	31	Fence wire	Cylinder	1.2	1.2	594.5	-0.4	-162.0	0.0	-1784.8	35.7	579.9	184.8	5.7	565.7
Perimeter fence	32	Fence wire	Cylinder	1.2	1.2	594.5	-0.4	-162.0	0.0	-1784.8	75.7	579.9	184.8	5.7	565.7
Perimeter fence	33	Fence wire	Cylinder	1.2	1.2	594.5	-0.4	-162.0	0.0	-1784.8	115.7	579.9	184.8	5.7	565.7
Perimeter fence	34	Fence wire	Cylinder	1.2	1.2	594.5	0.0	180.0	0.0	-1876.6	38.0	0.0	1.2	1.2	594.5
Perimeter fence	35	Fence wire	Cylinder	1.2	1.2	594.5	0.0	180.0	0.0	-1876.6	78.0	0.0	1.2	1.2	594.5
Perimeter fence	36	Fence wire	Cylinder	1.2	1.2	594.5	0.0	180.0	0.0	-1876.6	118.0	0.0	1.2	1.2	594.5
Perimeter fence	37	Fence wire	Cylinder	1.2	1.2	594.5	0.9	162.0	0.0	-1784.8	33.5	-579.9	184.8	10.1	565.7
Perimeter fence	38	Fence wire	Cylinder	1.2	1.2	594.5	0.9	162.0	0.0	-1784.8	73.5	-579.9	184.8	10.1	565.7
Perimeter fence	39	Fence wire	Cylinder	1.2	1.2	594.5	0.9	162.0	0.0	-1784.8	113.5	-579.9	184.8	10.1	565.7
Perimeter fence	40	Fence wire	Cylinder	1.2	1.2	594.5	-0.3	144.0	0.0	-1518.2	30.8	-1103.0	350.4	4.7	481.6
Perimeter fence	41	Fence wire	Cylinder	1.2	1.2	594.5	-0.3	144.0	0.0	-1518.2	70.8	-1103.0	350.4	4.7	481.6
Perimeter fence	42	Fence wire	Cylinder	1.2	1.2	594.5	-0.3	144.0	0.0	-1518.2	110.8	-1103.0	350.4	4.7	481.6
Perimeter fence	43	Fence wire	Cylinder	1.2	1.2	594.5	-0.5	126.0	0.0	-1103.0	35.0	-1518.2	481.6	6.2	350.4
Perimeter fence	44	Fence wire	Cylinder	1.2	1.2	594.5	-0.5	126.0	0.0	-1103.0	75.0	-1518.2	481.6	6.2	350.4
Perimeter fence	45	Fence wire	Cylinder	1.2	1.2	594.5	-0.5	126.0	0.0	-1103.0	115.0	-1518.2	481.6	6.2	350.4
Perimeter fence	46	Fence wire	Cylinder	1.2	1.2	594.5	0.4	108.0	0.0	-579.9	35.2	-1784.8	565.7	5.8	184.8
Perimeter fence	47	Fence wire	Cylinder	1.2	1.2	594.5	0.4	108.0	0.0	-579.9	75.2	-1784.8	565.7	5.8	184.8
Perimeter fence	48	Fence wire	Cylinder	1.2	1.2	594.5	0.4	108.0	0.0	-579.9	115.2	-1784.8	565.7	5.8	184.8
Perimeter fence	49	Fence wire	Cylinder	1.2	1.2	594.5	-0.1	90.0	0.0	0.0	33.5	-1876.6	594.5	2.5	1.2
Perimeter fence	50	Fence wire	Cylinder	1.2	1.2	594.5	-0.1	90.0	0.0	0.0	73.5	-1876.6	594.5	2.5	1.2
Perimeter fence	51	Fence wire	Cylinder	1.2	1.2	594.5	-0.1	90.0	0.0	0.0	113.5	-1876.6	594.5	2.5	1.2
Perimeter fence	52	Fence wire	Cylinder	1.2	1.2	594.5	0.0	72.0	0.0	579.9	34.4	-1784.8	565.7	1.6	184.8
Perimeter fence	53	Fence wire	Cylinder	1.2	1.2	594.5	0.0	72.0	0.0	579.9	74.4	-1784.8	565.7	1.6	184.8
Perimeter fence	54	Fence wire	Cylinder	1.2	1.2	594.5	0.0	72.0	0.0	579.9	114.4	-1784.8	565.7	1.6	184.8
Perimeter fence	55	Fence wire	Cylinder	1.2	1.2	594.5	-0.3	54.0	0.0	1103.0	36.3	-1518.2	481.6	4.6	350.4
Perimeter fence	56	Fence wire	Cylinder	1.2	1.2	594.5	-0.3	54.0	0.0	1103.0	76.3	-1518.2	481.6	4.6	350.4
Perimeter fence	57	Fence wire	Cylinder	1.2	1.2	594.5	-0.3	54.0	0.0	1103.0	116.3	-1518.2	481.6	4.6	350.4
Perimeter fence	58	Fence wire	Cylinder	1.2	1.2	594.9	2.2	36.0	0.0	1518.2	26.7	-1103.0	350.4	23.8	481.7
Perimeter fence	59	Fence wire	Cylinder	1.2	1.2	594.9	2.2	36.0	0.0	1518.2	66.7	-1103.0	350.4	23.8	481.7
Perimeter fence	60	Fence wire	Cylinder	1.2	1.2	594.9	2.2	36.0	0.0	1518.2	106.7	-1103.0	350.4	23.8	481.7
Perimeter fence	61	Fence wire	Cylinder	1.2	1.2	594.8	-1.9	18.0	0.0	1784.8	25.2	-579.9	184.8	20.7	565.8
Perimeter fence	62	Fence wire	Cylinder	1.2	1.2	594.8	-1.9	18.0	0.0	1784.8	65.2	-579.9	184.8	20.7	565.8
Perimeter fence	63	Fence wire	Cylinder	1.2	1.2	594.8	-1.9	18.0	0.0	1784.8	105.2	-579.9	184.8	20.7	565.8
Perimeter fence	64	Fence wire	Cylinder	1.2	1.2	594.5	0.6	0.0	0.0	1876.6	32.0	0.0	1.2	7.2	594.5
Perimeter fence	65	Fence wire	Cylinder	1.2	1.2	594.5	0.6	0.0	0.0	1876.6	72.0	0.0	1.2	7.2	594.5
Perimeter fence	66	Fence wire	Cylinder	1.2	1.2	594.5	0.6	0.0	0.0	1876.6	112.0	0.0	1.2	7.2	594.5
Perimeter fence	67	Fence wire	Cylinder	1.2	1.2	594.5	-0.3	-18.0	0.0	1784.8	30.5	579.9	184.8	4.3	565.7
Perimeter fence	68	Fence wire	Cylinder	1.2	1.2	594.5	-0.3	-18.0	0.0	1784.8	70.5	579.9	184.8	4.3	565.7
Perimeter fence	69	Fence wire	Cylinder	1.2	1.2	594.5	-0.3	-18.0	0.0	1784.8	110.5	579.9	184.8	4.3	565.7
Perimeter fence	70	Fence wire	Cylinder	1.2	1.2	594.5	-0.6	-36.0	0.0	1518.2	35.1	1103.0	350.4	7.3	481.6
Perimeter fence	71	Fence wire	Cylinder	1.2	1.2	594.5	-0.6	-36.0	0.0	1518.2	75.1	1103.0	350.4	7.3	481.6
Perimeter fence	72	Fence wire	Cylinder	1.2	1.2	594.5	-0.6	-36.0	0.0	1518.2	115.1	1103.0	350.4	7.3	481.6
Perimeter fence	73	Fence wire	Cylinder	1.2	1.2	594.5	0.8	-54.0	0.0	1103.0	34.1	1518.2	481.6	9.3	350.4
Perimeter fence	74	Fence wire	Cylinder	1.2	1.2	594.5	0.8	-54.0	0.0	1103.0	74.1	1518.2	481.6	9.3	350.4
Perimeter fence	75	Fence wire	Cylinder	1.2	1.2	594.5	0.8	-54.0	0.0	1103.0	114.1	1518.2	481.6	9.3	350.4
Perimeter fence	76	Fence wire	Cylinder	1.2	1.2	594.5	0.1	-72.0	0.0	579.9	29.4	1784.8	565.7	2.4	184.8
Perimeter fence	77	Fence wire	Cylinder	1.2	1.2	594.5	0.1	-72.0	0.0	579.9	69.4	1784.8	565.7	2.4	184.8

Perimeter fence	78	Fence wire	Cylinder	1.2	1.2	594.5	0.1	-72.0	0.0	579.9	109.4	1784.8	565.7	2.4	184.8
Perimeter fence	79	Gate stile	Cylinder	6.0	6.0	120.0	90.0	0.0	0.0	200.7	73.8	1611.4	7.7	120.0	7.7
Perimeter fence	80	Gate rail top	Cylinder	5.0	5.0	272.3	0.0	0.0	0.0	247.2	128.8	1739.3	97.8	5.0	257.5
Perimeter fence	81	Gate rail bottom	Cylinder	5.0	5.0	272.3	0.0	0.0	0.0	247.2	18.8	1739.3	97.8	5.0	257.5
Perimeter fence	82	Gate brace	Cylinder	4.0	4.0	293.6	-22.0	0.0	0.0	247.2	73.8	1739.3	97.4	113.7	258.6
Perimeter fence	83	Gate stile	Cylinder	6.0	6.0	120.0	90.0	0.0	0.0	-24.6	81.6	1803.6	7.3	120.0	7.3
Perimeter fence	84	Gate rail top	Cylinder	5.0	5.0	272.3	0.0	0.0	0.0	-156.1	136.6	1838.8	264.3	5.0	75.3
Perimeter fence	85	Gate rail bottom	Cylinder	5.0	5.0	272.3	0.0	0.0	0.0	-156.1	26.6	1838.8	264.3	5.0	75.3
Perimeter fence	86	Gate brace	Cylinder	4.0	4.0	293.6	-22.0	0.0	0.0	-156.1	81.6	1838.8	265.5	113.7	74.7
Perimeter fence	87	Warning sign	Cube	55.0	38.0	3.0	0.0	0.0	0.0	297.2	98.8	1888.6	55.0	38.0	3.0
Outer gate	2	Gate post	Cube	70.0	340.0	70.0	0.0	0.0	0.0	-535.0	170.0	20.0	70.0	340.0	70.0
Outer gate	3	Gate post cap	Pyramid	85.0	30.0	85.0	0.0	0.0	0.0	-535.0	352.5	20.0	85.0	30.0	85.0
Outer gate	4	Gate post	Cube	70.0	340.0	70.0	0.0	0.0	0.0	535.0	170.0	20.0	70.0	340.0	70.0
Outer gate	5	Gate post cap	Pyramid	85.0	30.0	85.0	0.0	0.0	0.0	535.0	352.5	20.0	85.0	30.0	85.0
Outer gate	6	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	415.8	9.0	-390.0	50.0	17.0	26.0
Outer gate	7	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	467.5	9.0	-390.0	50.0	17.0	26.0
Outer gate	8	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	519.2	9.0	-390.0	50.0	17.0	26.0
Outer gate	9	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	570.8	9.0	-390.0	50.0	17.0	26.0
Outer gate	10	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	622.5	9.0	-390.0	50.0	17.0	26.0
Outer gate	11	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	674.2	9.0	-390.0	50.0	17.0	26.0
Outer gate	12	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	441.7	24.0	-390.0	50.0	17.0	26.0
Outer gate	13	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	493.3	24.0	-390.0	50.0	17.0	26.0
Outer gate	14	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	545.0	24.0	-390.0	50.0	17.0	26.0
Outer gate	15	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	596.7	24.0	-390.0	50.0	17.0	26.0
Outer gate	16	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	648.3	24.0	-390.0	50.0	17.0	26.0
Outer gate	17	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	415.8	39.0	-390.0	50.0	17.0	26.0
Outer gate	18	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	467.5	39.0	-390.0	50.0	17.0	26.0
Outer gate	19	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	519.2	39.0	-390.0	50.0	17.0	26.0
Outer gate	20	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	570.8	39.0	-390.0	50.0	17.0	26.0
Outer gate	21	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	622.5	39.0	-390.0	50.0	17.0	26.0
Outer gate	22	Sandbag	Capsule	26.0	17.0	50.0	0.0	90.0	0.0	674.2	39.0	-390.0	50.0	17.0	26.0
Outer gate	23	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	9.0	-365.8	26.0	17.0	50.0
Outer gate	24	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	9.0	-317.5	26.0	17.0	50.0
Outer gate	25	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	9.0	-269.2	26.0	17.0	50.0
Outer gate	26	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	9.0	-220.8	26.0	17.0	50.0
Outer gate	27	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	9.0	-172.5	26.0	17.0	50.0
Outer gate	28	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	9.0	-124.2	26.0	17.0	50.0
Outer gate	29	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	24.0	-341.7	26.0	17.0	50.0
Outer gate	30	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	24.0	-293.3	26.0	17.0	50.0
Outer gate	31	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	24.0	-245.0	26.0	17.0	50.0
Outer gate	32	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	24.0	-196.7	26.0	17.0	50.0
Outer gate	33	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	24.0	-148.3	26.0	17.0	50.0
Outer gate	34	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	39.0	-365.8	26.0	17.0	50.0
Outer gate	35	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	39.0	-317.5	26.0	17.0	50.0
Outer gate	36	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	39.0	-269.2	26.0	17.0	50.0
Outer gate	37	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	39.0	-220.8	26.0	17.0	50.0
Outer gate	38	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	39.0	-172.5	26.0	17.0	50.0
Outer gate	39	Sandbag	Capsule	26.0	17.0	50.0	0.0	0.0	0.0	700.0	39.0	-124.2	26.0	17.0	50.0
Outer gate	40	Sign post	Cylinder	7.0	7.0	200.0	90.0	0.0	0.0	-360.0	100.0	-320.0	7.0	200.0	7.0
Outer gate	41	STOP sign	Cube	90.0	90.0	4.0	0.0	0.0	0.0	-360.0	175.0	-315.0	90.0	90.0	4.0
Outer gate	42	Sign post	Cylinder	7.0	7.0	200.0	90.0	0.0	0.0	370.0	100.0	-700.0	7.0	200.0	7.0

Outer gate	43	CHECKPOINT sign	Cube	120.0	80.0	4.0	0.0	0.0	0.0	370.0	175.0	-695.0	120.0	80.0	4.0
Outer gate	44	Sign post	Cylinder	7.0	7.0	200.0	90.0	0.0	0.0	-360.0	100.0	-990.0	7.0	200.0	7.0
Outer gate	45	SLOW hazard board	Cube	80.0	80.0	4.0	0.0	0.0	0.0	-360.0	175.0	-985.0	80.0	80.0	4.0
Outer gate	46	Gate lamp pole	Cylinder	12.0	12.0	540.0	90.0	0.0	0.0	-390.0	270.0	-240.0	12.0	540.0	12.0
Outer gate	47	Gate lamp arm	Cube	140.0	7.0	7.0	0.0	0.0	0.0	-320.0	535.0	-240.0	140.0	7.0	7.0
Outer gate	48	Gate lamp hood	Cube	42.0	4.0	34.0	0.0	0.0	0.0	-265.0	536.0	-240.0	42.0	4.0	34.0
Outer gate	49	Gate lamp	Sphere	20.0	20.0	8.0	90.0	0.0	0.0	-265.0	522.0	-240.0	20.0	8.0	20.0
Outer gate	50	Gate post marker	Sphere	10.0	10.0	5.0	0.0	0.0	0.0	-535.0	120.0	56.0	10.0	10.0	5.0
Outer gate	51	Gate post marker	Sphere	10.0	10.0	5.0	0.0	0.0	0.0	535.0	120.0	56.0	10.0	10.0	5.0
Camouflage net	2	Camo net panel	Cube	113.7	3.5	104.0	2.9	-1.1	-2.6	-187.5	387.0	-342.9	115.7	13.9	105.9
Camouflage net	3	Camo net panel	Cube	107.5	3.5	98.3	2.4	-0.1	-7.2	-187.5	381.6	-228.6	107.2	21.0	98.8
Camouflage net	4	Camo net panel	Cube	115.4	3.5	105.5	1.3	-2.7	-10.4	-187.5	377.9	-114.3	119.1	26.6	110.4
Camouflage net	5	Camo net panel	Cube	120.2	3.5	109.9	0.0	2.2	-11.5	-187.5	376.6	0.0	122.6	27.4	114.3
Camouflage net	6	Camo net panel	Cube	120.4	3.5	110.1	-1.3	3.5	-10.4	-187.5	377.9	114.3	125.6	27.6	116.8
Camouflage net	7	Camo net panel	Cube	111.6	3.5	102.0	-2.4	4.5	-7.2	-187.5	381.6	228.6	118.7	21.7	109.8
Camouflage net	8	Camo net panel	Cube	116.9	3.5	106.9	-2.9	5.7	-2.6	-187.5	387.0	342.9	127.0	14.2	117.7
Camouflage net	9	Camo net panel	Cube	118.9	3.5	108.7	7.1	0.3	-1.1	-62.5	382.8	-342.9	119.5	19.0	109.2
Camouflage net	10	Camo net panel	Cube	119.2	3.5	109.0	5.7	-5.2	-3.0	-62.5	369.8	-228.6	128.6	20.5	118.6
Camouflage net	11	Camo net panel	Cube	119.3	3.5	109.1	3.2	1.8	-4.3	-62.5	360.9	-114.3	122.6	18.5	113.3
Camouflage net	12	Camo net panel	Cube	109.6	3.5	100.2	-3.2	-0.2	-4.3	-62.5	360.9	114.3	109.9	17.3	101.1
Camouflage net	13	Camo net panel	Cube	112.6	3.5	103.0	-5.7	3.9	-3.0	-62.5	369.8	228.6	119.3	19.5	109.6
Camouflage net	14	Camo net panel	Cube	113.5	3.5	103.8	-7.1	-4.6	-1.1	-62.5	382.8	342.9	121.4	18.3	112.4
Camouflage net	15	Camo net panel	Cube	103.7	3.5	94.8	7.1	-5.2	1.1	62.5	382.8	-342.9	111.8	17.0	103.7
Camouflage net	16	Camo net panel	Cube	118.1	3.5	108.0	5.7	-1.2	3.0	62.5	369.8	-228.6	120.4	20.3	110.9
Camouflage net	17	Camo net panel	Cube	108.2	3.5	99.0	3.2	-1.9	4.3	62.5	360.9	-114.3	111.4	17.1	103.0
Camouflage net	18	Camo net panel	Cube	120.3	3.5	110.0	0.0	-2.1	4.8	62.5	357.7	0.0	124.1	13.6	114.3
Camouflage net	19	Camo net panel	Cube	107.1	3.5	97.9	-3.2	-3.5	4.3	62.5	360.9	114.3	112.8	17.0	103.8
Camouflage net	20	Camo net panel	Cube	110.3	3.5	100.8	-5.7	2.8	3.0	62.5	369.8	228.6	115.1	19.2	106.5
Camouflage net	21	Camo net panel	Cube	111.3	3.5	101.7	-7.1	4.0	1.1	62.5	382.8	342.9	118.0	18.0	109.1
Camouflage net	22	Camo net panel	Cube	102.8	3.5	94.0	2.9	3.4	2.6	187.5	387.0	-342.9	108.3	12.9	99.8
Camouflage net	23	Camo net panel	Cube	120.9	3.5	110.6	2.4	5.7	7.2	187.5	381.6	-228.6	130.8	23.2	121.4
Camouflage net	24	Camo net panel	Cube	115.4	3.5	105.5	1.3	4.9	10.4	187.5	377.9	-114.3	122.7	26.6	114.4
Camouflage net	25	Camo net panel	Cube	116.5	3.5	106.5	0.0	-3.9	11.5	187.5	376.6	0.0	121.9	26.6	114.1
Camouflage net	26	Camo net panel	Cube	104.6	3.5	95.7	-1.3	-3.6	10.4	187.5	377.9	114.3	109.4	24.5	101.6
Camouflage net	27	Camo net panel	Cube	112.8	3.5	103.1	-2.4	4.6	7.2	187.5	381.6	228.6	120.1	21.9	112.3
Camouflage net	28	Camo net panel	Cube	116.4	3.5	106.4	-2.9	4.9	2.6	187.5	387.0	342.9	125.0	14.2	116.1
Camouflage net	29	Net pole	Cylinder	10.0	10.0	400.0	90.0	0.0	0.0	-250.0	200.0	-400.0	10.0	400.0	10.0
Camouflage net	30	Net pole	Cylinder	10.0	10.0	400.0	90.0	0.0	0.0	-250.0	200.0	0.0	10.0	400.0	10.0
Camouflage net	31	Net pole	Cylinder	10.0	10.0	400.0	90.0	0.0	0.0	-250.0	200.0	400.0	10.0	400.0	10.0
Camouflage net	32	Net pole	Cylinder	10.0	10.0	400.0	90.0	0.0	0.0	250.0	200.0	-400.0	10.0	400.0	10.0
Camouflage net	33	Net pole	Cylinder	10.0	10.0	400.0	90.0	0.0	0.0	250.0	200.0	0.0	10.0	400.0	10.0
Camouflage net	34	Net pole	Cylinder	10.0	10.0	400.0	90.0	0.0	0.0	250.0	200.0	400.0	10.0	400.0	10.0
Camouflage net	35	Net edge rope	Cylinder	2.0	2.0	500.0	0.0	90.0	0.0	0.0	395.0	-400.0	500.0	2.0	2.0
Camouflage net	36	Net edge rope	Cylinder	2.0	2.0	800.0	0.0	0.0	0.0	250.0	395.0	0.0	2.0	2.0	800.0
Camouflage net	37	Net edge rope	Cylinder	2.0	2.0	500.0	0.0	-90.0	0.0	0.0	395.0	400.0	500.0	2.0	2.0
Camouflage net	38	Net edge rope	Cylinder	2.0	2.0	800.0	0.0	180.0	0.0	-250.0	395.0	0.0	2.0	2.0	800.0
Earth revetment	2	Revetment core	Cube	80.0	110.5	920.0	0.0	0.0	0.0	-265.0	55.3	0.0	80.0	110.5	920.0
Earth revetment	3	Revetment slope (outer)	Cube	55.0	136.5	920.0	0.0	0.0	-28.0	-327.0	58.5	0.0	112.6	146.3	920.0
Earth revetment	4	Revetment slope (inner)	Cube	35.0	123.5	920.0	0.0	0.0	10.0	-218.0	61.1	0.0	55.9	127.7	920.0
Earth revetment	5	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	-431.2	50.0	17.0	28.0
Earth revetment	6	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	-373.7	50.0	17.0	28.0

Earth revetment	7	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	-316.2	50.0	17.0	28.0
Earth revetment	8	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	-258.7	50.0	17.0	28.0
Earth revetment	9	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	-201.3	50.0	17.0	28.0
Earth revetment	10	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	-143.8	50.0	17.0	28.0
Earth revetment	11	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	-86.3	50.0	17.0	28.0
Earth revetment	12	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	-28.8	50.0	17.0	28.0
Earth revetment	13	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	28.8	50.0	17.0	28.0
Earth revetment	14	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	86.3	50.0	17.0	28.0
Earth revetment	15	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	143.8	50.0	17.0	28.0
Earth revetment	16	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	201.2	50.0	17.0	28.0
Earth revetment	17	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	258.7	50.0	17.0	28.0
Earth revetment	18	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	316.2	50.0	17.0	28.0
Earth revetment	19	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	373.7	50.0	17.0	28.0
Earth revetment	20	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	-265.0	117.5	431.2	50.0	17.0	28.0
Earth revetment	21	Revetment core	Cube	80.0	110.5	920.0	0.0	0.0	0.0	265.0	55.3	0.0	80.0	110.5	920.0
Earth revetment	22	Revetment slope (outer)	Cube	55.0	136.5	920.0	0.0	0.0	28.0	327.0	58.5	0.0	112.6	146.3	920.0
Earth revetment	23	Revetment slope (inner)	Cube	35.0	123.5	920.0	0.0	0.0	-10.0	218.0	61.1	0.0	55.9	127.7	920.0
Earth revetment	24	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	-431.2	50.0	17.0	28.0
Earth revetment	25	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	-373.7	50.0	17.0	28.0
Earth revetment	26	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	-316.2	50.0	17.0	28.0
Earth revetment	27	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	-258.7	50.0	17.0	28.0
Earth revetment	28	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	-201.3	50.0	17.0	28.0
Earth revetment	29	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	-143.8	50.0	17.0	28.0
Earth revetment	30	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	-86.3	50.0	17.0	28.0
Earth revetment	31	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	-28.8	50.0	17.0	28.0
Earth revetment	32	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	28.7	50.0	17.0	28.0
Earth revetment	33	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	86.3	50.0	17.0	28.0
Earth revetment	34	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	143.8	50.0	17.0	28.0
Earth revetment	35	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	201.3	50.0	17.0	28.0
Earth revetment	36	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	258.7	50.0	17.0	28.0
Earth revetment	37	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	316.2	50.0	17.0	28.0
Earth revetment	38	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	373.7	50.0	17.0	28.0
Earth revetment	39	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	90.0	0.0	265.0	117.5	431.2	50.0	17.0	28.0
Earth revetment	40	Revetment core	Cube	650.0	110.5	80.0	0.0	0.0	0.0	0.0	55.3	445.0	650.0	110.5	80.0
Earth revetment	41	Revetment slope (outer)	Cube	650.0	136.5	55.0	-28.0	0.0	0.0	0.0	58.5	507.0	650.0	146.3	112.6
Earth revetment	42	Revetment slope (inner)	Cube	550.0	123.5	35.0	10.0	0.0	0.0	0.0	61.1	398.0	550.0	127.7	55.9
Earth revetment	43	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	0.0	0.0	-295.5	117.5	445.0	28.0	17.0	50.0
Earth revetment	44	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	0.0	0.0	-236.4	117.5	445.0	28.0	17.0	50.0
Earth revetment	45	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	0.0	0.0	-177.3	117.5	445.0	28.0	17.0	50.0
Earth revetment	46	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	0.0	0.0	-118.2	117.5	445.0	28.0	17.0	50.0
Earth revetment	47	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	0.0	0.0	-59.1	117.5	445.0	28.0	17.0	50.0
Earth revetment	48	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	0.0	0.0	0.0	117.5	445.0	28.0	17.0	50.0
Earth revetment	49	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	0.0	0.0	59.1	117.5	445.0	28.0	17.0	50.0
Earth revetment	50	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	0.0	0.0	118.2	117.5	445.0	28.0	17.0	50.0
Earth revetment	51	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	0.0	0.0	177.3	117.5	445.0	28.0	17.0	50.0
Earth revetment	52	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	0.0	0.0	236.4	117.5	445.0	28.0	17.0	50.0
Earth revetment	53	Revetment sandbag	Capsule	28.0	17.0	50.0	0.0	0.0	0.0	295.5	117.5	445.0	28.0	17.0	50.0
Radar surround	2	Radar pad	Cube	540.0	4.0	540.0	0.0	0.0	0.0	0.0	2.0	0.0	540.0	4.0	540.0
Radar surround	3	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	330.0	100.0	330.0	8.0	200.0	8.0
Radar surround	4	Fence wire	Cylinder	1.2	1.2	220.0	0.0	180.0	0.0	330.0	40.0	220.0	1.2	1.2	220.0
Radar surround	5	Fence wire	Cylinder	1.2	1.2	220.0	0.0	180.0	0.0	330.0	100.0	220.0	1.2	1.2	220.0

Radar surround	6	Fence wire	Cylinder	1.2	1.2	220.0	0.0	180.0	0.0	330.0	170.0	220.0	1.2	1.2	220.0
Radar surround	7	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	330.0	100.0	110.0	8.0	200.0	8.0
Radar surround	8	Fence wire	Cylinder	1.2	1.2	220.0	0.0	180.0	0.0	330.0	40.0	0.0	1.2	1.2	220.0
Radar surround	9	Fence wire	Cylinder	1.2	1.2	220.0	0.0	180.0	0.0	330.0	100.0	0.0	1.2	1.2	220.0
Radar surround	10	Fence wire	Cylinder	1.2	1.2	220.0	0.0	180.0	0.0	330.0	170.0	0.0	1.2	1.2	220.0
Radar surround	11	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	330.0	100.0	-110.0	8.0	200.0	8.0
Radar surround	12	Fence wire	Cylinder	1.2	1.2	220.0	0.0	180.0	0.0	330.0	40.0	-220.0	1.2	1.2	220.0
Radar surround	13	Fence wire	Cylinder	1.2	1.2	220.0	0.0	180.0	0.0	330.0	100.0	-220.0	1.2	1.2	220.0
Radar surround	14	Fence wire	Cylinder	1.2	1.2	220.0	0.0	180.0	0.0	330.0	170.0	-220.0	1.2	1.2	220.0
Radar surround	15	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	330.0	100.0	-330.0	8.0	200.0	8.0
Radar surround	16	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	330.0	100.0	-330.0	8.0	200.0	8.0
Radar surround	17	Fence wire	Cylinder	1.2	1.2	220.0	0.0	-90.0	0.0	220.0	40.0	-330.0	220.0	1.2	1.2
Radar surround	18	Fence wire	Cylinder	1.2	1.2	220.0	0.0	-90.0	0.0	220.0	100.0	-330.0	220.0	1.2	1.2
Radar surround	19	Fence wire	Cylinder	1.2	1.2	220.0	0.0	-90.0	0.0	220.0	170.0	-330.0	220.0	1.2	1.2
Radar surround	20	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	110.0	100.0	-330.0	8.0	200.0	8.0
Radar surround	21	Fence wire	Cylinder	1.2	1.2	220.0	0.0	-90.0	0.0	0.0	40.0	-330.0	220.0	1.2	1.2
Radar surround	22	Fence wire	Cylinder	1.2	1.2	220.0	0.0	-90.0	0.0	0.0	100.0	-330.0	220.0	1.2	1.2
Radar surround	23	Fence wire	Cylinder	1.2	1.2	220.0	0.0	-90.0	0.0	0.0	170.0	-330.0	220.0	1.2	1.2
Radar surround	24	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	-110.0	100.0	-330.0	8.0	200.0	8.0
Radar surround	25	Fence wire	Cylinder	1.2	1.2	220.0	0.0	-90.0	0.0	-220.0	40.0	-330.0	220.0	1.2	1.2
Radar surround	26	Fence wire	Cylinder	1.2	1.2	220.0	0.0	-90.0	0.0	-220.0	100.0	-330.0	220.0	1.2	1.2
Radar surround	27	Fence wire	Cylinder	1.2	1.2	220.0	0.0	-90.0	0.0	-220.0	170.0	-330.0	220.0	1.2	1.2
Radar surround	28	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	-330.0	100.0	-330.0	8.0	200.0	8.0
Radar surround	29	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	-330.0	100.0	-330.0	8.0	200.0	8.0
Radar surround	30	Fence wire	Cylinder	1.2	1.2	220.0	0.0	0.0	0.0	-330.0	40.0	-220.0	1.2	1.2	220.0
Radar surround	31	Fence wire	Cylinder	1.2	1.2	220.0	0.0	0.0	0.0	-330.0	100.0	-220.0	1.2	1.2	220.0
Radar surround	32	Fence wire	Cylinder	1.2	1.2	220.0	0.0	0.0	0.0	-330.0	170.0	-220.0	1.2	1.2	220.0
Radar surround	33	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	-330.0	100.0	-110.0	8.0	200.0	8.0
Radar surround	34	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	-330.0	100.0	110.0	8.0	200.0	8.0
Radar surround	35	Fence wire	Cylinder	1.2	1.2	220.0	0.0	0.0	0.0	-330.0	40.0	220.0	1.2	1.2	220.0
Radar surround	36	Fence wire	Cylinder	1.2	1.2	220.0	0.0	0.0	0.0	-330.0	100.0	220.0	1.2	1.2	220.0
Radar surround	37	Fence wire	Cylinder	1.2	1.2	220.0	0.0	0.0	0.0	-330.0	170.0	220.0	1.2	1.2	220.0
Radar surround	38	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	-330.0	100.0	330.0	8.0	200.0	8.0
Radar surround	39	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	-330.0	100.0	330.0	8.0	200.0	8.0
Radar surround	40	Fence wire	Cylinder	1.2	1.2	220.0	0.0	90.0	0.0	-220.0	40.0	330.0	220.0	1.2	1.2
Radar surround	41	Fence wire	Cylinder	1.2	1.2	220.0	0.0	90.0	0.0	-220.0	100.0	330.0	220.0	1.2	1.2
Radar surround	42	Fence wire	Cylinder	1.2	1.2	220.0	0.0	90.0	0.0	-220.0	170.0	330.0	220.0	1.2	1.2
Radar surround	43	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	-110.0	100.0	330.0	8.0	200.0	8.0
Radar surround	44	Fence wire	Cylinder	1.2	1.2	220.0	0.0	90.0	0.0	0.0	40.0	330.0	220.0	1.2	1.2
Radar surround	45	Fence wire	Cylinder	1.2	1.2	220.0	0.0	90.0	0.0	0.0	100.0	330.0	220.0	1.2	1.2
Radar surround	46	Fence wire	Cylinder	1.2	1.2	220.0	0.0	90.0	0.0	0.0	170.0	330.0	220.0	1.2	1.2
Radar surround	47	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	110.0	100.0	330.0	8.0	200.0	8.0
Radar surround	48	Fence wire	Cylinder	1.2	1.2	220.0	0.0	90.0	0.0	220.0	40.0	330.0	220.0	1.2	1.2
Radar surround	49	Fence wire	Cylinder	1.2	1.2	220.0	0.0	90.0	0.0	220.0	100.0	330.0	220.0	1.2	1.2
Radar surround	50	Fence wire	Cylinder	1.2	1.2	220.0	0.0	90.0	0.0	220.0	170.0	330.0	220.0	1.2	1.2
Radar surround	51	Fence post	Cylinder	8.0	8.0	200.0	90.0	0.0	0.0	330.0	100.0	330.0	8.0	200.0	8.0
Radar surround	52	Radar warning sign	Cube	60.0	40.0	3.0	0.0	90.0	0.0	-335.0	160.0	-110.0	3.0	40.0	60.0
AA gun surround	2	AA pad	Cylinder	700.0	700.0	4.0	90.0	0.0	0.0	0.0	2.0	0.0	700.0	4.0	700.0
AA gun surround	3	Sandbag	Capsule	26.0	17.0	52.0	0.0	0.0	0.0	370.0	9.0	0.0	26.0	17.0	52.0
AA gun surround	4	Sandbag	Capsule	26.0	17.0	52.0	0.0	-16.4	0.0	355.0	9.0	104.2	39.6	17.0	57.2
AA gun surround	5	Sandbag	Capsule	26.0	17.0	52.0	0.0	-32.7	0.0	311.3	9.0	200.0	50.0	17.0	57.8

AA gun surround	6	Sandbag	Capsule	26.0	17.0	52.0	0.0	-49.1	0.0	242.3	9.0	279.6	56.3	17.0	53.7
AA gun surround	7	Sandbag	Capsule	26.0	17.0	52.0	0.0	-65.5	0.0	153.7	9.0	336.6	58.1	17.0	45.3
AA gun surround	8	Sandbag	Capsule	26.0	17.0	52.0	0.0	-81.8	0.0	52.7	9.0	366.2	55.2	17.0	33.1
AA gun surround	9	Sandbag	Capsule	26.0	17.0	52.0	0.0	-98.2	0.0	-52.7	9.0	366.2	55.2	17.0	33.1
AA gun surround	10	Sandbag	Capsule	26.0	17.0	52.0	0.0	180.0	0.0	-370.0	9.0	0.0	26.0	17.0	52.0
AA gun surround	11	Sandbag	Capsule	26.0	17.0	52.0	0.0	163.6	0.0	-355.0	9.0	-104.2	39.6	17.0	57.2
AA gun surround	12	Sandbag	Capsule	26.0	17.0	52.0	0.0	147.3	0.0	-311.3	9.0	-200.0	50.0	17.0	57.8
AA gun surround	13	Sandbag	Capsule	26.0	17.0	52.0	0.0	130.9	0.0	-242.3	9.0	-279.6	56.3	17.0	53.7
AA gun surround	14	Sandbag	Capsule	26.0	17.0	52.0	0.0	114.5	0.0	-153.7	9.0	-336.6	58.1	17.0	45.3
AA gun surround	15	Sandbag	Capsule	26.0	17.0	52.0	0.0	98.2	0.0	-52.7	9.0	-366.2	55.2	17.0	33.1
AA gun surround	16	Sandbag	Capsule	26.0	17.0	52.0	0.0	81.8	0.0	52.7	9.0	-366.2	55.2	17.0	33.1
AA gun surround	17	Sandbag	Capsule	26.0	17.0	52.0	0.0	65.5	0.0	153.7	9.0	-336.6	58.1	17.0	45.3
AA gun surround	18	Sandbag	Capsule	26.0	17.0	52.0	0.0	49.1	0.0	242.3	9.0	-279.6	56.3	17.0	53.7
AA gun surround	19	Sandbag	Capsule	26.0	17.0	52.0	0.0	32.7	0.0	311.3	9.0	-200.0	50.0	17.0	57.8
AA gun surround	20	Sandbag	Capsule	26.0	17.0	52.0	0.0	16.4	0.0	355.0	9.0	-104.2	39.6	17.0	57.2
AA gun surround	21	Sandbag	Capsule	26.0	17.0	52.0	0.0	-8.2	0.0	366.2	24.0	52.7	33.1	17.0	55.2
AA gun surround	22	Sandbag	Capsule	26.0	17.0	52.0	0.0	-24.5	0.0	336.6	24.0	153.7	45.3	17.0	58.1
AA gun surround	23	Sandbag	Capsule	26.0	17.0	52.0	0.0	-40.9	0.0	279.6	24.0	242.3	53.7	17.0	56.3
AA gun surround	24	Sandbag	Capsule	26.0	17.0	52.0	0.0	-57.3	0.0	200.0	24.0	311.3	57.8	17.0	50.0
AA gun surround	25	Sandbag	Capsule	26.0	17.0	52.0	0.0	-73.6	0.0	104.2	24.0	355.0	57.2	17.0	39.6
AA gun surround	26	Sandbag	Capsule	26.0	17.0	52.0	0.0	-90.0	0.0	0.0	24.0	370.0	52.0	17.0	26.0
AA gun surround	27	Sandbag	Capsule	26.0	17.0	52.0	0.0	-106.4	0.0	-104.2	24.0	355.0	57.2	17.0	39.6
AA gun surround	28	Sandbag	Capsule	26.0	17.0	52.0	0.0	-171.8	0.0	-366.2	24.0	52.7	33.1	17.0	55.2
AA gun surround	29	Sandbag	Capsule	26.0	17.0	52.0	0.0	171.8	0.0	-366.2	24.0	-52.7	33.1	17.0	55.2
AA gun surround	30	Sandbag	Capsule	26.0	17.0	52.0	0.0	155.5	0.0	-336.6	24.0	-153.7	45.3	17.0	58.1
AA gun surround	31	Sandbag	Capsule	26.0	17.0	52.0	0.0	139.1	0.0	-279.6	24.0	-242.3	53.7	17.0	56.3
AA gun surround	32	Sandbag	Capsule	26.0	17.0	52.0	0.0	122.7	0.0	-200.0	24.0	-311.3	57.8	17.0	50.0
AA gun surround	33	Sandbag	Capsule	26.0	17.0	52.0	0.0	106.4	0.0	-104.2	24.0	-355.0	57.2	17.0	39.6
AA gun surround	34	Sandbag	Capsule	26.0	17.0	52.0	0.0	90.0	0.0	0.0	24.0	-370.0	52.0	17.0	26.0
AA gun surround	35	Sandbag	Capsule	26.0	17.0	52.0	0.0	73.6	0.0	104.2	24.0	-355.0	57.2	17.0	39.6
AA gun surround	36	Sandbag	Capsule	26.0	17.0	52.0	0.0	57.3	0.0	200.0	24.0	-311.3	57.8	17.0	50.0
AA gun surround	37	Sandbag	Capsule	26.0	17.0	52.0	0.0	40.9	0.0	279.6	24.0	-242.3	53.7	17.0	56.3
AA gun surround	38	Sandbag	Capsule	26.0	17.0	52.0	0.0	24.5	0.0	336.6	24.0	-153.7	45.3	17.0	58.1
AA gun surround	39	Sandbag	Capsule	26.0	17.0	52.0	0.0	8.2	0.0	366.2	24.0	-52.7	33.1	17.0	55.2
AA gun surround	40	Ammo crate	Cube	70.0	40.0	40.0	0.0	0.0	0.0	-476.5	20.0	114.3	70.0	40.0	40.0
AA gun surround	41	Ammo crate	Cube	70.0	40.0	40.0	0.0	20.0	0.0	-478.8	20.0	-104.3	79.5	40.0	61.5
AA gun surround	42	Ammo crate	Cube	70.0	40.0	40.0	0.0	40.0	0.0	-385.7	20.0	-302.2	79.3	40.0	75.6
Sandbag wall	2	Sandbag	Capsule	26.0	17.0	50.0	0.0	-35.0	0.0	634.7	11.3	1580.3	50.0	17.0	55.9
Sandbag wall	3	Sandbag	Capsule	26.0	17.0	50.0	0.0	-55.0	0.0	600.3	16.6	1614.7	55.9	17.0	50.0
Sandbag wall	4	Sandbag	Capsule	26.0	17.0	50.0	0.0	-75.0	0.0	556.2	20.4	1635.2	55.0	17.0	38.1
Sandbag wall	5	Sandbag	Capsule	26.0	17.0	50.0	0.0	-95.0	0.0	507.8	20.7	1639.5	52.1	17.0	30.3
Sandbag wall	6	Sandbag	Capsule	26.0	17.0	50.0	0.0	-115.0	0.0	460.8	14.5	1626.9	56.3	17.0	44.7
Sandbag wall	7	Sandbag	Capsule	26.0	17.0	50.0	0.0	-135.0	0.0	421.0	2.4	1599.0	53.7	17.0	53.7
Sandbag wall	8	Sandbag	Capsule	26.0	17.0	50.0	0.0	-155.0	0.0	393.1	-8.2	1559.2	44.7	17.0	56.3
Sandbag wall	9	Sandbag	Capsule	26.0	17.0	50.0	0.0	-45.0	0.0	619.0	28.2	1599.0	53.7	17.0	53.7
Sandbag wall	10	Sandbag	Capsule	26.0	17.0	50.0	0.0	-65.0	0.0	579.2	32.7	1626.9	56.3	17.0	44.7
Sandbag wall	11	Sandbag	Capsule	26.0	17.0	50.0	0.0	-85.0	0.0	532.2	35.2	1639.5	52.1	17.0	30.3
Sandbag wall	12	Sandbag	Capsule	26.0	17.0	50.0	0.0	-105.0	0.0	483.8	32.6	1635.2	55.0	17.0	38.1
Sandbag wall	13	Sandbag	Capsule	26.0	17.0	50.0	0.0	-125.0	0.0	439.7	22.8	1614.7	55.9	17.0	50.0
Sandbag wall	14	Sandbag	Capsule	26.0	17.0	50.0	0.0	-145.0	0.0	405.3	10.5	1580.3	50.0	17.0	55.9
Sandbag wall	15	Sandbag	Capsule	26.0	17.0	50.0	0.0	-55.0	0.0	600.3	44.6	1614.7	55.9	17.0	50.0

Sandbag wall	16	Sandbag	Capsule	26.0	17.0	50.0	0.0	-75.0	0.0	556.2	48.4	1635.2	55.0	17.0	38.1
Sandbag wall	17	Sandbag	Capsule	26.0	17.0	50.0	0.0	-95.0	0.0	507.8	48.7	1639.5	52.1	17.0	30.3
Sandbag wall	18	Sandbag	Capsule	26.0	17.0	50.0	0.0	-115.0	0.0	460.8	42.5	1626.9	56.3	17.0	44.7
Sandbag wall	19	Sandbag	Capsule	26.0	17.0	50.0	0.0	-135.0	0.0	421.0	30.4	1599.0	53.7	17.0	53.7
Boom barrier	2	Barrier foundation	Cube	90.0	12.0	90.0	0.0	0.0	0.0	0.0	6.0	0.0	90.0	12.0	90.0
Boom barrier	3	Barrier housing	Cube	50.0	105.0	50.0	0.0	0.0	0.0	0.0	64.0	0.0	50.0	105.0	50.0
Boom barrier	4	Housing door	Cube	30.0	50.0	2.0	0.0	0.0	0.0	0.0	50.0	26.0	30.0	50.0	2.0
Boom barrier	5	Boom receiver	Cube	30.0	95.0	30.0	0.0	0.0	0.0	-545.0	47.5	0.0	30.0	95.0	30.0
Boom barrier	6	Boom cradle	Cube	36.0	8.0	20.0	0.0	0.0	0.0	-545.0	98.0	0.0	36.0	8.0	20.0
Boom barrier	7	Boom stripe (red)	Cube	50.0	12.0	10.0	0.0	0.0	0.0	-55.0	105.0	0.0	50.0	12.0	10.0
Boom barrier	8	Boom stripe (white)	Cube	50.0	12.0	10.0	0.0	0.0	0.0	-105.0	105.0	0.0	50.0	12.0	10.0
Boom barrier	9	Boom stripe (red)	Cube	50.0	12.0	10.0	0.0	0.0	0.0	-155.0	105.0	0.0	50.0	12.0	10.0
Boom barrier	10	Boom stripe (white)	Cube	50.0	12.0	10.0	0.0	0.0	0.0	-205.0	105.0	0.0	50.0	12.0	10.0
Boom barrier	11	Boom stripe (red)	Cube	50.0	12.0	10.0	0.0	0.0	0.0	-255.0	105.0	0.0	50.0	12.0	10.0
Boom barrier	12	Boom stripe (white)	Cube	50.0	12.0	10.0	0.0	0.0	0.0	-305.0	105.0	0.0	50.0	12.0	10.0
Boom barrier	13	Boom stripe (red)	Cube	50.0	12.0	10.0	0.0	0.0	0.0	-355.0	105.0	0.0	50.0	12.0	10.0
Boom barrier	14	Boom stripe (white)	Cube	50.0	12.0	10.0	0.0	0.0	0.0	-405.0	105.0	0.0	50.0	12.0	10.0
Boom barrier	15	Boom stripe (red)	Cube	50.0	12.0	10.0	0.0	0.0	0.0	-455.0	105.0	0.0	50.0	12.0	10.0
Boom barrier	16	Boom stripe (white)	Cube	50.0	12.0	10.0	0.0	0.0	0.0	-505.0	105.0	0.0	50.0	12.0	10.0
Boom barrier	17	Counterweight arm	Cube	90.0	8.0	8.0	0.0	0.0	0.0	35.0	105.0	0.0	90.0	8.0	8.0
Boom barrier	18	Counterweight	Cube	30.0	34.0	26.0	0.0	0.0	0.0	62.0	93.0	0.0	30.0	34.0	26.0
Boom barrier	19	Pivot bolt	Cylinder	12.0	12.0	60.0	0.0	0.0	0.0	0.0	105.0	0.0	12.0	12.0	60.0
Boom barrier	20	Boom tip lamp	Sphere	12.0	12.0	12.0	0.0	0.0	0.0	-530.0	117.0	0.0	12.0	12.0	12.0
Boom barrier	21	Warning lamp	Sphere	22.0	22.0	22.0	0.0	0.0	0.0	0.0	130.0	0.0	22.0	22.0	22.0
HESCO blocks	2	HESCO cell	Cube	103.0	140.0	106.0	0.0	0.0	0.0	-105.0	70.0	0.0	103.0	140.0	106.0
HESCO blocks	3	HESCO cell	Cube	103.0	140.0	106.0	0.0	0.0	0.0	0.0	70.0	0.0	103.0	140.0	106.0
HESCO blocks	4	HESCO cell	Cube	103.0	140.0	106.0	0.0	0.0	0.0	105.0	70.0	0.0	103.0	140.0	106.0
HESCO blocks	5	HESCO cell	Cube	103.0	140.0	106.0	0.0	0.0	0.0	-105.0	210.0	0.0	103.0	140.0	106.0
HESCO blocks	6	HESCO cell	Cube	103.0	140.0	106.0	0.0	0.0	0.0	0.0	210.0	0.0	103.0	140.0	106.0
HESCO blocks	7	HESCO cell	Cube	103.0	140.0	106.0	0.0	0.0	0.0	105.0	210.0	0.0	103.0	140.0	106.0
Tank trap	2	Trap beam	Cube	150.0	12.0	12.0	0.0	0.0	0.0	0.0	55.0	0.0	150.0	12.0	12.0
Tank trap	3	Trap beam	Cube	150.0	12.0	12.0	0.0	0.0	55.0	0.0	55.0	0.0	95.9	129.8	12.0
Tank trap	4	Trap beam	Cube	150.0	12.0	12.0	0.0	90.0	35.0	0.0	55.0	0.0	12.0	95.9	129.8
Jersey barrier	2	Jersey foot	Cube	60.0	30.0	300.0	0.0	0.0	0.0	0.0	15.0	0.0	60.0	30.0	300.0
Jersey barrier	3	Jersey body	Cube	30.0	55.0	296.0	0.0	0.0	0.0	0.0	57.5	0.0	30.0	55.0	296.0
Broadleaf tree	2	Trunk	Cylinder	22.0	22.0	190.0	90.0	0.0	0.0	0.0	95.0	0.0	22.0	190.0	22.0
Broadleaf tree	3	Limb	Cylinder	10.0	10.0	107.8	-45.9	83.1	0.0	37.2	183.7	4.5	82.8	84.4	19.8
Broadleaf tree	4	Limb	Cylinder	10.0	10.0	109.8	-46.9	-36.7	0.0	-22.4	185.1	30.1	57.2	87.0	71.9
Broadleaf tree	5	Limb	Cylinder	10.0	10.0	127.3	-53.9	-153.7	0.0	-16.6	196.4	-33.6	45.8	108.8	78.9
Broadleaf tree	6	Canopy clump	Sphere	137.6	117.0	137.6	0.0	133.9	0.0	34.1	325.9	-1.8	194.6	117.0	194.6
Broadleaf tree	7	Canopy clump	Sphere	96.7	82.2	96.7	0.0	111.5	0.0	-39.8	315.8	40.8	125.4	82.2	125.4
Broadleaf tree	8	Canopy clump	Sphere	101.1	85.9	101.1	0.0	273.7	0.0	-9.3	305.7	-70.2	107.3	85.9	107.3
Broadleaf tree	9	Canopy clump	Sphere	91.1	77.5	91.1	0.0	254.3	0.0	34.7	295.5	72.5	112.4	77.5	112.4
Broadleaf tree	10	Canopy clump	Sphere	124.3	105.7	124.3	0.0	137.1	0.0	-80.3	285.4	-33.6	175.7	105.7	175.7
Broadleaf tree	11	Canopy clump	Sphere	90.2	76.7	90.2	0.0	156.5	0.0	83.9	275.3	-36.7	118.7	76.7	118.7
Broadleaf tree	12	Canopy clump	Sphere	95.7	81.3	95.7	0.0	80.9	0.0	-16.1	265.1	92.8	109.6	81.3	109.6
Broadleaf tree	13	Canopy clump	Sphere	113.0	96.0	113.0	0.0	205.2	0.0	-23.9	255.0	-91.9	150.3	96.0	150.3
Broadleaf tree	14	Canopy clump	Sphere	101.5	86.3	101.5	0.0	264.0	0.0	84.6	244.9	41.3	111.6	86.3	111.6
Broadleaf tree	15	Canopy clump	Sphere	141.3	120.1	141.3	0.0	91.8	0.0	-81.2	234.7	42.3	145.7	120.1	145.7
Broadleaf tree	16	Canopy clump	Sphere	93.1	79.2	93.1	0.0	132.3	0.0	55.3	224.6	-67.3	131.6	79.2	131.6
Broadleaf tree	17	Canopy clump	Sphere	127.6	108.4	127.6	0.0	312.5	0.0	23.0	214.5	77.0	180.2	108.4	180.2

Broadleaf tree	18	Canopy low	Sphere	190.0	130.0	190.0	0.0	0.0	0.0	0.0	255.0	0.0	190.0	130.0	190.0
Broadleaf tree	19	Canopy top	Sphere	130.0	100.0	130.0	0.0	0.0	0.0	10.0	315.0	-10.0	130.0	100.0	130.0
Pine	2	Pine trunk	Cylinder	19.0	19.0	249.6	90.0	0.0	0.0	0.0	124.8	0.0	19.0	249.6	19.0
Pine	3	Pine crown	Cone	207.4	226.1	99.2	-90.3	224.8	-3.7	-0.5	109.6	2.9	305.3	100.3	307.1
Pine	4	Pine crown	Cone	162.1	168.3	78.9	-89.5	337.8	0.3	-0.1	146.8	1.2	213.4	80.3	217.2
Pine	5	Pine crown	Cone	147.7	149.3	76.6	-94.5	60.8	2.5	3.9	186.7	-2.3	204.5	88.5	201.7
Pine	6	Pine crown	Cone	104.3	98.1	51.1	-93.7	195.2	3.6	-1.7	212.9	-2.1	131.2	57.8	129.5
Pine	7	Pine crown	Cone	71.6	67.2	32.5	-91.6	54.7	-2.2	3.8	241.2	-2.8	97.7	34.5	98.3
Pine	8	Pine crown	Cone	41.4	45.2	75.6	-96.0	195.1	2.4	3.7	299.4	0.6	55.1	80.1	63.0
Cut branches	2	Cut branch stem	Cylinder	5.0	5.0	198.5	-74.6	-78.9	0.0	2.7	95.7	-147.3	57.4	192.7	16.0
Cut branches	3	Cut branch foliage	Sphere	49.6	166.5	49.6	0.0	79.7	15.1	2.7	95.7	-147.3	65.1	173.7	98.7
Cut branches	4	Cut branch tip	Cone	35.0	35.0	50.0	-90.0	0.0	15.1	-23.1	201.4	-142.2	42.9	50.0	42.9
Cut branches	5	Cut branch stem	Cylinder	5.0	5.0	220.0	-63.2	-101.5	0.0	-37.4	98.2	-19.0	102.5	198.6	25.5
Cut branches	6	Cut branch foliage	Sphere	64.4	184.0	64.4	0.0	52.5	26.3	-37.4	98.2	-19.0	135.9	193.5	149.7
Cut branches	7	Cut branch tip	Cone	35.0	35.0	50.0	-90.0	0.0	26.3	-85.9	206.4	-28.9	46.9	50.0	46.9
Cut branches	8	Cut branch stem	Cylinder	5.0	5.0	184.2	-74.4	-91.4	0.0	-15.8	88.7	148.3	54.3	178.8	6.3
Cut branches	9	Cut branch foliage	Sphere	57.0	154.8	57.0	0.0	70.3	15.6	-15.8	88.7	148.3	86.2	164.4	110.0
Cut branches	10	Cut branch tip	Cone	35.0	35.0	50.0	-90.0	0.0	15.6	-40.5	187.5	147.8	43.1	50.0	43.1
Cut branches	11	Brush clump	Sphere	80.6	48.4	80.6	0.0	286.9	0.0	45.0	12.1	-98.0	100.5	48.4	100.5
Cut branches	12	Brush clump	Sphere	76.6	46.0	76.6	0.0	217.5	0.0	45.0	11.5	0.0	107.4	46.0	107.4
Cut branches	13	Brush clump	Sphere	100.1	60.1	100.1	0.0	11.9	0.0	45.0	15.0	98.0	118.6	60.1	118.6
Bush	2	Bush	Sphere	89.6	62.7	89.6	0.0	80.3	0.0	5.5	34.7	8.9	103.4	62.7	103.4
Bush	3	Bush	Sphere	85.6	59.9	85.6	0.0	70.9	0.0	11.5	29.8	20.1	108.9	59.9	108.9
Bush	4	Bush	Sphere	77.5	54.3	77.5	0.0	113.9	0.0	-12.1	26.6	11.9	102.3	54.3	102.3
Bush	5	Bush	Sphere	62.4	43.7	62.4	0.0	98.5	0.0	-20.9	22.8	-12.3	71.0	43.7	71.0
Bush	6	Bush	Sphere	91.9	64.3	91.9	0.0	326.8	0.0	1.5	26.5	-4.8	127.2	64.3	127.2
Bush	7	Bush	Sphere	61.3	42.9	61.3	0.0	194.6	0.0	-24.4	27.7	5.8	74.8	42.9	74.8
Treeline pine	2	Pine crown	Cone	451.7	451.7	500.0	-90.0	0.0	0.0	0.0	250.0	0.0	451.7	500.0	451.7
Treeline pine	3	Pine crown	Cone	293.6	293.6	443.6	-90.0	30.0	0.0	0.0	584.7	0.0	401.0	443.6	401.0
Flag	2	Flagpole base	Cube	70.0	20.0	70.0	0.0	0.0	0.0	0.0	10.0	0.0	70.0	20.0	70.0
Flag	3	Flagpole lower	Cylinder	16.0	16.0	450.0	90.0	0.0	0.0	0.0	245.0	0.0	16.0	450.0	16.0
Flag	4	Flagpole upper	Cylinder	11.0	11.0	450.0	90.0	0.0	0.0	0.0	695.0	0.0	11.0	450.0	11.0
Flag	5	Flagpole finial	Sphere	24.0	24.0	24.0	0.0	0.0	0.0	0.0	932.0	0.0	24.0	24.0	24.0
Flag	6	Halyard	Cylinder	1.5	1.5	740.0	90.0	0.0	0.0	-10.0	520.0	12.0	1.5	740.0	1.5
Flag	7	Cleat	Cube	8.0	5.0	5.0	0.0	0.0	0.0	-10.0	160.0	15.0	8.0	5.0	5.0
Flag	8	Halyard	Cylinder	1.5	1.5	740.0	90.0	0.0	0.0	10.0	520.0	12.0	1.5	740.0	1.5
Flag	9	Cleat	Cube	8.0	5.0	5.0	0.0	0.0	0.0	10.0	160.0	15.0	8.0	5.0	5.0
Flag	10	Flag spot housing	Cylinder	34.0	34.0	30.0	-68.1	180.0	0.0	0.0	30.0	190.0	34.0	40.5	42.7
Flag	11	Flag spot lens	Sphere	30.0	30.0	5.0	-68.1	180.0	0.0	0.0	44.8	184.0	30.0	15.8	29.7
Flag	12	Flag spot stand	Cube	20.0	12.0	20.0	0.0	0.0	0.0	0.0	6.0	190.0	20.0	12.0	20.0
Flag	13	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	5.2	869.6	4.4	20.1	50.8	12.6
Flag	14	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	7.0	819.7	3.4	20.1	50.8	12.6
Flag	15	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	8.7	769.7	2.4	20.1	50.8	12.6
Flag	16	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-11.8	868.8	15.0	20.5	50.8	12.1
Flag	17	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-9.5	818.9	14.8	20.5	50.8	12.1
Flag	18	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-7.2	768.9	14.7	20.5	50.8	12.1
Flag	19	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-28.4	868.0	26.1	20.8	50.8	13.6
Flag	20	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-25.4	818.1	27.1	20.8	50.8	13.6
Flag	21	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-22.4	768.2	28.0	20.8	50.8	13.6
Flag	22	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-44.7	867.3	37.7	20.7	50.7	14.7
Flag	23	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-41.4	817.4	39.1	20.7	50.7	14.7

Flag	24	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-38.0	767.5	40.5	20.7	50.7	14.7
Flag	25	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-61.0	866.5	49.3	20.2	50.7	14.5
Flag	26	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-58.0	816.6	50.2	20.2	50.7	14.5
Flag	27	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-55.0	766.7	51.1	20.2	50.7	14.5
Flag	28	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-77.5	865.8	60.5	19.6	50.7	13.4
Flag	29	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-75.2	815.8	60.4	19.6	50.7	13.4
Flag	30	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-73.0	765.9	60.3	19.6	50.7	13.4
Flag	31	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-94.3	865.0	71.2	19.7	50.7	13.3
Flag	32	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-92.6	815.0	70.3	19.7	50.7	13.3
Flag	33	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-91.0	765.1	69.3	19.7	50.7	13.3
Flag	34	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-111.6	864.2	81.3	20.3	50.7	12.1
Flag	35	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-109.9	814.3	80.3	20.3	50.7	12.1
Flag	36	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-108.2	764.3	79.4	20.3	50.7	12.1
Flag	37	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-129.2	863.4	90.7	21.1	50.8	10.3
Flag	38	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-127.1	813.5	90.7	21.1	50.8	10.3
Flag	39	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-125.0	763.5	90.7	21.1	50.8	10.3
Flag	40	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-147.0	862.7	99.8	21.6	50.8	11.5
Flag	41	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-144.4	812.8	100.9	21.6	50.8	11.5
Flag	42	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-141.7	762.9	102.1	21.6	50.8	11.5
Flag	43	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-164.7	862.0	109.1	21.4	50.7	13.0
Flag	44	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-161.8	812.1	110.7	21.4	50.7	13.0
Flag	45	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-158.8	762.2	112.3	21.4	50.7	13.0
Flag	46	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-181.9	861.3	119.3	20.2	50.6	14.0
Flag	47	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-179.3	811.4	120.4	20.2	50.6	14.0
Flag	48	Flag cloth	Cube	20.0	50.0	2.0	0.0	0.0	0.0	-176.7	761.4	121.5	20.2	50.6	14.0
Windsock	2	Windsock base	Cube	50.0	8.0	50.0	0.0	0.0	0.0	0.0	4.0	0.0	50.0	8.0	50.0
Windsock	3	Windsock pole	Cylinder	14.0	14.0	300.0	90.0	0.0	0.0	0.0	150.0	0.0	14.0	300.0	14.0
Windsock	4	Windsock pole	Cylinder	10.0	10.0	310.0	90.0	0.0	0.0	0.0	455.0	0.0	10.0	310.0	10.0
Windsock	5	Mouth hoop	Cylinder	2.5	2.5	36.0	-60.0	-90.0	0.0	-23.5	615.6	13.4	18.8	32.4	12.1
Windsock	6	Mouth hoop	Cylinder	2.5	2.5	36.0	0.0	-90.0	0.0	0.0	631.2	0.0	32.5	2.5	20.0
Windsock	7	Mouth hoop	Cylinder	2.5	2.5	36.0	60.0	-90.0	0.0	23.5	615.6	-13.4	18.8	32.4	12.1
Windsock	8	Mouth hoop	Cylinder	2.5	2.5	36.0	60.0	90.0	0.0	23.5	584.4	-13.4	18.8	32.4	12.1
Windsock	9	Mouth hoop	Cylinder	2.5	2.5	36.0	0.0	90.0	0.0	0.0	568.8	0.0	32.5	2.5	20.0
Windsock	10	Mouth hoop	Cylinder	2.5	2.5	36.0	-60.0	90.0	0.0	-23.5	584.4	13.4	18.8	32.4	12.1
Windsock	11	Hoop spoke	Cylinder	2.5	2.5	31.0	-90.0	0.0	0.0	0.0	579.5	0.0	3.4	31.0	3.4
Windsock	12	Sock cone	Cylinder	72.0	72.0	55.0	0.0	0.0	0.0	-13.6	597.9	-23.8	92.4	76.0	88.0
Windsock	13	Sock cone	Cylinder	60.0	60.0	55.0	0.0	0.0	0.0	-39.5	592.3	-72.0	81.9	66.7	82.6
Windsock	14	Sock cone	Cylinder	48.0	48.0	55.0	0.0	0.0	0.0	-62.6	584.1	-121.2	69.3	57.2	76.2
Windsock	15	Sock cone	Cylinder	38.0	38.0	55.0	0.0	0.0	0.0	-82.8	573.1	-171.0	58.2	50.4	71.6
Windsock	16	Sock cone	Cylinder	28.0	28.0	55.0	0.0	0.0	0.0	-101.6	558.0	-220.4	48.4	44.5	67.0
Supply cache	2	Crate	Cube	60.0	60.0	60.0	0.0	0.0	0.0	0.0	30.0	0.0	60.0	60.0	60.0
Supply cache	3	Crate	Cube	60.0	60.0	60.0	0.0	12.0	0.0	65.0	30.0	5.0	71.2	60.0	71.2
Supply cache	4	Crate	Cube	55.0	50.0	55.0	0.0	-8.0	0.0	30.0	85.0	0.0	62.1	50.0	62.1
Supply cache	5	Crate stencil	Cube	34.0	0.6	12.0	0.0	-8.0	0.0	30.0	110.4	0.0	35.3	0.6	16.6
Supply cache	6	Fuel drum	Cylinder	40.0	40.0	60.0	90.0	0.0	0.0	-55.0	30.0	75.0	40.0	60.0	40.0
Supply cache	7	Fuel drum	Cylinder	40.0	40.0	60.0	90.0	0.0	0.0	-5.0	30.0	105.0	40.0	60.0	40.0
Boulder	2	Rock	Sphere	119.6	54.3	97.4	-4.8	186.5	23.1	0.0	11.9	0.0	141.4	104.5	114.4
Boulder	3	Rock	Sphere	56.5	24.6	41.4	9.3	191.2	5.8	29.4	1.3	-21.1	64.9	36.6	54.5
Boulder	4	Rock	Sphere	62.8	45.3	54.9	-13.3	91.4	5.4	-26.7	15.0	-23.4	66.6	62.3	68.3
Boulder	5	Rock	Sphere	62.3	40.0	51.5	16.2	183.7	1.5	-20.9	7.4	29.3	65.7	54.4	64.2
Lamp post	2	Lamp pole	Cylinder	9.0	9.0	320.0	90.0	0.0	0.0	0.0	160.0	0.0	9.0	320.0	9.0

Lamp post	3	Lamp arm	Cube	40.0	5.0	5.0	0.0	0.0	0.0	-20.0	315.0	0.0	40.0	5.0	5.0
Lamp post	4	Lamp head	Cube	34.0	8.0	20.0	0.0	0.0	0.0	-42.0	312.0	0.0	34.0	8.0	20.0
Lamp post	5	Dim red lamp	Sphere	16.0	16.0	5.0	90.0	0.0	0.0	-42.0	305.0	0.0	16.0	5.0	16.0
Track sign	2	Sign post	Cylinder	7.0	7.0	190.0	90.0	0.0	0.0	0.0	95.0	0.0	7.0	190.0	7.0
Track sign	3	Sign board	Cube	80.0	55.0	3.0	0.0	0.0	0.0	0.0	175.0	4.0	80.0	55.0	3.0
Track sign	4	Sign arrow	Cube	45.0	8.0	1.0	0.0	0.0	0.0	0.0	175.0	6.5	45.0	8.0	1.0
Track sign	5	Sign arrow head	Cube	16.0	16.0	1.0	0.0	0.0	45.0	22.0	175.0	6.5	22.6	22.6	1.0
Fire barrel	2	Barrel	Cylinder	50.0	50.0	78.0	90.0	0.0	0.0	0.0	39.0	0.0	50.0	78.0	50.0
Fire barrel	3	Fire embers	Sphere	42.0	16.0	42.0	0.0	0.0	0.0	0.0	80.0	0.0	42.0	16.0	42.0
Sky clock (sun)	2	Sun	Sphere	900.0	900.0	900.0	0.0	0.0	0.0	-7739.6	-27559.8	8975.2	900.0	900.0	900.0
Moon	2	Moon	Sphere	1400.0	1400.0	1400.0	0.0	0.0	0.0	0.0	0.0	0.0	1400.0	1400.0	1400.0
Terrain disc	2	Ground disc	Terrain disc	100.0	100.0	100.0	0.0	0.0	0.0	0.0	0.0	0.0	7200.0		7200.0
Far ground	2	Far ground	Cube	40000.0	17600.0	10.0	90.0	0.0	0.0	0.0	0.0	0.0	40000.0	17600.0	10.0
Distant hills	2	Hill	Sphere	4425.2	1206.8	2600.0	0.0	-86.4	0.0	5604.0	-65.0	-354.7	2874.3	1206.8	4580.6
Distant hills	3	Hill	Sphere	4277.8	1405.5	2600.0	0.0	-112.5	0.0	5288.8	-65.0	2190.8	4039.2	1405.5	4947.1
Distant hills	4	Hill	Sphere	5454.6	1131.5	2600.0	0.0	-155.7	0.0	2363.1	-65.0	5221.6	6041.4	1131.5	4617.7
Distant hills	5	Hill	Sphere	4278.0	1580.8	2600.0	0.0	-177.9	0.0	212.2	-65.0	5813.3	4370.0	1580.8	2754.3
Distant hills	6	Hill	Sphere	4624.0	1529.9	2600.0	0.0	149.0	0.0	-3441.9	-65.0	5724.0	5302.6	1529.9	4611.1
Distant hills	7	Hill	Sphere	4743.2	1162.4	2600.0	0.0	119.0	0.0	-4987.3	-65.0	2768.5	4575.3	1162.4	5409.0
Distant hills	8	Hill	Sphere	4212.9	1014.8	2600.0	0.0	91.6	0.0	-5679.5	-65.0	158.6	2716.6	1014.8	4283.9
Distant hills	9	Hill	Sphere	4613.2	1055.2	2600.0	0.0	67.9	0.0	-5112.3	-65.0	-2080.2	4147.0	1055.2	5252.9
Distant hills	10	Hill	Sphere	4965.7	1105.6	2600.0	0.0	25.0	0.0	-2449.8	-65.0	-5252.2	5599.3	1105.6	4455.3
Distant hills	11	Hill	Sphere	4226.7	1621.7	2600.0	0.0	-3.2	0.0	377.6	-65.0	-6849.3	4363.4	1621.7	2828.7
Distant hills	12	Hill	Sphere	5575.1	1642.3	2600.0	0.0	-26.4	0.0	2702.8	-65.0	-5433.9	6149.7	1642.3	4810.8
Distant hills	13	Hill	Sphere	5085.7	1320.6	2600.0	0.0	-62.6	0.0	4987.7	-65.0	-2583.3	4647.7	1320.6	5711.7
Distant hills	14	Far hill	Sphere	9380.1	3525.0	4000.0	0.0	-105.0	0.0	9462.7	-65.0	2535.0	6291.0	3525.0	10095.7
Distant hills	15	Far hill	Sphere	9804.4	3391.7	4000.0	0.0	-154.7	0.0	3805.5	-65.0	8047.6	10573.3	3391.7	7807.3
Distant hills	16	Far hill	Sphere	9022.5	2531.2	4000.0	0.0	177.4	0.0	-409.0	-65.0	8853.8	9197.5	2531.2	4412.1
Distant hills	17	Far hill	Sphere	7315.8	3447.5	4000.0	0.0	123.3	0.0	-7161.6	-65.0	4695.7	7356.5	3447.5	8311.3
Distant hills	18	Far hill	Sphere	9322.2	2491.4	4000.0	0.0	82.7	0.0	-8617.4	-65.0	-1103.5	5151.7	2491.4	9754.8
Distant hills	19	Far hill	Sphere	7683.5	4076.1	4000.0	0.0	54.0	0.0	-7118.4	-65.0	-5175.7	7753.7	4076.1	8566.8
Distant hills	20	Far hill	Sphere	7559.2	4065.1	4000.0	0.0	10.1	0.0	-1705.4	-65.0	-9604.3	8142.1	4065.1	5260.0
Distant hills	21	Far hill	Sphere	8990.7	2866.7	4000.0	0.0	-23.3	0.0	3503.4	-65.0	-8148.2	9839.6	2866.7	7226.0
Distant hills	22	Far hill	Sphere	8850.0	3264.2	4000.0	0.0	-74.7	0.0	8434.5	-65.0	-2300.2	6187.6	3264.2	9590.6
Far radio tower	2	Tower leg	Cylinder	30.0	30.0	3807.6	-86.4	-135.0	0.0	115.0	1900.0	115.0	212.4	3801.9	212.4
Far radio tower	3	Tower leg	Cylinder	30.0	30.0	3807.6	-86.4	-45.0	0.0	115.0	1900.0	-115.0	212.4	3801.9	212.4
Far radio tower	4	Tower leg	Cylinder	30.0	30.0	3807.6	-86.4	45.0	0.0	-115.0	1900.0	-115.0	212.4	3801.9	212.4
Far radio tower	5	Tower leg	Cylinder	30.0	30.0	3807.6	-86.4	135.0	0.0	-115.0	1900.0	115.0	212.4	3801.9	212.4
Far radio tower	6	Tower brace	Cylinder	15.0	15.0	1729.0	-79.5	-166.0	0.0	153.0	1050.0	38.0	94.2	1702.7	324.0
Far radio tower	7	Tower brace	Cylinder	15.0	15.0	1729.0	-79.5	-76.0	0.0	38.0	1050.0	-153.0	324.0	1702.7	94.2
Far radio tower	8	Tower brace	Cylinder	15.0	15.0	1729.0	-79.5	14.0	0.0	-153.0	1050.0	-38.0	94.2	1702.7	324.0
Far radio tower	9	Tower brace	Cylinder	15.0	15.0	1729.0	-79.5	104.0	0.0	-38.0	1050.0	153.0	324.0	1702.7	94.2
Far radio tower	10	Tower antenna	Cylinder	12.0	12.0	400.0	90.0	0.0	0.0	0.0	4000.0	0.0	12.0	400.0	12.0
Far radio tower	11	Aviation light	Sphere	60.0	60.0	60.0	0.0	0.0	0.0	0.0	1100.0	0.0	60.0	60.0	60.0
Far radio tower	12	Aviation light	Sphere	60.0	60.0	60.0	0.0	0.0	0.0	0.0	2200.0	0.0	60.0	60.0	60.0
Far radio tower	13	Aviation light	Sphere	60.0	60.0	60.0	0.0	0.0	0.0	0.0	4210.0	0.0	60.0	60.0	60.0
Road	2	Asphalt	Cube	440.0	6.0	1550.0	0.0	0.0	0.0	0.0	3.0	2725.0	440.0	6.0	1550.0
Road	3	Edge line	Cube	12.0	1.2	1510.0	0.0	0.0	0.0	-195.0	6.6	2725.0	12.0	1.2	1510.0
Road	4	Edge line	Cube	12.0	1.2	1510.0	0.0	0.0	0.0	195.0	6.6	2725.0	12.0	1.2	1510.0
Road	5	Centre dash	Cube	12.0	1.2	200.0	0.0	0.0	0.0	0.0	6.6	2180.0	12.0	1.2	200.0
Road	6	Centre dash	Cube	12.0	1.2	200.0	0.0	0.0	0.0	0.0	6.6	2580.0	12.0	1.2	200.0

Road	7	Centre dash	Cube	12.0	1.2	200.0	0.0	0.0	0.0	0.0	6.6	2980.0	12.0	1.2	200.0
Road	8	Stop line	Cube	390.0	1.2	40.0	0.0	0.0	0.0	0.0	6.6	3000.0	390.0	1.2	40.0
Tyre tracks	2	Tyre track	Cube	24.0	1.2	244.7	0.0	-95.1	0.0	-184.0	0.6	2071.8	245.9	1.2	45.6
Tyre tracks	3	Tyre track	Cube	24.0	1.2	244.7	0.0	-95.1	0.0	-194.6	0.6	2191.4	245.9	1.2	45.6
Tyre tracks	4	Tyre track	Cube	24.0	1.2	244.7	0.0	-101.2	0.0	-404.9	0.6	2040.2	244.7	1.2	71.2
Tyre tracks	5	Tyre track	Cube	24.0	1.2	244.7	0.0	-101.2	0.0	-428.3	0.6	2157.9	244.7	1.2	71.2
Tyre tracks	6	Tyre track	Cube	24.0	1.2	244.7	0.0	-107.4	0.0	-621.1	0.6	1985.1	240.7	1.2	96.0
Tyre tracks	7	Tyre track	Cube	24.0	1.2	244.7	0.0	-107.4	0.0	-657.0	0.6	2099.6	240.7	1.2	96.0
Tyre tracks	8	Tyre track	Cube	24.0	1.2	244.7	0.0	-113.5	0.0	-830.2	0.6	1907.1	233.9	1.2	119.7
Tyre tracks	9	Tyre track	Cube	24.0	1.2	244.7	0.0	-113.5	0.0	-878.1	0.6	2017.1	233.9	1.2	119.7
Tyre tracks	10	Tyre track	Cube	24.0	1.2	244.7	0.0	-119.7	0.0	-1029.8	0.6	1807.2	224.5	1.2	142.0
Tyre tracks	11	Tyre track	Cube	24.0	1.2	244.7	0.0	-119.7	0.0	-1089.2	0.6	1911.5	224.5	1.2	142.0
Tyre tracks	12	Tyre track	Cube	24.0	1.2	244.7	0.0	-125.8	0.0	-1217.4	0.6	1686.5	212.5	1.2	162.7
Tyre tracks	13	Tyre track	Cube	24.0	1.2	244.7	0.0	-125.8	0.0	-1287.7	0.6	1783.8	212.5	1.2	162.7
Tyre tracks	14	Tyre track	Cube	24.0	1.2	244.7	0.0	-132.0	0.0	-1391.1	0.6	1546.3	198.0	1.2	181.5
Tyre tracks	15	Tyre track	Cube	24.0	1.2	244.7	0.0	-132.0	0.0	-1471.4	0.6	1635.6	198.0	1.2	181.5
Tyre tracks	16	Tyre track	Cube	24.0	1.2	244.7	0.0	-138.1	0.0	-1548.8	0.6	1388.4	181.2	1.2	198.2
Tyre tracks	17	Tyre track	Cube	24.0	1.2	244.7	0.0	-138.1	0.0	-1638.1	0.6	1468.5	181.2	1.2	198.2
Tyre tracks	18	Tyre track	Cube	24.0	1.2	244.7	0.0	-144.3	0.0	-1688.6	0.6	1214.5	162.4	1.2	212.7
Tyre tracks	19	Tyre track	Cube	24.0	1.2	244.7	0.0	-144.3	0.0	-1786.0	0.6	1284.6	162.4	1.2	212.7
Tyre tracks	20	Tyre track	Cube	24.0	1.2	244.7	0.0	-150.4	0.0	-1809.0	0.6	1026.6	141.6	1.2	224.7
Tyre tracks	21	Tyre track	Cube	24.0	1.2	244.7	0.0	-150.4	0.0	-1913.4	0.6	1085.8	141.6	1.2	224.7
Tyre tracks	22	Tyre track	Cube	24.0	1.2	244.7	0.0	-156.6	0.0	-1908.6	0.6	826.9	119.3	1.2	234.1
Tyre tracks	23	Tyre track	Cube	24.0	1.2	244.7	0.0	-156.6	0.0	-2018.7	0.6	874.6	119.3	1.2	234.1
Tyre tracks	24	Tyre track	Cube	24.0	1.2	244.7	0.0	-162.7	0.0	-1986.2	0.6	617.7	95.6	1.2	240.8
Tyre tracks	25	Tyre track	Cube	24.0	1.2	244.7	0.0	-162.7	0.0	-2100.8	0.6	653.3	95.6	1.2	240.8
Tyre tracks	26	Tyre track	Cube	24.0	1.2	244.7	0.0	-168.9	0.0	-2040.9	0.6	401.3	70.8	1.2	244.7
Tyre tracks	27	Tyre track	Cube	24.0	1.2	244.7	0.0	-168.9	0.0	-2158.7	0.6	424.5	70.8	1.2	244.7
Tyre tracks	28	Tyre track	Cube	24.0	1.2	244.7	0.0	-175.0	0.0	-2072.2	0.6	180.4	45.1	1.2	245.9
Tyre tracks	29	Tyre track	Cube	24.0	1.2	244.7	0.0	-175.0	0.0	-2191.7	0.6	190.8	45.1	1.2	245.9
Tyre tracks	30	Tyre track	Cube	24.0	1.2	244.7	0.0	178.8	0.0	-2079.6	0.6	-42.7	29.0	1.2	245.1
Tyre tracks	31	Tyre track	Cube	24.0	1.2	244.7	0.0	178.8	0.0	-2199.5	0.6	-45.1	29.0	1.2	245.1
Tyre tracks	32	Tyre track	Cube	24.0	1.2	244.7	0.0	172.7	0.0	-2063.0	0.6	-265.2	55.0	1.2	245.8
Tyre tracks	33	Tyre track	Cube	24.0	1.2	244.7	0.0	172.7	0.0	-2182.0	0.6	-280.5	55.0	1.2	245.8
Tyre tracks	34	Tyre track	Cube	24.0	1.2	244.7	0.0	166.5	0.0	-2022.7	0.6	-484.7	80.4	1.2	243.6
Tyre tracks	35	Tyre track	Cube	24.0	1.2	244.7	0.0	166.5	0.0	-2139.4	0.6	-512.6	80.4	1.2	243.6
Tyre tracks	36	Tyre track	Cube	24.0	1.2	244.7	0.0	160.4	0.0	-1959.2	0.6	-698.6	104.8	1.2	238.5
Tyre tracks	37	Tyre track	Cube	24.0	1.2	244.7	0.0	160.4	0.0	-2072.2	0.6	-738.9	104.8	1.2	238.5
Tyre tracks	38	Tyre track	Cube	24.0	1.2	244.7	0.0	154.2	0.0	-1873.1	0.6	-904.5	128.0	1.2	230.8
Tyre tracks	39	Tyre track	Cube	24.0	1.2	244.7	0.0	154.2	0.0	-1981.1	0.6	-956.6	128.0	1.2	230.8
Tyre tracks	40	Tyre track	Cube	24.0	1.2	244.7	0.0	148.1	0.0	-1765.4	0.6	-1099.9	149.8	1.2	220.4
Tyre tracks	41	Tyre track	Cube	24.0	1.2	244.7	0.0	148.1	0.0	-1867.2	0.6	-1163.4	149.8	1.2	220.4
Tyre tracks	42	Tyre track	Cube	24.0	1.2	239.1	0.0	-85.0	0.0	181.3	0.6	2072.1	240.3	1.2	44.7
Tyre tracks	43	Tyre track	Cube	24.0	1.2	239.1	0.0	-85.0	0.0	191.7	0.6	2191.6	240.3	1.2	44.7
Tyre tracks	44	Tyre track	Cube	24.0	1.2	239.1	0.0	-79.0	0.0	396.9	0.6	2041.8	239.3	1.2	69.2
Tyre tracks	45	Tyre track	Cube	24.0	1.2	239.1	0.0	-79.0	0.0	419.8	0.6	2159.6	239.3	1.2	69.2
Tyre tracks	46	Tyre track	Cube	24.0	1.2	239.1	0.0	-73.0	0.0	608.1	0.6	1989.1	235.7	1.2	92.9
Tyre tracks	47	Tyre track	Cube	24.0	1.2	239.1	0.0	-73.0	0.0	643.2	0.6	2103.9	235.7	1.2	92.9
Tyre tracks	48	Tyre track	Cube	24.0	1.2	239.1	0.0	-67.0	0.0	812.7	0.6	1914.6	229.5	1.2	115.5
Tyre tracks	49	Tyre track	Cube	24.0	1.2	239.1	0.0	-67.0	0.0	859.6	0.6	2025.1	229.5	1.2	115.5
Tyre tracks	50	Tyre track	Cube	24.0	1.2	239.1	0.0	-61.0	0.0	1008.4	0.6	1819.2	220.8	1.2	136.9

Tyre tracks	51	Tyre track	Cube	24.0	1.2	239.1	0.0	-61.0	0.0	1066.6	0.6	1924.2	220.8	1.2	136.9
Tank spot marking	2	Spot line	Cube	600.0	1.2	12.0	0.0	0.0	0.0	0.0	1.2	450.0	600.0	1.2	12.0
Tank spot marking	3	Spot line	Cube	600.0	1.2	12.0	0.0	0.0	0.0	0.0	1.2	-450.0	600.0	1.2	12.0
Tank spot marking	4	Spot line	Cube	12.0	1.2	900.0	0.0	0.0	0.0	300.0	1.2	0.0	12.0	1.2	900.0
Tank spot marking	5	Spot line	Cube	12.0	1.2	900.0	0.0	0.0	0.0	-300.0	1.2	0.0	12.0	1.2	900.0

components	2804

World_Coordinates: every component of the scene (m, degrees), at t = 0

Legend: yellow = input from the code; white = formula.

Source: scene export: components[]. S, R, T are the part's own scale, rotation (pitch x, yaw y, roll z) and translation in its parent's frame; M = T Ry Rx Rz S; world = parent world x M. Centre = last column of the world matrix; extent = box of the unit primitive under the world matrix.

drawn parts (not groups)	5474	*scene export says*	5474	agree
groups	595			
components	6069			

id	object	design	component	primitive	parent	depth	S x	S y	S z	R x	R y	R z	T x	T y	T z
TER-00	Terrain	Terrain disc	Terrain	Group		0	1.000	1.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000
TER-01	Terrain	Terrain disc	Ground disc	Terrain disc	TER-00	1	1.000	1.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000
BKG-00	Backdrop		Backdrop	Group		0	1.000	1.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000
BKG-01	Backdrop	Far ground	Far ground	Cube	BKG-00	1	400.000	176.000	0.100	90.000	0.000	0.000	0.000	-0.400	-112.000
BKG-02	Backdrop	Far ground	Far ground	Cube	BKG-00	1	176.000	48.000	0.100	90.000	0.000	0.000	-112.000	-0.400	0.000
BKG-03	Backdrop	Far ground	Far ground	Cube	BKG-00	1	400.000	176.000	0.100	90.000	0.000	0.000	0.000	-0.400	112.000
BKG-04	Backdrop	Far ground	Far ground	Cube	BKG-00	1	176.000	48.000	0.100	90.000	0.000	0.000	112.000	-0.400	0.000
BKG-05	Backdrop	Distant hills	Distant hills	Group	BKG-00	1	1.000	1.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000
BKG-06	Backdrop	Distant hills	Hill	Sphere	BKG-05	2	44.252	12.068	26.000	0.000	-86.379	0.000	56.040	-0.650	-3.547
BKG-07	Backdrop	Distant hills	Hill	Sphere	BKG-05	2	42.778	14.055	26.000	0.000	-112.501	0.000	52.889	-0.650	21.908
BKG-08	Backdrop	Distant hills	Hill	Sphere	BKG-05	2	54.546	11.315	26.000	0.000	-155.650	0.000	23.631	-0.650	52.216
BKG-09	Backdrop	Distant hills	Hill	Sphere	BKG-05	2	42.780	15.808	26.000	0.000	-177.910	0.000	2.122	-0.650	58.133
BKG-10	Backdrop	Distant hills	Hill	Sphere	BKG-05	2	46.240	15.299	26.000	0.000	148.981	0.000	-34.419	-0.650	57.240
BKG-11	Backdrop	Distant hills	Hill	Sphere	BKG-05	2	47.432	11.624	26.000	0.000	119.035	0.000	-49.873	-0.650	27.685
BKG-12	Backdrop	Distant hills	Hill	Sphere	BKG-05	2	42.129	10.148	26.000	0.000	91.600	0.000	-56.795	-0.650	1.587
BKG-13	Backdrop	Distant hills	Hill	Sphere	BKG-05	2	46.132	10.552	26.000	0.000	67.859	0.000	-51.123	-0.650	-20.802
BKG-14	Backdrop	Distant hills	Hill	Sphere	BKG-05	2	49.657	11.056	26.000	0.000	25.006	0.000	-24.498	-0.650	-52.522
BKG-15	Backdrop	Distant hills	Hill	Sphere	BKG-05	2	42.267	16.217	26.000	0.000	-3.156	0.000	3.776	-0.650	-68.493
BKG-16	Backdrop	Distant hills	Hill	Sphere	BKG-05	2	55.751	16.423	26.000	0.000	-26.446	0.000	27.028	-0.650	-54.339
BKG-17	Backdrop	Distant hills	Hill	Sphere	BKG-05	2	50.857	13.206	26.000	0.000	-62.618	0.000	49.877	-0.650	-25.833
BKG-18	Backdrop	Distant hills	Far hill	Sphere	BKG-05	2	93.802	35.250	40.000	0.000	-104.997	0.000	94.627	-0.650	25.350
BKG-19	Backdrop	Distant hills	Far hill	Sphere	BKG-05	2	98.044	33.917	40.000	0.000	-154.692	0.000	38.055	-0.650	80.477
BKG-20	Backdrop	Distant hills	Far hill	Sphere	BKG-05	2	90.225	25.312	40.000	0.000	177.355	0.000	-4.090	-0.650	88.538
BKG-21	Backdrop	Distant hills	Far hill	Sphere	BKG-05	2	73.158	34.475	40.000	0.000	123.252	0.000	-71.616	-0.650	46.957
BKG-22	Backdrop	Distant hills	Far hill	Sphere	BKG-05	2	93.222	24.914	40.000	0.000	82.702	0.000	-86.174	-0.650	-11.036
BKG-23	Backdrop	Distant hills	Far hill	Sphere	BKG-05	2	76.835	40.761	40.000	0.000	53.979	0.000	-71.184	-0.650	-51.757
BKG-24	Backdrop	Distant hills	Far hill	Sphere	BKG-05	2	75.592	40.651	40.000	0.000	10.069	0.000	-17.054	-0.650	-96.043
BKG-25	Backdrop	Distant hills	Far hill	Sphere	BKG-05	2	89.907	28.667	40.000	0.000	-23.266	0.000	35.034	-0.650	-81.482
BKG-26	Backdrop	Distant hills	Far hill	Sphere	BKG-05	2	88.500	32.642	40.000	0.000	-74.745	0.000	84.345	-0.650	-23.003
BKG-27	Backdrop		Treeline	Group	BKG-00	1	1.000	1.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000
BKG-28	Backdrop	Treeline pine	Treeline pine	Group	BKG-27	2	1.000	1.000	1.000	0.000	186.124	0.000	39.108	-0.350	-1.164
BKG-29	Backdrop	Treeline pine	Pine crown	Cone	BKG-28	3	4.517	4.517	5.001	-90.000	0.000	0.000	0.000	2.500	0.000
BKG-30	Backdrop	Treeline pine	Pine crown	Cone	BKG-28	3	2.936	2.936	4.436	-90.000	30.000	0.000	0.000	5.847	0.000
BKG-31	Backdrop	Treeline pine	Treeline pine	Group	BKG-27	2	1.000	1.000	1.000	0.000	236.187	0.000	41.445	-0.350	6.828
BKG-32	Backdrop	Treeline pine	Pine crown	Cone	BKG-31	3	5.174	5.174	5.728	-90.000	0.000	0.000	0.000	2.864	0.000
BKG-33	Backdrop	Treeline pine	Pine crown	Cone	BKG-31	3	3.363	3.363	5.081	-90.000	30.000	0.000	0.000	6.698	0.000
BKG-34	Backdrop	Treeline pine	Treeline pine	Group	BKG-27	2	1.000	1.000	1.000	0.000	161.266	0.000	37.783	-0.350	8.617
BKG-35	Backdrop	Treeline pine	Pine crown	Cone	BKG-34	3	3.886	3.886	4.302	-90.000	0.000	0.000	0.000	2.151	0.000
BKG-36	Backdrop	Treeline pine	Pine crown	Cone	BKG-34	3	2.526	2.526	3.817	-90.000	30.000	0.000	0.000	5.031	0.000
BKG-37	Backdrop	Treeline pine	Treeline pine	Group	BKG-27	2	1.000	1.000	1.000	0.000	206.361	0.000	38.339	-0.350	17.181
BKG-38	Backdrop	Treeline pine	Pine crown	Cone	BKG-37	3	5.415	5.415	5.996	-90.000	0.000	0.000	0.000	2.998	0.000

Layout: sites, things that can be operated, lights and the walk space (m)

Legend: yellow = input from the code; white = formula.

Source: scene export: layout (src/objects/BaseLayout.h), controllables, lights, practical_lights, walkspace

A. Sites of the base

name	value
base_radius	36
perimeter_radius	34
apron_level	-0.300
bunker_floor_y	-3.400
bunker_ceiling_y	-0.800
bunker_roof_top_y	-0.450
ramp_half_width	1.500
ramp_top_z	13
ramp_bottom_z	6
blast_door_z	4
bunker_min_x	-5.500
bunker_max_x	5.500
bunker_min_z	-8.500
bunker_max_z	4
outer_gate.x	0
outer_gate.y	-0.300
outer_gate.z	34
helipad.x	20.300
helipad.y	-0.300
helipad.z	17
helipad_radius	5
radar_site.x	29
radar_site.y	-0.300
radar_site.z	-3
aa_gun_site.x	21
aa_gun_site.y	-0.300
aa_gun_site.z	-19
motor_pool.x	-22
motor_pool.y	-0.300
motor_pool.z	-16
barracks.x	-27
barracks.y	-0.300
barracks.z	11
flagpole.x	-4.500
flagpole.y	-0.300
flagpole.z	27
fuel_depot.x	-6
fuel_depot.y	-0.300
fuel_depot.z	-28.500
road_half_width	2.200
rover_count	3

B. Things that can be operated

name	object	x	y	z	parts	can be controlled	key	controls
UAV-1 DRONE	Autonomous UAV	0.00	7.50	12.50	79	yes	0	W/S fly A/D yaw PGUP/PGDN climb ARROWS gimbal ENTER launch \ gim
MBT-1 TANK	Main Battle Tank	-22.00	-0.30	-16.00	255	yes	5	W/S drive A/D steer Shift boost arrows/mouse turret+gun ENTER fire C s
MH-6 HELICOPTER	Scout Helicopter	20.30	1.12	17.00	91	yes	6	W/S fly A/D turn PGUP/PGDN climb SHIFT boost ARROWS aim searchligh
AA-1 GATLING	AA Gatling Gun	21.00	-0.30	-19.00	78	yes	7	arrows/mouse aim ENTER/Ctrl hold to fire C sight
AIR-DEFENCE RADAR	Air-defence radar	29.00	5.60	-3.00	108	yes	8	ARROWS/mouse aim (sector centre / azimuth, tilt) \ mode ENTER transmit
BLAST DOORS	Command Bunker	0.00	-2.10	3.83	32	yes	9	ENTER or \ open / close the blast doors (about 4 s; they also open for soldi
SAM-1 LAUNCHER	SAM launcher	25.10	-0.30	6.70	112	yes	-	\ stow/ready arrows/mouse turntable + elevation ENTER launch (reload 8
MLRS-1 ROCKET TRUCK	MLRS rocket truck	-11.50	-0.30	22.50	136	yes	-	\ stow/ready arrows/mouse mount + elevation ENTER ripple fire (12 tubes
MBT-2 (PARKED)	Main Battle Tank 2	9.50	-0.30	-23.80	255	no	-	(parked in its revetment)
MBT-3 (PARKED)	Main Battle Tank 3	-25.00	-0.30	-3.50	255	no	-	(parked in its revetment)
AMMO BUNKER	Ammunition bunker	10.50	-0.30	26.20	43	no	-	(static scenery)
CAMP NETS	Camp camouflage	25.10	0.00	6.70	640	no	-	(static scenery)
GATE BARRIER	Boom barrier	2.90	0.75	31.60	20	yes	-	ENTER / \ raise or lower the boom
WINDSOCK	Windsock	12.50	5.70	22.50	15	no	-	(scenery)
FLAG	Flag	-4.50	8.60	27.00	47	no	-	(scenery)
SOLDIER 1 SENTRY	Soldier 1	-0.80	-0.30	21.00	75	yes	-	WASD move Shift sprint Space jump Alt crouch mouse look LMB fire RM
SOLDIER 2 WATCH	Soldier 2	5.20	-0.22	14.00	75	yes	-	WASD move Shift sprint Space jump Alt crouch mouse look LMB fire RM
SOLDIER 3 GUNNER	Soldier 3	-10.00	0.07	10.50	75	yes	-	WASD move Shift sprint Space jump Alt crouch mouse look LMB fire RM
SOLDIER 4 MECHANIC	Soldier 4	-4.00	-3.40	0.30	75	yes	-	WASD move Shift sprint Space jump Alt crouch mouse look LMB fire RM
SOLDIER 5 LOGISTICS	Soldier 5	-3.00	-3.40	1.65	75	yes	-	WASD move Shift sprint Space jump Alt crouch mouse look LMB fire RM
SOLDIER 6 TECHNICIAN	Soldier 6	0.00	-3.40	-5.25	75	yes	-	WASD move Shift sprint Space jump Alt crouch mouse look LMB fire RM
SOLDIER 7 PATROL	Soldier 7	28.50	-0.30	-10.40	75	yes	-	WASD move Shift sprint Space jump Alt crouch mouse look LMB fire RM
SOLDIER 8 RUNNER	Soldier 8	-4.00	-3.40	2.50	75	yes	-	WASD move Shift sprint Space jump Alt crouch mouse look LMB fire RM
SKY / TIME OF DAY	Sky Clock	-77.40	-275.60	89.75	1	yes	-	W/S scrub the clock \ auto / manual ENTER next phase
BARRACKS DOOR	Barracks hall	23.50	0.82	-8.64	27	yes	-	ENTER or \ open / close the door (E on foot)
GENERATOR ROOM DOOR	Command Bunker	1.77	-2.40	1.21	27	yes	-	ENTER or \ open / close the door (E on foot)
that can be controlled						20		

C. Scene lights

name	value
moon.direction.x	-0.560
moon.direction.y	-0.387
moon.direction.z	0.733
moon.color.x	0.550
moon.color.y	0.620
moon.color.z	0.800
moon.intensity	0.950
beacon.position.x	-3.600
beacon.position.y	4.299
beacon.position.z	1.600
beacon.color.x	1
beacon.color.y	0.250
beacon.color.z	0.100
beacon.intensity	1.300
beacon.intensity_min	0.600
beacon.intensity_max	2.000
beacon.linear	0.350
beacon.quadratic	0.450

spot.position.x	0.000
spot.position.y	7.356
spot.position.z	12.370
spot.direction.x	0.000
spot.direction.y	-0.766
spot.direction.z	-0.643
spot.color.x	1
spot.color.y	0.950
spot.color.z	0.820
spot.intensity	5
spot.inner_deg	9
spot.outer_deg	14
spot.linear	0.050
spot.quadratic	0.030
ambient.x	0.060
ambient.y	0.065
ambient.z	0.085

D. Practical lights (lamps)

name	object	x	y	z	red	green	blue	intensity	range (m)	cone (deg)
Aviation light	Backdrop	-76.60	10.65	64.28	1.00	0.08	0.05	1.60	4.00	0.00
Aviation light	Backdrop	-76.60	21.65	64.28	1.00	0.08	0.05	1.60	4.00	0.00
Aviation light	Backdrop	-76.60	41.75	64.28	1.00	0.08	0.05	1.60	4.00	0.00
Aviation light	Environment	-13.00	9.05	6.00	1.00	0.08	0.05	1.60	4.00	0.00
Aviation light	Environment	-13.00	4.60	6.00	1.00	0.08	0.05	1.60	4.00	0.00
Dim red lamp	Environment	3.18	2.80	13.00	1.00	0.20	0.08	1.10	6.00	50.00
Fire embers	Environment	5.30	0.59	15.35	1.00	0.42	0.10	1.50	5.00	0.00
Landing floodlight	Command Bunker	1.15	-0.85	5.60	1.00	0.12	0.05	1.00	6.00	55.00
Wall lamp	Command Bunker	-1.30	-1.22	5.00	1.00	0.14	0.05	0.90	4.00	0.00
Wall lamp	Command Bunker	1.30	-0.47	7.60	1.00	0.14	0.05	0.90	4.00	0.00
Wall lamp	Command Bunker	-1.30	-0.53	10.20	1.00	0.14	0.05	0.90	4.00	0.00
Wall lamp	Command Bunker	1.30	-1.22	5.00	1.00	0.14	0.05	0.90	4.00	0.00
Gate lamp	Command Bunker	-0.75	-0.65	4.52	1.00	0.12	0.05	1.00	6.00	55.00
Gate lamp	Command Bunker	0.75	-0.65	4.52	1.00	0.12	0.05	1.00	6.00	55.00
Beacon lamp A	Command Bunker	-1.00	-0.54	4.10	1.00	0.55	0.05	3.00	7.00	0.00
Beacon lamp A	Command Bunker	1.10	-0.54	4.00	1.00	0.55	0.05	3.00	7.00	0.00
Status lamp	Command Bunker	-1.40	-1.70	4.09	1.00	0.05	0.03	0.70	3.50	0.00
Tunnel tube (light)	Command Bunker	0.00	-0.87	1.00	0.85	0.95	1.00	3.20	9.00	0.00
Tunnel tube (light)	Command Bunker	0.00	-0.87	-0.90	0.85	0.95	1.00	3.20	9.00	0.00
Emergency lamp	Command Bunker	1.35	-1.25	1.00	1.00	0.05	0.03	1.40	4.50	0.00
CIC tube (light)	Command Bunker	-3.20	-0.87	-3.20	0.85	0.95	1.00	3.20	9.00	0.00
CIC tube (light)	Command Bunker	3.20	-0.87	-7.40	0.85	0.95	1.00	3.20	9.00	0.00
CIC tube (light)	Command Bunker	0.00	-0.87	-5.40	0.85	0.95	1.00	3.20	9.00	0.00
Emergency lamp	Command Bunker	-5.05	-1.25	-2.40	1.00	0.05	0.03	1.40	4.50	0.00
Task lamp	Command Bunker	3.85	-2.26	-3.60	1.00	0.60	0.22	1.60	4.50	0.00
Task lamp	Command Bunker	-2.55	-2.26	-5.80	1.00	0.60	0.22	1.60	4.50	0.00
Bunk room tube (light)	Command Bunker	-3.50	-0.87	1.00	0.85	0.95	1.00	3.20	9.00	0.00
Plant room tube (light)	Command Bunker	3.50	-0.87	0.60	0.85	0.95	1.00	3.20	9.00	0.00
Strobe	Autonomous UAV	0.00	7.52	12.64	1.00	1.00	1.00	2.00	3.50	0.00
Strobe	Autonomous UAV	0.00	7.55	-12.64	1.00	1.00	1.00	2.00	3.50	0.00

Gate lamp	Base	-2.65	4.92	31.60	1.00	0.22	0.08	1.30	8.00	55.00
Perimeter light	Base	15.54	-0.23	17.95	0.08	0.25	1.00	0.50	4.00	0.00
Pad floodlight lens	Base	14.23	3.83	20.92	1.00	0.46	0.20	2.00	13.00	38.00
Floodlight lens	Base	-22.00	5.00	-8.26	1.00	0.50	0.22	2.40	13.00	42.00
Hut door lamp	Base	-20.70	1.83	8.98	1.00	0.10	0.05	0.80	5.00	60.00
Hall door lamp	Barracks hall	23.50	2.30	-8.46	1.00	0.78	0.52	1.20	6.00	0.00
Ceiling tube (light)	Barracks hall	20.60	2.44	-11.80	1.00	0.95	0.86	2.40	7.50	0.00
Ceiling tube (light)	Barracks hall	23.30	2.44	-11.80	1.00	0.95	0.86	2.40	7.50	0.00
Ceiling tube (light)	Barracks hall	26.00	2.44	-11.80	1.00	0.95	0.86	2.40	7.50	0.00
Lit vent	Air-defence radar	23.17	1.60	-2.40	1.00	0.62	0.20	0.80	5.00	0.00
Aviation light	Air-defence radar	29.00	8.75	-3.00	1.00	0.05	0.03	2.50	14.00	0.00
Door lamp	Ammunition bunker	7.75	1.63	26.20	0.05	0.85	0.60	0.60	4.00	60.00
Warning lamp	Boom barrier	2.90	1.00	31.60	1.00	0.15	0.03	1.00	4.50	0.00
lamps	43									

E. Walk space: floors

#		x0	z0	x1	z1	y0	y1
1		-1.50	6.00	1.50	13.00	-3.40	-0.13
2		-1.50	4.00	1.50	6.00	-3.40	-3.40
3		-1.50	3.65	1.50	4.00	-3.40	-3.40
4		-1.50	-1.80	1.50	3.65	-3.40	-3.40
5		-0.90	-2.10	0.90	-1.80	-3.40	-3.40
6		-5.20	-8.20	5.20	-2.10	-3.40	-3.40
7		-5.20	-1.80	-1.80	3.65	-3.40	-3.40
8		-1.80	0.90	-1.50	1.90	-3.40	-3.40
9		1.80	-1.80	5.20	3.65	-3.40	-3.40
10		1.50	0.60	1.80	1.80	-3.40	-3.40
11		4.00	-6.90	4.90	-6.60	-3.40	-3.40
12		3.90	-8.20	5.20	-6.90	-3.40	-3.40
13		18.82	-14.78	27.38	-8.82	-0.18	-0.18
14		22.85	-8.82	24.15	-8.60	-0.18	-0.18

F. Walk space: walls (boxes)

#		min x	min y	min z	max x	max y	max z
1		-1.85	-3.70	3.65	-1.50	0.28	13.45
2		1.50	-3.70	3.65	1.85	0.28	13.45
3		-5.50	-3.40	3.65	-1.30	-0.80	4.00
4		1.30	-3.40	3.65	5.50	-0.80	4.00
5		-5.50	-3.40	-8.50	-5.20	-0.80	4.00
6		5.20	-3.40	-6.60	5.50	-0.80	4.00
7		-5.50	-3.40	-8.50	3.60	-0.80	-8.20
8		-1.80	-3.40	-1.80	-1.50	-0.80	0.90
9		-1.80	-3.40	1.90	-1.50	-0.80	3.65
10		1.50	-3.40	-1.80	1.80	-0.80	0.60
11		1.50	-3.40	1.80	1.80	-0.80	3.65
12		-5.50	-3.40	-2.10	-0.90	-0.80	-1.80
13		0.90	-3.40	-2.10	5.50	-0.80	-1.80
14		3.60	-3.40	-8.50	3.90	0.08	-6.60
15		5.20	-3.40	-8.50	5.50	0.08	-6.60

16	3.90	-3.40	-8.50	5.20	0.08	-8.20
17	3.90	-3.40	-6.90	4.00	0.08	-6.60
18	4.90	-3.40	-6.90	5.20	0.08	-6.60
19	-1.20	-3.40	-7.00	1.20	-2.55	-5.80
20	-4.00	-3.40	-3.75	-2.40	-2.65	-3.05
21	2.40	-3.40	-3.75	4.00	-2.65	-3.05
22	-4.00	-3.40	-5.95	-2.40	-2.65	-5.25
23	2.20	-3.40	-5.75	3.80	-2.65	-5.05
24	-5.20	-3.40	-4.00	-4.60	-1.40	-3.20
25	-5.20	-3.40	-5.00	-4.60	-1.40	-4.20
26	-5.20	-3.40	-6.00	-4.60	-1.40	-5.20
27	-5.15	-3.40	-0.20	-4.25	-1.70	0.80
28	-5.15	-3.40	2.00	-4.25	-1.70	3.00
29	-4.00	-3.40	-1.80	-2.20	-2.10	-1.55
30	-4.90	-3.40	3.15	-3.00	-1.50	3.65
31	-3.50	-3.40	0.50	-2.50	-2.65	1.40
32	2.30	-3.40	2.00	5.00	-3.22	3.30
33	4.70	-3.40	1.00	5.20	-1.50	1.80
34	4.70	-3.40	0.00	5.20	-1.50	0.80
35	2.30	-3.40	-1.50	4.50	-2.05	-0.50
36	3.65	-0.52	-8.45	5.45	0.28	-6.65

Algorithms: every constant of the vision, tracking, squad, flight, intruder, mission and garrison code

Legend: yellow = input from the code; white = formula.

Source: A: constants read from the source files (file and line given). B: the values the running program reports in its scene export.

A. Constants in the source

file	line	name	value	comment in the source
src/ai/DroneController.h	15	kCvWidth	320	320 x 240 LWIR core (e.g. FLIR Boson 320 class)
src/ai/DroneController.h	16	kCvHeight	240	
src/ai/DroneController.h	17	kCvFovYDeg	40.0	
src/ai/DroneController.h	40	kLockTime	1.5	s of continuous track before a launch
src/ai/DroneController.h	41	kLaunchInterval	4.0	s between rounds
src/ai/DroneController.h	42	kWreckIgnoreRadius	2.5	
src/ai/DroneController.h	42	kWreckIgnoreTime	30.0	
src/ai/DroneController.h	43	kFriendlyRadius	4.5	
src/ai/Squad.h	57	kSquadLeadLock	0.8	
src/ai/Squad.h	57	kSquadWingLock	0.5	
src/ai/Squad.h	57	kSquadSoloLock	1.3	
src/ai/Squad.h	58	kSquadWingWait	2.5	s the lead waits for the wing's lock before a solo strike
src/ai/Squad.h	59	kSquadWingWaitFree	8.0	... when the wing is in the pair (not detached, not away): one intruder gets both rounds
src/ai/Squad.h	63	kSquadSplitDist	14.0	
src/ai/Squad.h	63	kSquadRejoinDist	9.0	
src/ai/Squad.h	66	kSquadSplitThreat	10.0	
src/ai/Squad.h	67	kSquadStandOff	6.5	
src/ai/Squad.h	67	kSquadEngageAlt	5.0	
src/ai/Squad.h	68	kSquadPatrolAlt	7.5	
src/ai/Squad.h	69	kSquadReacquireTime	12.0	s of searching round the last known position
src/ai/Squad.h	70	kSquadBlobHeight	0.2	m above the terrain of a rover's hot body (blob ray stops there)
src/ai/Squad.h	84	kDetourMinGap	15.0	s between investigate detours
src/ai/Squad.h	84	kDetourMaxGap	35.0	s between investigate detours
src/ai/Squad.h	85	kDetourMinTime	6.0	s spent on one
src/ai/Squad.h	85	kDetourMaxTime	9.0	s spent on one
src/ai/Squad.h	99	kPadClimbOut	4.0	m above the pad: the climb-out hands over to the patrol here
src/ai/Squad.h	100	kPadApproachAlt	5.0	m above the pad: hover point before the descent
src/ai/Squad.h	101	kRotorSpoolUp	2.0	s from still to full speed and back
src/ai/Squad.h	101	kRotorSpoolDown	3.0	s from still to full speed and back
src/ai/Squad.h	102	kTakeOffStagger	2.5	s between the two UAVs' take-offs
src/ai/Squad.h	103	kTouchdownSpeed	0.25	m/s
src/ai/Squad.h	103	kMaxDescent	1.4	m/s
src/ai/Squad.h	103	kMaxClimb	1.6	m/s
src/ai/Squad.h	111	kCount	2	
src/ai/HideController.h	21	kCellSize	0.5	
src/ai/HideController.h	22	kGridCells	80	80 x 80 cells over [-20, 20] m
src/ai/HideController.h	23	kIrSensorCount	5	
src/ai/HideController.h	24	kEvadeRadius	5.0	a UAV this close (horizontal) makes an infiltrating rover hide
src/ai/HideController.h	25	kEvadeTime	4.0	s of hiding before infiltrating again
src/ai/HideController.cpp	14	kFreeRadius	18.3	stay inside the berm / perimeter fence (r = 19 m)
src/ai/HideController.cpp	15	kLosBuckets	20	2 m line-of-sight buckets over the grid square
src/ai/HideController.cpp	16	kInflate	0.3	planning margin around obstacles
src/ai/HideController.cpp	17	kBodyRadius	0.15	physical clearance used by the motion check
src/ai/HideController.cpp	18	kSensorForward	0.31	sensor bar distance ahead of the robot centre
src/ai/HideController.cpp	19	kSensorSpacing	0.06	

src/ai/HideController.cpp	20	kLineHalfWidth	0.05	virtual line is 10 cm wide
src/ai/HideController.cpp	21	kMaxSpeed	0.75	m/s
src/ai/HideController.cpp	22	kKp	3.2	PD steering gains
src/ai/HideController.cpp	22	kKd	0.25	PD steering gains
src/ai/HideController.cpp	23	kReplanPeriod	0.4	s
src/ai/HideController.cpp	24	kInfiltrateReplan	1.6	s between infiltration re-plans
src/ai/HideController.cpp	25	kWheelAccel	2.0	m/s^2
src/scene/Mission.h	55	kSpawnJitter	1.2	spawn radius kSpawnRadius +- this
src/scene/Mission.h	56	kNorthGapDeg	18.0	no spawns within this of the gate road (north)
src/scene/Mission.h	57	kViaSectorDeg	70.0	via points within this of the spawn bearing
src/scene/Mission.h	58	kViaMinR	4.0	
src/scene/Mission.h	58	kViaMaxR	15.0	
src/scene/Mission.h	59	kMaxVia	3	
src/scene/Mission.h	60	kSwitchProbability	0.3	
src/scene/Mission.h	61	kSwitchMinS	8.0	
src/scene/Mission.h	61	kSwitchMaxS	22.0	
src/scene/Mission.h	63	kKillScore	100	
src/scene/Mission.h	63	kCombinedBonus	50	
src/scene/Mission.h	64	kBreachDamage	10.0	
src/scene/Mission.h	65	kSpawnInterval	3.5	
src/scene/Mission.h	65	kWaveBreak	6.0	
src/scene/Mission.h	65	kFirstWaveDelay	0.0	
src/scene/Scene.h	74	kWreckBurnTime	15.0	s of fire and smoke from the wreck
src/scene/Scene.h	75	kWreckClearTime	25.0	s until the wreck is removed
src/scene/Scene.h	76	kRespawnDelay	3.0	s between removal and the slot being free again
src/scene/Scene.h	77	kUavReloadTime	6.0	s until a launched rail round is replaced
src/scene/Scene.h	78	kRoverHealth	100.0	armoured intruder
src/scene/Scene.h	79	kRoundDamage	30.0	damage of a power-1 blast at the centre, half at the edge of its radius
src/scene/Scene.h	80	kDroneOwner	0	Combat owner id of UAV 0 (UAV 1 = owner 10)
src/scene/Scene.h	82	kExitRadius	17.5	v6: a withdrawing rover leaves through a free cell at this radius
src/scene/Scene.h	83	kRetreatTimeout	90.0	s: a rover that cannot reach its exit is withdrawn anyway
src/scene/Scene.h	278	kMaxRiflemen	32	
src/scene/Scene.h	279	kRifleRange	26.0	m: guards engage tracked intruders closer than this
src/scene/Scene.h	289	kRoverDriveSpeed	1.1	m/s, m/s, rad/s
src/scene/Scene.h	289	kRoverBoostSpeed	1.8	m/s, m/s, rad/s
src/scene/Scene.h	289	kRoverTurnRate	2.2	m/s, m/s, rad/s
src/scene/Garrison.h	45	kDayShiftStart	6.4	
src/scene/Garrison.h	45	kDayShiftEnd	17.6	
src/scene/Garrison.h	50	kCells	72	
src/scene/Garrison.h	51	kHalf	36.0	
src/scene/Garrison.h	51	kCell	1.0	
src/scene/Garrison.h	52	kPlanClearance	0.4	free cell: centre this far from every obstacle
src/scene/Garrison.h	53	kBodyClearance	0.28	every spline sample at least this far
src/scene/Garrison.h	54	kEdgeClearance	0.36	A* edge (centre to centre) clearance
src/scene/Garrison.h	55	kCap	1.5	clearance values are capped here
src/scene/Garrison.h	205	kSoldiers	8	the duty roster (indices 0..7); v8 residents follow from index 8
src/scene/Garrison.h	206	kMaxResidents	6	
src/scene/Combat.h	53	kMissileMaxSpeed	60.0	
src/scene/Combat.h	53	kMissileRampTime	0.4	
src/scene/Combat.h	53	kMissileLife	6.0	
src/scene/Combat.h	54	kMissileTurnRate	6.0	rad/s, limited manoeuvrability

src/scene/Combat.h	55	kMissileFuse	0.5	proximity fuse radius, m
src/scene/Combat.h	56	kShellSpeed	120.0	
src/scene/Combat.h	56	kTracerSpeed	1000.0	
src/scene/Combat.h	56	kTracerLife	2.5	
src/scene/Combat.h	57	kDamageRadiusPerPower	1.6	blast damage radius = 1.6 * power (m)
src/scene/Combat.h	58	kShakeRange	25.0	
src/scene/Combat.h	58	kShakeDecay	0.8	
src/scene/DayCycle.h	35	kNightBelowDeg	-6.0	
src/scene/DayCycle.h	35	kDayAboveDeg	4.0	
src/scene/DayCycle.h	35	kPhaseHysteresisDeg	0.25	
src/scene/DayCycle.h	36	kBeamsBelowDeg	-2.0	
src/scene/DayCycle.h	39	kSunIntensity	3.2	
src/scene/DayCycle.h	39	kSunKeyAboveDeg	-1.0	
src/scene/DayCycle.h	42	kDayExposureKeyLog2	-2.8	mean log2 luminance of the reference noon frame
src/scene/DayCycle.h	106	kSolarGain	0.12	
src/scene/DayCycle.h	106	kPassiveLo	0.55	
src/scene/DayCycle.h	106	kPassiveHi	0.75	
src/scene/DayCycle.h	112	kCloudAltitude	120.0	
src/scene/DayCycle.h	112	kCloudCoverage	0.45	
src/scene/DayCycle.h	114	kCloudWindX	3.2	drift, m/s
src/scene/DayCycle.h	114	kCloudWindZ	1.4	drift, m/s
src/scene/DayCycle.h	115	kCloudLo	0.395	fBm -> density soft threshold (coverage ~0.45)
src/scene/DayCycle.h	115	kCloudHi	0.545	fBm -> density soft threshold (coverage ~0.45)
src/scene/DayCycle.h	116	kCloudOpacity	0.6	sunlight removed under the thickest cloud
src/scene/Gait.h	17	kAnkleHeight	0.08	ankle joint above the sole (m)

B. Values reported by the program (scene export)

group		name	value
v5.flight_model		launch_speed	8
v5.flight_model		accel	130
v5.flight_model		ramp_time	0.4
v5.flight_model		cruise_speed	60
v5.flight_model		fuse_m	0.5
v5.flight_model		source	Squad.cpp roundFlightTime comment
v6.day_cycle.config		latitude_deg	22.8
v6.day_cycle.config		declination_deg	-4.6
v6.day_cycle.config		day_sec_per_hour	10
v6.day_cycle.config		night_sec_per_hour	13.3
v6.day_cycle.config		twilight_slow	1.5
v6.day_cycle.config		scrub_hours_per_sec	2
v6.day_cycle.config		shadow_quant_deg	1
v6.day_cycle.config		cycle_seconds	290.607
v6.day_cycle.config		night_below_deg	-6
v6.day_cycle.config		day_above_deg	4
v6.day_cycle.config		beams_below_deg	-2
v6.day_cycle.config		sun_intensity	3.2
v6.day_cycle.config		day_exposure_key_log2	-2.8
v6.day_cycle.config		solar_gain	0.12
v6.day_cycle.config		cloud_altitude_m	120
v6.day_cycle.config		cloud_coverage	0.45

v6.day_cycle.config	sunrise	6.1292
v6.day_cycle.config	sunset	17.8708
terrain	radius	36
terrain	compound_radius	20
terrain	rings	130
terrain	slices	384

Film: shot list of the full-length film and the numbers it states

Legend: yellow = input from the code; white = formula.

Source: report/data/film.json, written by tools/film/build.py plan from tools/film/acts/.py (narration lengths set the times)*

length (s)	2375.8	39.6	*minutes*
beats	150		

start (s)	act	chapter	picture	source	title on screen	length (s)
0.0		ThermalSense	card	Hi, this is Shakhoyat		18.2
18.2		ThermalSense	clip	f3:f_night_wide		7.9
26.1		ThermalSense	clip	f3:f_night_follow		7.3
33.4		ThermalSense	clip	f3:f_tank2		6.5
39.8		ThermalSense	card	ThermalSense		14.0
53.8		ThermalSense	plate	workbook	The workbook behind the project	20.3
74.1	Orientation	The outpost	plate	map	The outpost, from above	19.8
94.0	Orientation	The outpost	live	tour0	An open world: fly anywhere	13.0
107.0	Orientation	The outpost	live	tour0	An open world: fly anywhere	7.1
114.1	Orientation	The outpost	live	tour0	Menu and help	12.6
126.7	Orientation	The outpost	plate	roadmap	The plan of the video	32.6
159.3	Geometry	Primitives: from a triangle to sixteen shapes	card	Part 1 | The graphics lab		7.4
166.6	Geometry	Primitives: from a triangle to sixteen shapes	plate	triangle	Vertices, indices, triangles	28.2
194.8	Geometry	Primitives: from a triangle to sixteen shapes	plate	sphere	Tessellation: how round is round enough	25.0
219.9	Geometry	Primitives: from a triangle to sixteen shapes	plate	primgrid	The sixteen primitives	30.3
250.1	Geometry	Primitives: from a triangle to sixteen shapes	live	prims	F: which primitive made each part	17.0
267.1	Geometry	Primitives: from a triangle to sixteen shapes	live	prims	G: explode into parts	7.0
274.1	Geometry	Primitives: from a triangle to sixteen shapes	live	prims	X: wireframe	8.0
282.1	Geometry	Primitives: from a triangle to sixteen shapes	still	pictures	A drone, part by part	13.7
295.8	Geometry	Transformations and the scene graph	plate	trs	M = T R S, and why the order matters	27.3
323.2	Geometry	Transformations and the scene graph	plate	graph	The scene graph: parent times child	21.7
344.8	Geometry	Transformations and the scene graph	live	graph	H: the axes of the nodes	11.6
356.4	Geometry	Transformations and the scene graph	live	graph	A hierarchy in motion	7.8
364.2	Geometry	Transformations and the scene graph	live	graph	A soldier: 75 parts, 7 levels	13.1
377.3	Viewing	Viewing and projection	plate	lookat	The view matrix	26.0
403.3	Viewing	Viewing and projection	plate	project	Perspective: divide by the distance	25.1
428.5	Viewing	Viewing and projection	live	proj	Ctrl+P: the projection	9.7
438.1	Viewing	Viewing and projection	live	proj	Ctrl+P: the projection	21.9
460.1	Viewing	Clipping and hidden surfaces	plate	clipping	Clipping a line to the window	33.5
493.6	Viewing	Clipping and hidden surfaces	plate	polyclip	Clipping a polygon, culling an object	23.9
517.5	Viewing	Clipping and hidden surfaces	plate	zbuffer	The depth buffer	17.0
534.5	Viewing	Clipping and hidden surfaces	live	depth	Z: the depth test	11.7
546.2	Viewing	Clipping and hidden surfaces	live	depth	F: the depth buffer	7.6
553.8	Viewing	Rasterisation and colour interpolation	plate	raster	Edge functions fill a triangle	24.0
577.8	Viewing	Rasterisation and colour interpolation	still	pictures	Colour interpolation, with the correction for perspective	19.5
597.3	Viewing	Rasterisation and colour interpolation	live	explorer	Frame Explorer: one frame in 16 stages	13.6
610.9	Viewing	Rasterisation and colour interpolation	live	explorer	Frame Explorer: rasterisation	8.0
618.9	Viewing	Rasterisation and colour interpolation	live	explorer	Frame Explorer: depth, shadow, fog, tone curve	17.0
635.9	Viewing	Rasterisation and colour interpolation	still	pictures	Frame Explorer: image laboratory	10.4
646.3	Light	Illumination	plate	phong	Ambient + diffuse + specular	38.4
684.7	Light	Illumination	plate	tsmath	The vector mathematics, written by hand	18.1
702.8	Light	Illumination	live	terms	Ctrl+L: one term at a time	6.4

709.2	Light	Illumination	live	terms	Ctrl+L: one term at a time	10.0
719.2	Light	Illumination	live	lights	Keys 1 to 4: the lights	20.0
739.2	Light	Shading: flat, Gouraud, Phong	still	pictures	Flat, Gouraud, Phong	22.3
761.5	Light	Shading: flat, Gouraud, Phong	live	shade	O: the shading model	16.5
778.0	Light	Texture mapping	plate	texture	Procedural texture: layers of noise	24.7
802.7	Light	Texture mapping	still	pictures	Relief without more triangles	10.0
812.7	Light	Ray tracing	plate	rays	Tracing a ray	35.7
848.4	Light	Ray tracing	live	rt	; the render mode	6.4
854.8	Light	Ray tracing	live	rt	; the render mode	17.3
872.1	Light	Ray tracing	live	rt	' the ray debugger	9.0
881.1	Rendering	Materials that behave like the real thing	card	Part 2 | Beyond the lab		11.6
892.7	Rendering	Materials that behave like the real thing	still	pictures	From Blinn-Phong to a microfacet model	23.8
916.5	Rendering	Materials that behave like the real thing	live	pbr	B: Blinn-Phong or microfacet	11.8
928.4	Rendering	Shadows and ambient occlusion	plate	shadowmap	Shadow maps with soft edges	27.6
956.0	Rendering	Shadows and ambient occlusion	live	shadows	Shadows and ambient occlusion	11.6
967.6	Rendering	The camera as an eye	plate	tonemap	From light to a pixel value	18.5
986.1	Rendering	The camera as an eye	live	air	Fog, bloom, and the first renderer	15.9
1001.9	Rendering	The camera as an eye	plate	taa	Temporal anti-aliasing	29.9
1031.8	Rendering	The camera as an eye	still	pictures	Screen-space reflections against traced ones	22.7
1054.6	Rendering	Scale and speed	plate	instancing	One blade, drawn 450,000 times	36.9
1091.5	Rendering	Scale and speed	live	veg	F5: quality LOW, HIGH, ULTRA, EPIC	15.9
1107.4	Rendering	Scale and speed	plate	bench	Frame time per preset	7.3
1114.8	World	Day, night and weather	plate	sun	One clock drives the light	28.9
1143.6	World	Day, night and weather	live	daylapse	A day in 20 seconds	21.0
1164.6	World	Day, night and weather	live	weather	R: rain J: fireflies	13.0
1177.6	World	Sound synthesis	plate	sound	Sound synthesis	30.2
1207.9	Control	Buildings you can walk into	card	Part 3 | Take control		8.8
1216.6	Control	Buildings you can walk into	plate	walkspace	The walk space	16.0
1232.6	Control	Buildings you can walk into	clip	f3:f_bunker	A soldier under my control	7.0
1239.6	Control	Buildings you can walk into	clip	f3:f_bunker	Tunnel and command room	6.5
1246.1	Control	Buildings you can walk into	clip	f3:f_gen	E: open a door	6.2
1252.4	Control	Buildings you can walk into	clip	f3:f_hall	The barracks hall	7.5
1259.9	Control	Buildings you can walk into	clip	walk	W A S D: walk Shift: sprint V: first or third person	15.8
1275.6	Control	Take control of anything	plate	control	Select, control, sight, release	22.2
1297.8	Control	Take control of anything	clip	f3:f_tank2	The tank	5.5
1303.3	Control	Take control of anything	clip	f3:f_tank2	Drive: W A S D aim: arrows	8.5
1311.8	Control	Take control of anything	clip	f3:f_tank2	V: gunner's sight Enter: fire	7.4
1319.2	Control	Take control of anything	clip	f3:f_aa	Anti-aircraft gun	5.1
1324.4	Control	Take control of anything	clip	f3:f_sam	Missile launcher	4.4
1328.8	Control	Take control of anything	live	operate	8: the radar	9.2
1337.9	Control	Take control of anything	live	operate	The gate barrier	7.0
1344.9	Control	Take control of anything	live	operate	9: the blast doors	7.5
1352.4	Control	Take control of anything	live	heli	The helicopter	17.5
1369.9	Control	Take control of anything	live	uavfly	A drone: chase, nose camera, thermal camera	17.0
1386.9	Control	Take control of anything	live	rovdrive	Driving an intruder	10.0
1396.9	Control	Jump into anything that moves	plate	picking	A click becomes a ray	24.5
1421.5	Control	Jump into anything that moves	live	povjump	Click a soldier	7.0
1428.5	Control	Jump into anything that moves	live	povjump	Through his eyes	7.5
1436.0	Control	Jump into anything that moves	live	povjump	V: third person	6.0
1442.0	Control	Jump into anything that moves	live	povjump	F4: the list of points of view	8.0
1450.0	Control	Jump into anything that moves	live	povjump	Insert / Delete, or the Travel menu: any point of view	10.7

1460.6	Control	Jump into anything that moves	clip	v7a	K: the cinematic director	10.8
1471.4	Control	Jump into anything that moves	clip	v7a	Riding a round	6.8
1478.2	Control	Jump into anything that moves	clip	v7a	Into an intruder	5.8
1484.0	Control	Jump into anything that moves	clip	v7a	A sentry, first person	6.4
1490.4	Control	The garrison's day	plate	astar	A soldier finds his way	25.5
1515.9	Control	The garrison's day	plate	gait	Walking: the foot decides, the knee follows	22.7
1538.6	Control	The garrison's day	live	garrison	The day shift	13.0
1551.6	Control	The garrison's day	clip	f3:f_al_sam	On alert	6.7
1558.4	Control	The garrison's day	clip	f3:f_al_mlrs	On alert	6.6
1565.0	Control	The garrison's day	clip	f3:f_al_tank	On alert	4.7
1569.6	Control	The garrison's day	plate	tick	A fixed tick, and a session you can replay	28.4
1598.0	Vision	The thermal camera	card	Part 4 | Machine vision		12.4
1610.4	Vision	The thermal camera	plate	sensor	A camera that sees heat, with its faults	35.3
1645.7	Vision	The thermal camera	live	thermal	F: thermal view T: palette	9.0
1654.7	Vision	The thermal camera	live	thermal	The drone's camera: 320 x 240, recalibration, three zoom st	25.2
1679.9	Vision	From a heat picture to a target	plate	cvsteps	Five steps, every frame	30.2
1710.1	Vision	From a heat picture to a target	plate	otsu	Otsu's method: let the picture choose its own threshold	26.2
1736.3	Vision	From a heat picture to a target	plate	classify	Target, person or clutter	27.2
1763.5	Vision	From a heat picture to a target	still	pictures	Soldiers are marked as persons	9.1
1772.5	Vision	Tracking	plate	kalman	Predict, then correct	36.6
1809.2	Vision	Tracking	plate	assoc	Gating, matching, two views	23.8
1833.0	Vision	Tracking	clip	n_flir	Tracks coast behind cover	12.2
1845.2	Vision	Tracking	plate	tracklog	One track's uncertainty, measured	11.8
1857.0	Agents	The drones	card	Part 5 | The agents		12.7
1869.7	Agents	The drones	plate	duty	A drone's day	19.5
1889.2	Agents	The drones	clip	f2:uav_takeoff	Take-off at dusk	7.0
1896.2	Agents	The drones	clip	n_wide	Night patrol	5.0
1901.2	Agents	One drone, or both?	plate	decide1	Lead and wing	27.1
1928.3	Agents	One drone, or both?	plate	tot	Time on target	26.6
1954.9	Agents	One drone, or both?	clip	n_follow	Combined strike	8.5
1963.4	Agents	One drone, or both?	plate	decide2	Two intruders: split, or stay together?	22.3
1985.7	Agents	One drone, or both?	clip	n_follow	Single strike	8.7
1994.4	Agents	One drone, or both?	clip	n_uav1	The guards fire too	8.7
2003.1	Agents	The intruder: choosing a route	plate	rgrid	The intruder's map	15.7
2018.8	Agents	The intruder: choosing a route	plate	rcost	What a cell costs	20.3
2039.1	Agents	The intruder: choosing a route	plate	rflood	Flood fill from the goal	25.3
2064.5	Agents	The intruder: choosing a route	live	rovplan	M: the flood field and the route	9.0
2073.5	Agents	The intruder: following the line	plate	lfr	Five sensors, two wheels	33.2
2106.7	Agents	The intruder: following the line	live	rovclose	A rover on its line	14.6
2121.3	Agents	The intruder: following the line	live	rovclose	The rover's camera	6.0
2127.3	Agents	The intruder: hearing a drone and hiding	plate	revade	Hiding from a drone	27.0
2154.2	Agents	The intruder: hearing a drone and hiding	split	split screen	One moment, three views	20.6
2174.8	Agents	The game, and the blast	plate	game	Night Watch: the rules	20.0
2194.8	Agents	The game, and the blast	live	hud	F11: the game display U: the drone's feed	15.9
2210.6	Agents	The game, and the blast	live	dayops	Shift+F2: day operations	10.0
2220.6	Agents	The game, and the blast	plate	particles	An explosion is particles	20.8
2241.5	Agents	The game, and the blast	clip	n_wide	Blast, fire, smoke	6.0
2247.5	Agents	The game, and the blast	clip	f2:uav_landing	Dawn	7.0
2254.5	Engineering	Hard problems	plate	hard	Four hard problems	28.8
2283.3	Engineering	How it is checked	plate	proof	How it is checked	17.6
2300.9	Engineering	How it is checked	still	pictures	One workbook for every number	17.4

2318.2		One night at the outpost	clip	n_wide		5.5
2323.7		One night at the outpost	clip	n_follow		7.0
2330.7		One night at the outpost	clip	n_follow		7.0
2337.7		One night at the outpost	clip	n_wide		4.5
2342.2		One night at the outpost	plate	numbers		13.5
2355.8		One night at the outpost	card	“If student A and student B both use	AI for a project,show me how creative you can be,and how	20.1

sum of the lengths		2375.8

Numbers the narration states, with their source

		what	value		source
		generators (primitives)	16		scene export: primitives
		drawn parts	5474		scene export: leaf_count
		designs	62		scene export: designs
		things that can be operated	20		scene export: controllables (possessable)
		lamps (practical lights)	43		scene export: practical_lights
		slices of the sphere	36		scene export: primitives
		corners of the sphere	614		scene export: primitives
		triangles of the sphere	1224		scene export: primitives
		parts of a drone	158		scene export: objects
		parts of a tank	255		scene export: objects
		parts of a soldier	75		scene export: objects
		latitude (degrees north)	22.8		scene export: v6.day_cycle.config
		thermal image width (pixels)	320		src/ai/DroneController.h:15 kCvWidth
		thermal image height (pixels)	240		src/ai/DroneController.h:16 kCvHeight
		lead's lock time (s)	0.8		src/ai/Squad.h:57 kSquadLeadLock
		wing's lock time (s)	0.5		src/ai/Squad.h:57 kSquadWingLock
		lead waits for a free wing (s)	8.0		src/ai/Squad.h:59 kSquadWingWaitFree
		split distance (m)	14.0		src/ai/Squad.h:63 kSquadSplitDist
		rejoin distance (m)	9.0		src/ai/Squad.h:63 kSquadRejoinDist
		urgency of a second intruder (m)	10.0		src/ai/Squad.h:66 kSquadSplitThreat
		size of a cell (m)	0.5		src/ai/HideController.h:21 kCellSize
		cells per side of the rover's grid	80		src/ai/HideController.h:22 kGridCells
		line sensors of a rover	5		src/ai/HideController.h:23 kIrSensorCount
		a drone this close makes a rover hide (m)	5.0		src/ai/HideController.h:24 kEvadeRadius
		time in cover (s)	4.0		src/ai/HideController.h:25 kEvadeTime
		line follower: proportional gain	3.2		src/ai/HideController.cpp:22 kKp
		line follower: derivative gain	0.25		src/ai/HideController.cpp:22 kKd
		score of a kill	100		src/scene/Mission.h:63 kKillScore
		bonus of a combined kill	50		src/scene/Mission.h:63 kCombinedBonus
		base integrity lost per breach (%)	10.0		src/scene/Mission.h:64 kBreachDamage
		armour of a rover	100.0		src/scene/Scene.h:78 kRoverHealth
		damage of a round at its centre	30.0		src/scene/Scene.h:79 kRoundDamage